

(12) **United States Patent**
Becker et al.

(10) **Patent No.:** **US 12,406,596 B2**

(45) **Date of Patent:** ***Sep. 2, 2025**

(54) **STICK WELDING ELECTRODE HOLDER SYSTEMS AND METHODS**

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(73) Assignee: **ILLINOIS TOOL WORKS INC.,**
Glenview, IL (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/752,135**

(22) Filed: **Jun. 24, 2024**

(65) **Prior Publication Data**
US 2024/0346948 A1 Oct. 17, 2024

Related U.S. Application Data

(63) Continuation of application No. 18/099,002, filed on Jan. 19, 2023, now Pat. No. 12,020,586, which is a continuation of application No. 16/814,152, filed on Mar. 10, 2020, now Pat. No. 11,594,148, which is a (Continued)

(51) **Int. Cl.**
G09B 19/24 (2006.01)
B23K 9/28 (2006.01)
G09B 9/00 (2006.01)
G09B 19/00 (2006.01)

(52) **U.S. Cl.**
CPC **G09B 19/003** (2013.01); **B23K 9/282** (2013.01); **B23K 9/287** (2013.01); **G09B 9/00** (2013.01); **G09B 19/24** (2013.01)

(58) **Field of Classification Search**
CPC G09B 19/24; B23K 9/09; B23K 9/095; B23K 9/0953
See application file for complete search history.

(56) **References Cited**

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Primary Examiner — Robert J Utama
(74) *Attorney, Agent, or Firm* — McAndrews, Held & Malloy, Ltd.

(57) **ABSTRACT**

A method for providing a welding score, comprising determining an initial position of a welding operation, determining a terminal position of the welding operation, determining a welding path between the initial position and the terminal position, and determining the welding score based on the welding path, wherein the welding score takes into account one or both of a time used for welding and a length of the welding path. A system comprising a welding torch, and a computer configured to: determine an initial position of a welding operation, determine a terminal position of the welding operation, determine a welding path between the initial position and the terminal position, determine a welding score, wherein the welding score takes into account a time used for welding and/or a length of the welding path, and comparing the welding score to a score threshold.

20 Claims, 68 Drawing Sheets

The diagram illustrates a **WELDING TRAINING SYSTEM (10)** and its networked environment. The system is composed of several interconnected modules:

- Internal System Components:**
 - TRAINING STAND (32):** Includes a **WORKPIECE (82)** and is connected to a **COMPUTER (18)**.
 - COMPUTER (18):** Contains a **PROCESSOR (20)**, **MEMORY (22)**, and **STORAGE (24)**.
 - SENSING DEVICE(S) (12, 16):** One is connected to the training stand, and another is connected to the computer.
 - OPERATOR ID SYSTEM (47):** Includes a **RESETTABLE IDENTIFIER (45)**, **BIOMETRIC IDENTIFIER (46)**, and **PHYSICAL IDENTIFIER (49)**, all connected to the computer.
 - WELDING TOOL (14):** Connected to the computer and a **WELDING POWER SUPPLY (28)**.
 - WIRE FEEDER (30):** Connected to the welding power supply and the welding tool.
 - HELMET (41):** Includes a **DISPLAY (32)** and **SENSING DEVICE(S) (16)**, connected to the computer.
- External Networked Components:**
 - EXTERNAL DISPLAY (34):** Connected to the computer.
 - PRODUCTION WELDING SYSTEM (42):** Connected to the computer via a **NETWORK DEVICE (36)**.
 - INTERNET / NETWORK (38):** A central hub connecting the training system to external systems.
 - EXTERNAL WELDING TRAINING SYSTEM (40):** Connected to the network.
 - REMOTE COMPUTER (44):** Connected to the network.
 - DATA STORAGE SYSTEM (318):** Connected to the network.

Related U.S. Application Data

continuation of application No. 15/211,743, filed on
Jul. 15, 2016, now Pat. No. 10,593,230.

- (60) Provisional application No. 62/204,191, filed on Aug.
12, 2015.

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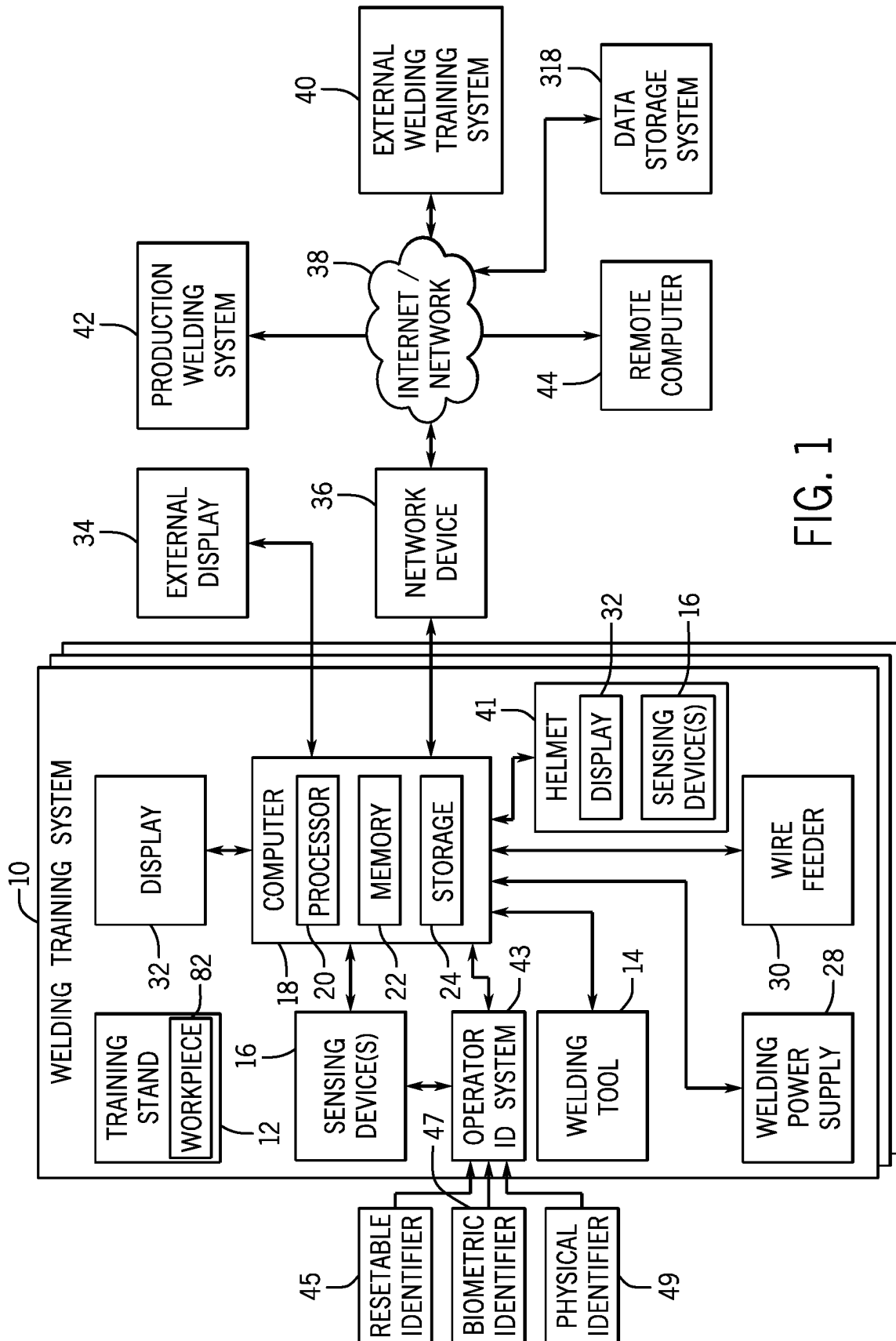
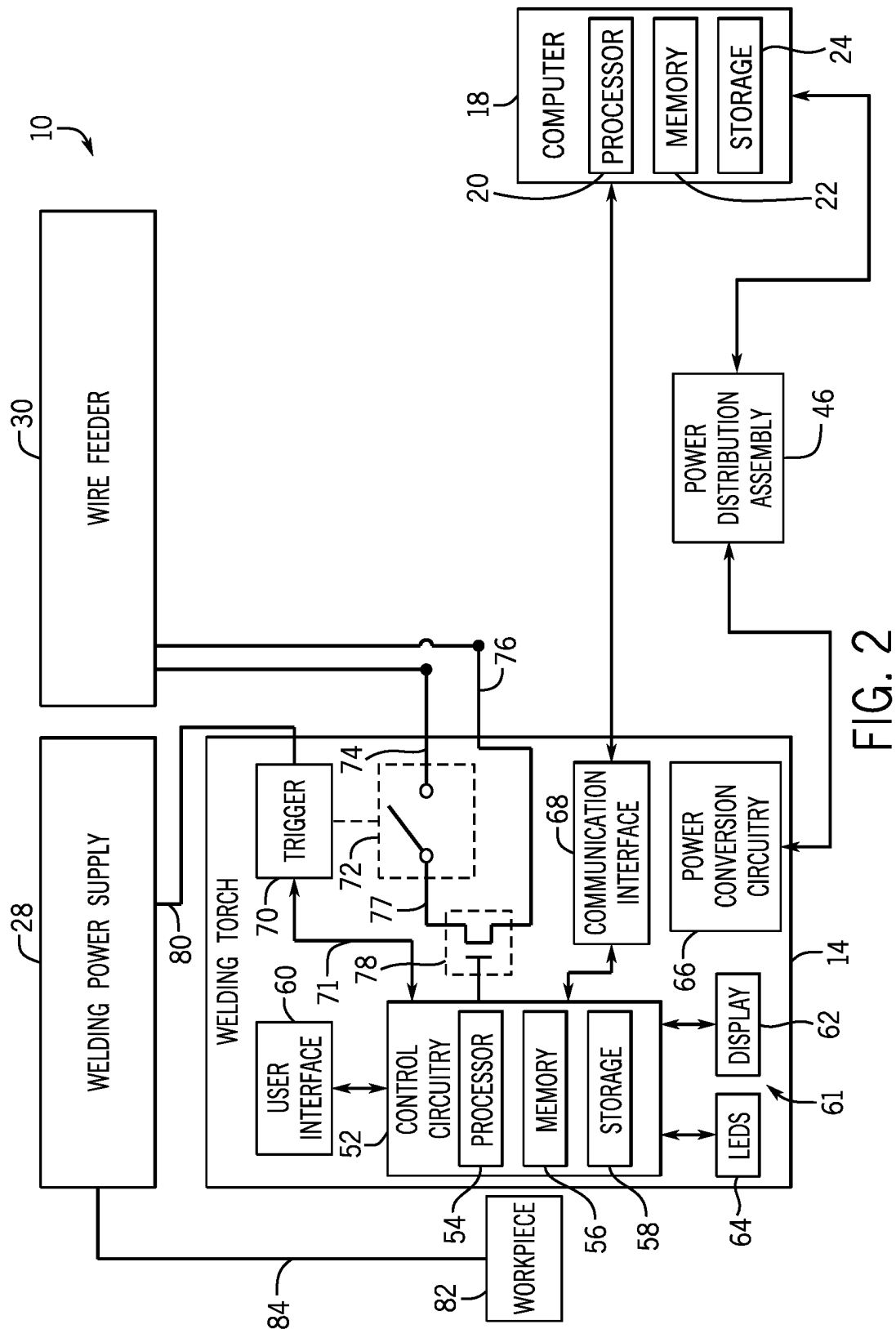
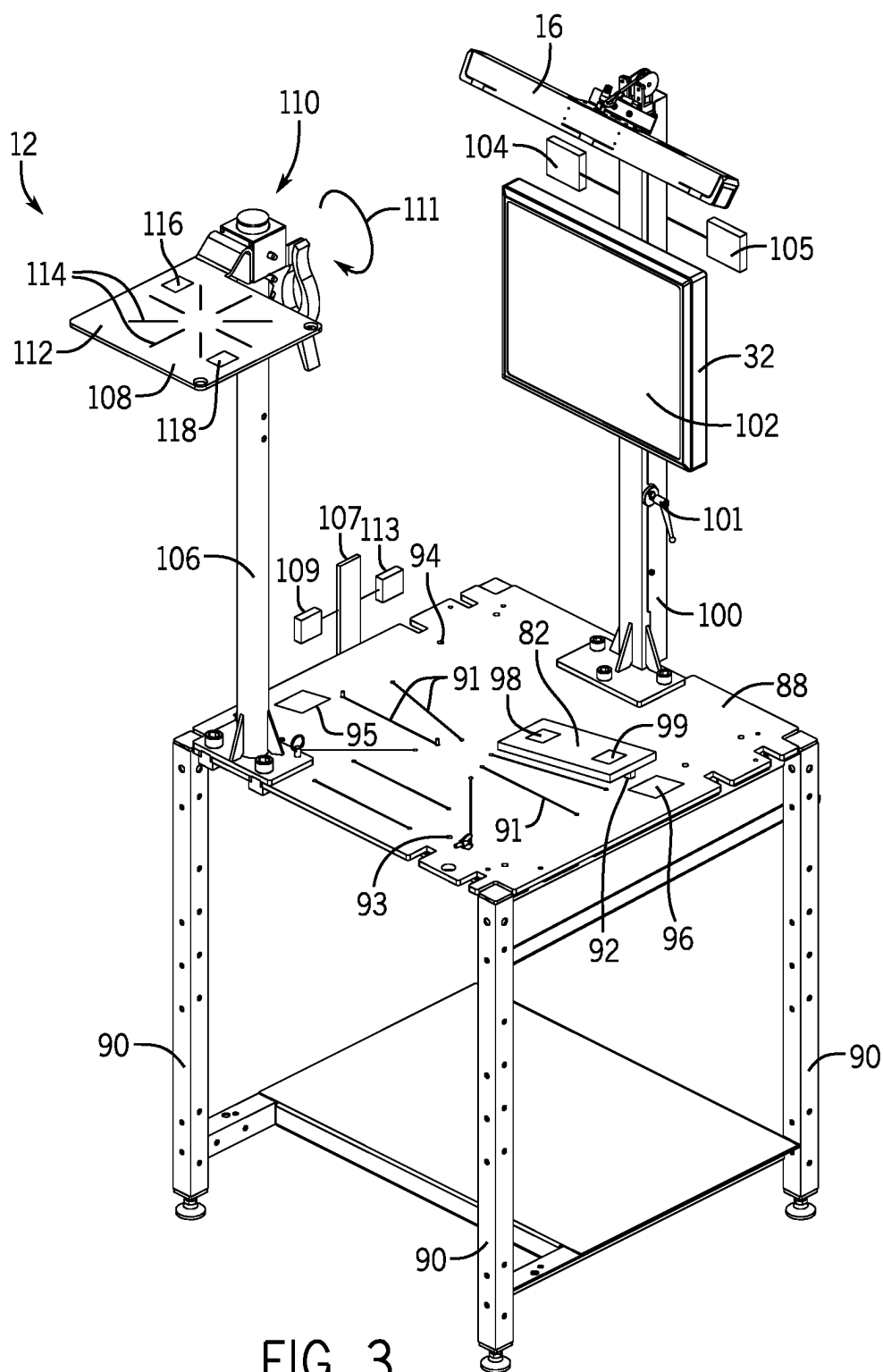


FIG. 1





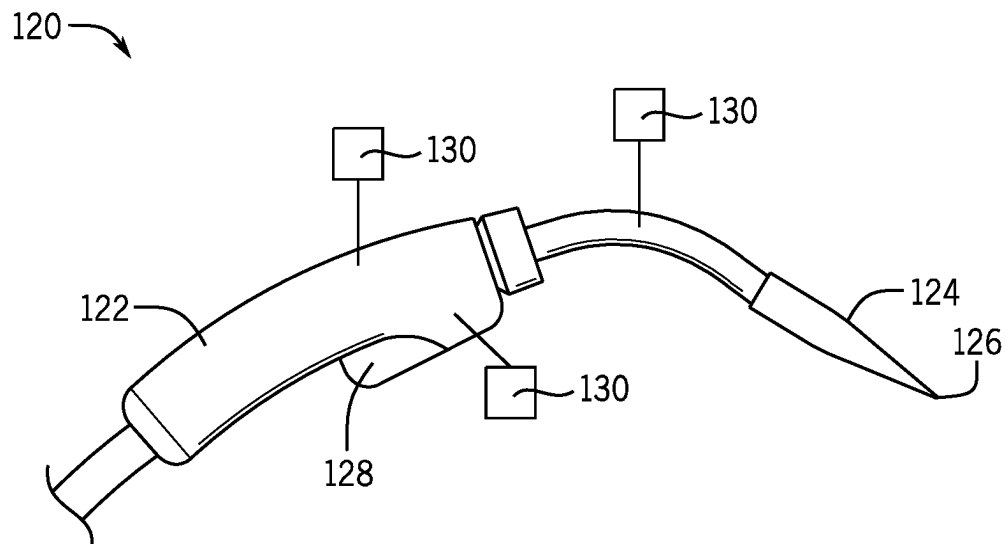
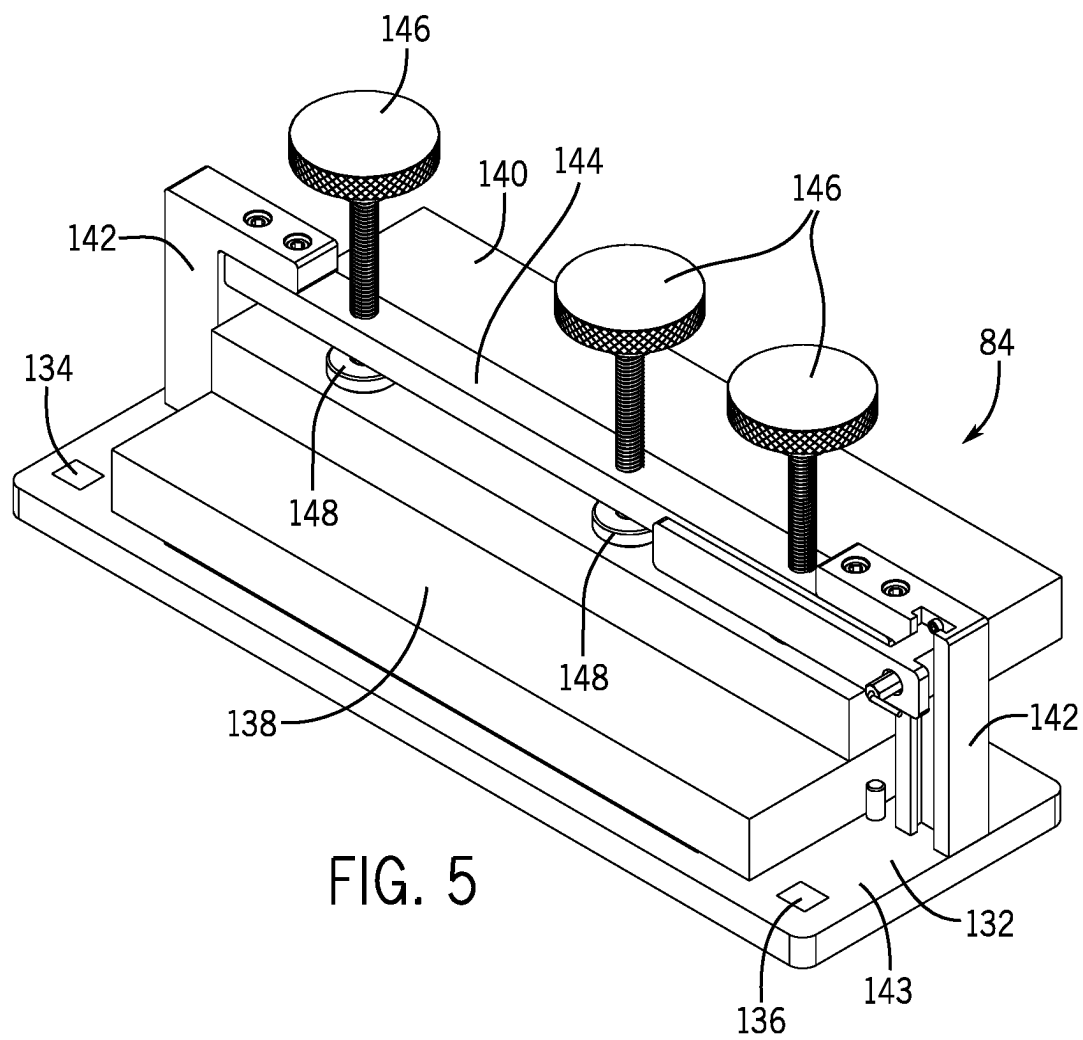
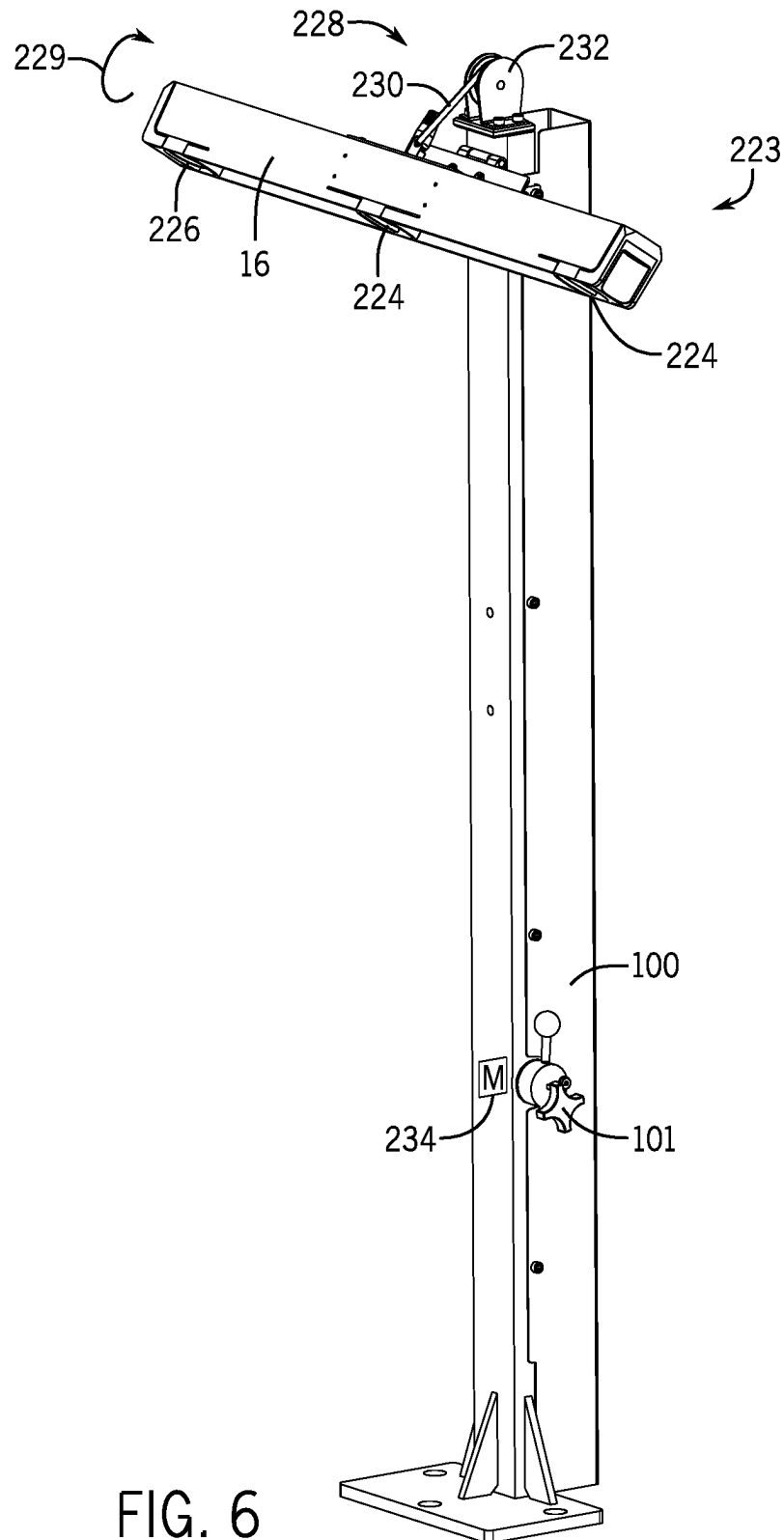


FIG. 4





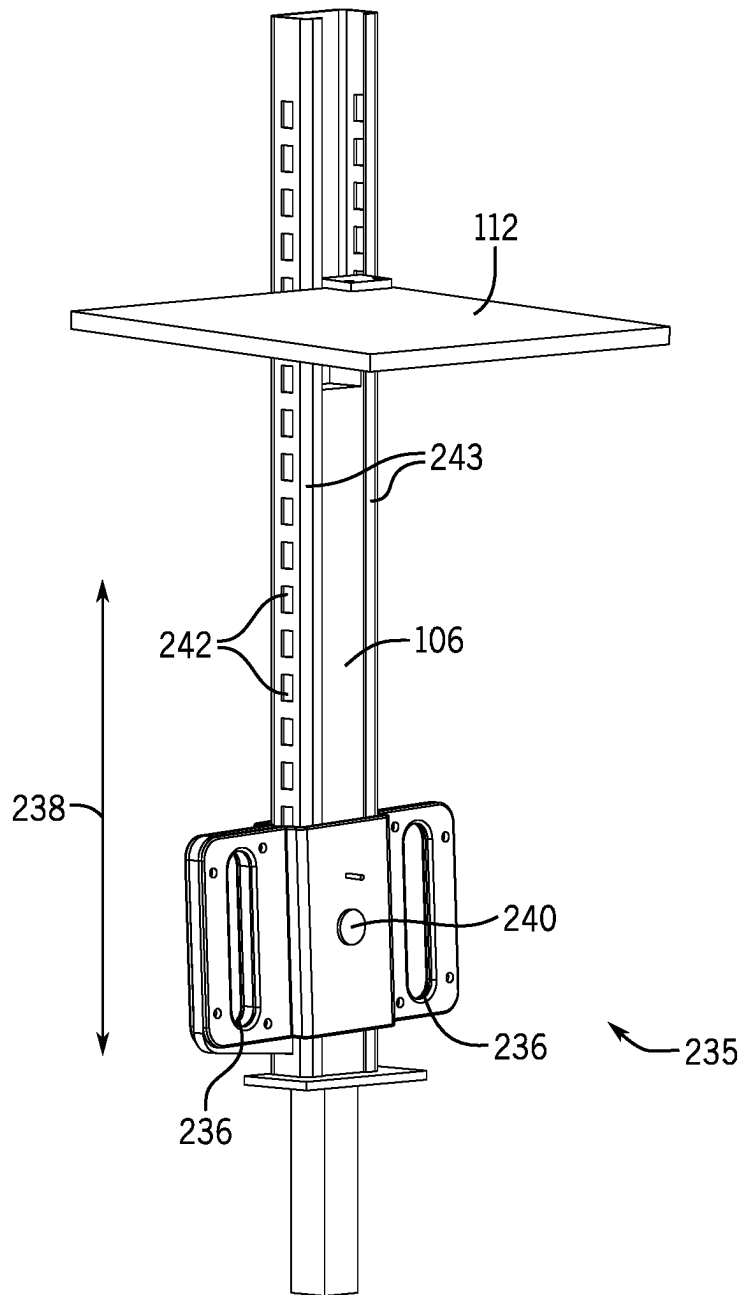


FIG. 7

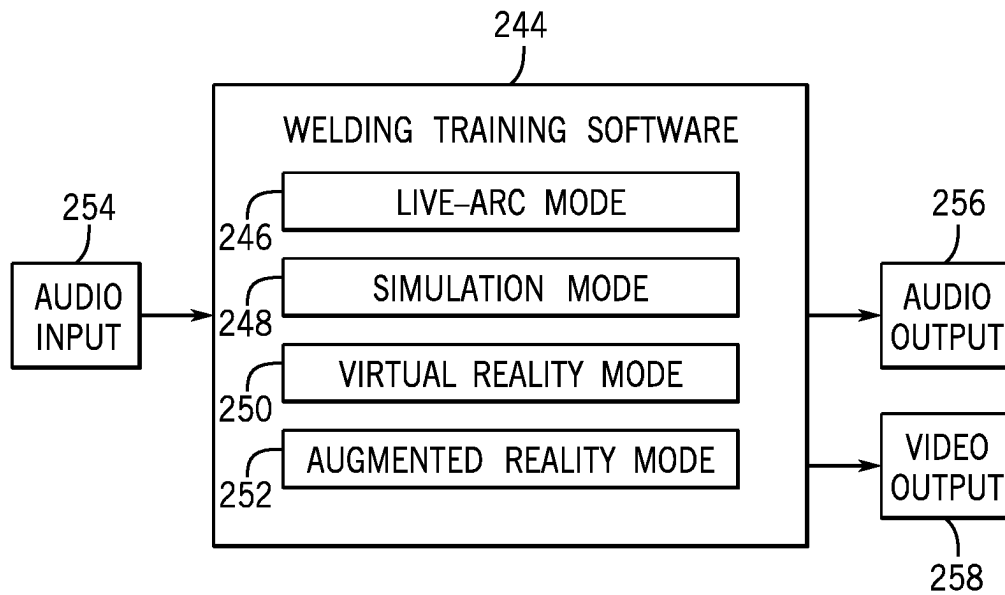


FIG. 8

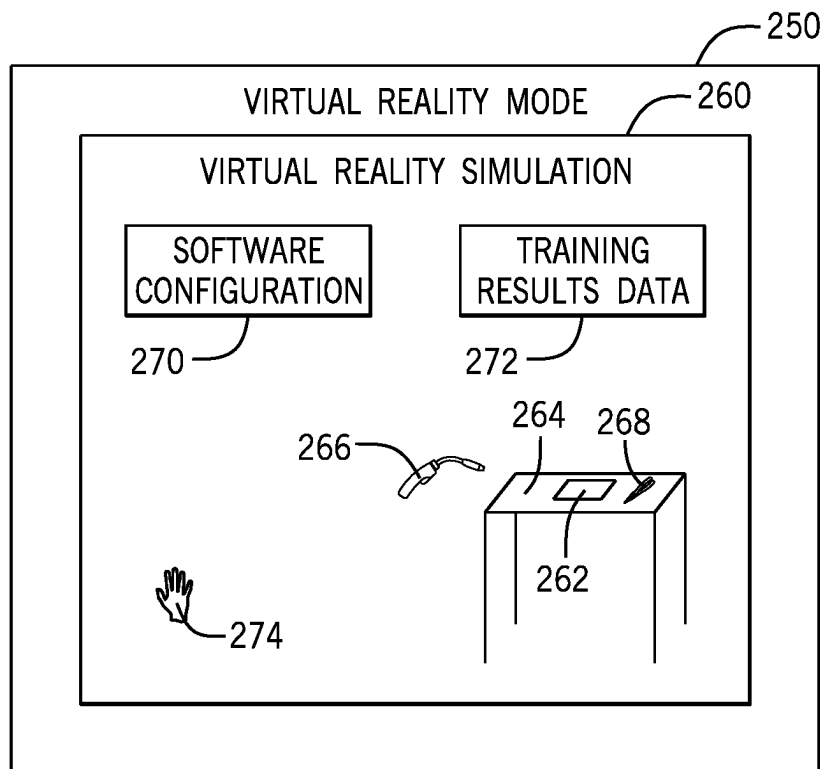


FIG. 9

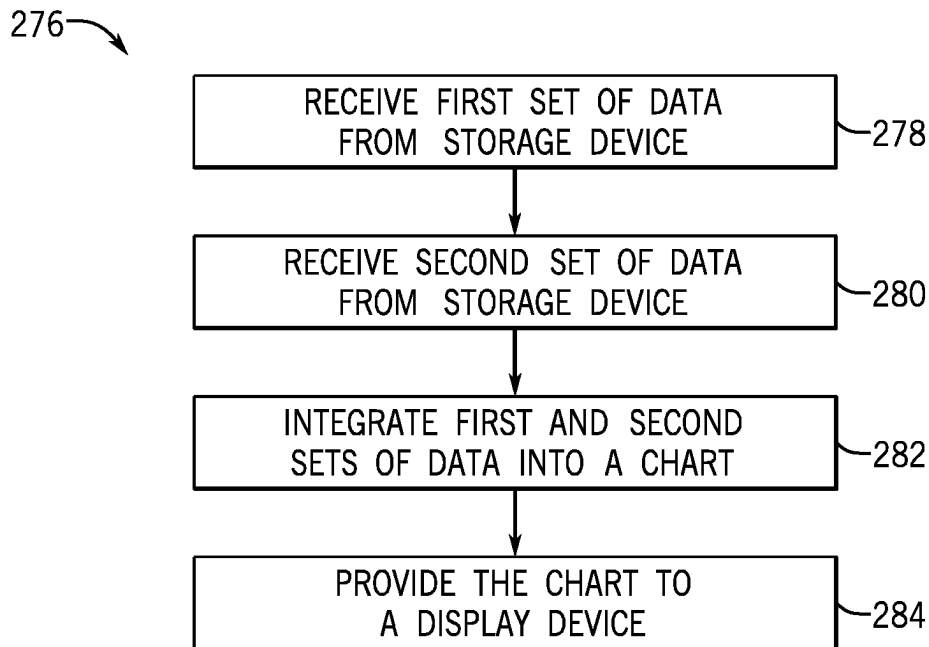


FIG. 10

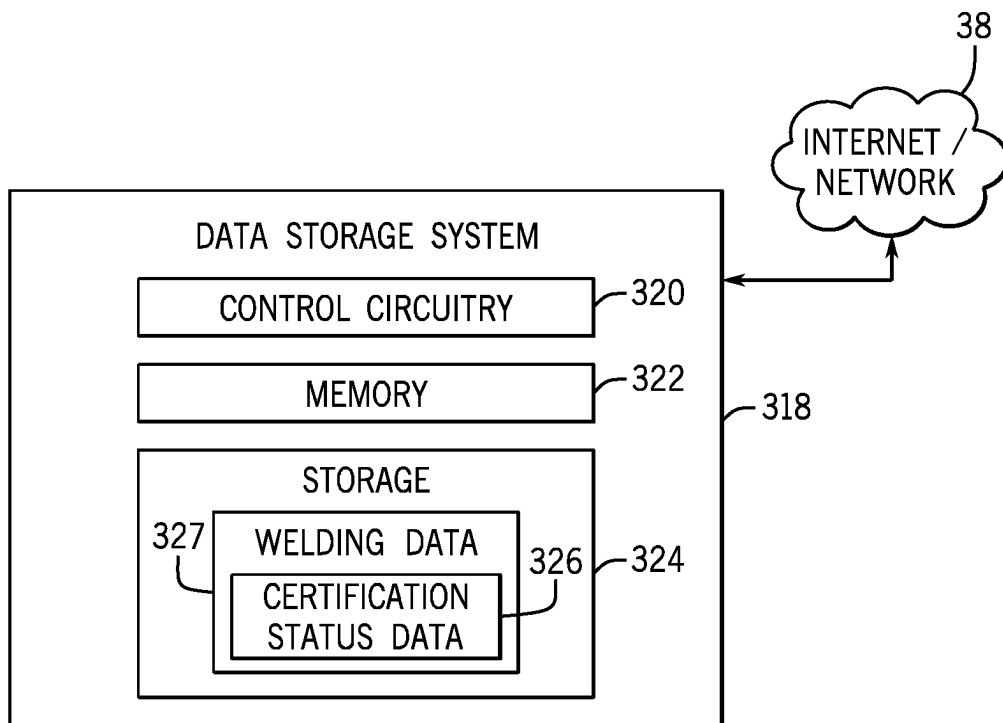
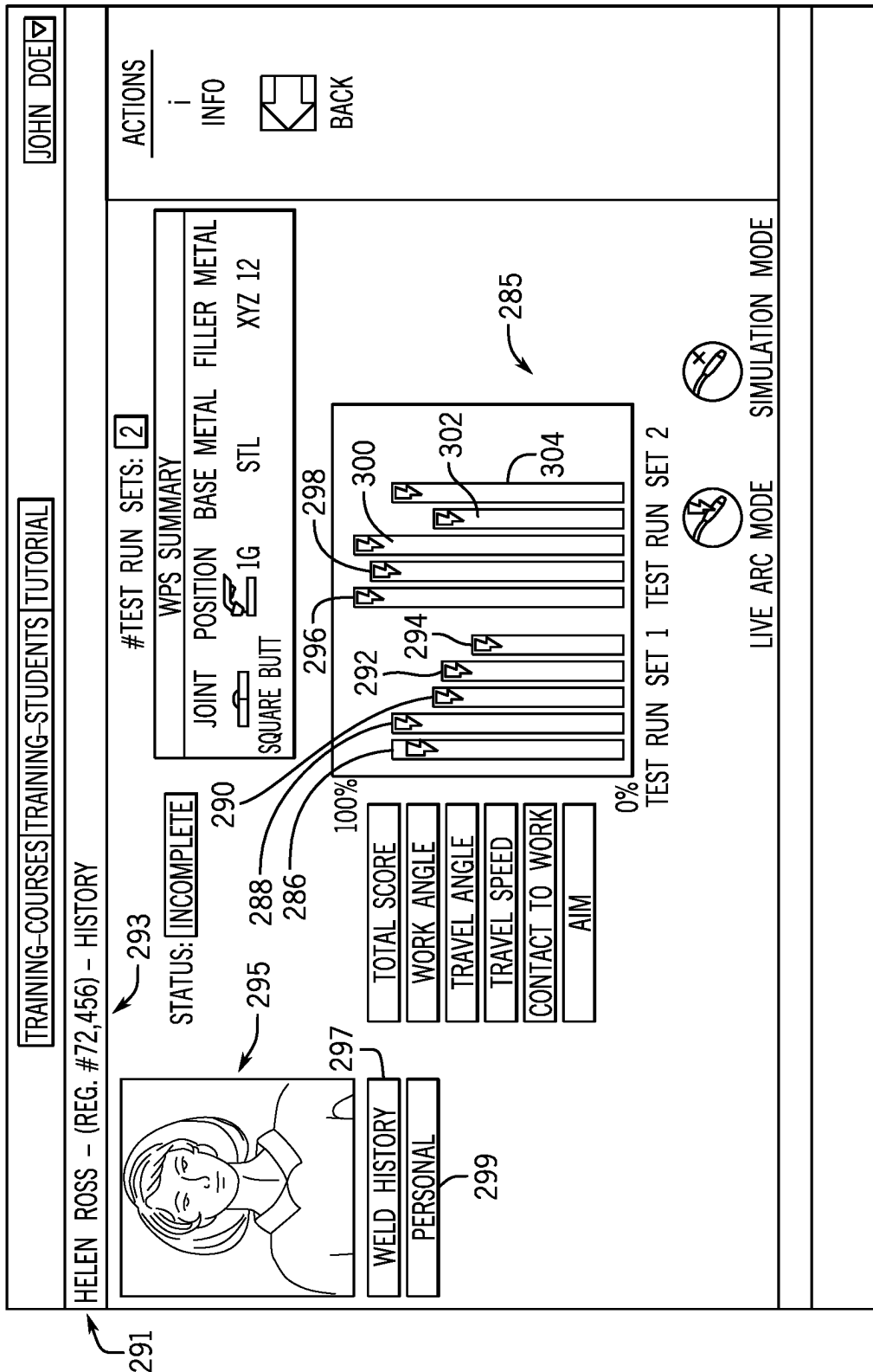


FIG. 13



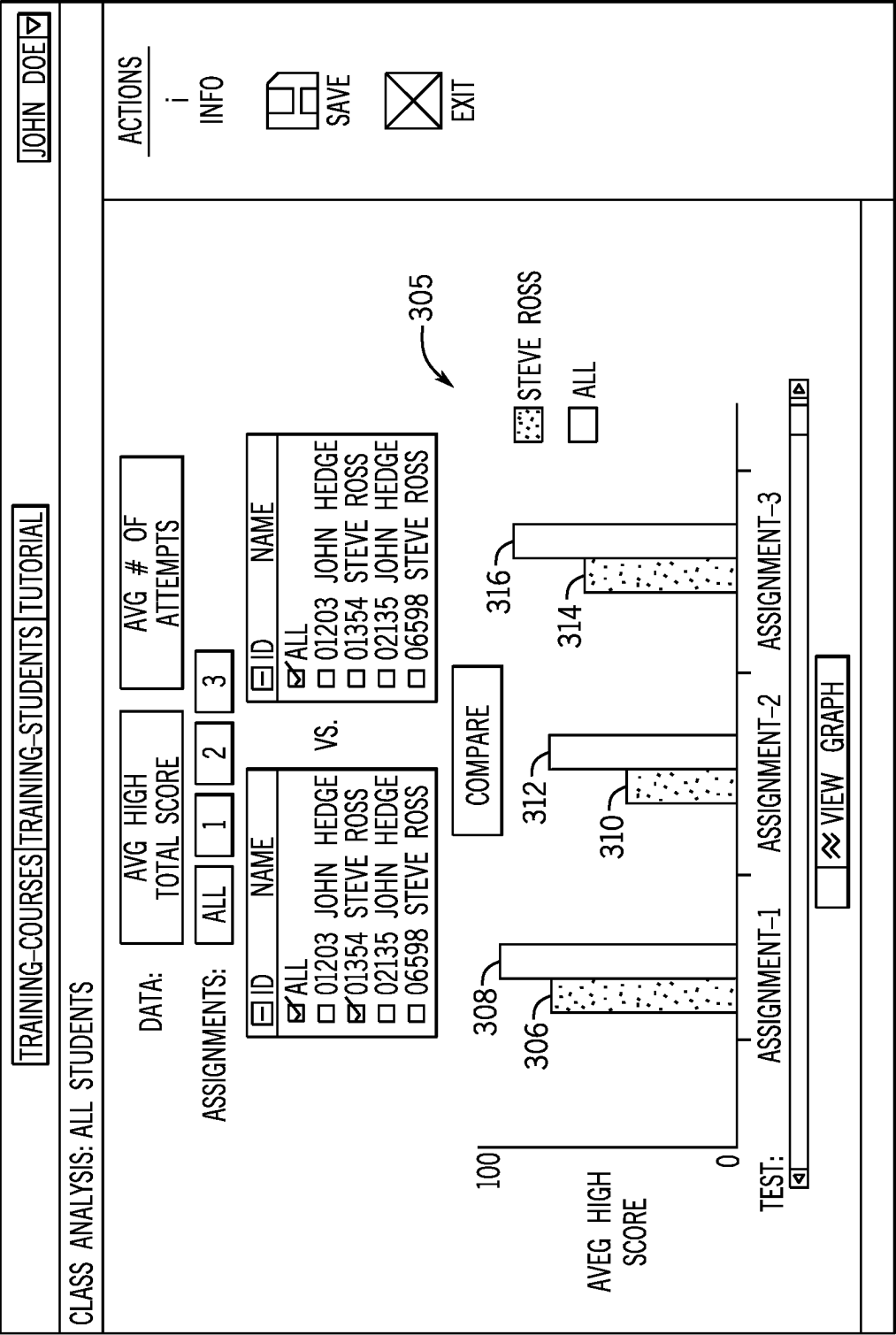


FIG. 12

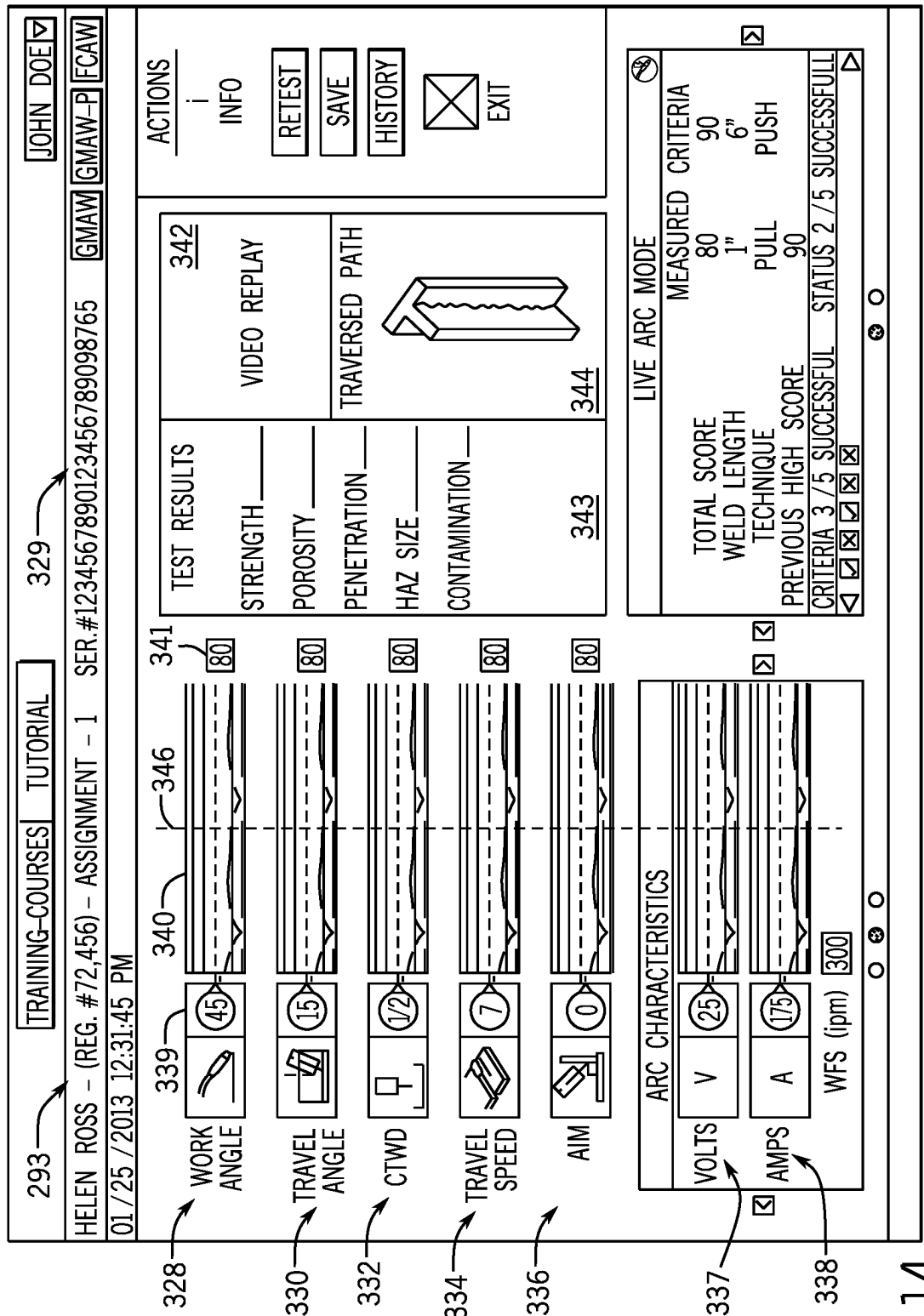


FIG. 14

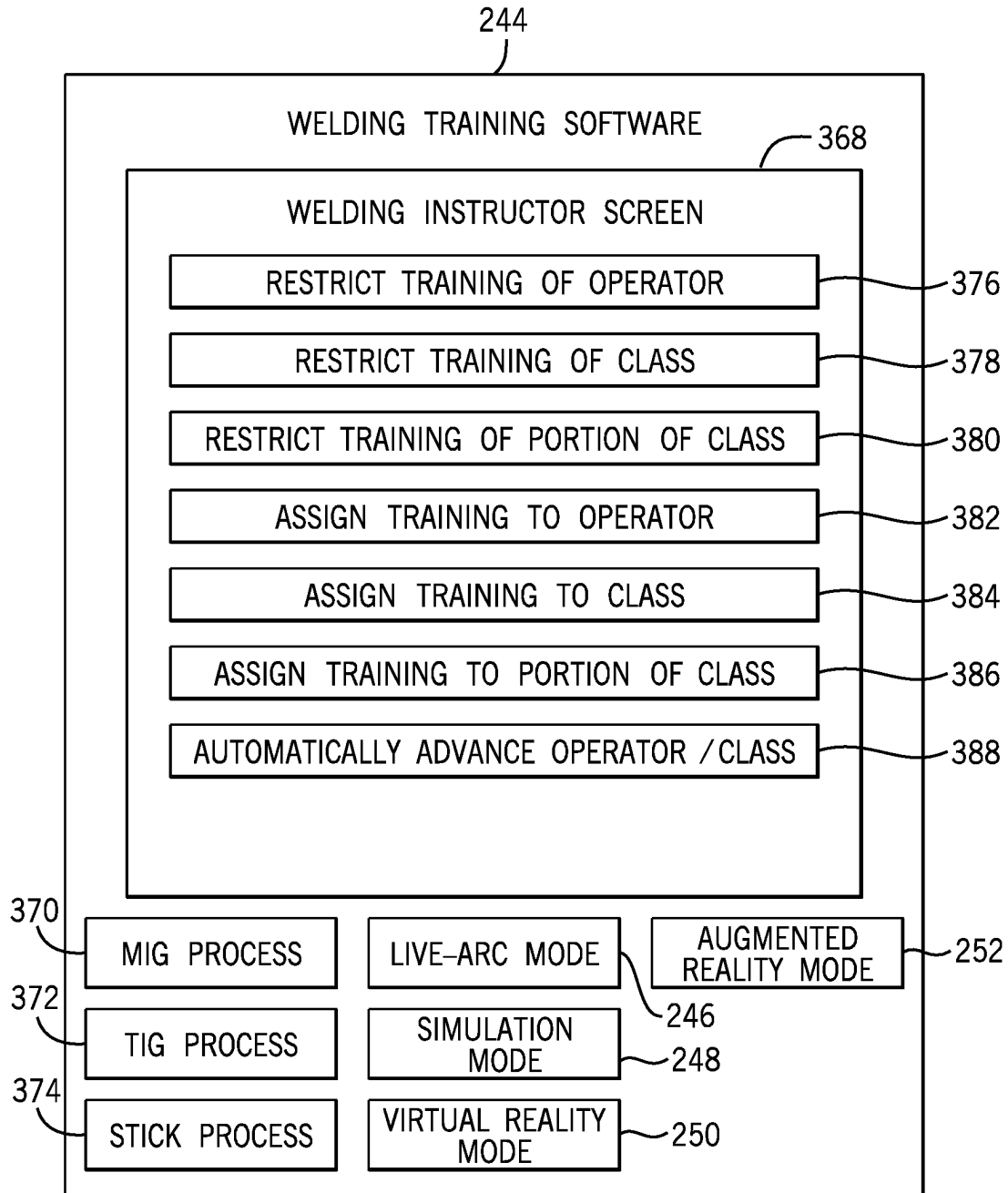


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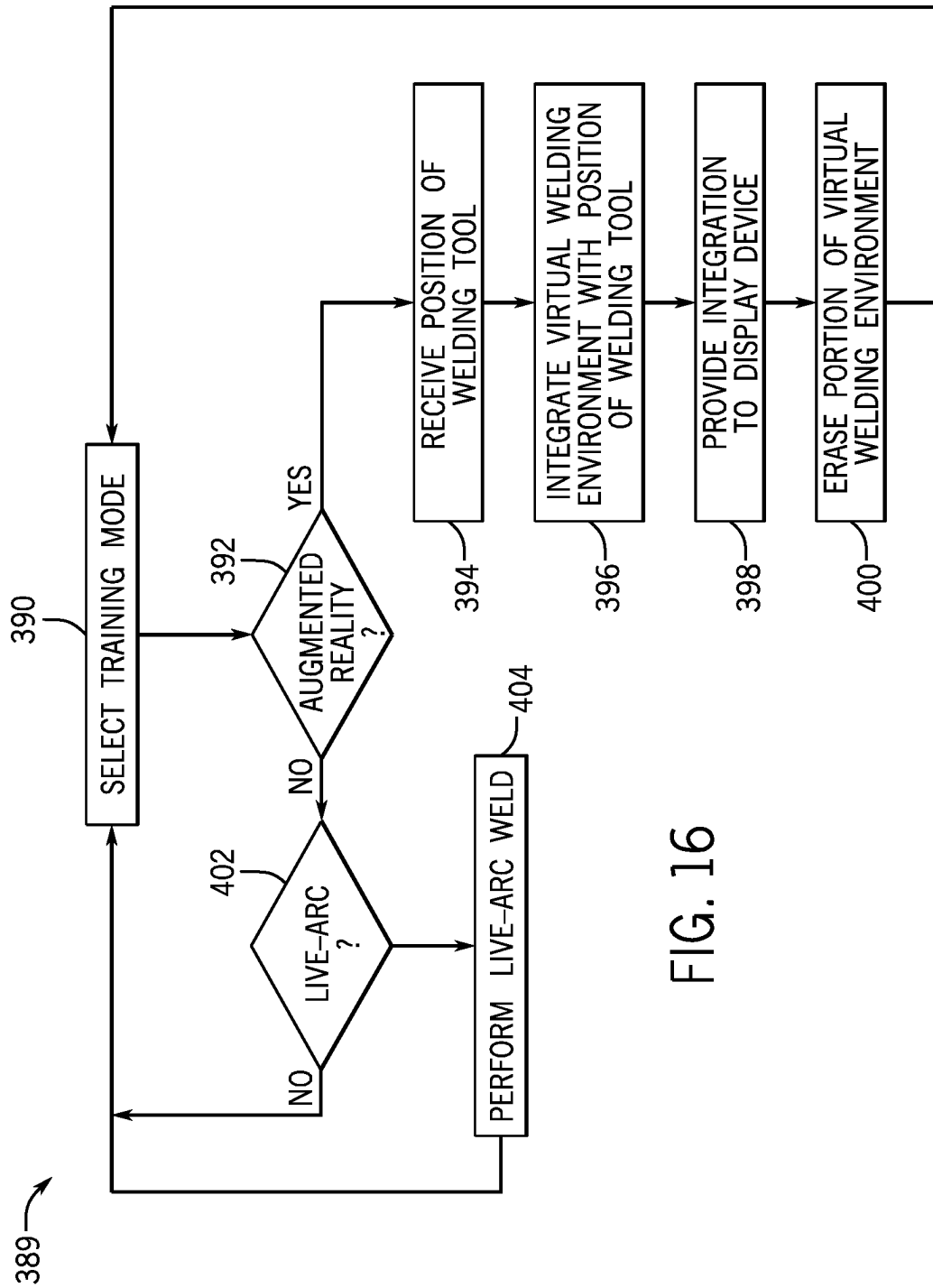


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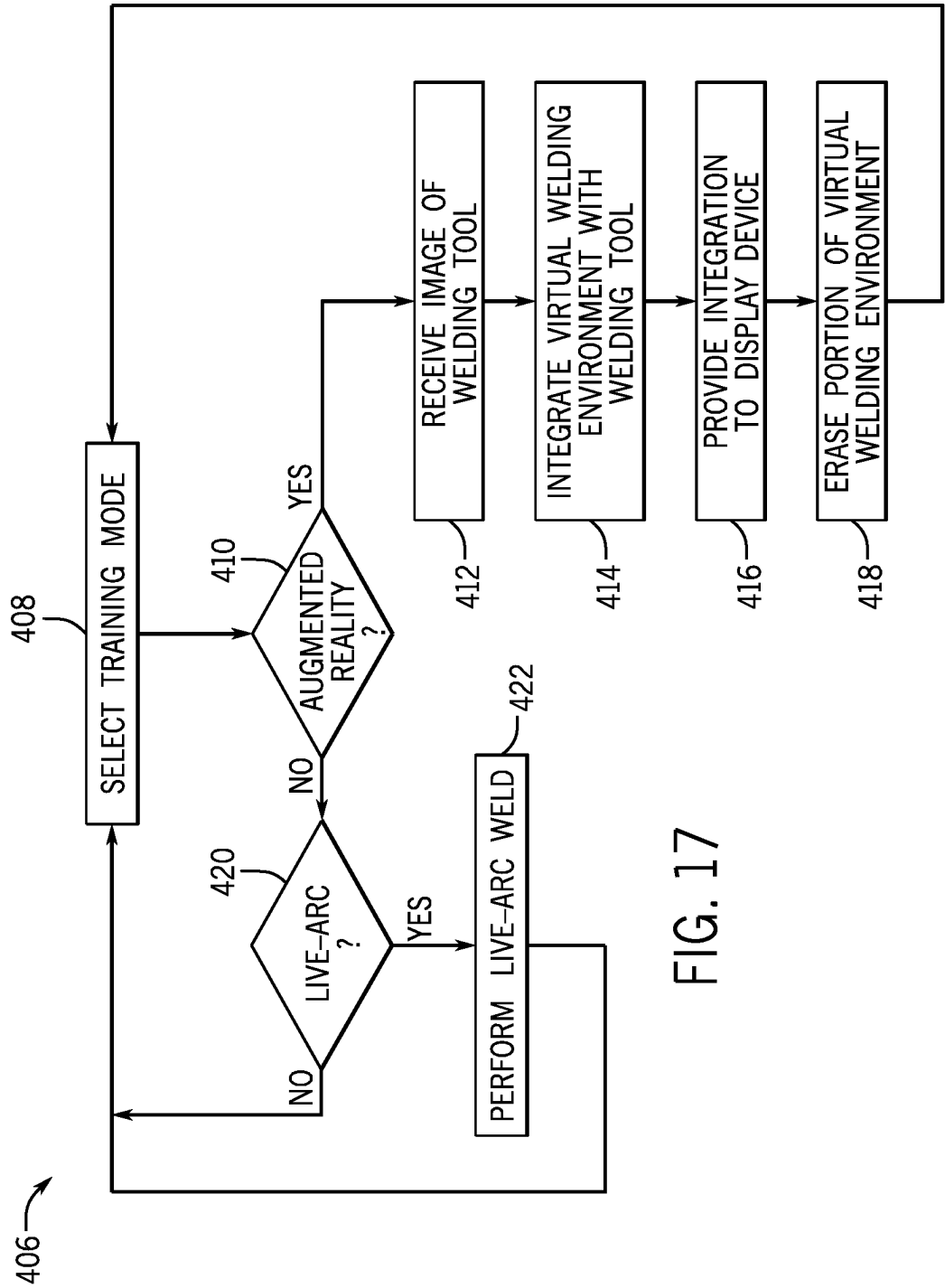


FIG. 17

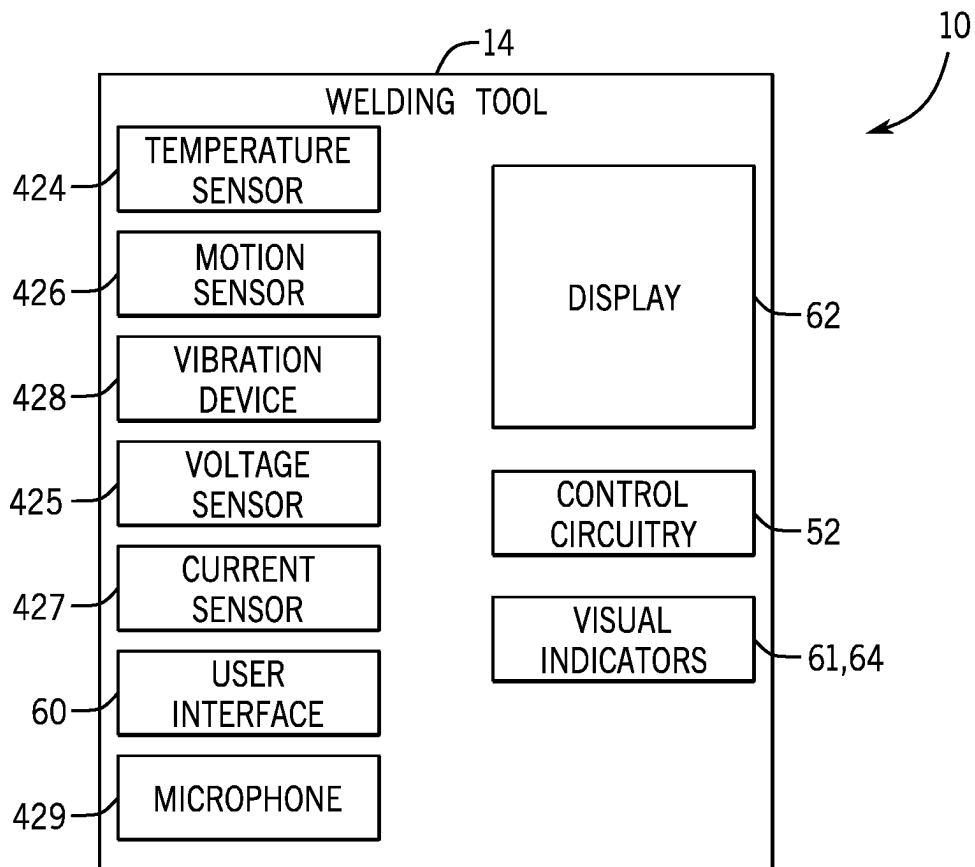


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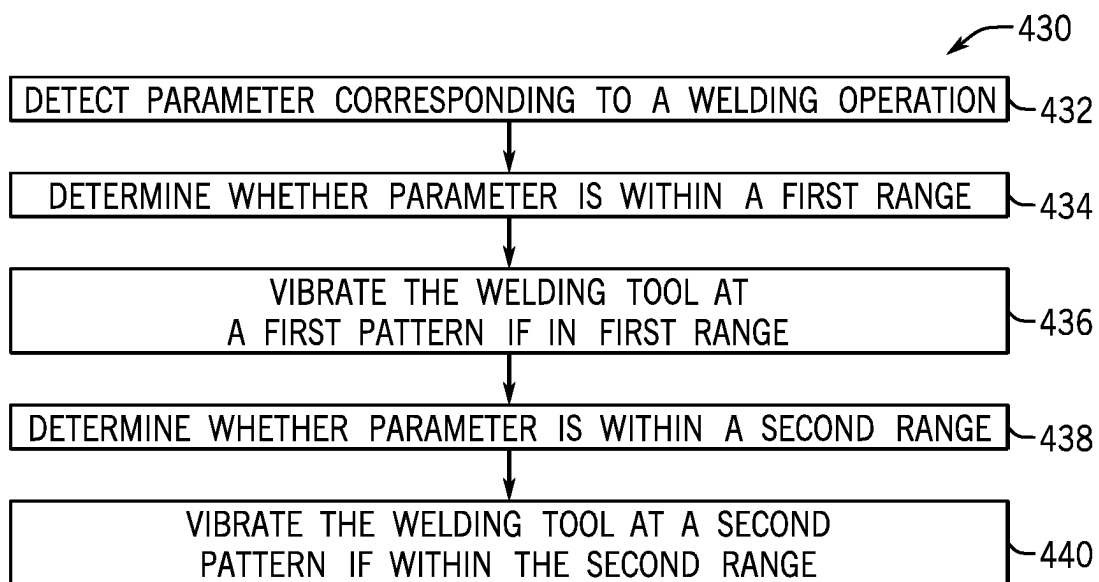


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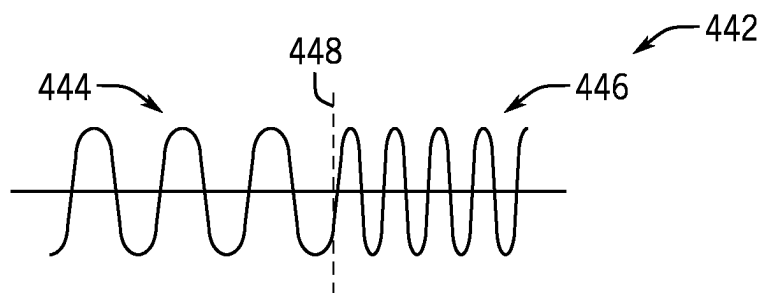


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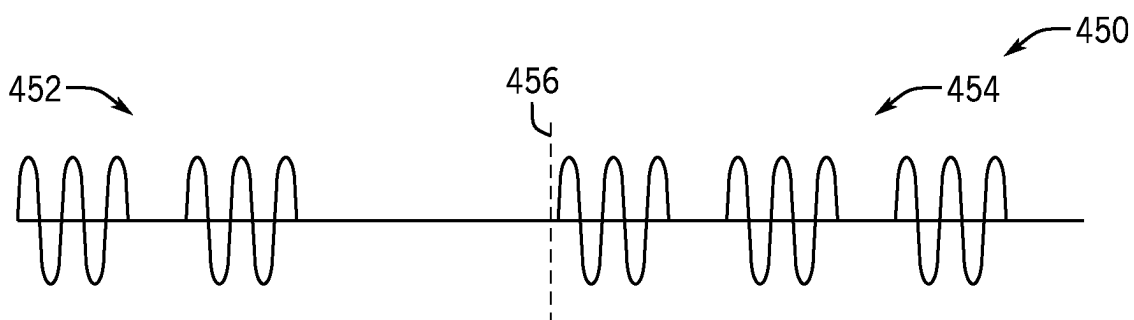


FIG. 21

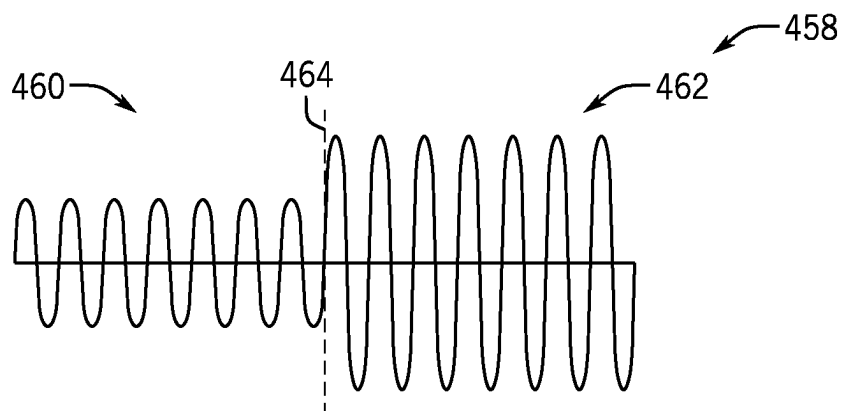
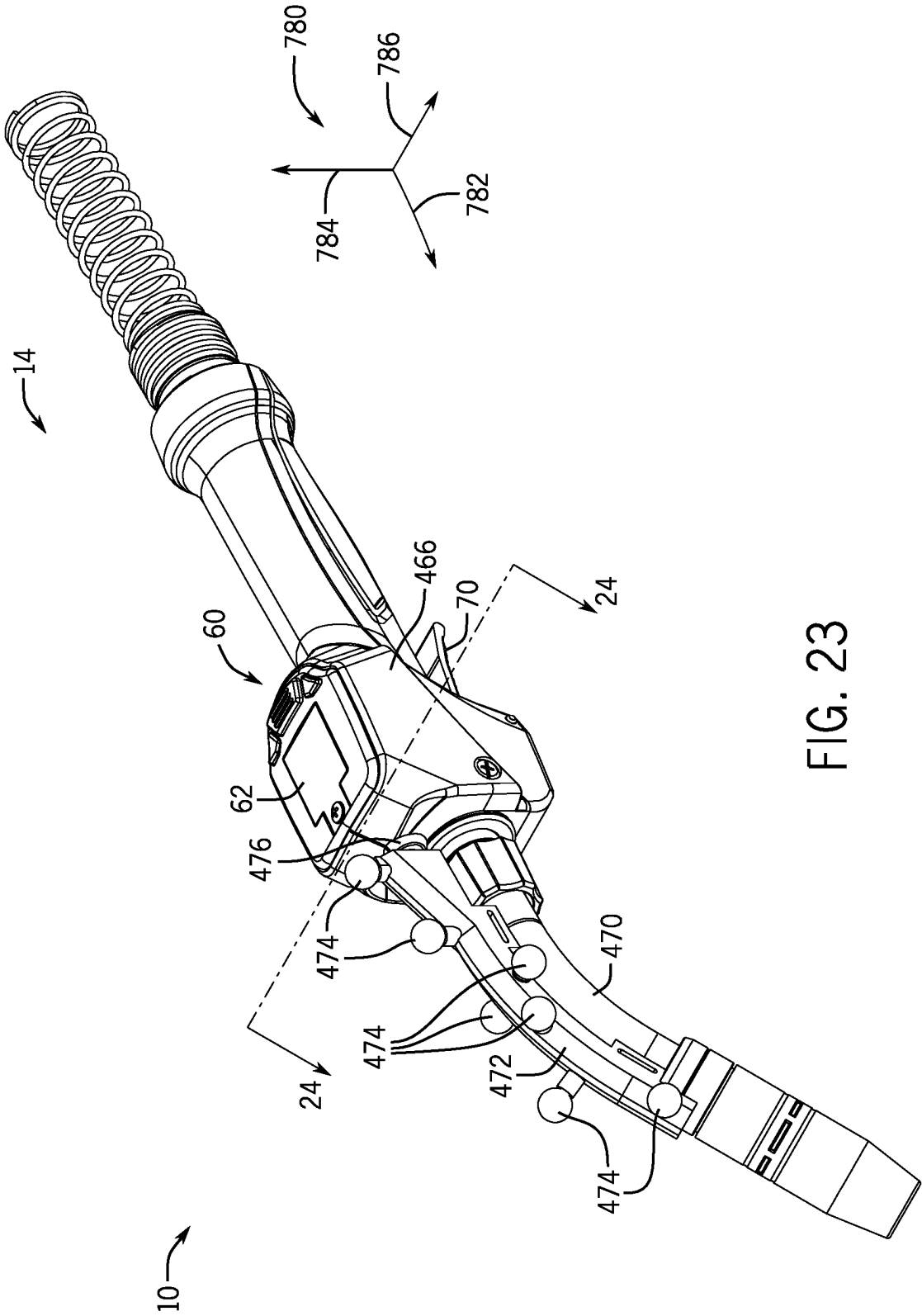


FIG. 22



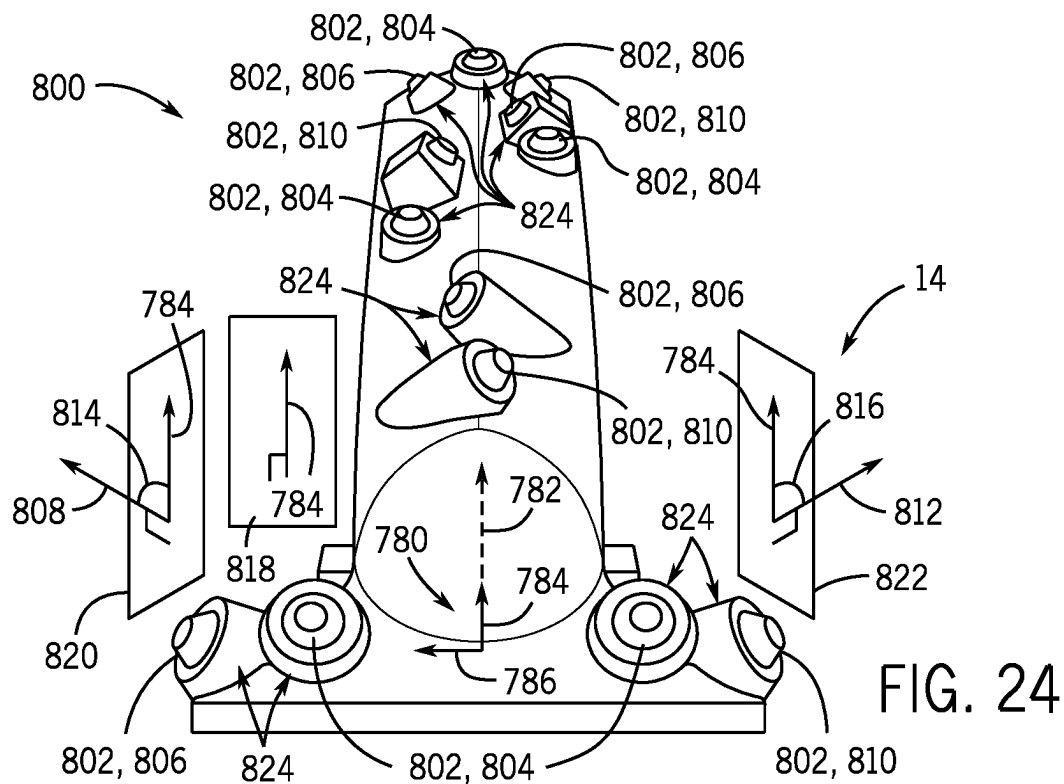


FIG. 24

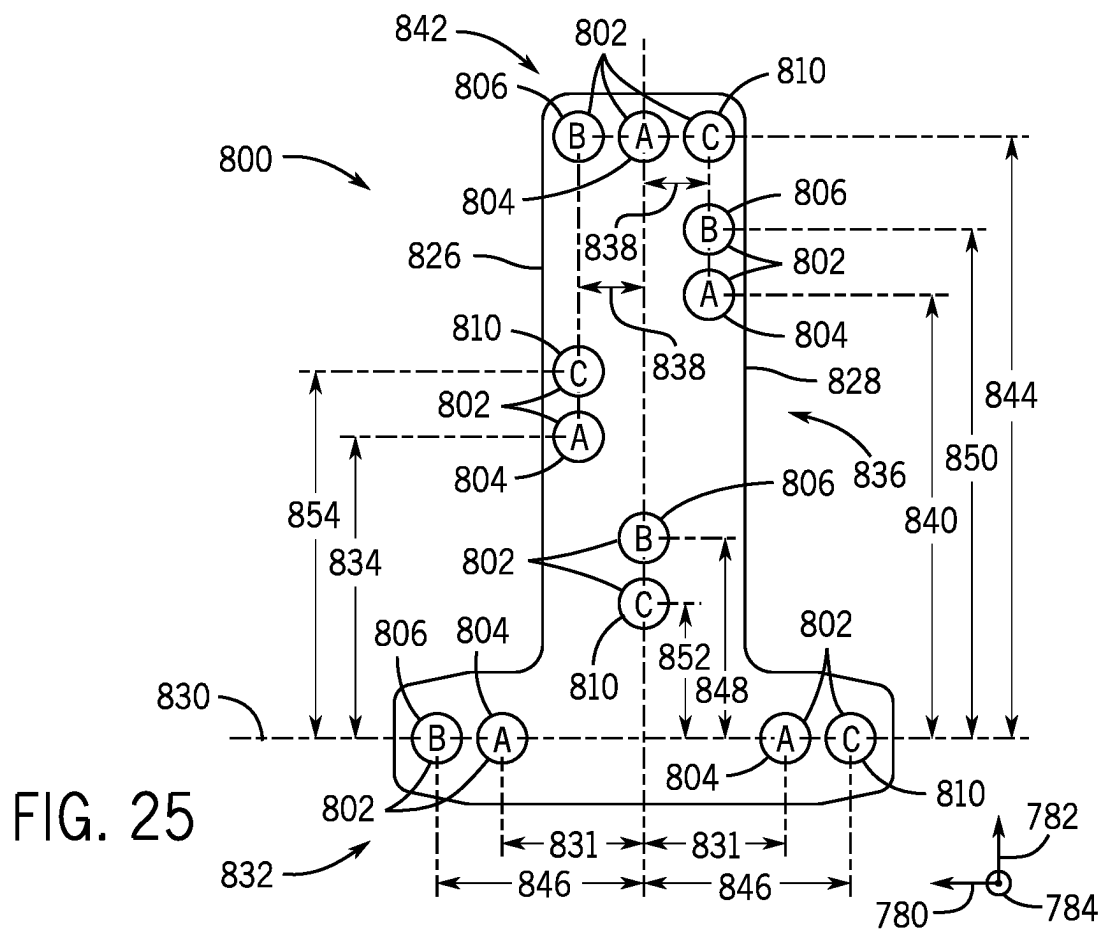


FIG. 25

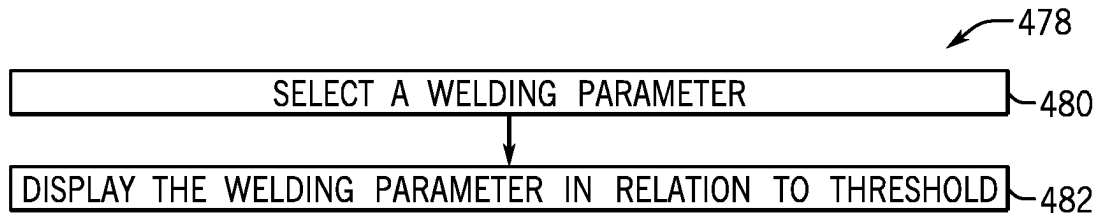


FIG. 26

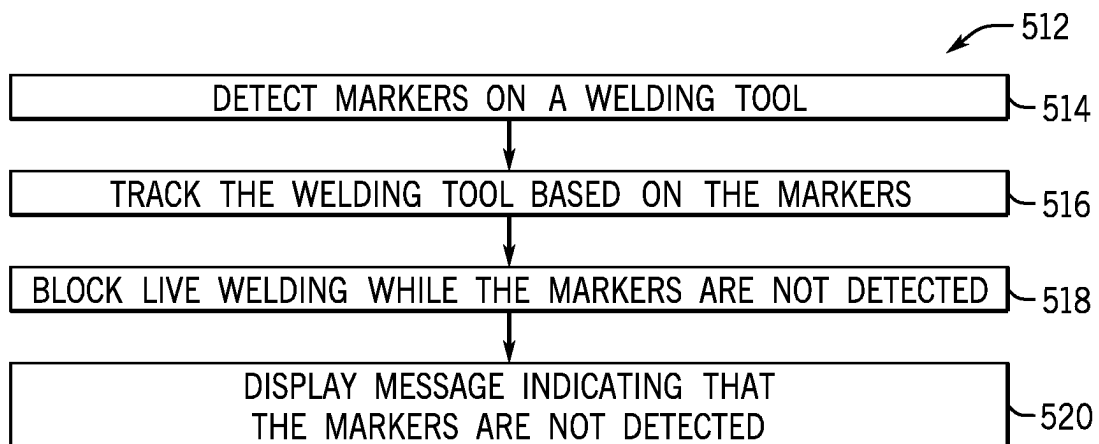


FIG. 28

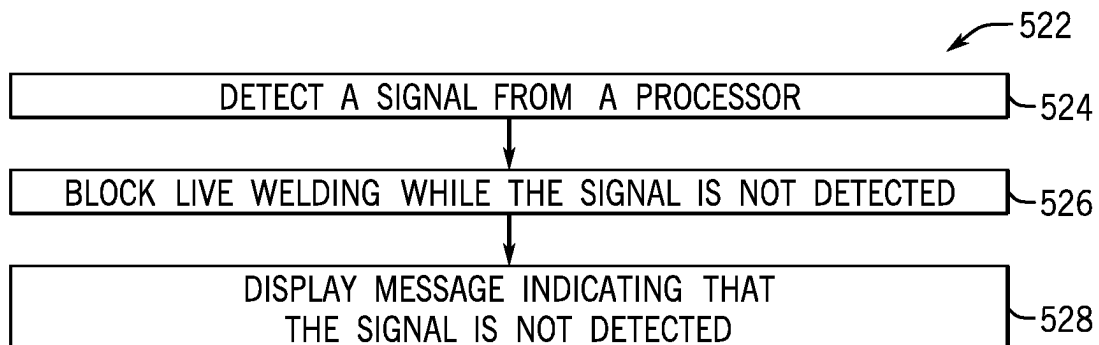


FIG. 29

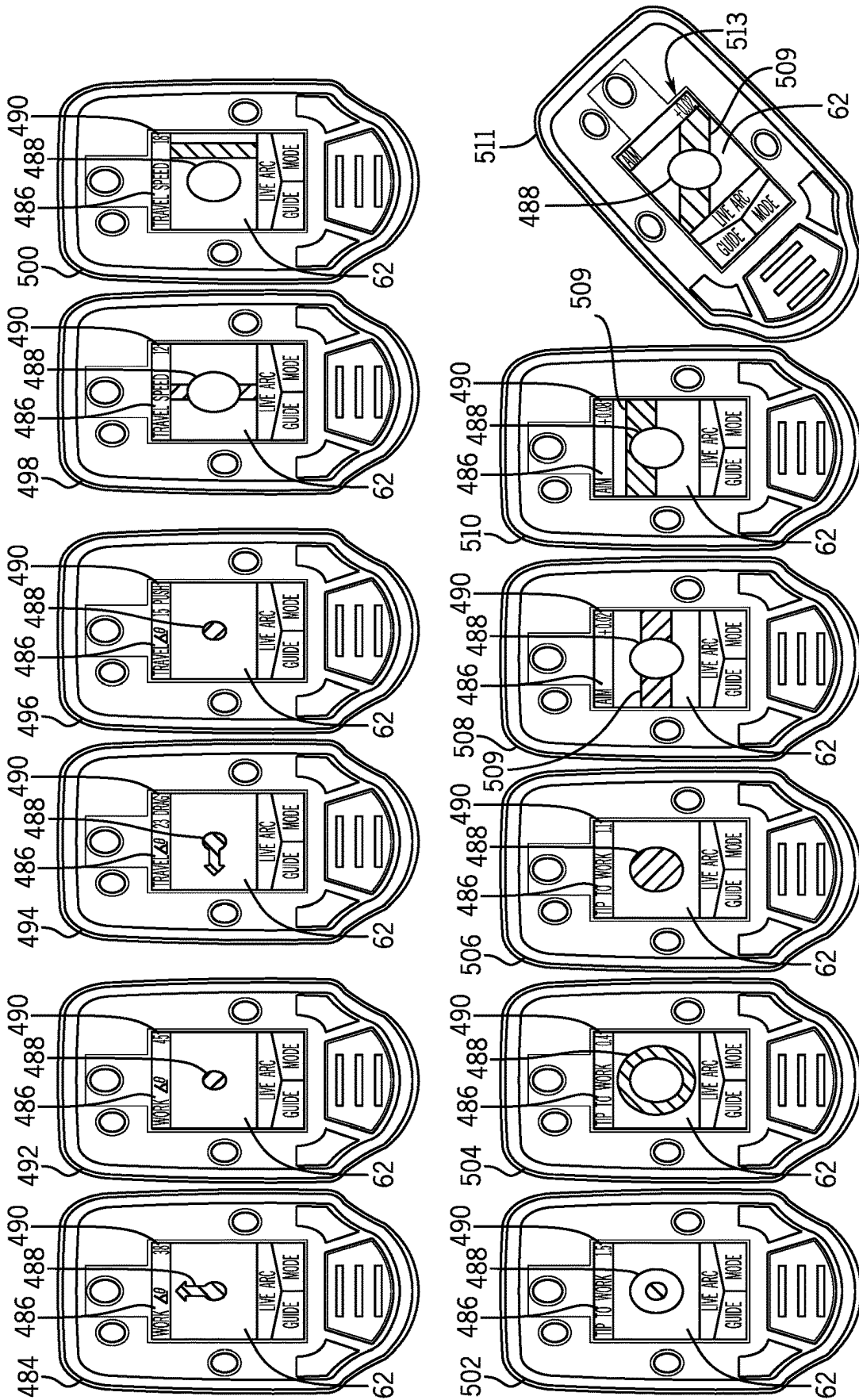


FIG. 27

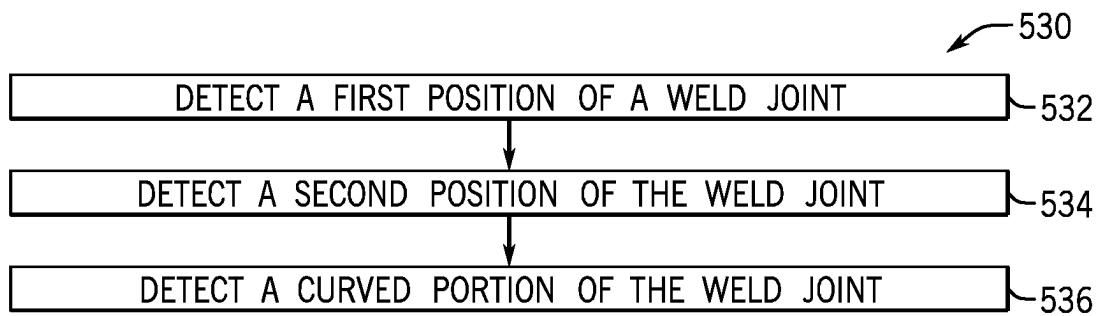


FIG. 30

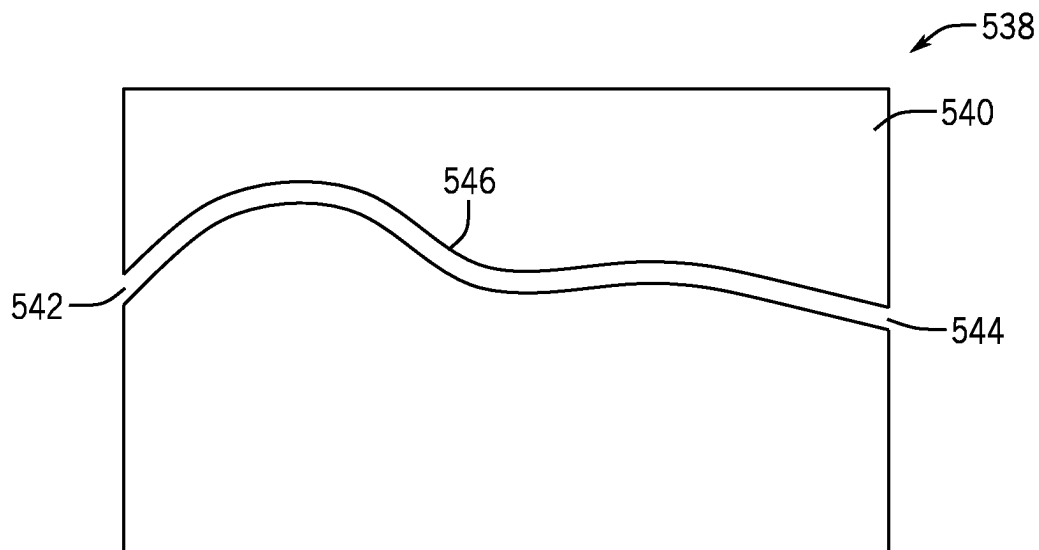


FIG. 31

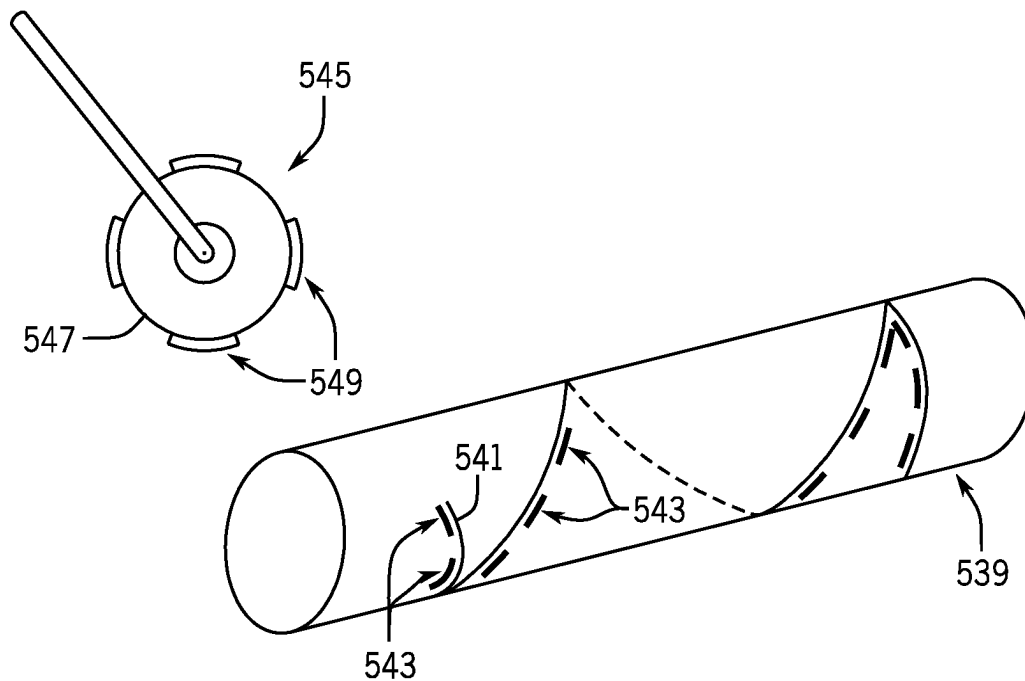


FIG. 32

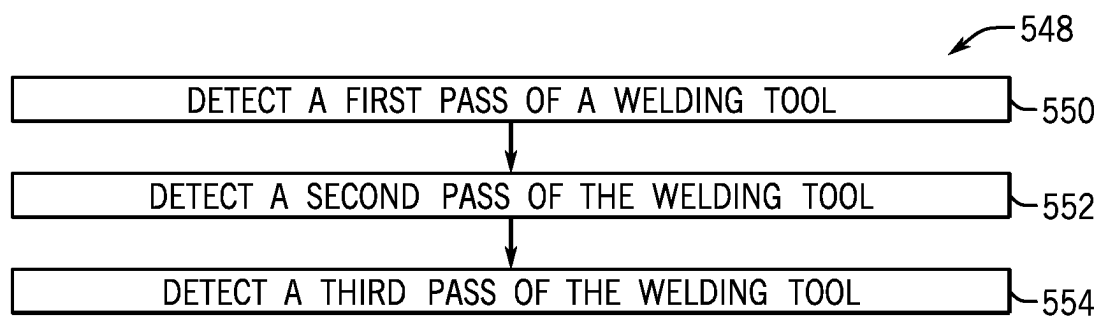


FIG. 33

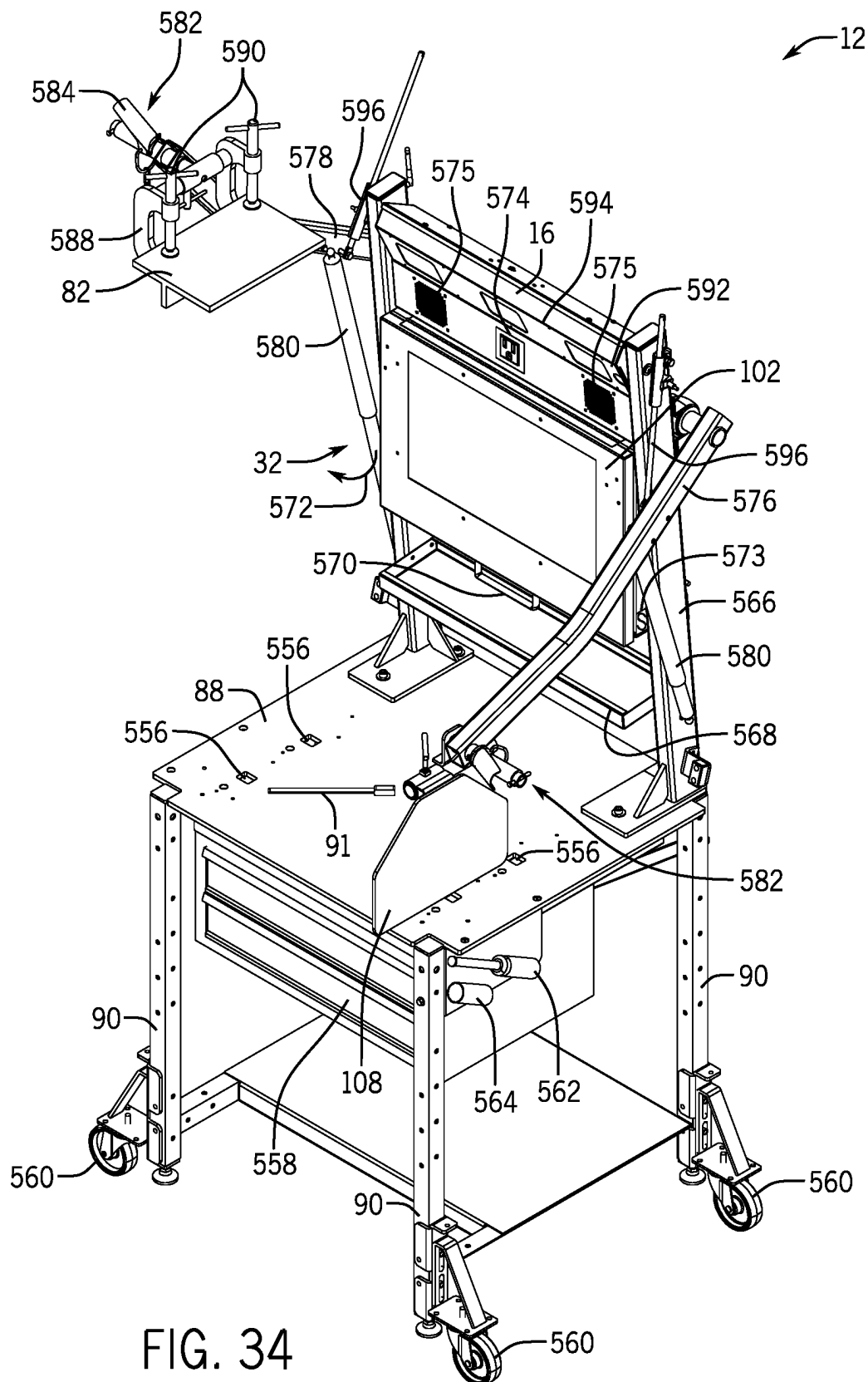


FIG. 34

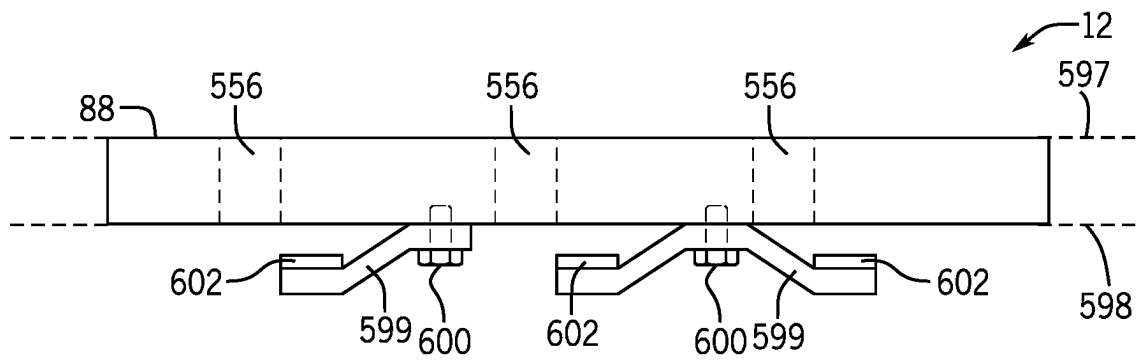


FIG. 35

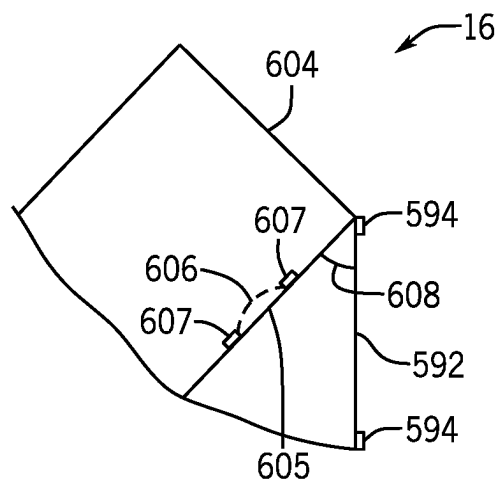
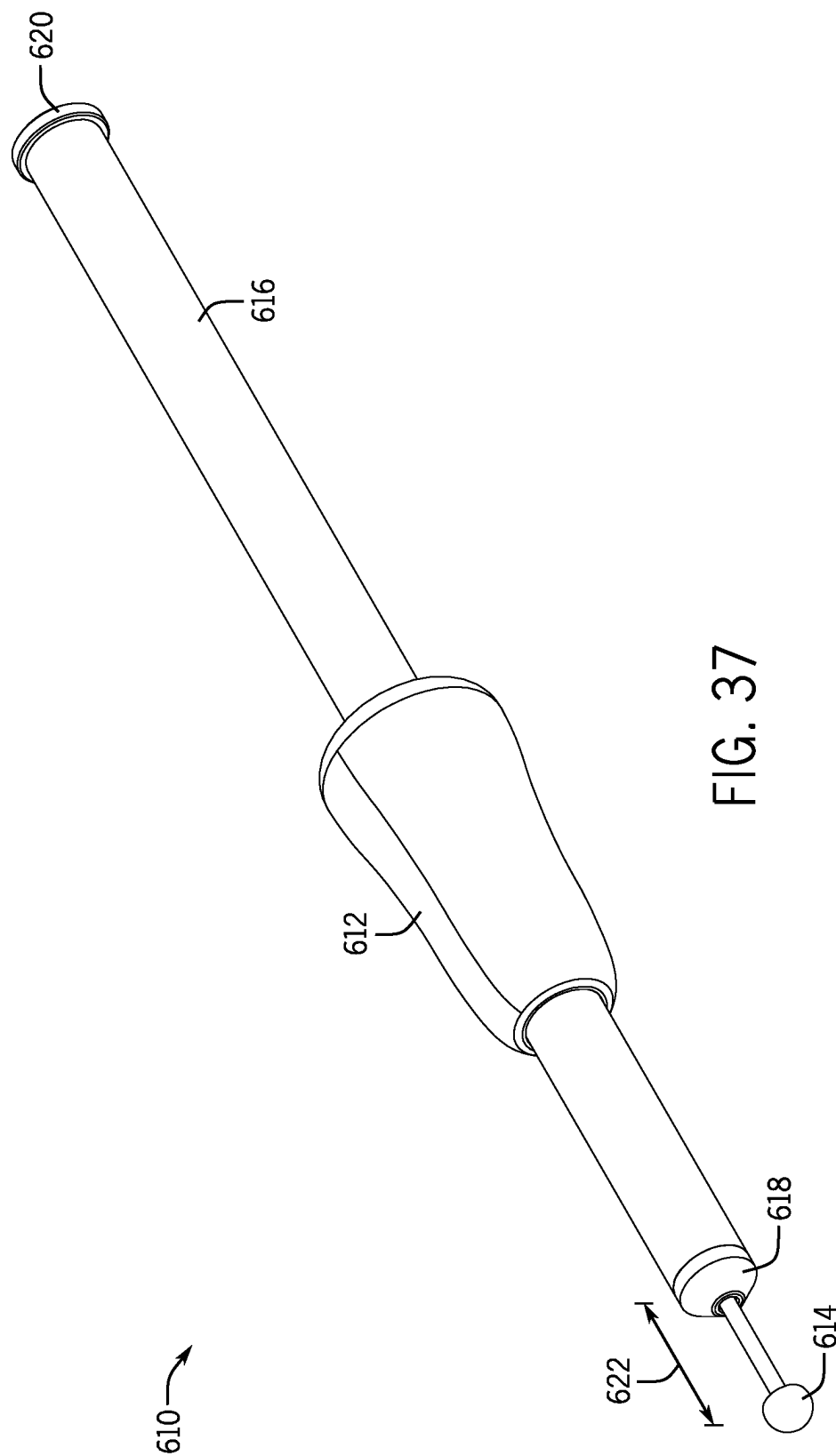
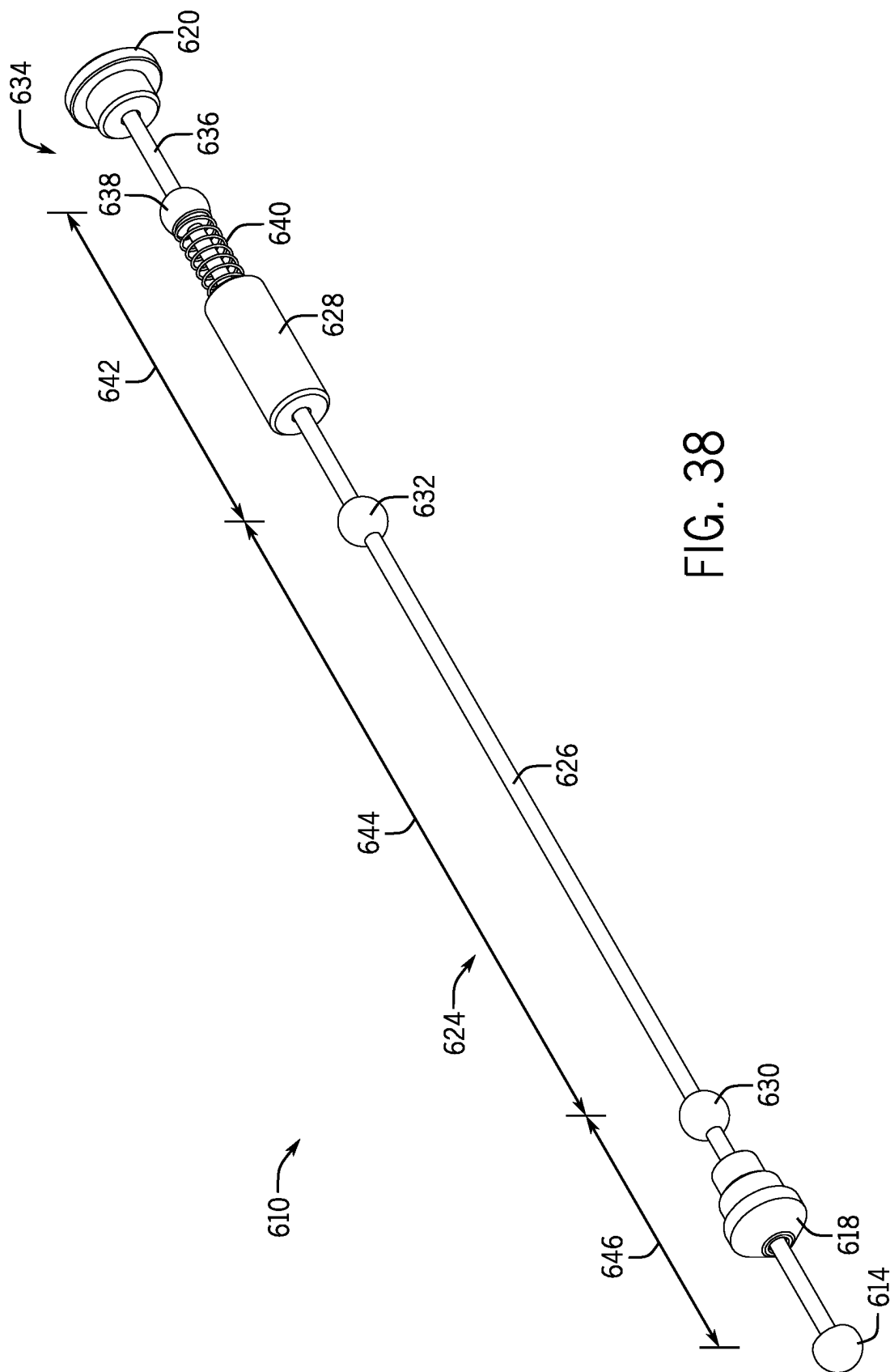


FIG. 36





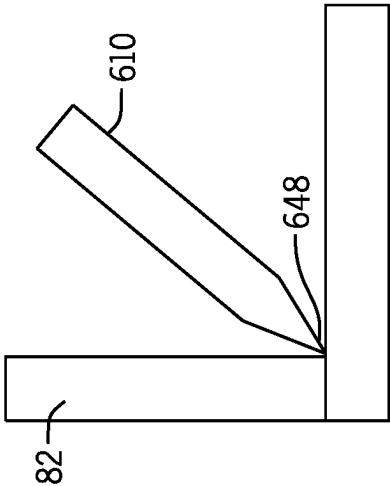


FIG. 39

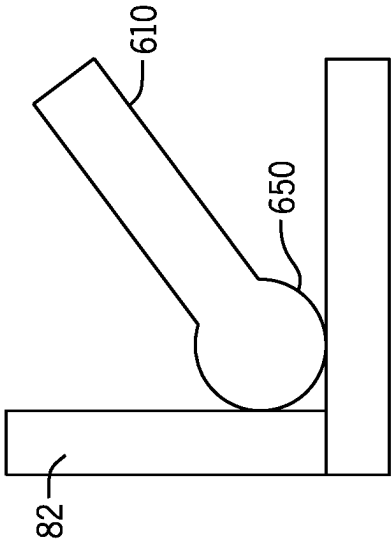


FIG. 40

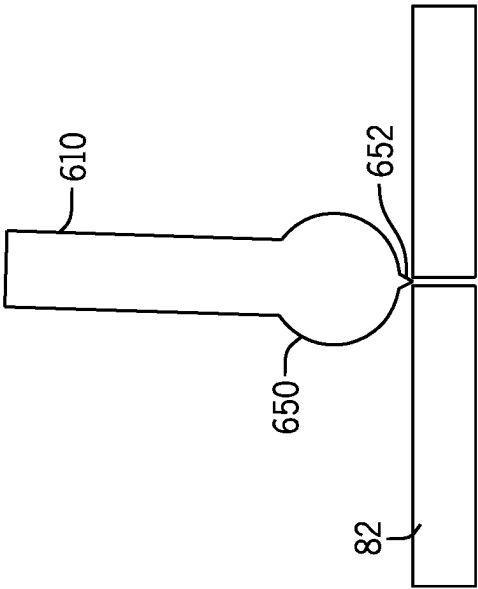


FIG. 41

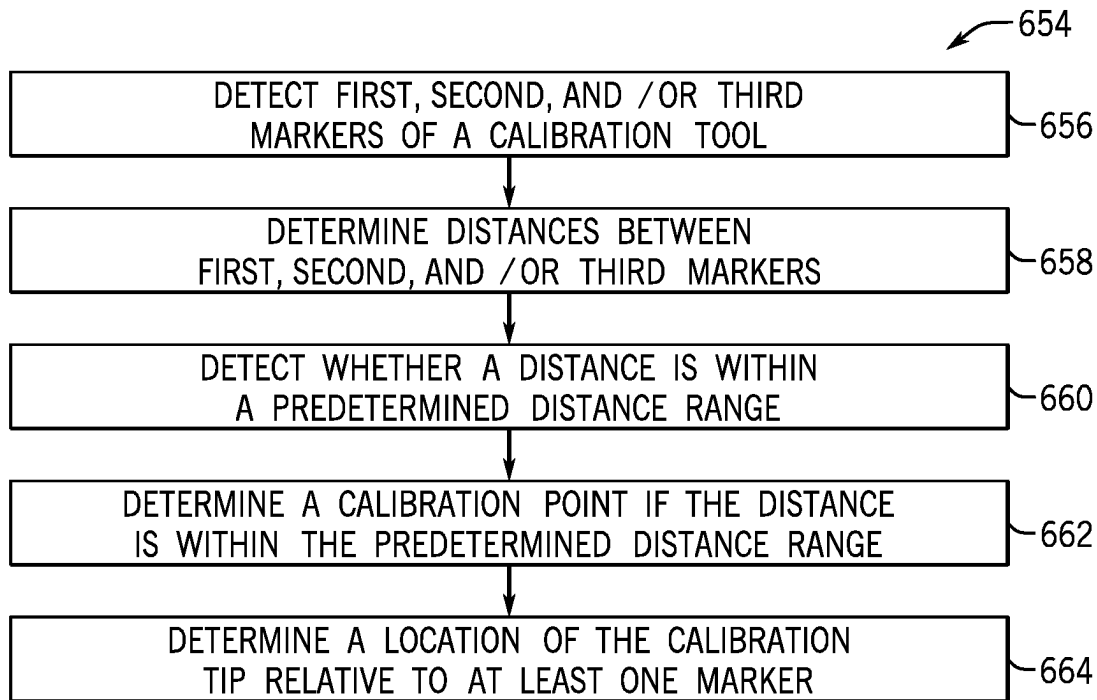


FIG. 42

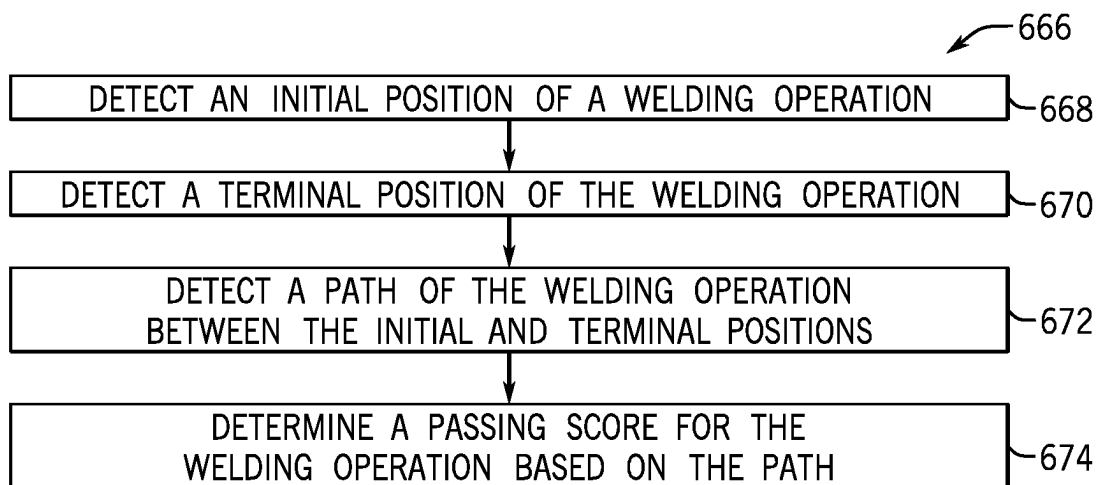


FIG. 43

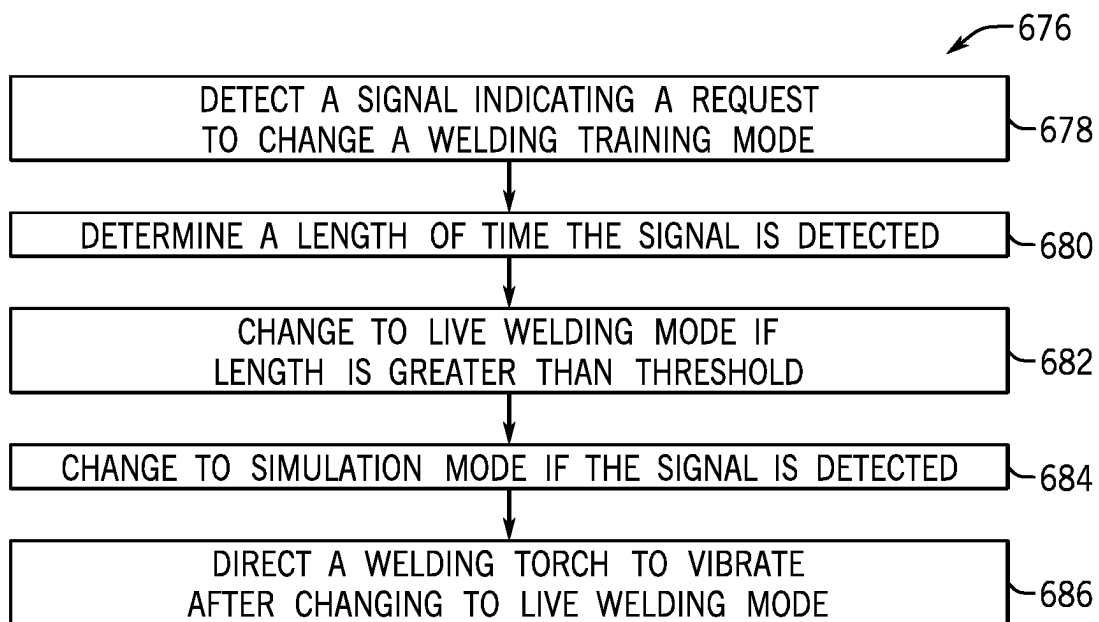


FIG. 44

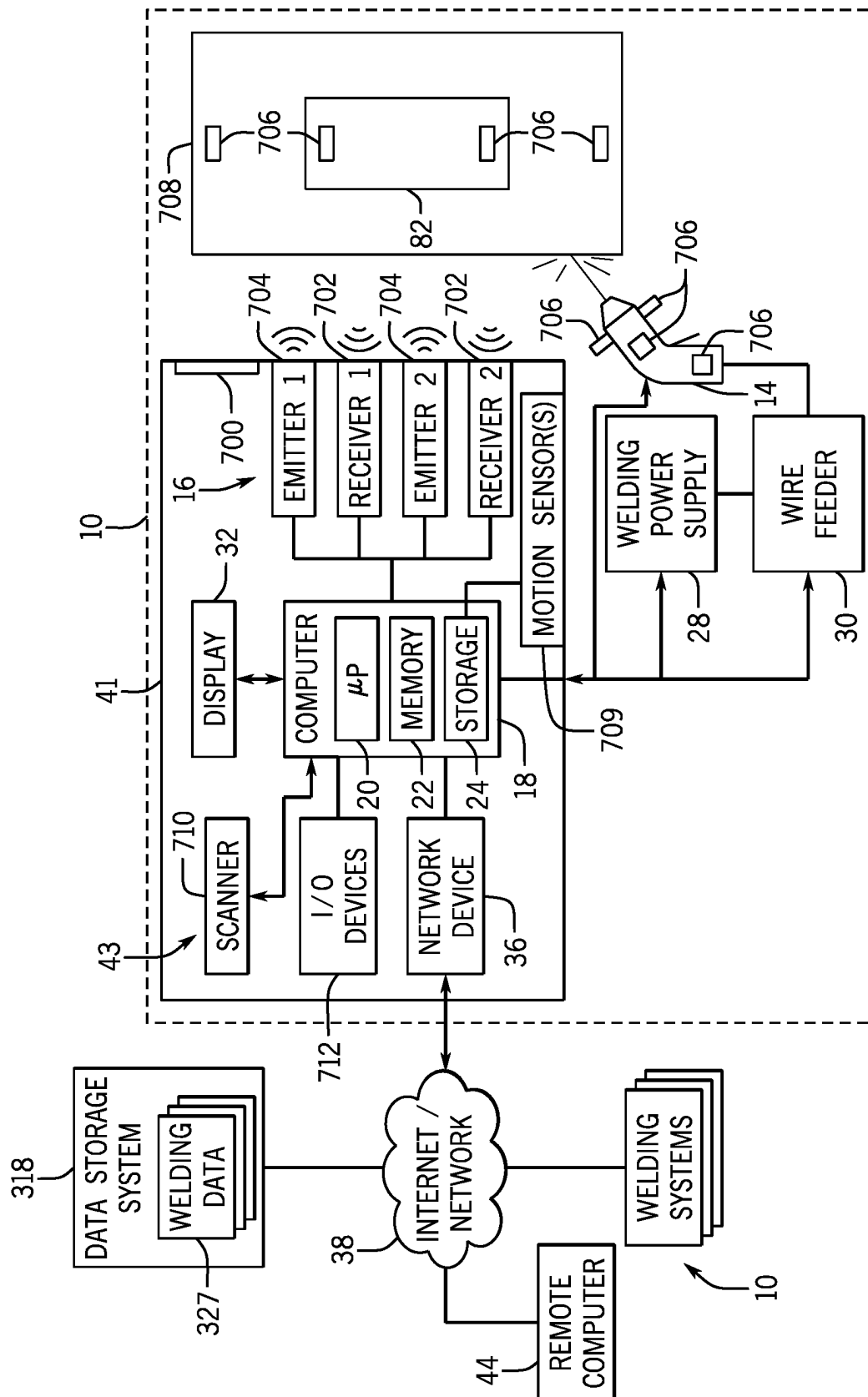


FIG. 45

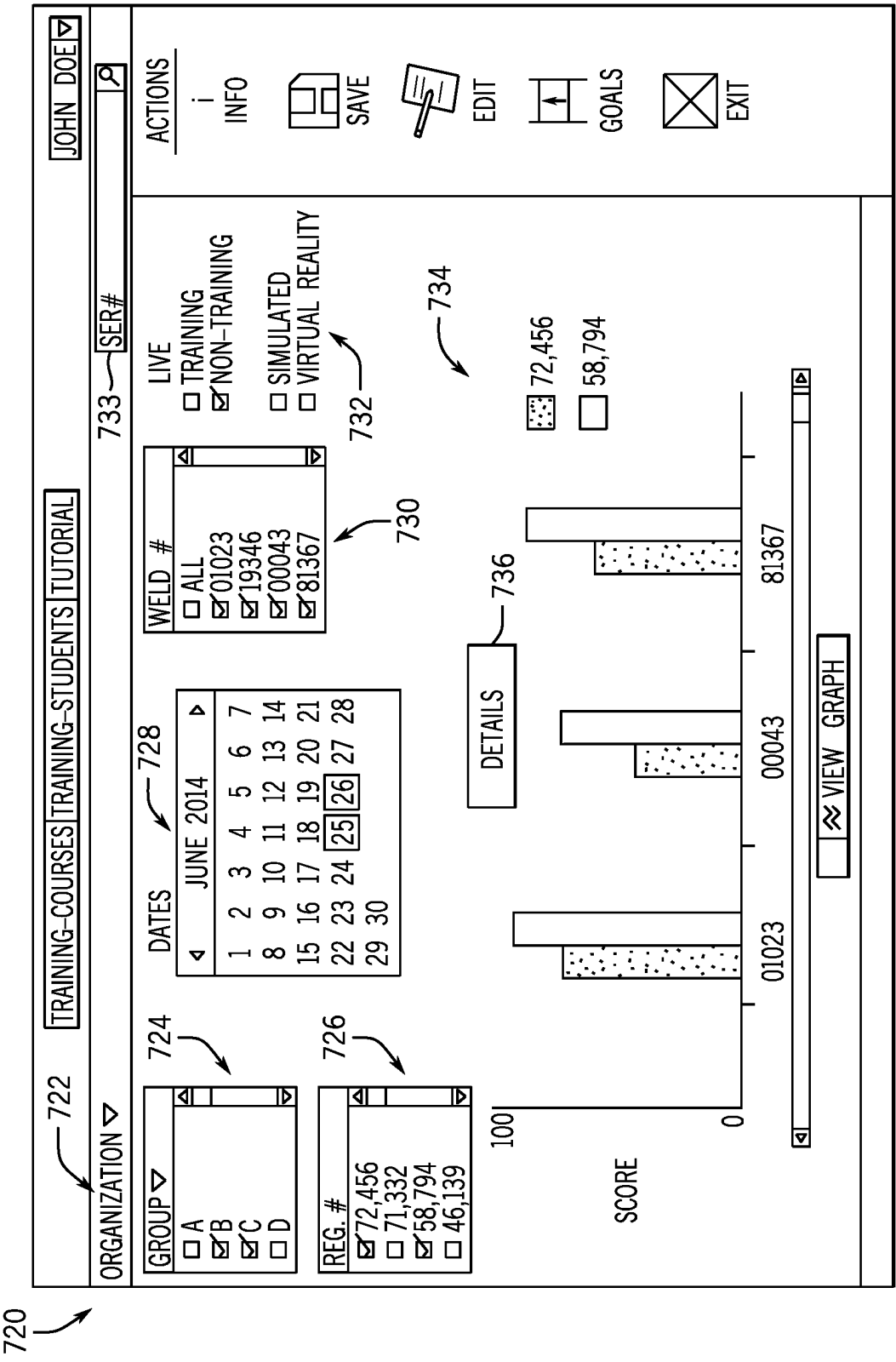


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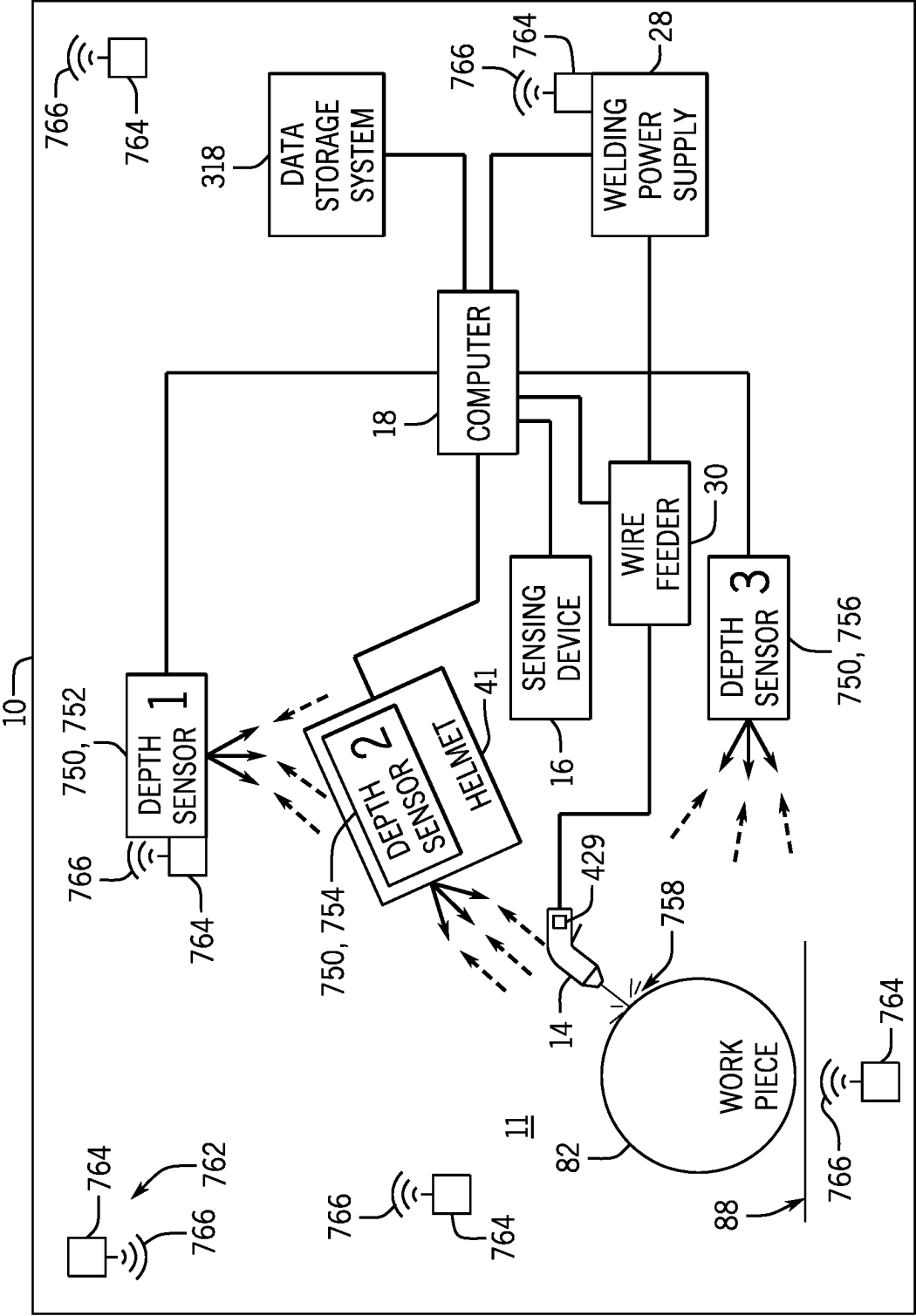


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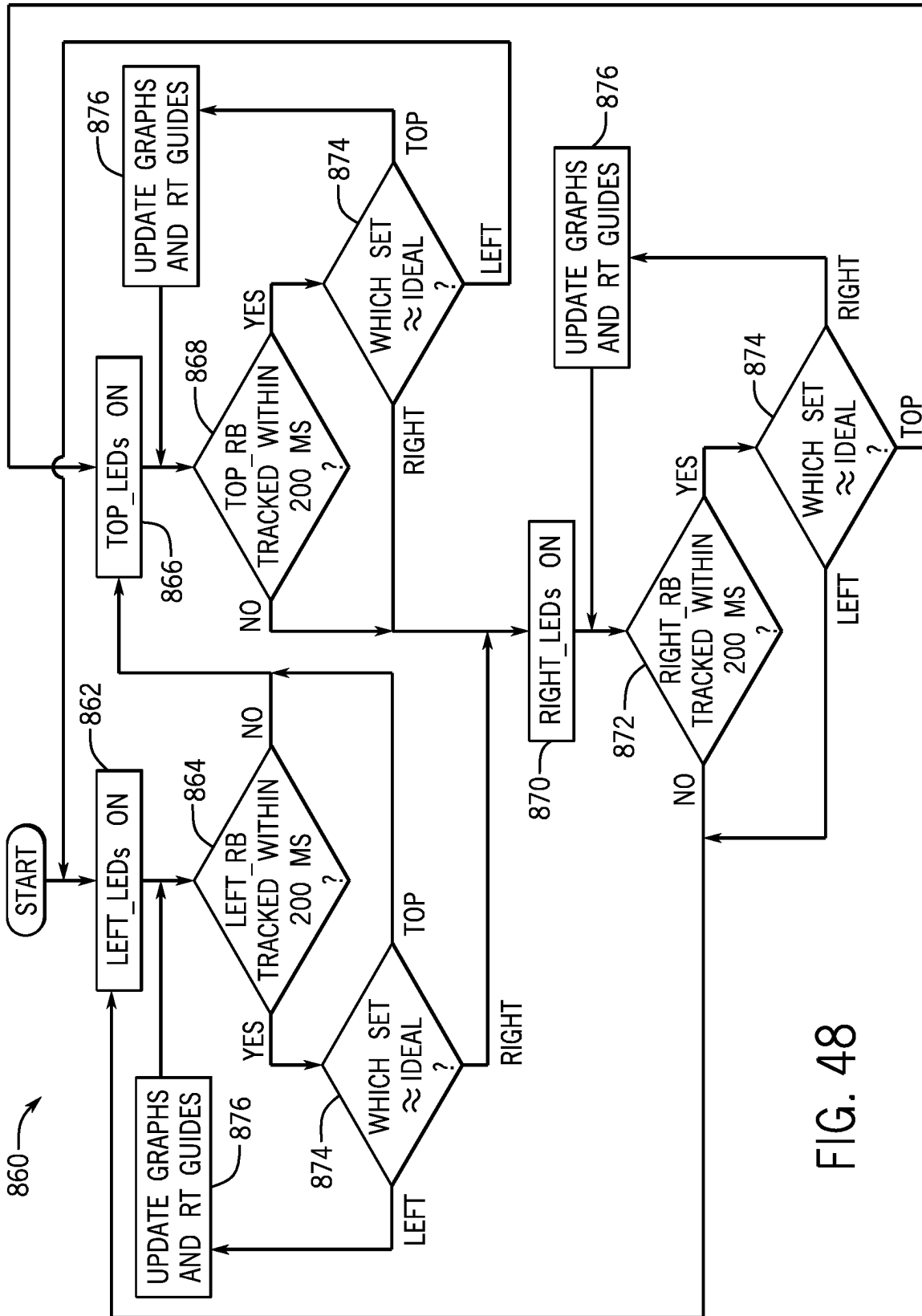


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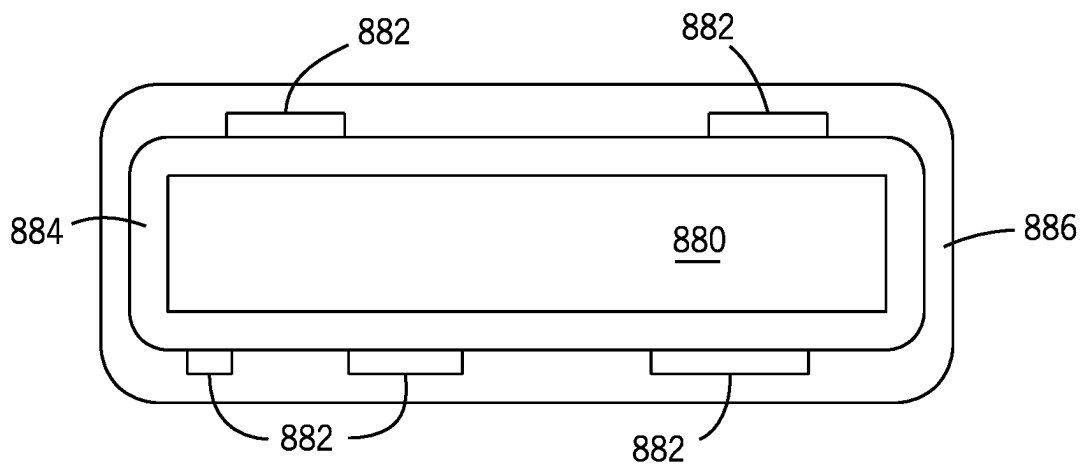


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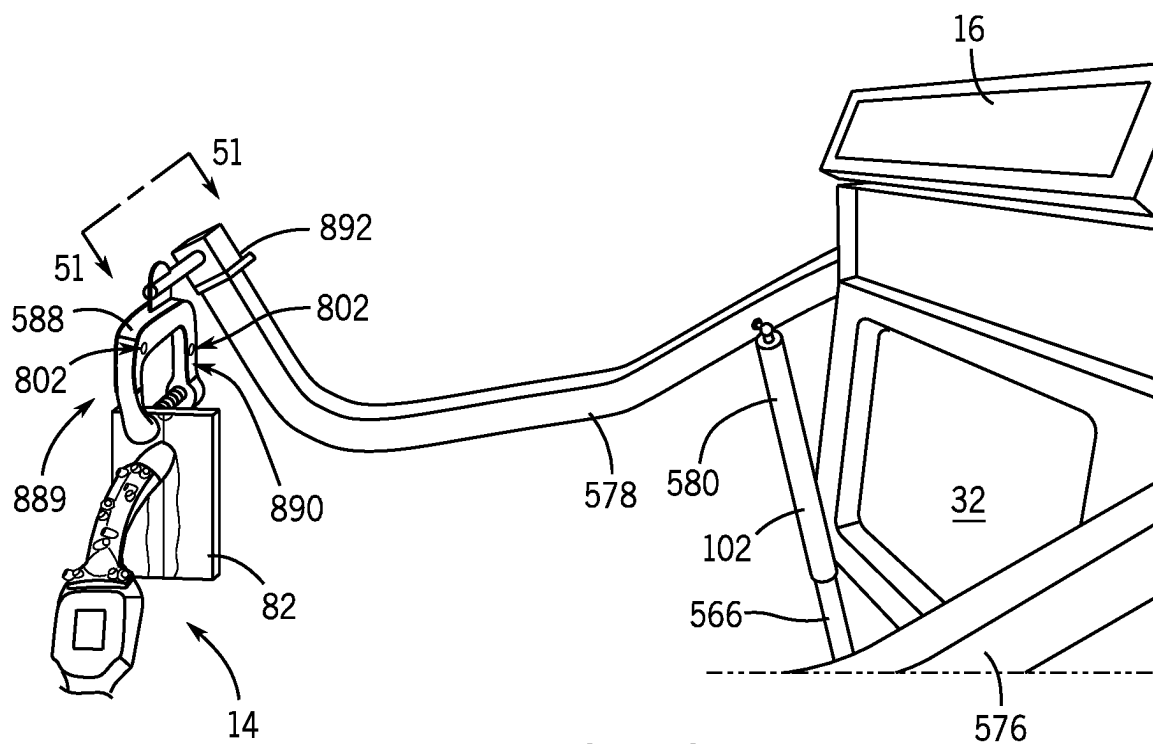


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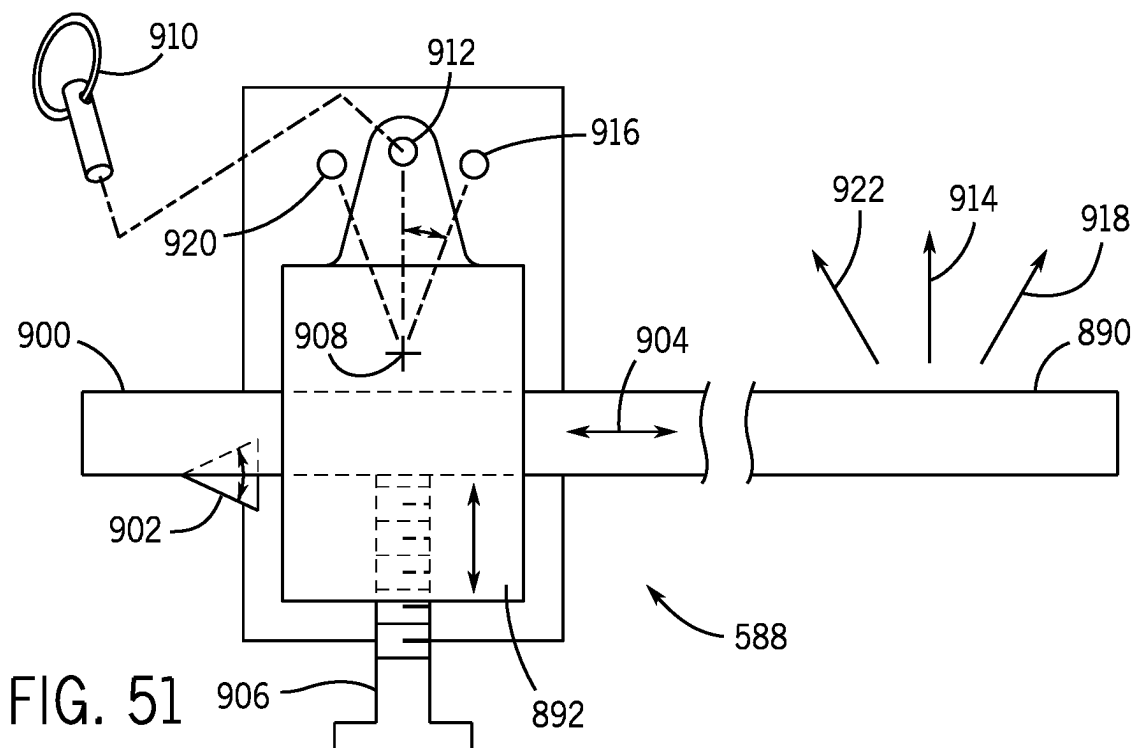
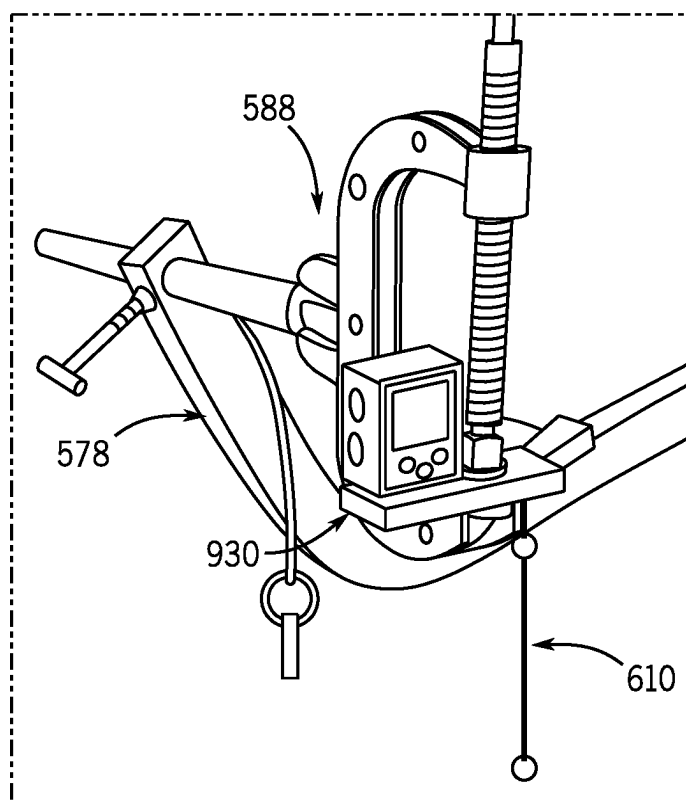


FIG. 52



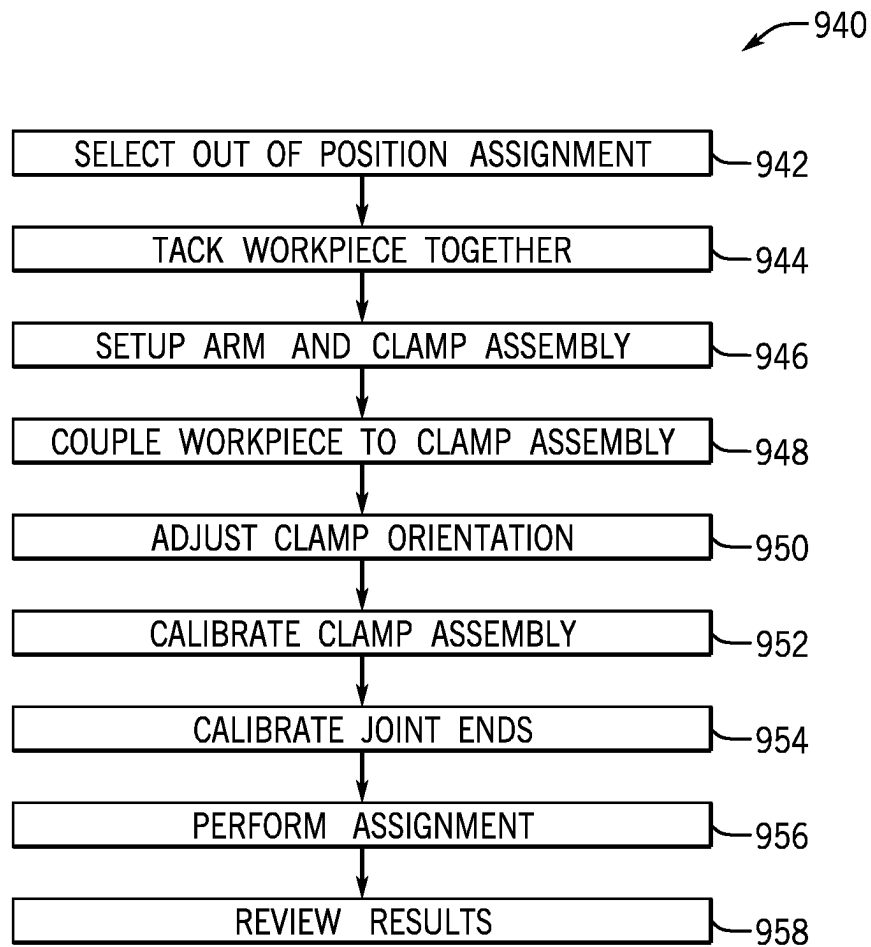


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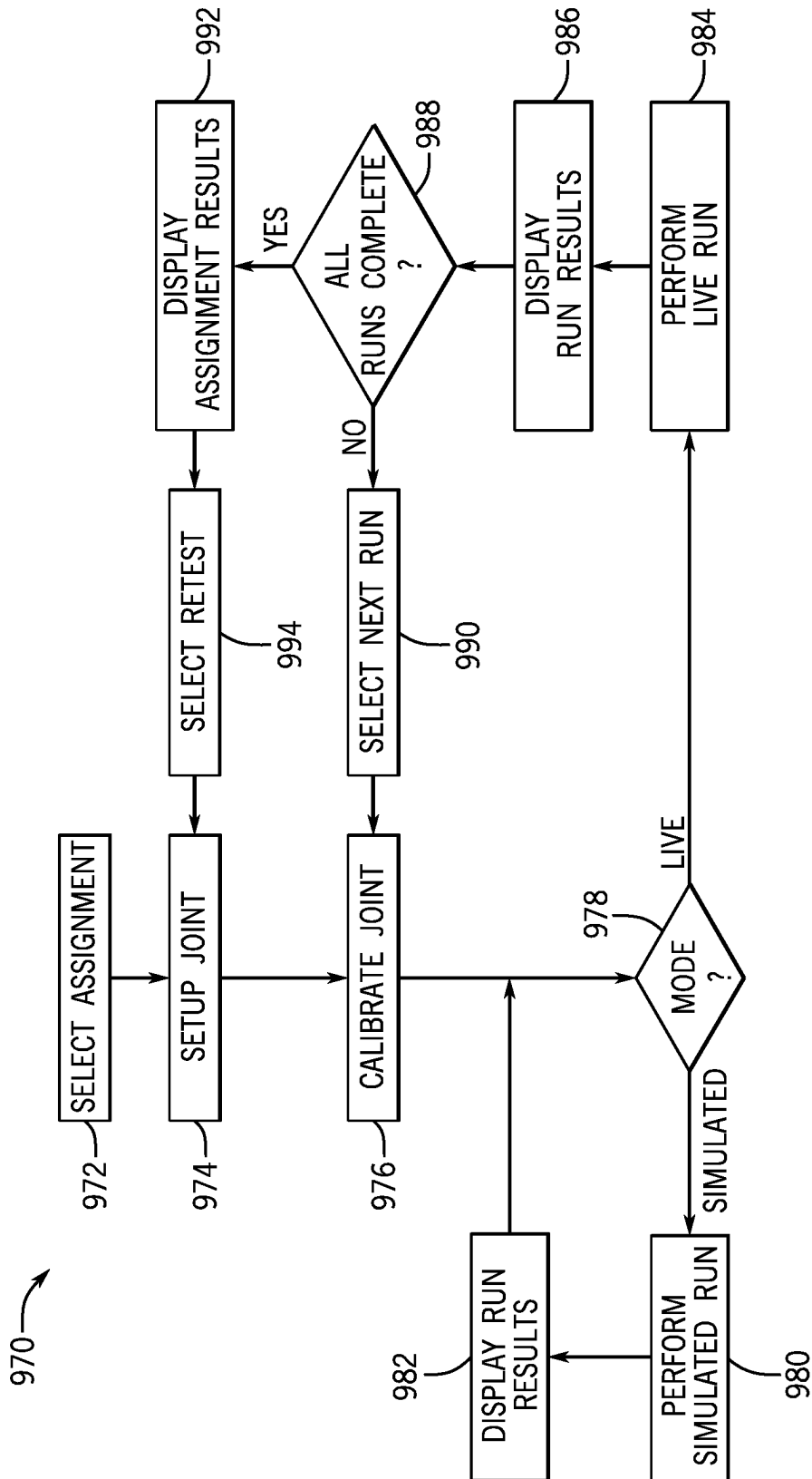


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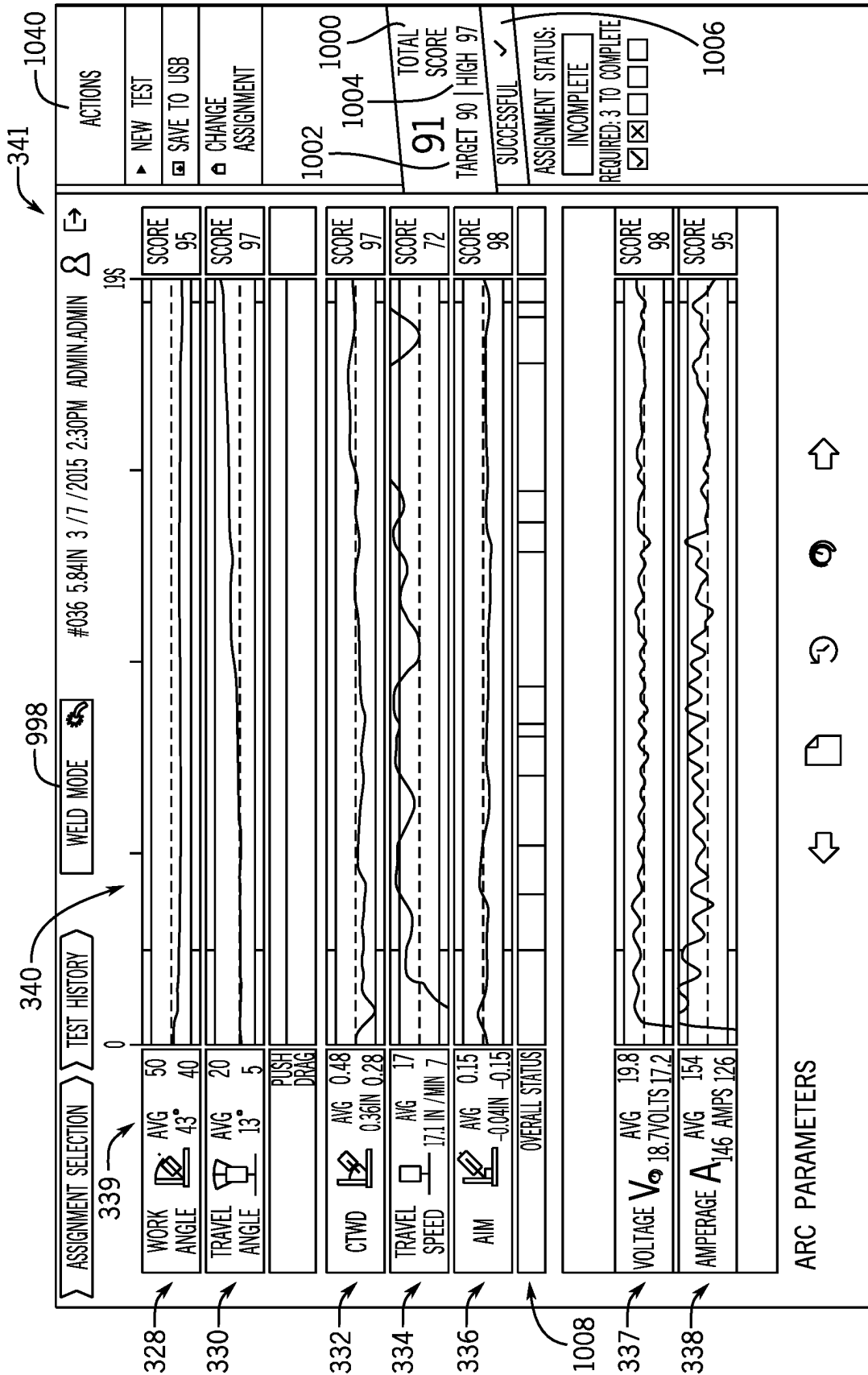


FIG. 55

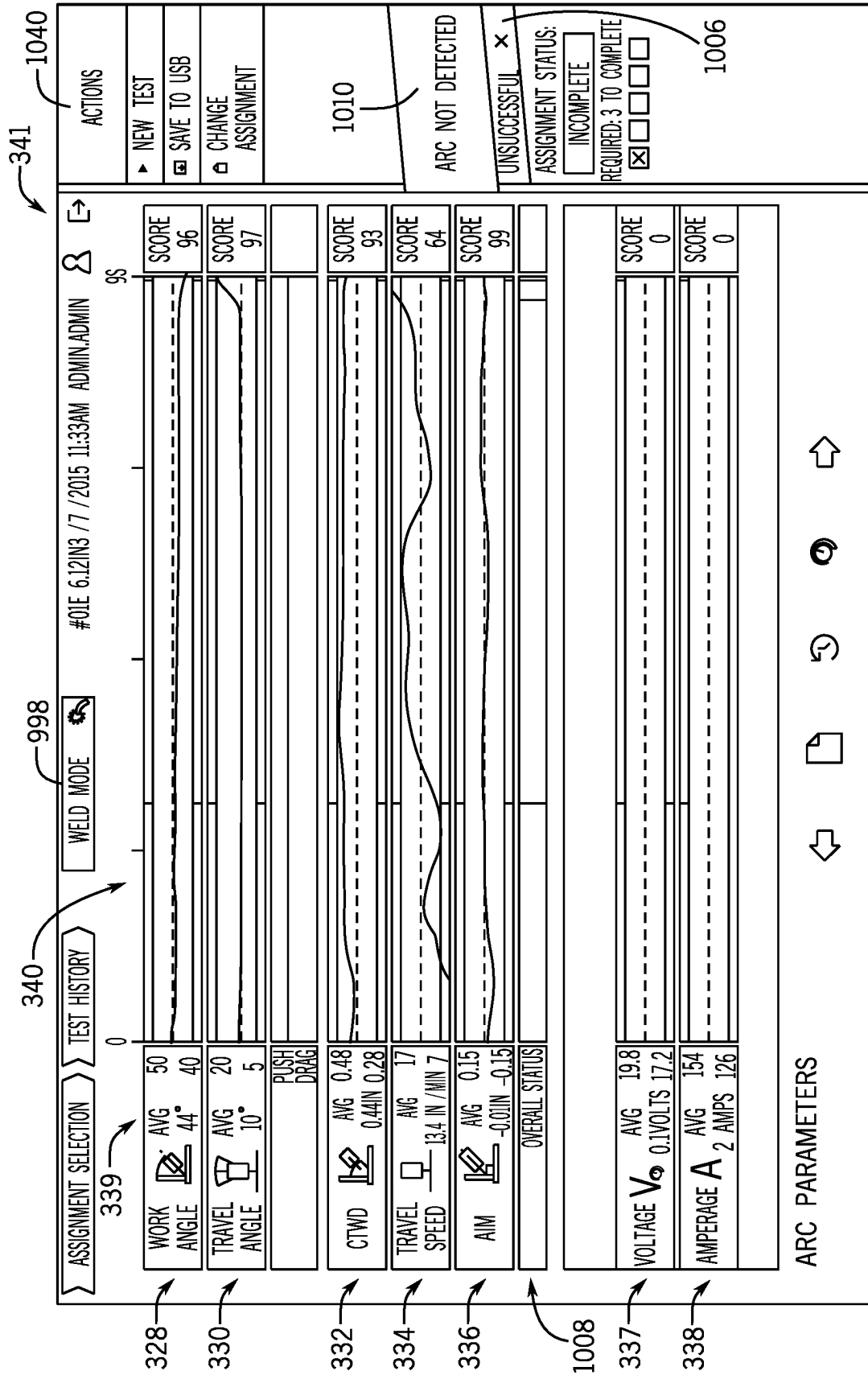


FIG. 56

1016 HOME >> ASSIGNMENT MANAGEMENT > ASSIGNMENT DEVELOPMENT > ADMIN ADMIN

1018 ASSIGNMENT NAME ✓

1020 JOINT DESIGN ✓

1022 BASE METALS ✓

1024 FILLER METALS SHIELDING ✓

1026 POSITION ELECTRICAL CHAR. ✓

1028 PREHEAT POSTWELD HEAT TR. ✓

1014 WELDING PROCEDURE 1 PASS ✓

1030 COMPLETION CRITERIA LENGTH REQUIREMENTS ✓

1036 REAL-TIME FEEDBACK ✓

1038 TOTAL SCORE 90

1040 ACTIONS

1032

1042

1044

1012

FIG. 57

HOME

ASSIGNMENT MANAGEMENT

ASSIGNMENT DEVELOPMENT

XLR-8 3F

ADMIN ADMIN

1016

ASSIGNMENT NAME

1018

JOINT DESIGN

1020

BASE METALS

1022

FILLER METALS
SHIELDING

1024

POSITION
ELECTRICAL CHAR.

1026

PREHEAT
POSTWELD HEAT TR.

1028

WELDING PROCEDURE
1 PASS

1014

COMPLETION CRITERIA
LENGTH REQUIREMENTS

1030

REAL-TIME
FEEDBACK

PREV

NEXT

WELDING PROCEDURE (PASS 1 OF 1)

PROCESS

FILLER METALS

PATTERN

VERTICAL PROGRESSION

COMMENTS

FCAW-G

E7T-8D H8

.072 INCHES

STRINGER

WEAVE

UP

DOWN

MIN.

MAX.

19

21.5

24

149

160

171

175

215

240

37

45

53

-30 DRAG

-20 DRAG

-10 DRAG

0.9

1

1.1

1

5

9

-0.15

0

0.15

WELDING POWER SOURCE SETTINGS

TORCH TECHNIQUE PARAMETERS

VOLTS

WIREFEED SPEED

AMPS

WORK ANGLE

TRAVEL ANGLE

CTWD

TRAVEL SPEED

AIM

APPLY WIDE RANGE DEFAULT VALUES

*INDICATES REQUIRED PARAMETER

WIDING RANGE

12%

12%

24

21.5

19

VOLTS

VOLTS

21.5

VOLTS

1

2

3

4

5

6

7

8

9

0

.

-

CLEAR

PARAMETER REQUIREMENTS

-DECIMAL GREATER THAN 0

-NO LESS THAN VOLTS MIN VALUE

-NO GREATER THAN VOLTS MAX VALUE

ACTIONS

SAVE AND EXIT

SAVE AND TEST

SAVE

CANCEL

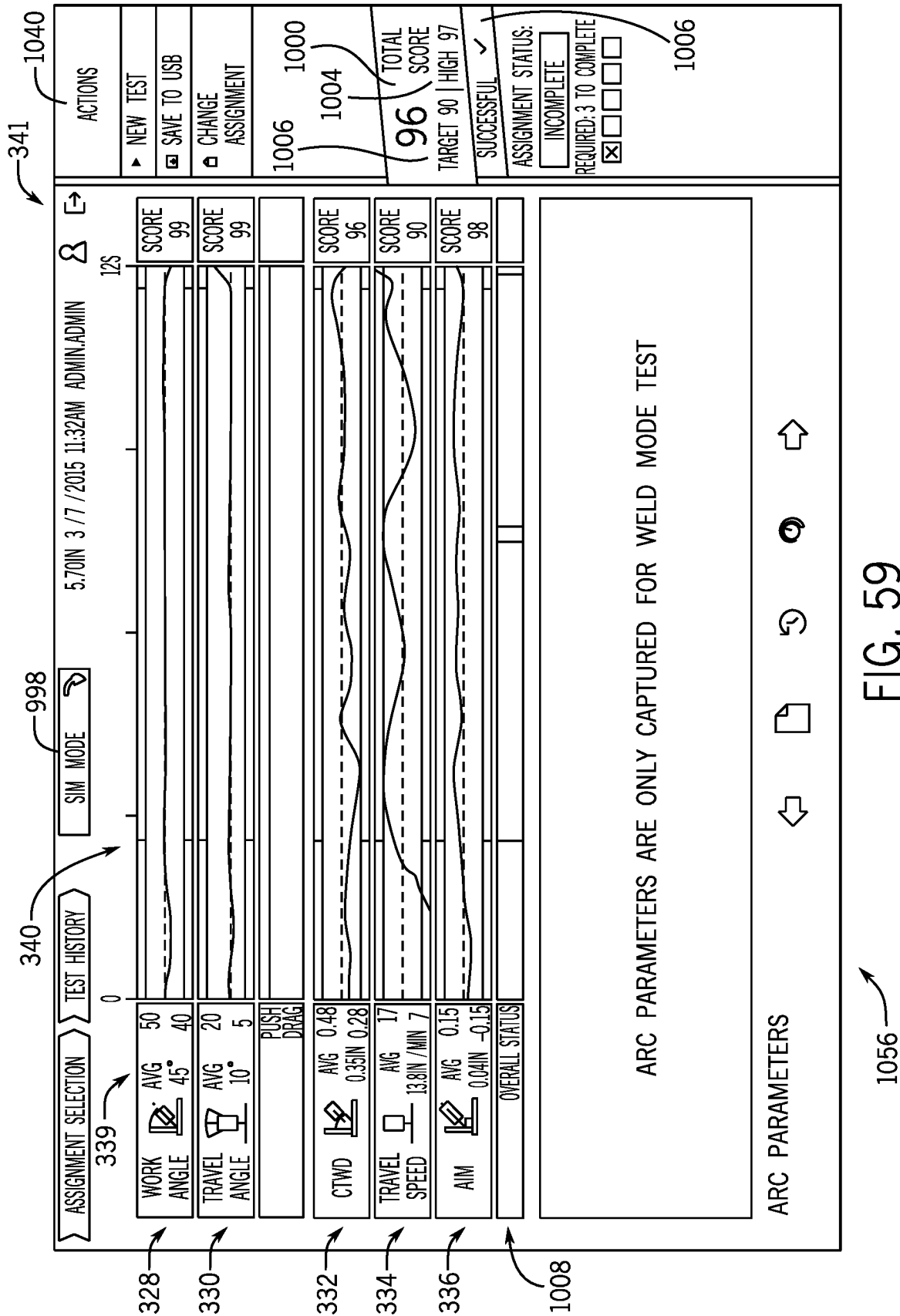
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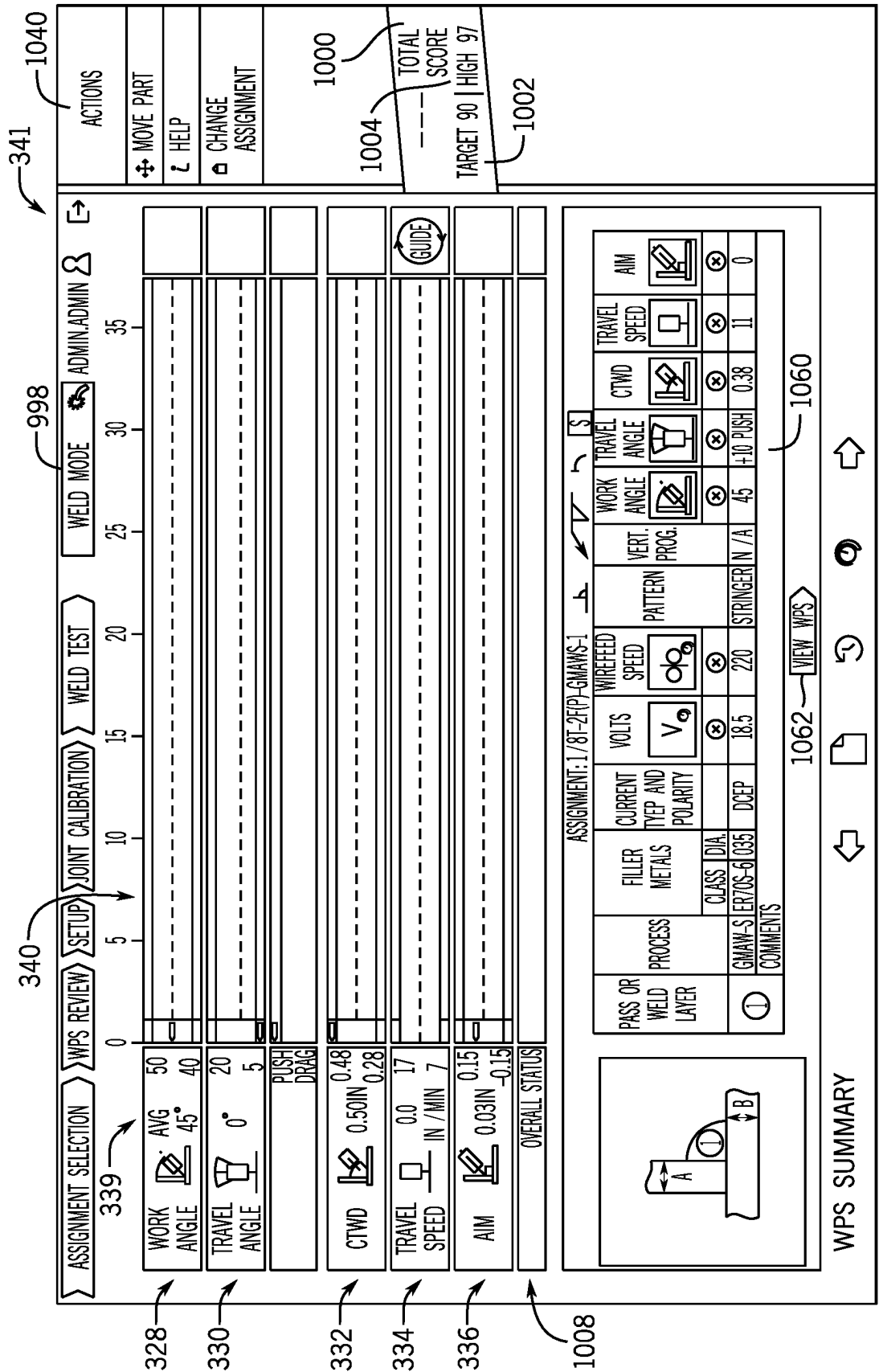
1048

1050

1052

FIG. 58



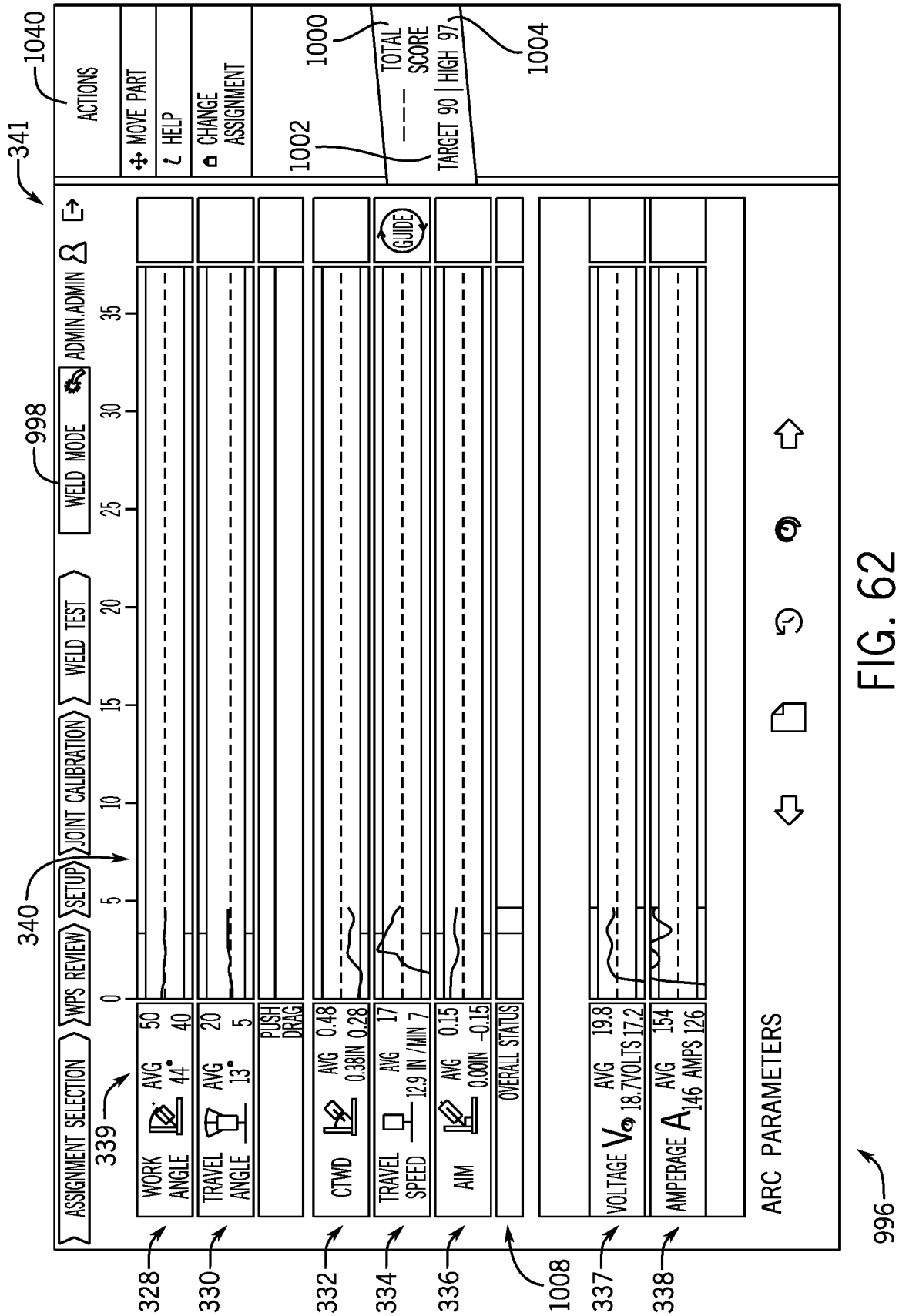


1040

ASSIGNMENT SELECTION				WPS REVIEW				ADMIN ADMIN				ACTIONS					
JOIN DESIGN USED				XLR-8 3F				POSITION				☐ NEXT ☒ SAVE TO USB ☐ BACK ☐ CHANGE ASSIGNMENT					
JOINT TYPE				T				POSITION OF GROOVE				IN / A					
BACKING				NO				POSITION OF FILLET				3F					
ROOT OPENING				0.00000 IN				ELECTRICAL CHARACTERISTICS				1000					
BASE METAL A				BASE METAL B				TRANSFER MODE				DCEP					
PLATE / PIPE				PLATE				CURRENT									
THICKNESS				1/2 IN				TECHNIQUE									
METAL TYPE				STEEL				STRINGER OR WAVE BEAD				STRINGER					
MATERIAL SPEC.				A1008				MULTIPLE OF SINGLE PASS (PER SIDE) SINGLE									
TYPE / GRADE				SS GRADE 50				PREHEAT									
FILLER METALS								WHEN BASE METAL IS BELOW				90					
AWS CLASSIFICATION				E71T-8JD H8				PREHEAT TEMP. MIN.				32°F (0°C), PREHEAT TO A					
OTHER				HOBART FABSHIELD XLR-8				MIN. OF 70°F (20°C).									
SHIELDING								GENERAL COMMENTS									
GAS COMPOSITION				N / A				WELD SIZE									
FLOW RATE				0 SCFH				3 / 8 IN.									
CONTACT TIP SETBACK				GAS CUP SIZE													
WELDING PROCEDURE												COMPLETION CRITERIA:					
PASS OR WELD LAYER		WELDING PROCESS		WELDING POWER SOURCE SETTINGS		TORCH TECHNIQUE PARAMETERS		WORK ANGLE		TRAVEL ANGLE		CTWD		TRAVEL SPEED		AIM	
		FILLER METALS		VOLTS		WIREFEED SPEED		AMPS		VERT. PROG.		PATTERN		DEG		IN / MIN	
		CLASS		DIA.		MIN		MAX		MIN		MAX		MIN		MAX	
1		FCW-G		E71T-8...		.072		19		21.5		24		149		160	
		COMMENTS:		UP		37		45		53		-30		-20		-10	
				STRINGER		1		1.1		1		5		9		0	
				0.15													
VIEW FULL WPS																	

FIG. 61

1064



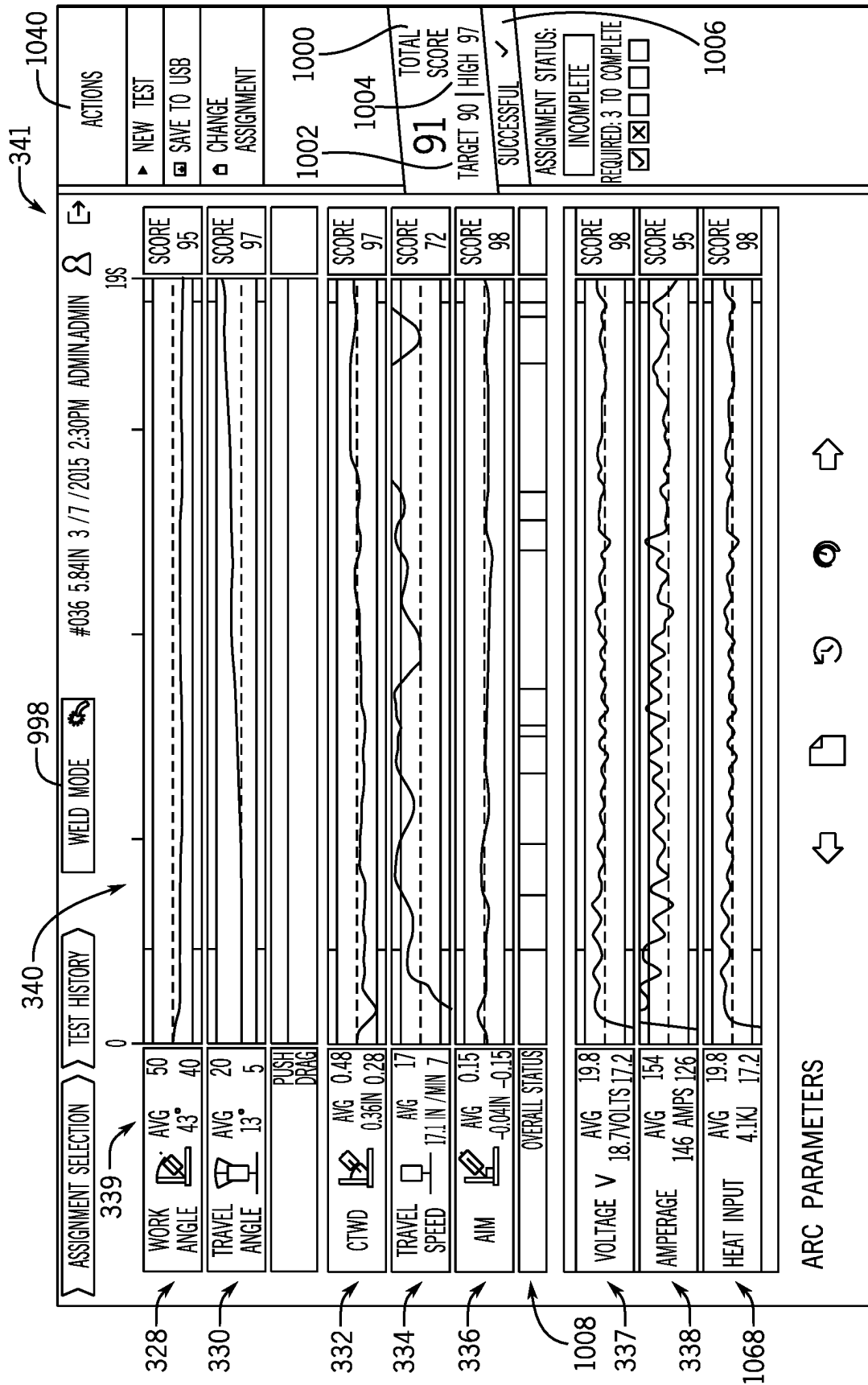


FIG. 63

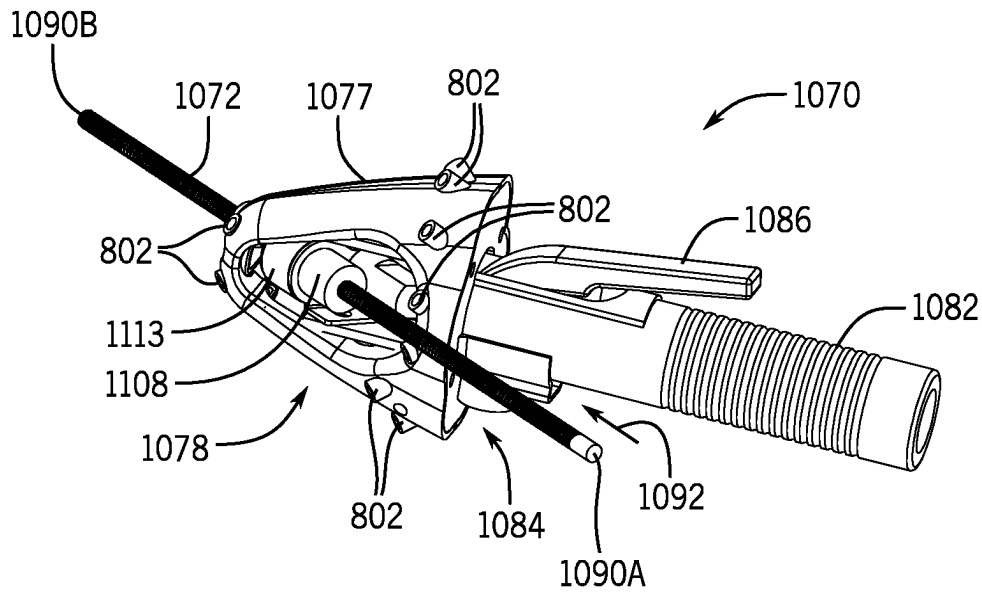


FIG. 64A

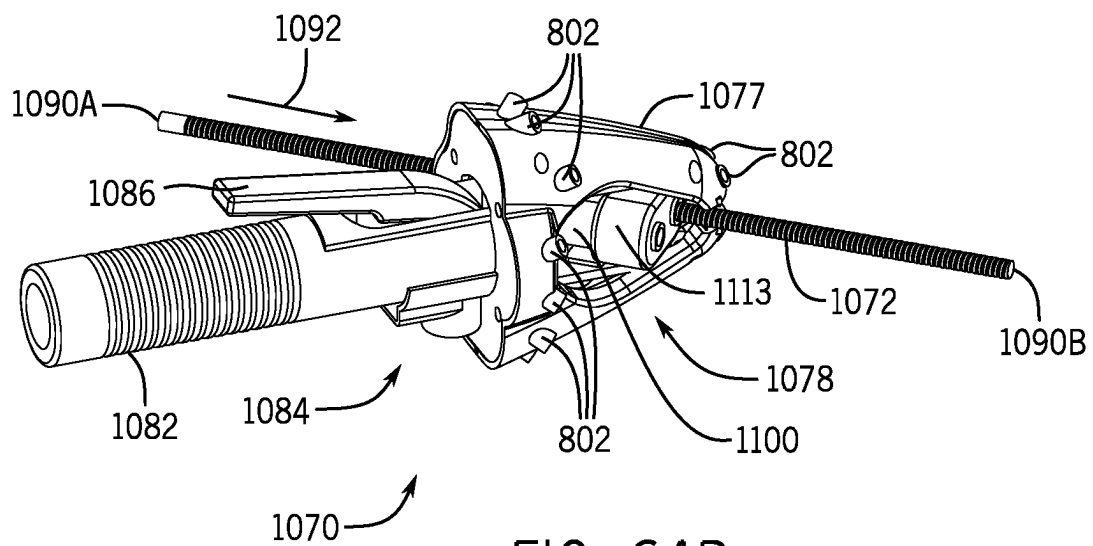


FIG. 64B

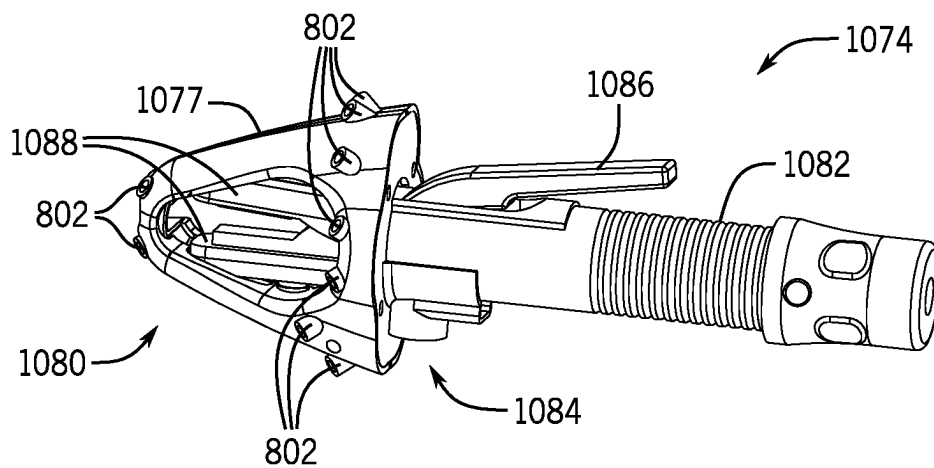


FIG. 65A

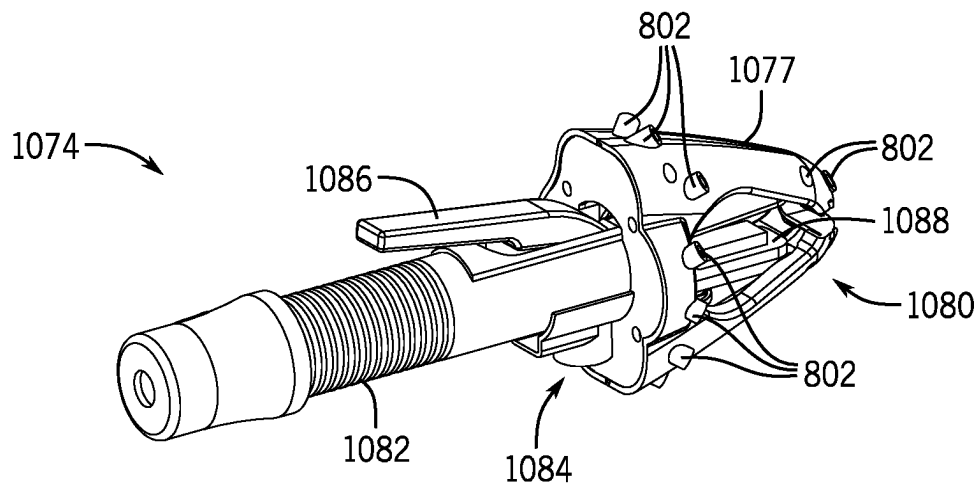


FIG. 65B

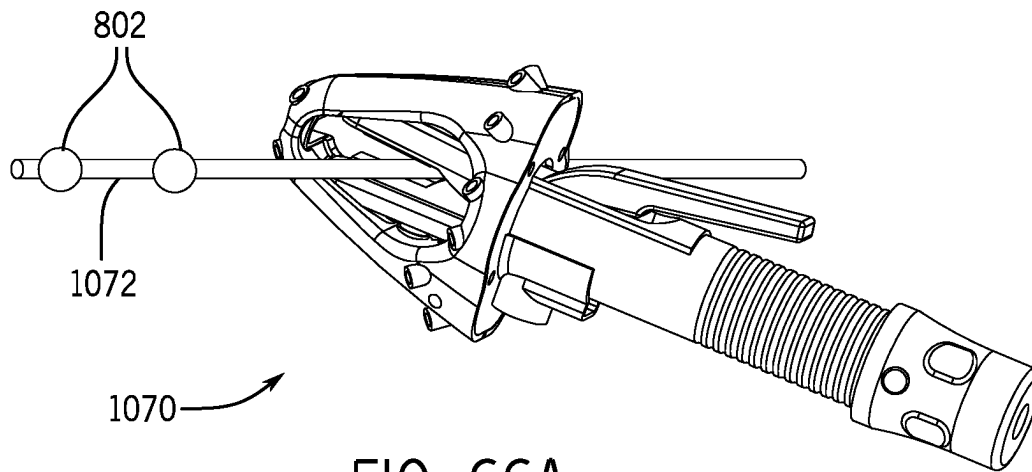


FIG. 66A

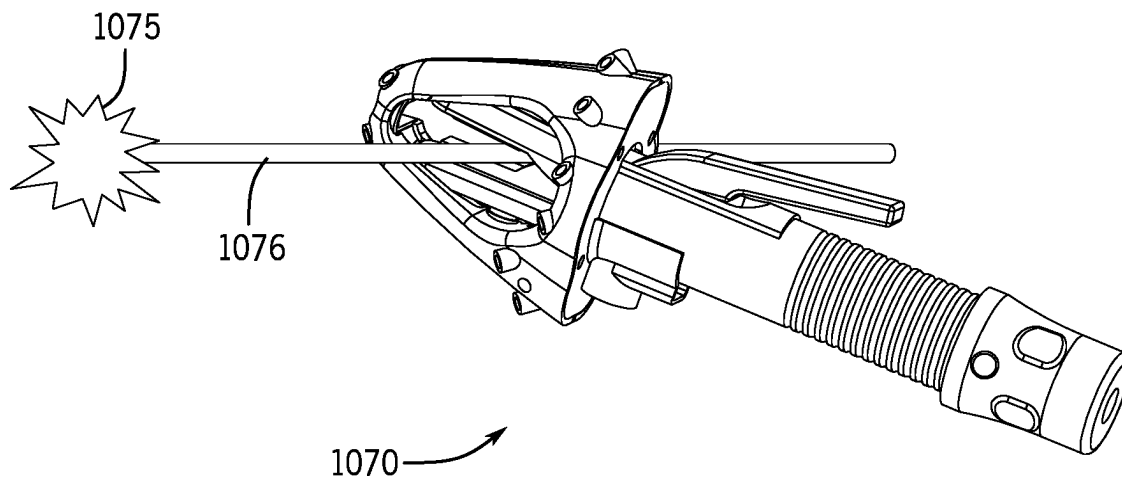


FIG. 66B

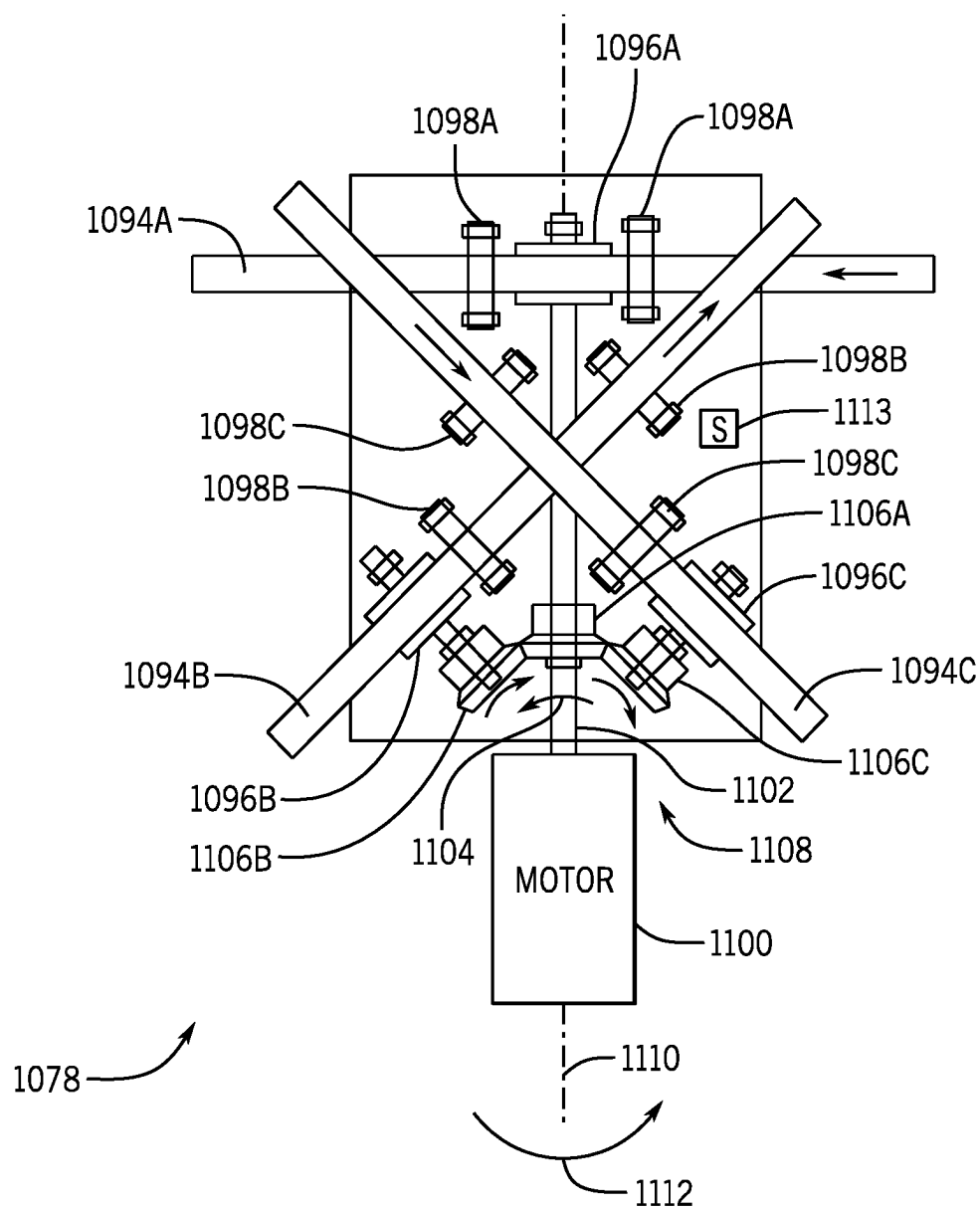


FIG. 67

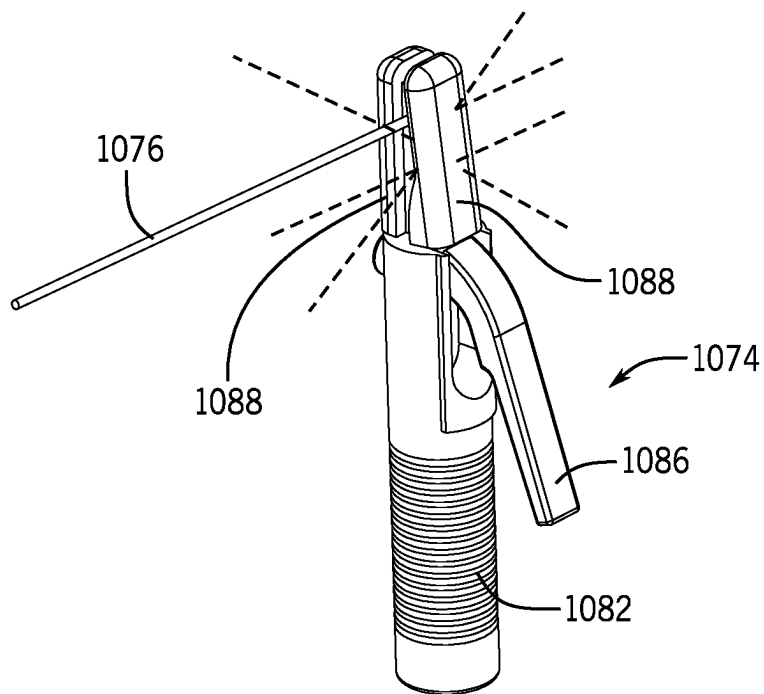


FIG. 68A

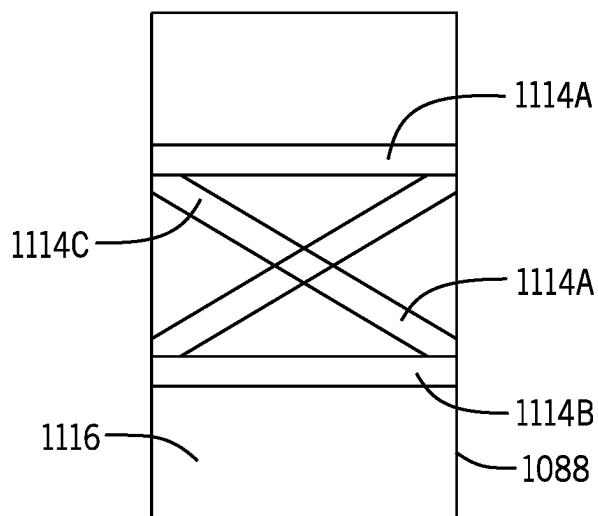


FIG. 68B

1117

ADMIN ADMIN

ASSIGNMENT SELECTION > WPS REVIEW > SETUP > JOINT CALIBRATION > WELD TEST

SELECT THE NUMBER CORRESPONDING TO THE SLOT USED

1 2 3 4

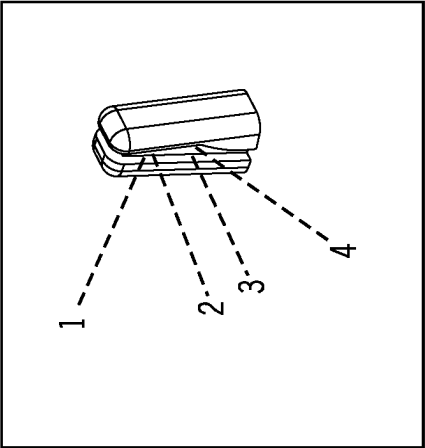
1

2

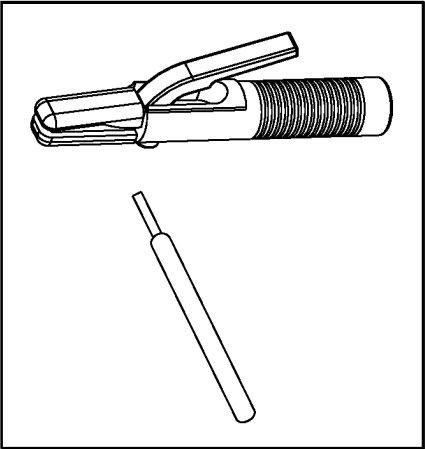
3

4

INSERT THE ELECTRODE INTO THE STINGER IN THE DESIRED SLOT



1 2 3 4



ACTIONS

▶ NEXT

◀ BACK

CHANGE ASSIGNMENT

FIG. 68C

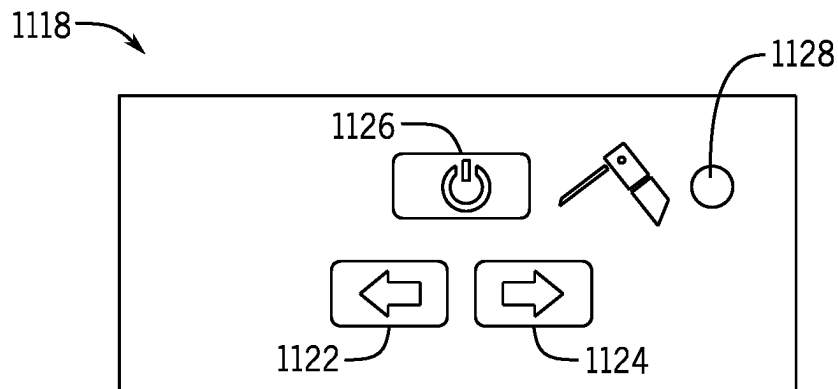


FIG. 69A

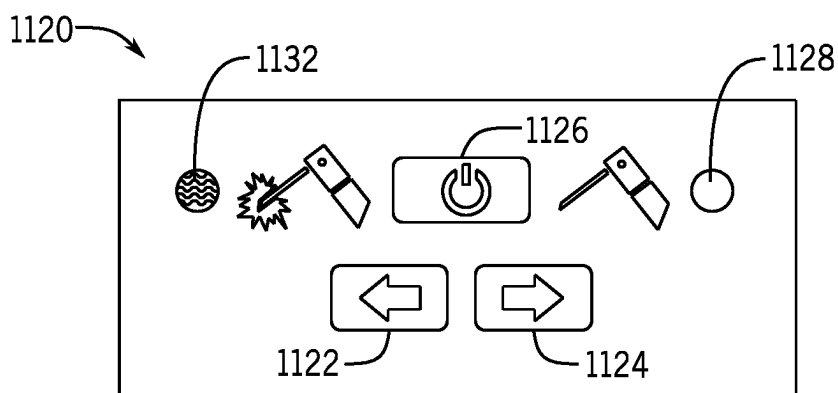


FIG. 69B

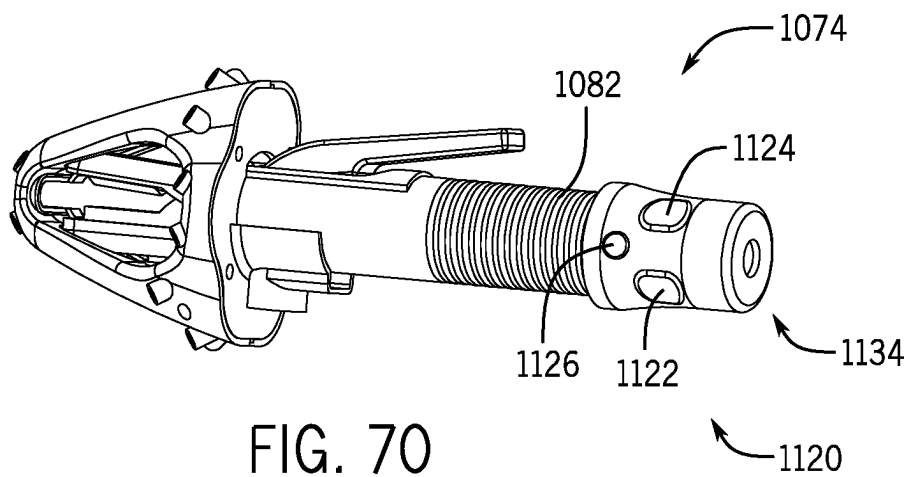
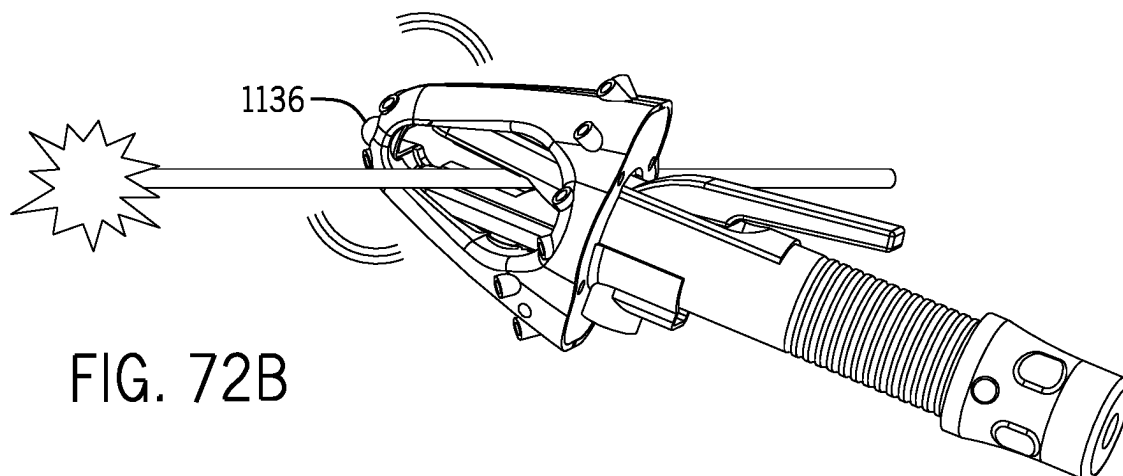
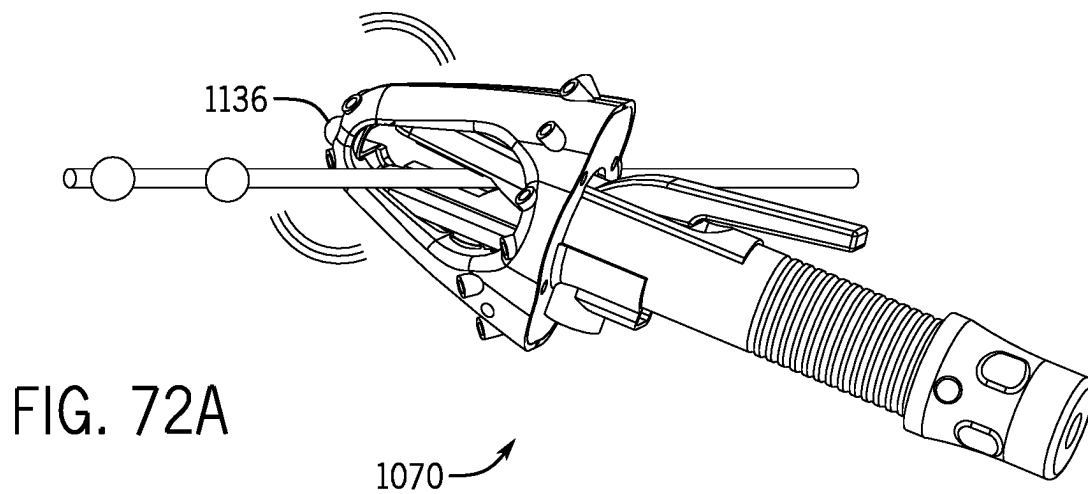
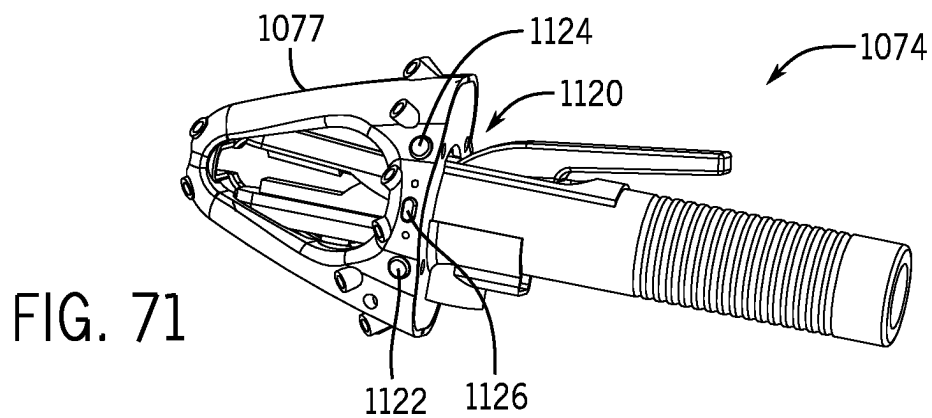
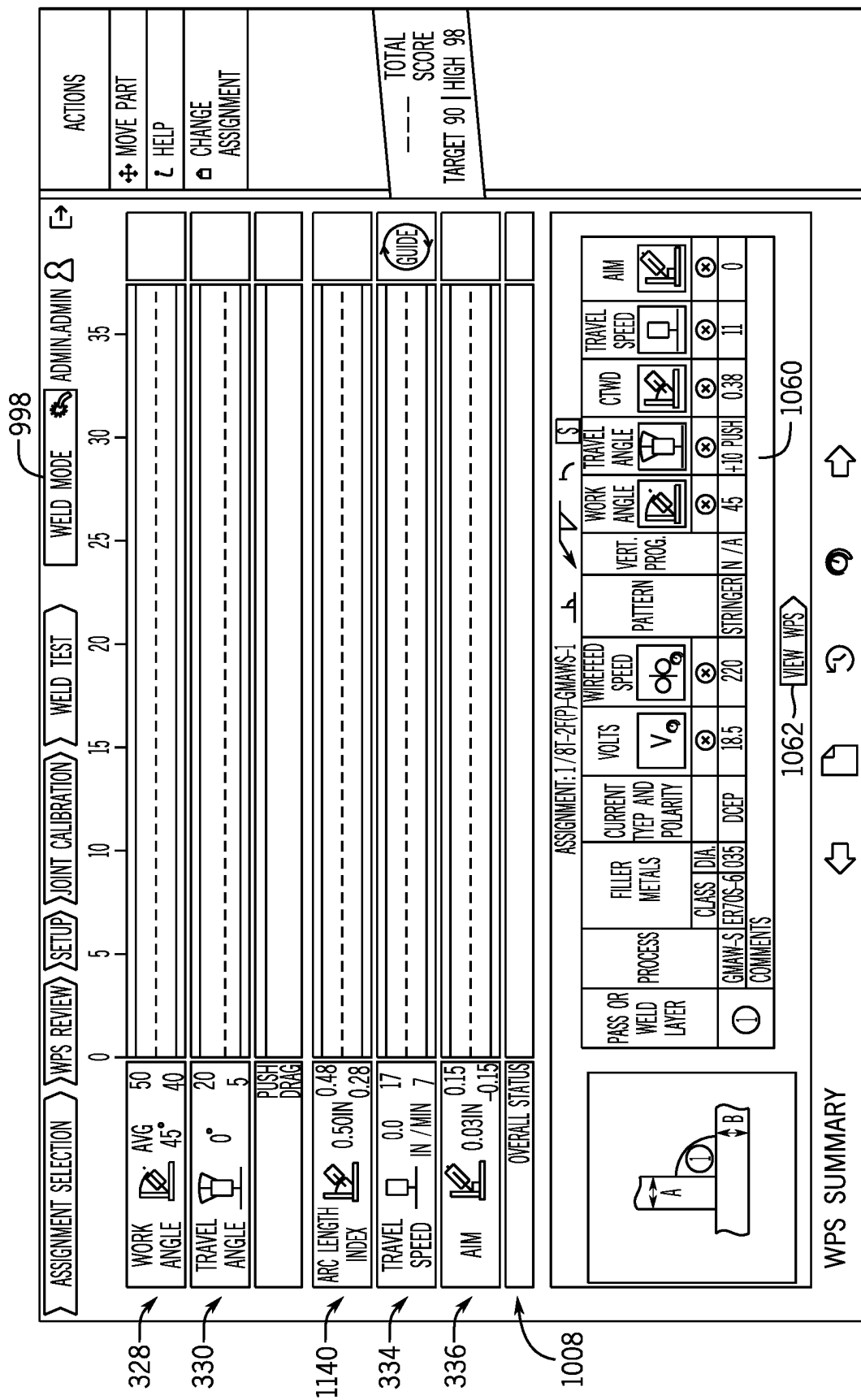
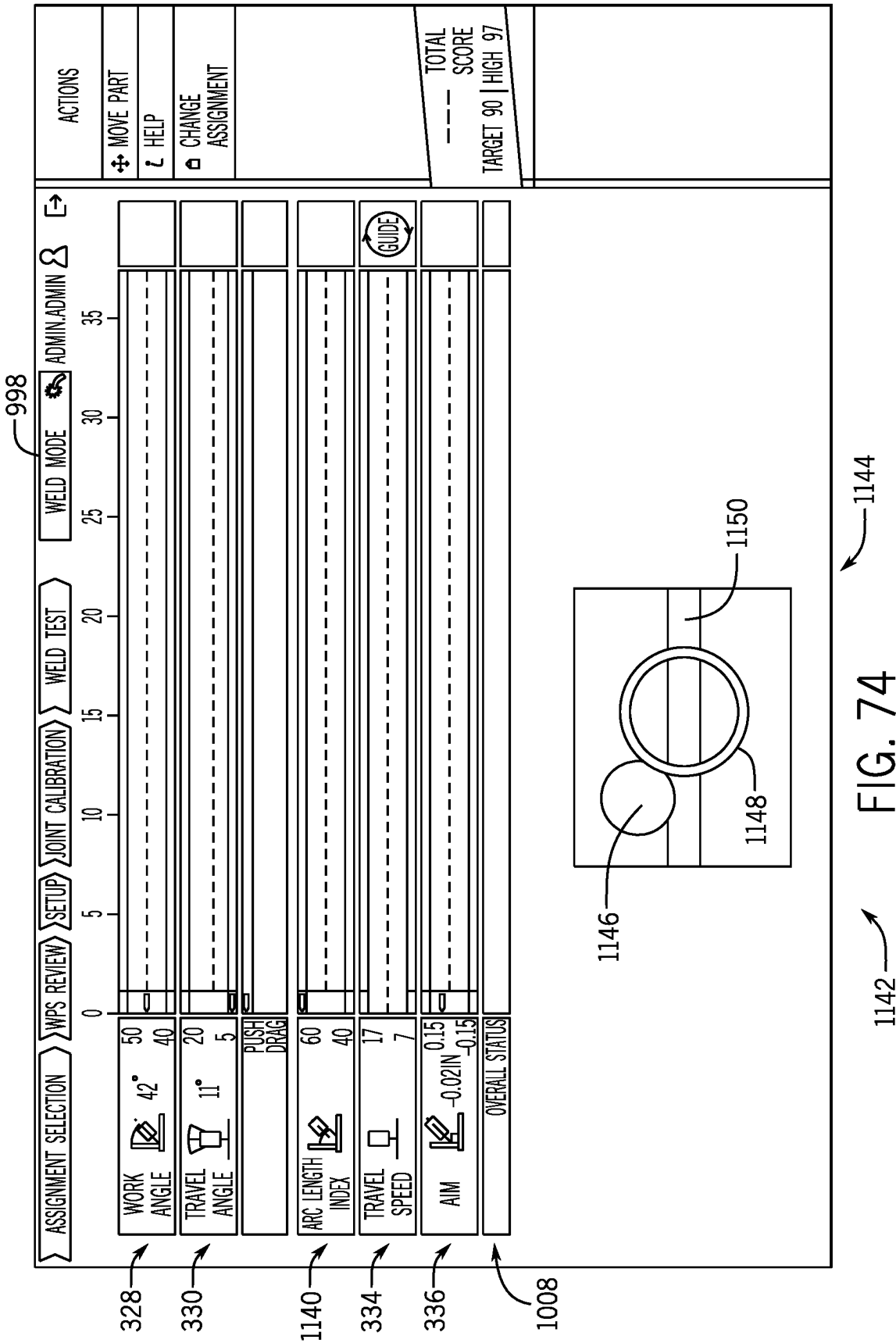


FIG. 70







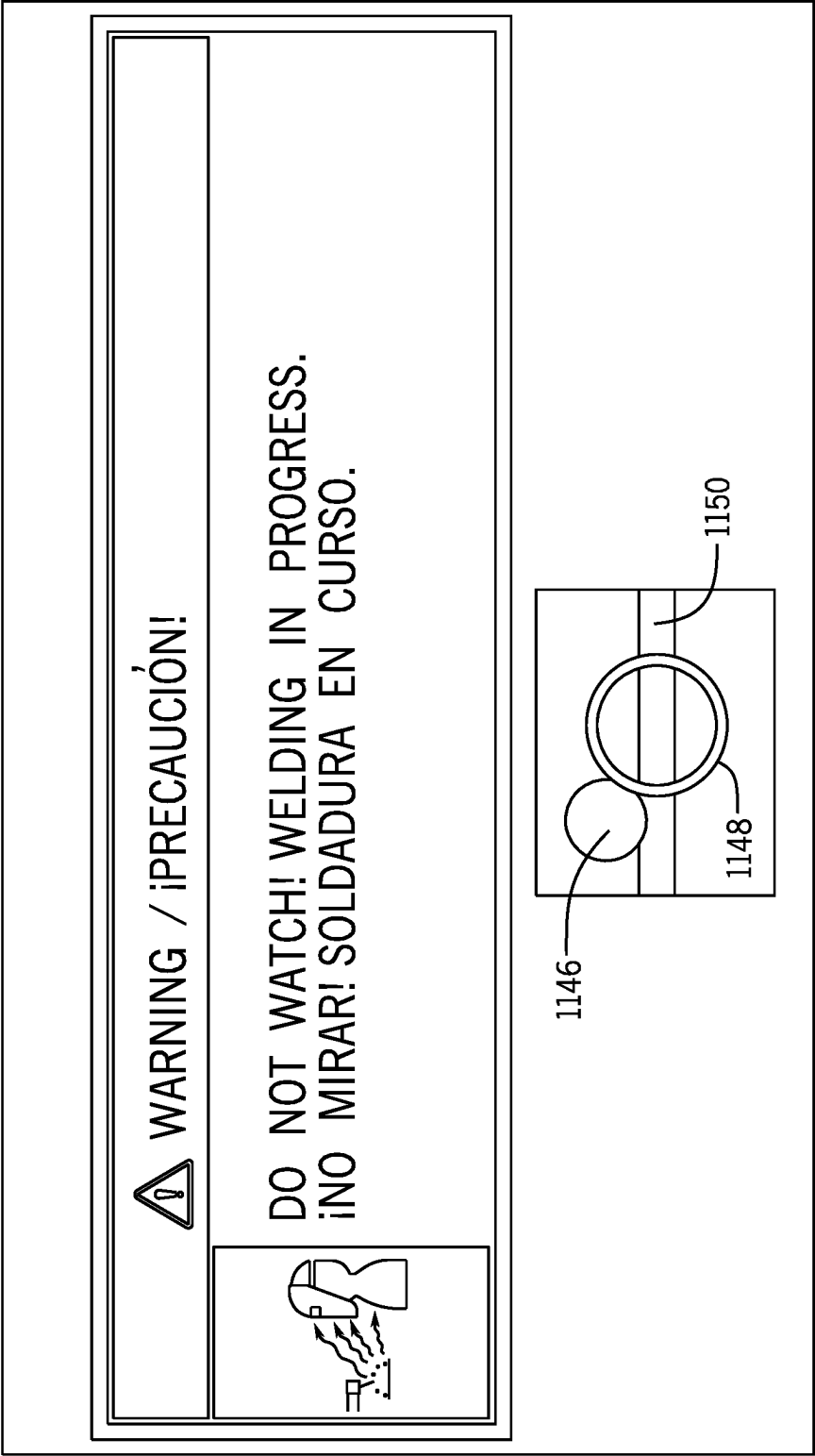
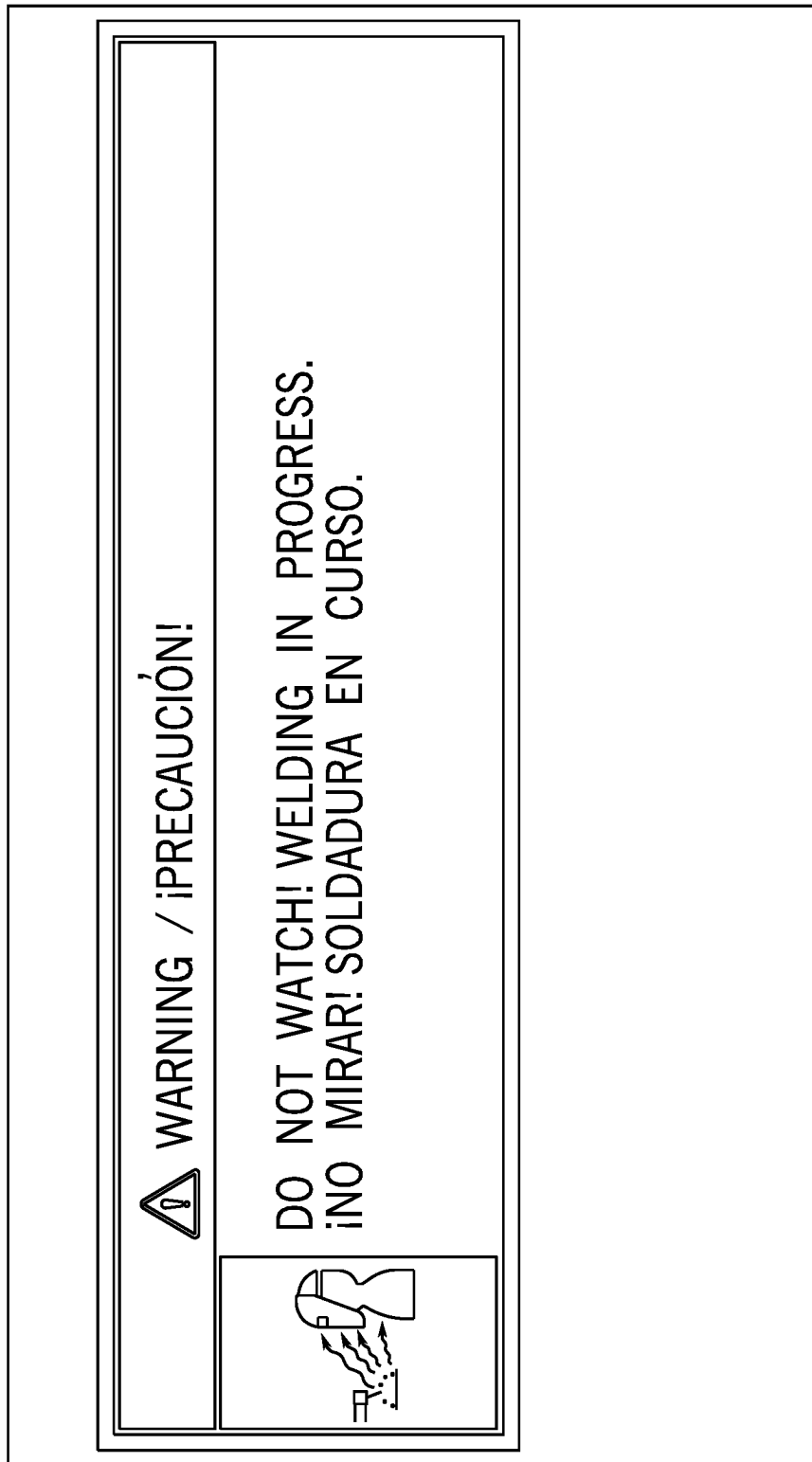
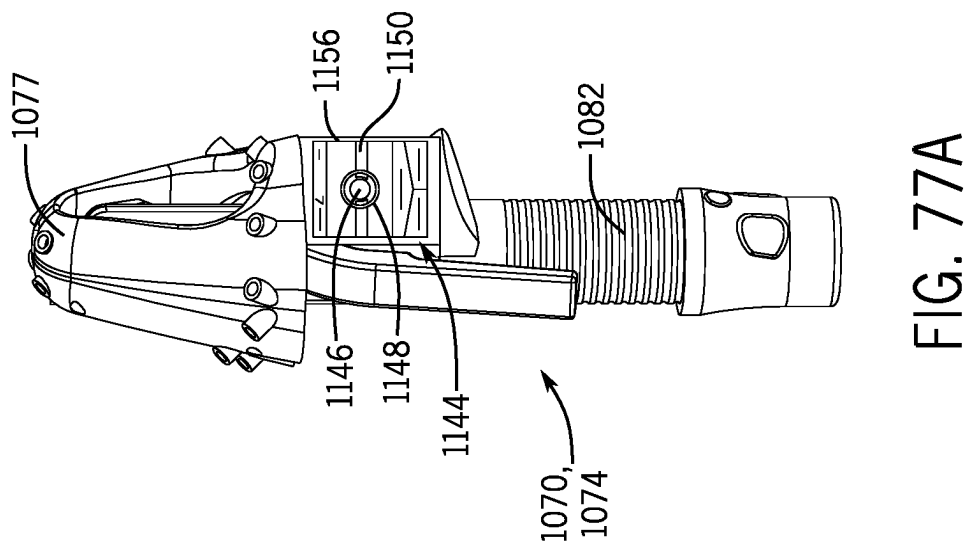
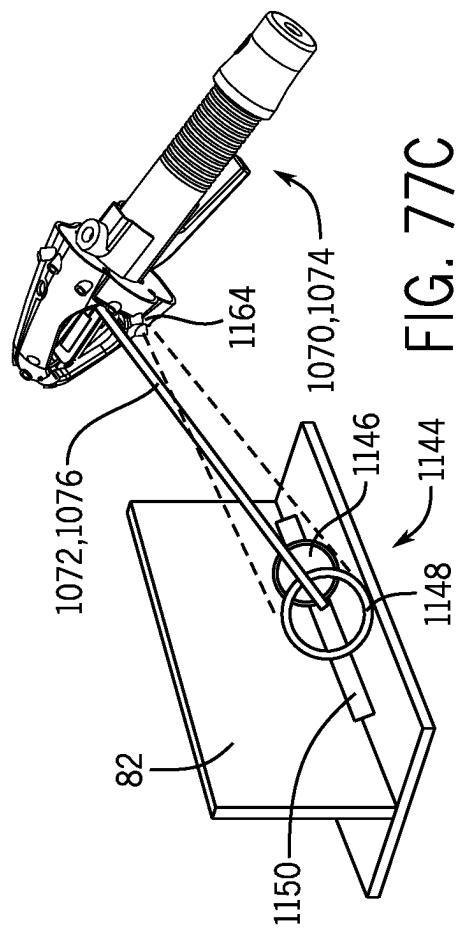
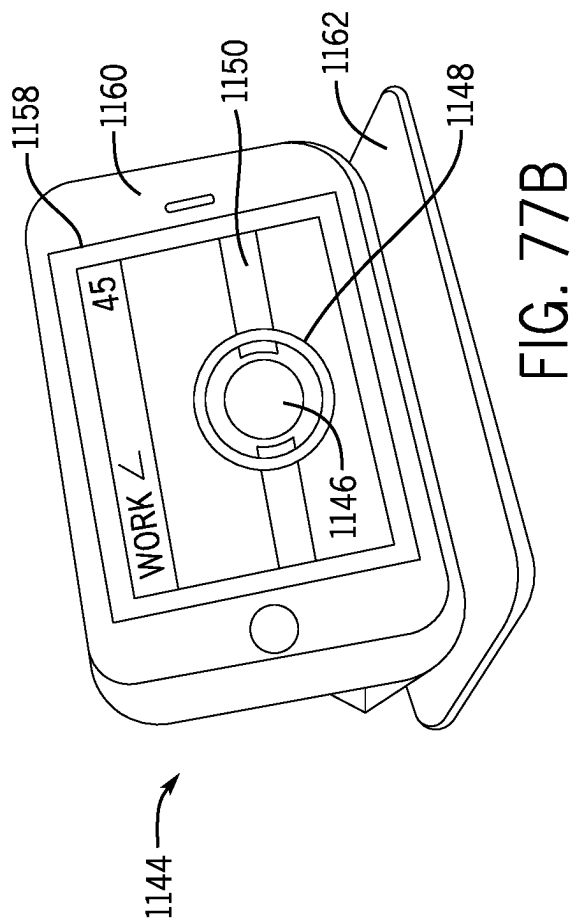


FIG. 75



1154 —→

FIG. 76



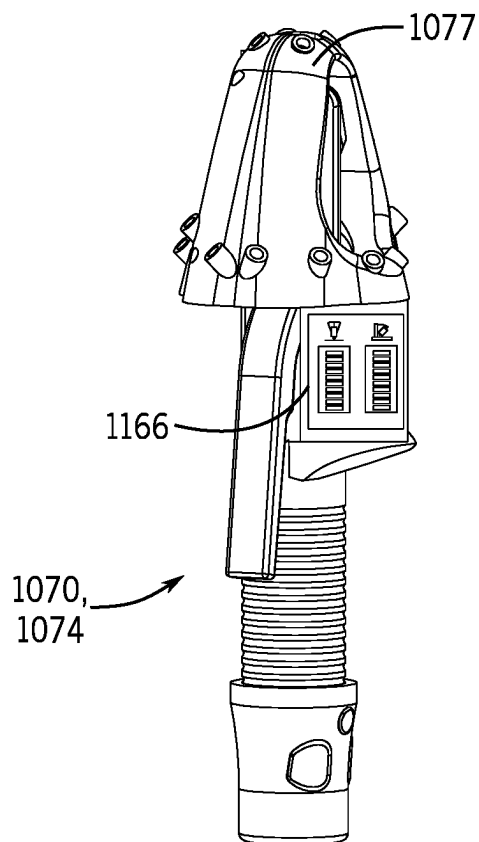


FIG. 78

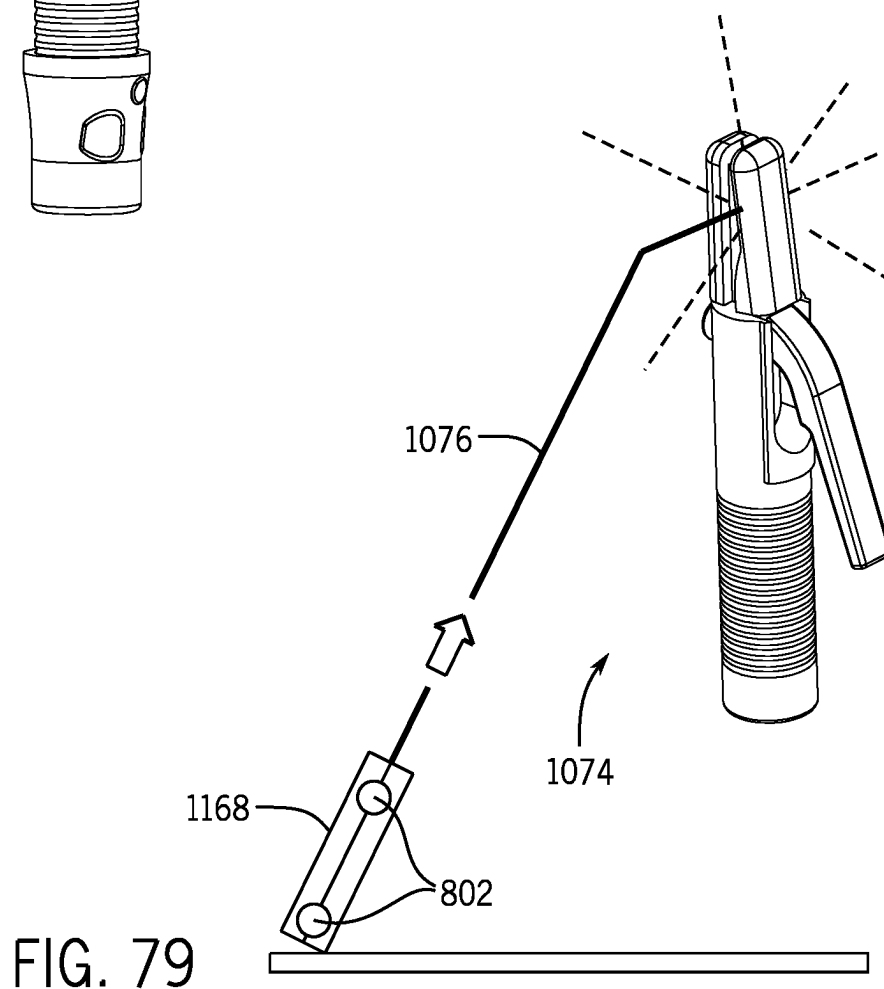


FIG. 79

1184

ASSIGNMENT SELECTION

WPS REVIEW

SETUP

JOINT CALIBRATION

WELD TEST

ADMIN ADMIN

SMARTGUN

SMARTSTINGER

1180

1176

1178

HIDE

PROCESS

1174

1170

1178

POSITION

FLAT

HORIZONTAL

VERTICAL

OVERHEAD

JOINT TYPE

BUTT JOINT

LAP JOINT

T JOINT

BASE METAL / THICKNESS

VIEWING 1-6 OF 18

1/8T-2FD-GMAWS-1

1/8T-2FP-GMAWS-1

1/8LAP-2FD-GMAWS-1

1/8LAP-2FP-GMAWS-1

1/8BUTT-1G(P)-GMAWS-1

3/16T-2FD-GMAWS-1

TOUCH TO VIEW MORE ASSIGNMENTS

NAME

PROCESS

JOINT TYPE

WELD TYPE

POSITION

HISTORY

1/8T-2FD-GMAWS-1

GMAW-S

T

FILLET

HORIZONTAL

1/8 INCH

1/8T-2FP-GMAWS-1

GMAW-S

T

FILLET

HORIZONTAL

1/8 INCH

1/8LAP-2FD-GMAWS-1

GMAW-S

LAP

FILLET

HORIZONTAL

1/8 INCH

1/8LAP-2FP-GMAWS-1

GMAW-S

LAP

FILLET

HORIZONTAL

1/8 INCH

1/8BUTT-1G(P)-GMAWS-1

GMAW-S

BUTT

SQUARE GROOVE

FLAT

1/8 INCH

3/16T-2FD-GMAWS-1

GMAW-S

T

FILLET

HORIZONTAL

3/16 INCH

FIG. 80A

1184

ASSIGNMENT SELECTION

WPS REVIEW

SETUP

JOINT CALIBRATION

WELD TEST

ADMIN ADMIN

SMARTGUN

SMARTSTINGER

1178

FILTERS

SHOW

HIDE

ELECTRODE CLASS

E6010 E6013 E7018

JOINT TYPE

BUTT JOINT LAP JOINT T JOINT

POSITION

HORIZONTAL VERTICAL

NAME

ELECTRODE CLASS

JOINT TYPE

WELD TYPE

POSITION

BASE METAL / THICKNESS

HISTORY

VIEWING 1-3 OF 3

1/8T-2F(P)-SMAW-1

E6010

T

FILLET

HORIZONTAL

1/8 INCH

1/8LAP-2F(D)-SMAW-1

E6013

LAP

FILLET

HORIZONTAL

1/8 INCH

1/8BUTT-3G(DWN)-SMAW-1

E7018

BUTT

SQUARE GROOVE

VERTICAL

1/8 INCH

ACTIONS

SYSTEM ADMIN

ASSIGNMENT MGMT

8 USER MANAGEMENT

DASHBOARD

1174

1172

FIG. 80B

HOME

SYSTEM ADMIN

SMARTGUN

SMARTSTINGER

ADMIN ADMIN

↔

1 IS THIS GUN NEW TO THIS LINEARC SYSTEM?

YES

NO

ONLY SELECT TO RESTORE COMMUNICATION WITH GUN

RESET GUN

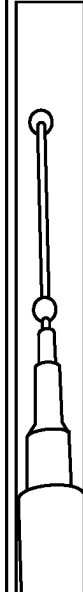
2 IF NECESSARY, CONFIRM AND UPDATE GUN AXIS CALIBRATION TOOL DIMENSIONS. USE HIGH PRECISION'S CALIPERS

18.1100

MM

SAVE DIMENSIONS

3 ATTACH AXIS CALIBRATION TIP AS SHOWN. ENSURE IT IS SCREWED IN ALL THE WAY

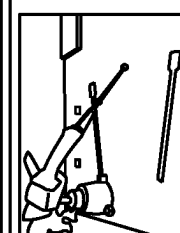
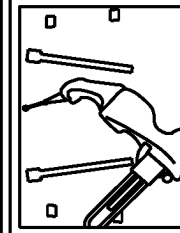


4 ENSURE ARMS ARE LOWERED FULLY INTO BACK POSITION. MOUNT THE GUN FIRMLY IN THE VISE AND POSITION ON THE TABLE AS SHOWN. ENSURE THE GUN IS COMPLETELY STILL. TOUCH CALIBRATE.

LEFT SIDE

RIGHT SIDE

TOP SIDE

LED AIM ANGLE: UNTRACKED DEGREES CALIBRATE LEFT

MOUNT THE GUN FIRMLY IN THE VISE AND POSITION ON THE TABLE AS SHOWN. THE LEFT-FACING LED SHOULD BE AIMED AT THE CAMERA SENSOR. ENSURE THE GUN IS COMPLETELY STILL.

OPTION FOR ADVANCE USERS. WITH THE GUN AXIS CALIBRATION TOOL STILL ATTACHED, PLACE THE GUN IN DIFFERENT POSITIONS AND ORIENTATIONS AROUND THE TABLE TO VERIFY THE CALIBRATION.

VERIFY GUN CALIBRATION

RESTORE SYSTEM DEFAULTS

RESTORE SYSTEM DEFAULTS

CAMERA SETTINGS

CLAMP CALIBRATION

CAMERA-TO-ARM LINKAGE ADJUSTMENT

TABLE CALIBRATION

JOINT CALIBRATION TOOL

TRACKING OPTIMIZATION

WELDING TOOL CALIBRATION

DATA BACKUP & RESTORATION

SOFTWARE UPDATES

SETTINGS

ACTIONS

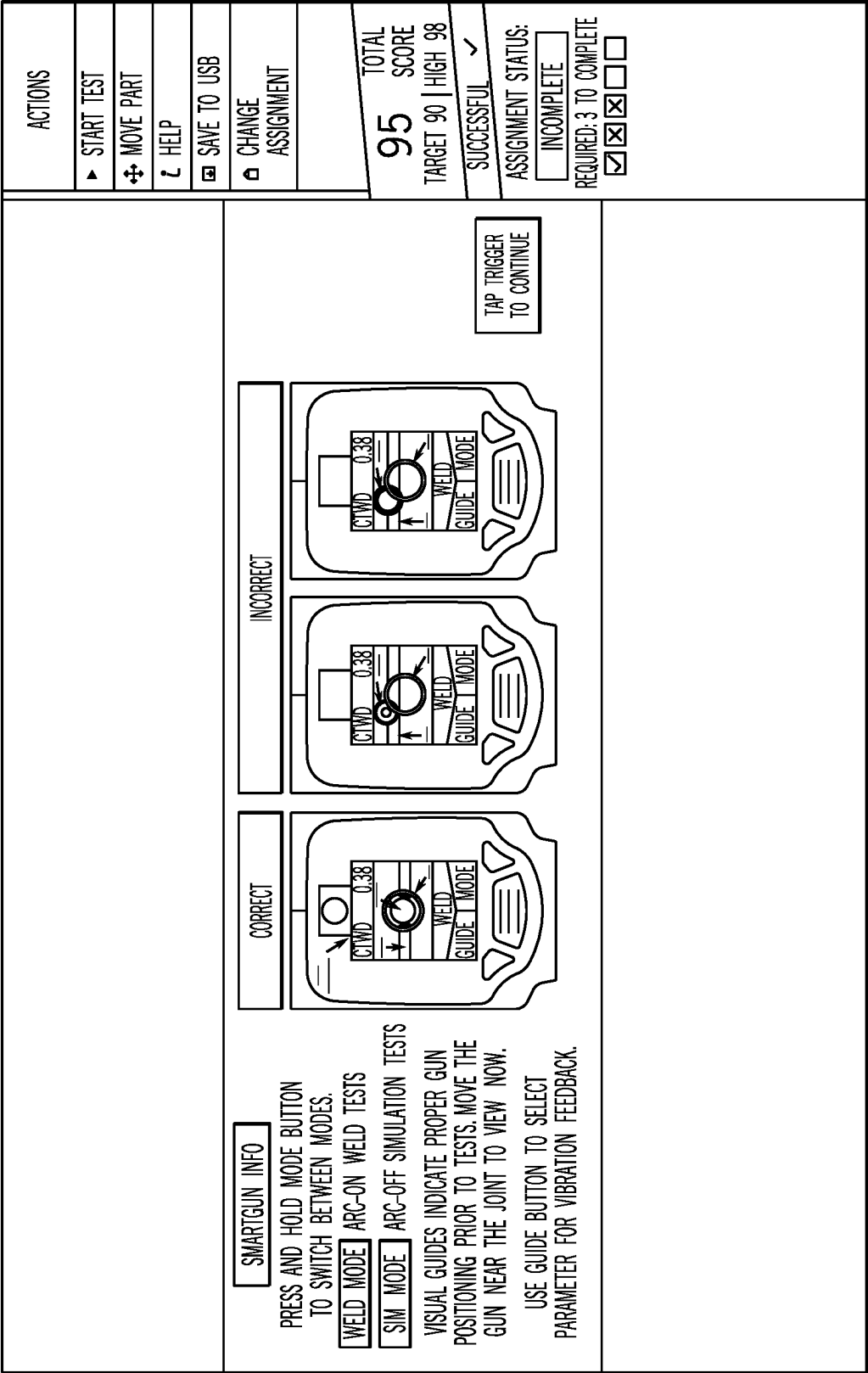
SHUT DOWN

RESTART APPLICATION

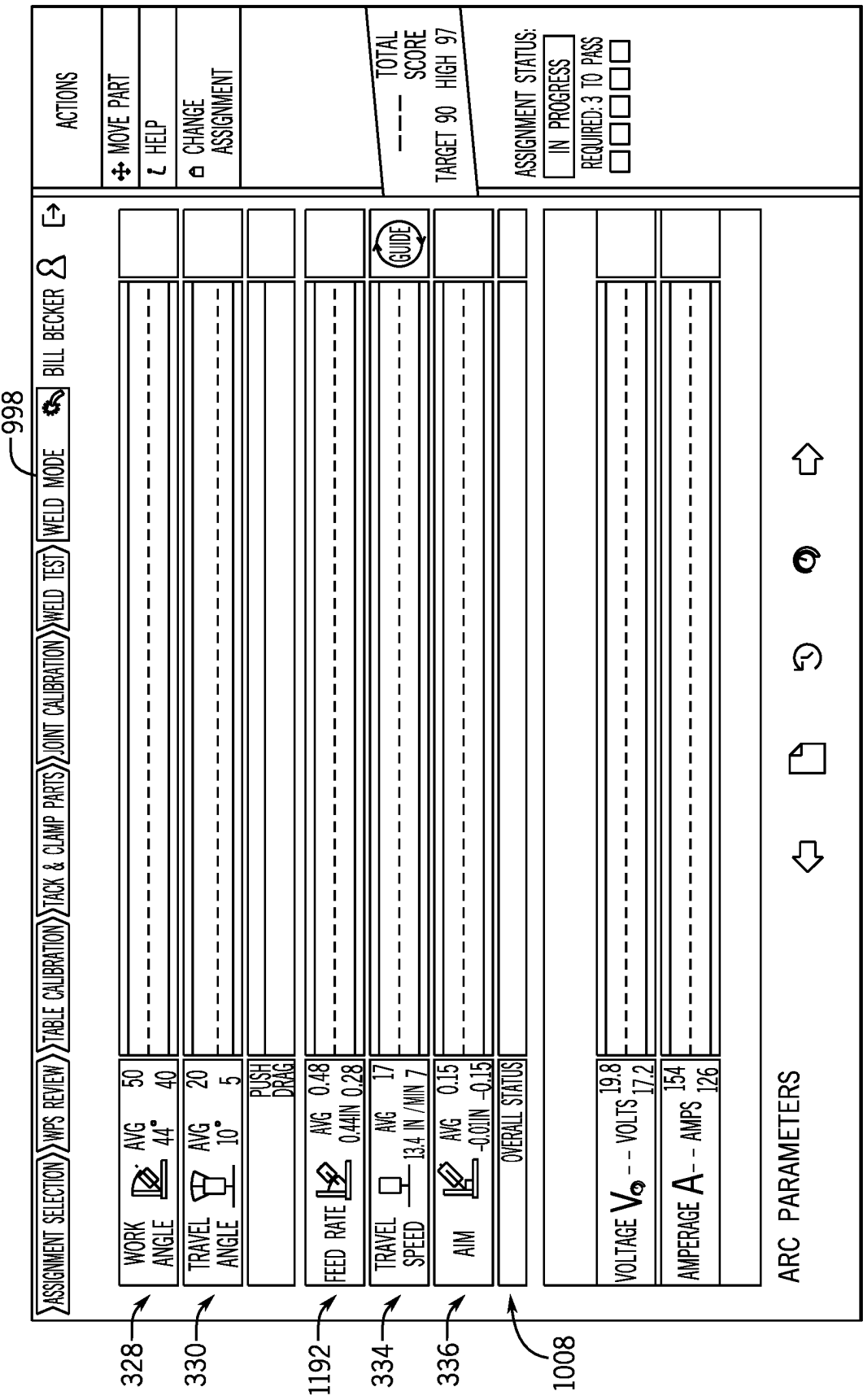
EXIT APPLICATION

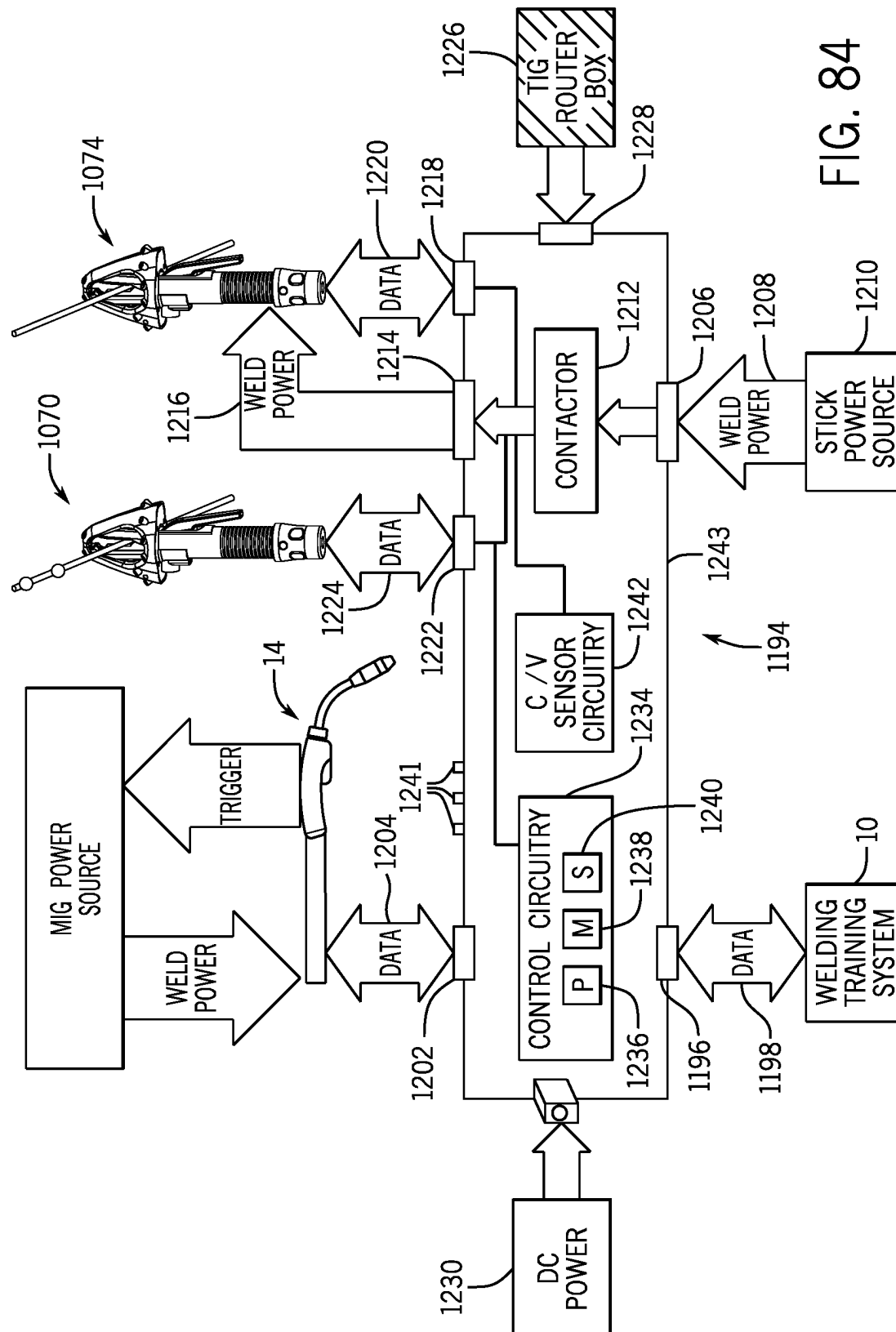
HOME

1186 — FIG. 81



1188 — FIG. 82





1244	1246	1248	1250	1252	1254	1256	1258	1260	1262	1264	1266
SCREEN COVER	PRE /MID TEST?	TRACKED?	NEAR /FAR	ARC DETECTED?	CONTACTOR	GRAPHS	VISUAL GUIDES /STATUS LED	SOUND FX	WARNING (PRIMARY SCREEN ONLY) (IMPLEMENT 5 SECOND LATCH)	START TEST?	END TEST?
CLOSED	PRE	N	-	-	OPEN	N	N	-	N	N	-
CLOSED	PRE	Y	FAR	-	OPEN	N	N	-	N	N	-
CLOSED	PRE	Y	NEAR	N	CLOSED	ARROWS	Y	-	Y	N	-
CLOSED	PRE	Y	NEAR	Y	CLOSED	GRAPHS	N	-	Y	Y	-
CLOSED	MID	N	-	N	OPEN	END TEST ANIMATION	N	-	N	-	Y
CLOSED	MID	N	-	Y	CLOSED	GRAPHS	N	LOST TRACKING	Y	-	N
CLOSED	MID	Y	FAR	N	OPEN	ANIMATION	N	-	N	-	Y
CLOSED	MID	Y	FAR	Y	CLOSED	GRAPHS	N	-	Y	-	N
CLOSED	MID	Y	NEAR	N	OPEN	END TEST ANIMATION	N	-	N	-	Y
CLOSED	MID	Y	NEAR	Y	CLOSED	GRAPHS	N	-	Y	-	N
OPEN	PRE	N	-	-	OPEN	N	N	-	N	N	-
OPEN	PRE	Y	FAR	-	OPEN	N	N	-	N	N	-
OPEN	PRE	Y	NEAR	-	OPEN	MESSAGE	N	INFO SOUND	N	N	-
OPEN	MID	-	-	-	OPEN	END TEST ANIMATION	N	-	N	-	Y

FIG. 85

1

STICK WELDING ELECTRODE HOLDER SYSTEMS AND METHODS

RELATED APPLICATIONS

This application claims priority to, and is a continuation of, U.S. patent application Ser. No. 18/099,002, filed Jan. 19, 2023, which is a continuation of U.S. patent application Ser. No. 16/814,152, filed Mar. 10, 2020, which is a continuation of U.S. patent application Ser. No. 15/211,743, filed Jul. 15, 2016, which is a Non-provisional U.S. patent application of U.S. Provisional Application No. 62/204,191, entitled “Stick Welding Electrode Holder Systems and Methods,” filed Aug. 12, 2015, which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND OF THE INVENTION

The present disclosure relates generally to welding and, more particularly, to a welding system that may be used for monitoring a weld environment and managing welding data associated with shielded metal arc welding (SMAW) electrode holders in the weld environment, such as welding data collected from the weld environment during and/or preceding welding.

Welding is a process that has increasingly become utilized in various industries and applications. Such processes may be automated in certain contexts, although a large number of applications continue to exist for manual welding operations. In both cases, such welding operations rely on a variety of types of equipment to ensure the supply of welding consumables (e.g., wire feed, shielding gas, etc.) is provided to the weld in appropriate amounts at the desired time.

In preparation for performing manual welding operations, welding operators may be trained using a welding system (e.g., a welding training system). The welding system may be designed to train welding operators with the proper techniques for performing various welding operations. Certain welding systems may use various training methods. As may be appreciated, these training systems may be expensive to acquire and operate. Accordingly, welding training institutions may only acquire a limited number of such training systems. Furthermore, certain welding systems may not adequately train welding operators to perform high quality welds.

DRAWINGS

These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

FIG. 1 is a block diagram of an embodiment of a welding system in accordance with aspects of the present disclosure;

FIG. 2 is a block diagram of an embodiment of portions of the welding system of FIG. 1 in accordance with aspects of the present disclosure;

FIG. 3 is a perspective view of an embodiment of the welding stand of FIG. 1 in accordance with aspects of the present disclosure;

FIG. 4 is a perspective view of an embodiment of a calibration device in accordance with aspects of the present disclosure;

2

FIG. 5 is a perspective view of an embodiment of a fixture assembly in accordance with aspects of the present disclosure;

FIG. 6 is a perspective view of an embodiment of a vertical arm assembly of the welding stand of FIG. 1 in accordance with aspects of the present disclosure;

FIG. 7 is a perspective view of an embodiment of an overhead welding arm assembly in accordance with aspects of the present disclosure;

FIG. 8 is a block diagram of an embodiment of welding software having multiple training modes in accordance with aspects of the present disclosure;

FIG. 9 is a block diagram of an embodiment of a virtually reality mode of welding software in accordance with aspects of the present disclosure;

FIG. 10 is an embodiment of a method for integrating training results data in accordance with aspects of the present disclosure;

FIG. 11 is an embodiment of a chart illustrating multiple sets of welding data for a welding operator in accordance with aspects of the present disclosure;

FIG. 12 is an embodiment of a chart illustrating welding data for a welder compared to welding data for a class in accordance with aspects of the present disclosure;

FIG. 13 is a block diagram of an embodiment of a data storage system (e.g., cloud storage system) for storing certification status data in accordance with aspects of the present disclosure;

FIG. 14 is an embodiment of a screen illustrating data corresponding to a weld in accordance with aspects of the present disclosure;

FIG. 15 is a block diagram of an embodiment of a welding instructor screen of welding software in accordance with aspects of the present disclosure;

FIG. 16 is an embodiment of a method for weld training using augmented reality in accordance with aspects of the present disclosure;

FIG. 17 is an embodiment of another method for weld training using augmented reality in accordance with aspects of the present disclosure;

FIG. 18 is a block diagram of an embodiment of a welding tool in accordance with aspects of the present disclosure;

FIG. 19 is an embodiment of a method for providing vibration feedback to a welding operator using a welding tool in accordance with aspects of the present disclosure;

FIG. 20 is a graph of an embodiment of two patterns each including a different frequency for providing vibration feedback to a welding operator in accordance with aspects of the present disclosure;

FIG. 21 is a graph of an embodiment of two patterns each including a different modulation for providing vibration feedback to a welding operator in accordance with aspects of the present disclosure;

FIG. 22 is a graph of an embodiment of two patterns each including a different amplitude for providing vibration feedback to a welding operator in accordance with aspects of the present disclosure;

FIG. 23 is a perspective view of an embodiment of a welding tool having spherical markers that may be used for tracking the welding tool in accordance with aspects of the present disclosure;

FIG. 24 is perspective view of an embodiment of the welding tool, taken along line 24-24 of FIG. 23 in accordance with aspects of the present disclosure;

FIG. 25 is a top view of an embodiment of the welding tool and visual markers in accordance with aspects of the present disclosure;

3

FIG. 26 is an embodiment of a method for displaying on a display of a welding tool a welding parameter in relation to a threshold in accordance with aspects of the present disclosure;

FIG. 27 is an embodiment of a set of screenshots of a display of a welding tool for showing a welding parameter in relation to a threshold in accordance with aspects of the present disclosure;

FIG. 28 is an embodiment of a method for tracking a welding tool in a welding system using at least four markers in accordance with aspects of the present disclosure;

FIG. 29 is an embodiment of a method for detecting the ability for a processor to communicate with a welding tool in accordance with aspects of the present disclosure;

FIG. 30 is an embodiment of a method for calibrating a curved weld joint that may be used with a welding system in accordance with aspects of the present disclosure;

FIG. 31 is a diagram of an embodiment of a curved weld joint in accordance with aspects of the present disclosure;

FIG. 32 is a diagram of an embodiment of a curved weld joint and a marking tool in accordance with aspects of the present disclosure;

FIG. 33 is an embodiment of a method for tracking a multi-pass welding operation in accordance with aspects of the present disclosure;

FIG. 34 is a perspective view of an embodiment of a welding stand in accordance with aspects of the present disclosure;

FIG. 35 is a cross-sectional view of an embodiment of a welding surface of the welding stand of FIG. 34 in accordance with aspects of the present disclosure;

FIG. 36 is a cross-sectional view of an embodiment of a sensing device having a removable cover in accordance with aspects of the present disclosure;

FIG. 37 is a perspective view of an embodiment of a calibration tool in accordance with aspects of the present disclosure;

FIG. 38 is a perspective view of the calibration tool of FIG. 37 having an outer cover removed in accordance with aspects of the present disclosure;

FIG. 39 is a side view of an embodiment of a pointed tip of a calibration tool in accordance with aspects of the present disclosure;

FIG. 40 is a side view of an embodiment of a rounded tip of a calibration tool in accordance with aspects of the present disclosure;

FIG. 41 is a side view of an embodiment of a rounded tip of a calibration tool having a small pointed tip in accordance with aspects of the present disclosure;

FIG. 42 is an embodiment of a method for detecting a calibration point in accordance with aspects of the present disclosure;

FIG. 43 is an embodiment of a method for determining a welding score based on a welding path in accordance with aspects of the present disclosure;

FIG. 44 is an embodiment of a method for transitioning between welding modes using a user interface of a welding tool in accordance with aspects of the present disclosure;

FIG. 45 is an embodiment of a remote welding training system in accordance with aspects of the present disclosure;

FIG. 46 is an embodiment of a dashboard page with welding data from different operators, in accordance with aspects of the present disclosure;

FIG. 47 is an embodiment of a welding system with depth sensors and a local positioning system, in accordance with aspects of the present disclosure;

4

FIG. 48 is an embodiment of a method of controlling visual markers of the welding tool to track the movement and position of the welding tool, in accordance with aspects of the present disclosure;

FIG. 49 is a cross-sectional view of a base component with visual markers, in accordance with aspects of the present disclosure;

FIG. 50 is a perspective view of an embodiment of the arms and clamp assembly of the welding stand, in accordance with aspects of the present disclosure;

FIG. 51 is a top view of an embodiment of a mount of the clamp assembly of FIG. 50, taken along line 51-51, in accordance with aspects of the present disclosure;

FIG. 52 is perspective view of an embodiment of a calibration block coupled to the clamp assembly of FIG. 50, in accordance with aspects of the present disclosure;

FIG. 53 is an embodiment of a method for the set up of the arms of the welding stand for an out of position welding assignment, in accordance with aspects of the present disclosure;

FIG. 54 is an embodiment of a method for the selection and execution of a multi-pass welding assignment with the welding system, in accordance with aspects of the present disclosure;

FIG. 55 is an embodiment of a screen illustrating data, including arc parameters, corresponding to a weld in accordance with aspects of the present disclosure;

FIG. 56 is an embodiment of a screen illustrating data corresponding to a weld test for which an arc has not been detected in accordance with aspects of the present disclosure;

FIG. 57 is an embodiment of a screen illustrating assignment development routines in accordance with aspects of the present disclosure;

FIG. 58 is an embodiment of a screen illustrating properties relating to a welding procedure in accordance with aspects of the present disclosure;

FIG. 59 is an embodiment of a screen illustrating data corresponding to a simulated weld in accordance with aspects of the present disclosure;

FIG. 60 is an embodiment of a screen illustrating data corresponding to a weld prior to initiation of the weld in accordance with aspects of the present disclosure;

FIG. 61 is an embodiment of a screen illustrating a summary of weld test parameters in accordance with aspects of the present disclosure;

FIG. 62 is an embodiment of a screen illustrating data, including arc parameters, corresponding to a weld during a weld test in accordance with aspects of the present disclosure;

FIG. 63 is an embodiment of a screen illustrating data, including heat input, corresponding to a weld in accordance with aspects of the present disclosure;

FIGS. 64A and 64B illustrate an embodiment of a simulation stick welding electrode holder in accordance with aspects of the present disclosure;

FIGS. 65A and 65B illustrate an embodiment of an actual stick welding electrode holder in accordance with aspects of the present disclosure;

FIGS. 66A and 66B illustrate embodiments of a simulation stick welding electrode holder and an actual stick welding electrode holder, respectively, in accordance with aspects of the present disclosure;

FIG. 67 is an embodiment of a stick electrode holding assembly of a simulation stick welding electrode holder in accordance with aspects of the present disclosure;

5

FIG. 68A is an embodiment of an actual stick welding electrode holder having a plurality of discrete stick electrode holding slots in accordance with aspects of the present disclosure;

FIG. 68B is an embodiment of a jaw of the actual stick welding electrode holder of FIG. 68A, illustrating the plurality of discrete stick electrode holding slots in accordance with aspects of the present disclosure;

FIG. 68C is an embodiment of an on-screen prompt relating to the use of the plurality of discrete stick electrode holding slots of FIG. 68B, in accordance with aspects of the present disclosure;

FIGS. 69A and 69B illustrate embodiments of button panels of a simulation stick welding electrode holder and an actual stick welding electrode holder, respectively, in accordance with aspects of the present disclosure;

FIG. 70 is an embodiment of a stick welding electrode holder having a button panel on a handle in accordance with aspects of the present disclosure;

FIG. 71 is an embodiment of a stick welding electrode holder having a button panel on an outer support structure in accordance with aspects of the present disclosure;

FIGS. 72A and 72B illustrate embodiments of status indicators of a simulation stick welding electrode holder and an actual stick welding electrode holder, respectively, in accordance with aspects of the present disclosure;

FIG. 73 is an embodiment of a screen illustrating parameters corresponding to a stick welding process in accordance with aspects of the present disclosure;

FIG. 74 is an embodiment of a screen illustrating a targeting graphic (e.g., visual guides) for a stick welding process in accordance with aspects of the present disclosure;

FIG. 75 is an embodiment of a screen illustrating the targeting graphic (e.g., visual guides) just prior to initiation of the stick welding process in accordance with aspects of the present disclosure;

FIG. 76 is an embodiment of a screen illustrating removal of the targeting graphic (e.g., visual guides) during performance of the stick welding process in accordance with aspects of the present disclosure;

FIG. 77A is an embodiment of a stick welding electrode holder having the targeting graphic (e.g., visual guides) in accordance with aspects of the present disclosure;

FIG. 77B is an embodiment of a handheld device having the targeting graphic (e.g., visual guides) in accordance with aspects of the present disclosure;

FIG. 77C is an embodiment of a stick welding electrode holder having a projection system configured to project the targeting graphic (e.g., visual guides) onto a workpiece in accordance with aspects of the present disclosure;

FIG. 78 is an embodiment of a stick welding electrode holder having graphical range indicators in accordance with aspects of the present disclosure;

FIG. 79 is an embodiment of a position calibration device configured to slip onto a tip of a stick welding electrode in accordance with aspects of the present disclosure;

FIGS. 80A and 80B are embodiments of screens illustrating assignment lists for an actual stick welding process and a simulated stick welding process, respectively, in accordance with aspects of the present disclosure;

FIG. 81 is an embodiment of a screen illustrating a calibration procedure for a stick welding electrode holder in accordance with aspects of the present disclosure;

FIG. 82 is an embodiment of a screen illustrating additional help screens in accordance with aspects of the present disclosure;

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FIG. 83 is an embodiment of a screen illustrating parameters corresponding to a stick welding process (including feed rate) in accordance with aspects of the present disclosure;

FIG. 84 is a schematic diagram of an embodiment of a connection box for use with the welding training system in accordance with aspects of the present disclosure; and

FIG. 85 is a tabular summary of an embodiment of a state machine for the connection box of FIG. 84 in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

FIG. 1 is a block diagram of an embodiment of one or more welding systems 10. As used herein, a welding system may include any suitable welding related system, including, but not limited to, a welding training system, a live welding system, a remote welding training system (e.g., helmet training system), a simulated welding system, a virtual reality welding system, and so forth. For example, the welding system 10 may include, but is not limited to, a LiveArc™ Welding Performance Management System, which is a weld training system available from Miller Electric of Appleton, WI. The welding system 10 may include a welding stand 12 for providing support for various training devices. For example, the welding stand 12 may be configured to support a welding surface, a workpiece 82, a fixture, one or more training arms, and so forth. The welding system 10 includes one or more welding tools 14 that may be used by a welding operator (e.g., welding student) to perform welding operations (e.g., training operations). As described in greater detail below, in certain embodiments, the welding tool(s) 14 may be configured with a user interface configured to receive inputs from the welding operator, control circuitry configured to process the inputs, and a communication interface configured to provide the inputs to another device. Furthermore, in certain embodiments, the welding tool 14 may include one or more displays and/or indicators to provide data to the welding operator. In certain embodiments, the welding tool 14 may be a fully functional welding torch or electrode holder capable of generating a live arc between welding wire or a welding electrode and a workpiece 82. In contrast, in other embodiments, the welding tool 14 may be a simulation welding torch or electrode holder that is not capable of generating a live arc between welding wire or a welding electrode and a workpiece 82, but rather may be configured to simulate the generation of a live arc between welding wire or a welding electrode and a workpiece 82.

Moreover, in certain embodiments, the welding system 10 includes one or more sensing devices 16 (e.g., sensor, sensing assembly, and so forth) used to sense a position of one or more welding devices and/or to sense an orientation of one or more welding devices. For example, the sensing device 16 may be used to sense a position and/or an orientation of the welding stand 12, the welding tool 14, a welding surface, the workpiece 82, a fixture, one or more training arms, the operator, an identification token, and so forth. The one or more sensing devices 16 may include any suitable sensing device, such as a motion sensing device or a motion tracking device. Furthermore, the one or more sensing devices 16 may include one or more cameras, such as one or more infrared cameras, one or more visible spectrum cameras, one or more high dynamic range (HDR) cameras, and so forth. Additionally, or in the alternative, the one or more sensing devices 16 may include one or more depth sensors to determine relative distances between the

respective depth sensors and an object (e.g., welding tool **14**, workpiece **82**, operator, and so forth). The one or more sensing devices **16** may be positioned in various locations about the welding environment of the welding system **10**, thereby enabling some sensing devices **16** to monitor the welding environment (e.g., track movement of an object) when other sensing devices **16** are obscured. For example, a sensing device **16** (e.g., camera, depth sensor) integrated with a welding helmet **41** may facilitate tracking the position, orientation, and/or movement of the welding tool **14** relative to the workpiece **82** when the welding tool **14** is at least partially obscured from other sensing devices **16** by the workpiece **82** or the operator. Furthermore, a sensing device **16** (e.g., accelerometer) integrated with the welding tool **14** may facilitate tracking the position, orientation, and/or movement of the welding tool **14** relative to the workpiece **82** when the welding tool **14** is at least partially obscured from other sensing devices **16** (e.g., cameras, depth sensors) by the workpiece **82** or the operator.

The one or more sensing devices **16** are communicatively coupled to a computer **18**. The one or more sensing devices **16** are configured to provide data (e.g., image data, acoustic data, sensed data, six degrees of freedom (6DOF) data, etc.) to the computer **18**. Furthermore, the one or more sensing devices **16** may be configured to receive data (e.g., configuration data, setup data, commands, register settings, etc.) from the computer **18**. The computer **18** includes one or more processors **20**, memory devices **22**, and storage devices **24**. The computer **18** may include, but is not limited to, a desktop, a laptop, a tablet, a mobile device, a wearable computer, or any combination thereof. The processor(s) **20** may be used to execute software, such as welding software, image processing software, sensing device software, and so forth. Moreover, the processor(s) **20** may include one or more microprocessors, such as one or more “general purpose” microprocessors, one or more special-purpose microprocessors and/or application specific integrated circuits (ASICs), or some combination thereof. For example, the processor(s) **20** may include one or more reduced instruction set (RISC) processors.

The storage device(s) **24** (e.g., nonvolatile storage) may include ROM, flash memory, a hard drive, or any other suitable optical, magnetic, or solid-state storage medium, or a combination thereof. The storage device(s) **24** may store data (e.g., data corresponding to a welding operation, video and/or parameter data corresponding to a welding operation, data corresponding to an identity and/or a registration number of the operator, data corresponding to past operator performance, etc.), instructions (e.g., software or firmware for the welding system, the one or more sensing devices **16**, etc.), and any other suitable data. As will be appreciated, data that corresponds to a welding operation may include a video recording of the welding operation, a simulated video, an orientation of the welding tool **14**, a position of the welding tool **14**, a work angle, a travel angle, a distance between a contact tip of the welding tool **14** and a workpiece, a travel speed, an aim, a voltage, a current, a traversed path, a discontinuity analysis, welding device settings, and so forth.

The memory device(s) **22** may include a volatile memory, such as random access memory (RAM), and/or a nonvolatile memory, such as read-only memory (ROM). The memory device(s) **22** may store a variety of information and may be used for various purposes. For example, the memory device(s) **22** may store processor-executable instructions (e.g., firmware or software) for the processor(s) **20** to execute, such as instructions for a welding training simulation, for the

one or more sensing devices **16**, and/or for an operator identification system **43**. In addition, a variety of control regimes for various welding processes, along with associated settings and parameters may be stored in the storage device(s) **24** and/or memory device(s) **22**, along with code configured to provide a specific output (e.g., initiate wire feed, enable gas flow, capture welding current data, detect short circuit parameters, determine amount of spatter, etc.) during operation. The welding power supply **28** may be used to provide welding power to a live-arc welding operation, and the wire feeder **30** may be used to provide welding wire to the live-arc welding operation.

In certain embodiments, the welding system **10** includes a display **32** for displaying data and/or screens associated with welding (e.g., to display data corresponding to a welding software). For example, the display **32** may provide a graphical user interface to a welding operator (e.g., welding instructor, welding student). The graphical user interface may provide various screens to enable the welding instructor to organize a class, provide assignments to the class, analyze assignments performed by the class, provide assignments to an individual, analyze assignments performed by the individual, add, change, and/or delete parameters for a welding assignment, and so forth. Furthermore, the graphical user interface may provide various screens to enable a welding operator (e.g., welding student) to perform a welding assignment, view results from prior welding assignments, and so forth. In certain embodiments, the display **32** may be a touch screen display configured to receive touch inputs, and to provide data corresponding to the touch inputs to the computer **18**.

In certain embodiments, an external display **34** may be coupled to the computer **18** to enable an individual located remotely from the welding system **10** to view data corresponding to the welding system **10**. Furthermore, in certain embodiments, a network device **36** may be coupled to the computer **18** to enable the computer **18** to communicate with other devices connected to the Internet or another network **38** (e.g., for providing test results to another device and/or for receiving test results from another device). For example, the network device **36** may enable the computer **18** to communicate with an external welding system **40**, a production welding system **42**, a remote computer **44**, and/or a data storage system (e.g., cloud storage system) **318**. As may be appreciated, the welding system **10** described herein may be used to train welding students in a cost effective manner. In some embodiments, the one or more welding systems **10** may include a helmet **41** having a display **32** and one or more sensing devices **16**, such as optical or acoustic sensing devices. As described in detail below, the helmet **41** is communicatively coupled to the computer **18**, and the helmet **41** may facilitate welding training and/or welding monitoring without the welding stand **12**. In some embodiments, the one or more sensing devices **16** integrated with the helmet **41** may facilitate welding training and/or welding monitoring without separate sensing devices **16** external to the helmet **41**. Furthermore, the welding system **10** is configured to integrate real welding with simulated welding in a manner that prepares welding students for high quality production welding.

In certain embodiments, an operator identification system **43** may be coupled to the computer **18** to enable an operator utilizing the welding system **10** to be identified. The operator identification system **43** utilizes one or more types of operator information (e.g. identifiers) to identify the operator. Operator information may include, but is not limited to, a resettable identifier **45** (e.g., password, motion sequence,

operator-performed action), a biometric identifier 47 (e.g., retinal scan, fingerprint, palm print, facial profile, voice profile, inherent operator trait), information based at least in part on a biometric identifier 47, a token 49 (e.g., key, key fob, radio frequency identification (RFID) tag, passcard, barcode, physical identifier), or any combination thereof. Additionally, or in the alternative, an instructor or manager may provide an input to the operator identification system 43 to verify the identity of the operator, thereby authorizing the operator for the welding session (e.g., welding assignment) and the associated weld data. That is, the identification of an operator may involve one or more steps, such as operator identification via information received from the operator, and operator verification via information received from the instructor and/or manager of the operator. In some embodiments, the operator identification system 43 may utilize the one or more sensing devices 16 to facilitate operator identification. For example, a camera or microphone of the welding system 10 may receive the biometric identifier 47. Moreover, the operator identification system 43 may have an input device 51 (e.g., keypad, touch screen, retinal scanner, fingerprint sensor, camera, microphone, barcode scanner, radio transceiver, and so forth) configured to receive the one or more types of operator identification information.

The operator identification system 43 may identify the operator prior to performing a weld process (e.g., live process, training process, simulated process, virtual reality process) or after performing the weld process. In some embodiments, the operator identification system 43 may enable or lock out an operator from utilizing the welding system 10 based on the one or more identifiers received via the input device 51. For example, the operator identification system 43 may lock out a first operator (e.g., student) from utilizing the welding system 10 until the operator identification system 43 receives a first input from the first operator that may identify the first operator. In some embodiments, the welding system 10 may enable the first operator to perform a welding session with the welding system 10 without verification of the identity of the first operator; however, the welding system 10 may store and/or transmit the welding data associated with such a welding session only upon verification of the identity of the first operator based at least in part on a second input from a second operator (e.g., instructor, administrator). That is, the operator identification system 43 may disable the storage or transmission of the welding data associated with a welding session until the identity of the first operator that performed the welding session is verified by the second operator. Moreover, some embodiments of the welding system 10 may lock out the first operator from utilizing the welding system until a second input is received from the second operator that verifies the identity of the first operator, which was preliminarily determined based on the first input from the first operator. In some embodiments, the operator identification system 43 may identify the operator during a weld process, such as via an identifying characteristic of an operator during the weld process. For example, a first operator may hold the welding tool 14 differently than a second operator, and a sensing device 16 (e.g., camera) coupled to the operator identification system 43 may facilitate distinguishing the first operator from the second operator. Additionally, or in the alternative, the operator identification system 43 may include a sensor (e.g., fingerprint scanner, camera, microphone) on the welding tool 14 and/or the helmet 41. In some embodiments, an instructor and/or a manager may confirm upon completion of a weld process that the identified operator performed the weld process.

The operator identification system 43 may communicate with the computer 18 to determine the identity of the operator utilizing the received identification information. In some embodiments, the computer 18 may communicate with the network 38 and/or a remote computer 44 to determine the identity of the operator. The computer 18 may control the display 32 to display at least some of the information associated with the operator upon identification of the operator. For example, the display 32 may present the name, a photo, registration number, experience level, or any combination thereof. In some embodiments, the operator identification system 43 may be utilized with one or more welding systems 10.

The computer 18 may receive welding data (e.g., welding parameters, arc parameters) corresponding to a welding session (e.g., welding assignment) during and/or after the respective welding session is performed by the operator. The computer 18 may receive the welding data from the network 38, the one or more sensing devices 16, the welding tool 14, the welding power supply 28, the wire feeder 30, or the helmet 41, or any combination thereof. Additionally, or in the alternative, the computer 18 may associate the received welding data with the identity of the operator, such as via a registration number unique to the operator, the operator's name, and/or a photograph of the operator. Moreover, the computer 18 may transmit the associated welding data and identity of the operator (e.g., registration number) to a data storage system within the welding system 10 or located remotely via the network 38. Association of the welding data with the identity of the operator (e.g., via the registration number) enables significantly more than the collection of unassociated welding data from operators. That is, association of the welding data with a registration number unique to the operator enables someone (e.g., the operator, instructor, manager) that is either local or remote from the operator to track the performance, progress, and skills of the operator over time via the registration number.

FIG. 2 is a block diagram of an embodiment of portions of the welding system 10 of FIG. 1. As illustrated in certain embodiments, a power distribution assembly 46 provides power to the welding tool 14 and the computer 18. Moreover, the welding tool 14 includes control circuitry 52 configured to control the operation of the welding tool 14. In the illustrated embodiment, the control circuitry 52 includes one or more processors 54, memory devices 56, and storage devices 58. In other embodiments, the control circuitry 52 may not include the processors 54, the memory devices 56, and/or the storage devices 58. The processor(s) 54 may be used to execute software, such as welding tool software. Moreover, the processor(s) 54 may be similar to the processor(s) 20 described previously. Furthermore, the memory device(s) 56 may be similar to the memory device(s) 22, and the storage device(s) 58 may be similar to the storage device(s) 24.

In certain embodiments, the welding tool 14 includes a user interface 60 to enable a welding operator (e.g., welding student, welding instructor, etc.) to interact with the welding tool 14 and/or to provide inputs to the welding tool 14. For example, the user interface 60 may include buttons, switches, touch screens, touchpads, scanners, and so forth. The inputs provided to the welding tool 14 by the welding operator may be provided to the computer 18. For example, the inputs provided to the welding tool 14 may be used to control welding software being executed by the computer 18. As such, the welding operator may use the user interface 60 on the welding tool 14 to navigate the welding software screens, setup procedures, data analysis, welding courses,

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make selections within the welding software, configure the welding software, and so forth. Thus, the welding operator can use the welding tool 14 to control the welding software (e.g., the welding operator does not have to put down the welding tool 14 to use a different input device). In certain embodiments the welding tool 14 also includes visual indicators 61, such as a display 62 and LEDs 64. The visual indicators 61 may be configured to indicate or display data and/or images corresponding to a weld, welding training, and/or welding software. For example, the visual indicators 61 may be configured to indicate a welding tool orientation, a welding tool travel speed, a welding tool position, a contact tip to workpiece distance, an aim of the welding tool 14, training information for the welding operator, and so forth. Moreover, the visual indicators 61 may be configured to provide visual indications before a weld, during a weld, and/or after a weld. In certain embodiments, the LEDs 64 may illuminate to facilitate their detection by the one or more sensing devices 16. In such embodiments, the LEDs 64 may be positioned to enable the one or more sensing devices 16 to determine a position and/or an orientation of the welding tool 14 based on a spatial position of the LEDs 64.

Returning to FIG. 2, in certain embodiments, the welding tool 14 includes power conversion circuitry 66 configured to receive power from the power distribution assembly 46 the computer 18, or another device, and to convert the received power for powering the welding tool 14. In certain embodiments, the welding tool 14 may receive power that is already converted and/or does not utilize power conversion. Moreover, in some embodiments, the welding tool 14 may be powered by a battery or any suitable powering mechanism. In certain embodiments, the welding tool 14 also includes a communication interface 68 (e.g., RS-232 driver) to facilitate communication between the welding tool 14 and the computer 18.

In embodiments where the welding tool 14 is a welding torch, the welding tool 14 may include a trigger 70 configured to mechanically actuate a trigger switch 72 between an open position (as illustrated) and a closed position. The trigger 70 provides a conductor 71 to carry a signal to the control circuitry 52 to indicate whether the trigger switch 72 is in the open position or the closed position. The wire feeder 30, the welding power supply 28, and/or the computer 18 may determine whether there is continuity through the welding tool 14 across a first trigger conductor 74 and a second trigger conductor 76. The trigger switch 72 is electrically coupled between the first trigger conductor 74 and the second trigger conductor 76. Continuity across the first trigger conductor 74 and the second trigger conductor 76 may be determined by applying a voltage across the conductors 74 and 76, applying a current across the conductors 74 and 76, measuring a resistance across the conductors 74 and 76, and so forth. In certain embodiments, portions of the first trigger conductor 74 and/or portions of the second trigger conductor 76 may be disposed within a connector of the welding tool 14. Furthermore, in certain embodiments, the arrangement of switches and/or conductors within the welding tool 14 may be different than illustrated in FIG. 2.

The welding power supply 28 may determine whether to enable welding power to flow through the welding tool 14 based on whether there is continuity across the conductors 74 and 76. For example, the welding power supply 28 may enable welding power to flow through the welding tool 14 while there is continuity across the conductors 74 and 76, and the welding power supply 28 may block welding power from flowing through the welding tool 14 while there is an open circuit across the conductors 74 and 76. Furthermore,

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the wire feeder 30 may provide welding wire to the welding tool 14 while there is continuity across the conductors 74 and 76, and may block welding wire from being provided to the welding tool 14 while there is an open circuit across the conductors 74 and 76. Moreover, the computer 18 may use the continuity across the conductors 74 and 76 and/or the position of the trigger 70 or trigger switch 72 to start and/or stop a welding operation, a welding simulation, data recording, and so forth.

With the trigger switch 72 in the open position, there is an open circuit across the conductors 74 and 76, thus, the open position of the trigger switch 72 blocks electron flow between the conductors 74 and 76. Accordingly, the welding power supply 28 may block welding power from flowing through the welding tool 14 and the wire feeder 30 may block welding wire from being provided to the welding tool 14. Pressing the trigger 70 directs the trigger switch 72 to the closed position where the trigger switch 72 remains as long as the trigger 70 is pressed. With the trigger switch 72 in the closed position, there is continuity between the first trigger conductor 74 and a conductor 77 electrically connected to the trigger switch 72 and a training switch 78.

The training switch 78 is electrically coupled between the first trigger conductor 74 and the second trigger conductor 76. Moreover, the training switch 78 is electrically controlled by the control circuitry 52 to an open position or to a closed position. In certain embodiments, the training switch 78 may be any suitable electrically controlled switch, such as a transistor, relay, etc. The control circuitry 52 may selectively control the training switch 78 to the open position or to the closed position. For example, while welding software of the welding system 10 is operating in a live-arc mode, the control circuitry 52 may be configured to control the training switch 78 to the closed position to enable a live welding arc while the trigger 70 is pressed. In contrast, while welding software of the welding system 10 is operating in any mode other than the live-arc mode (e.g., simulation, virtual reality, augmented reality, etc.), the control circuitry 52 may be configured to control the training switch 78 to the open position to block a live welding arc (by blocking electron flow between the conductors 74 and 76).

In certain embodiments, the training switch 78 may default to the open position, thereby establishing an open circuit across the conductors 74 and 76. As may be appreciated, while the training switch 78 is in the open position, there will be an open circuit across the conductors 74 and 76 regardless of the position of the trigger switch 72 (e.g., electron flow between the conductors 74 and 76 is blocked by the open position of the training switch 78). However, while the training switch 78 is controlled to the closed position, and the trigger switch 72 is in the closed position, conductivity is established between the conductors 74 and 76 (e.g., electron flow between the conductors 74 and 76 is enabled). Accordingly, the welding power supply 28 may enable welding power to flow through the welding tool 14 only while the training switch 78 is in the closed position and while the trigger switch 72 is in the closed position. For example, welding power may flow from the welding power supply 28, through a weld cable 80, the welding tool 14, a workpiece 82, and return to the welding power supply 28 via a work cable 84 (e.g., electrode-negative, or straight polarity). Conversely, welding power may flow from the welding power supply 28, through the work cable 84, the workpiece 82, the welding tool 14, and return to the welding power supply 28 via the weld cable 80 (e.g., electrode-positive, or reverse polarity).

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As may be appreciated, the training switch **78** may be physically located in any suitable portion of the welding system **10**, such as the computer **18**, and so forth. Furthermore, in certain embodiments, the functionality of the training switch **78** may be replaced by any suitable hardware and/or software in the welding system **10**.

FIG. **3** is a perspective view of an embodiment of the welding stand **12** of FIG. **1**. The welding stand **12** includes a welding surface **88** on which live welds (e.g., real welds, actual welds) and/or simulated welds may be performed. Legs **90** provide support to the welding surface **88**. In certain embodiments, the welding surface **88** may include slots **91** to aid a welding operator in positioning and orienting the workpiece **82**. In certain embodiments, the position and orientation of the workpiece **82** may be provided to welding software of the welding system **10** to calibrate the welding system **10**. For example, a welding operator may provide an indication to the welding software identifying which slot **91** of the welding surface **88** the workpiece **82** is aligned with. Furthermore, a predefined welding assignment may direct the welding operator to align the workpiece **82** with a particular slot **91**. In certain embodiments, the workpiece **82** may include an extension **92** configured to extend into one or more of the slots **91** for alignment of the workpiece **82** with the one or more slots **91**. As may be appreciated, each of the slots **91** may be positioned at a location corresponding to a respective location defined in the welding software.

In certain embodiments, the welding surface **88** includes a first aperture **93** and a second aperture **94**. The first and second apertures **93** and **94** may be used together to determine a position and/or an orientation of the welding surface **88**. As may be appreciated, in certain embodiments at least three apertures may be used to determine the position and/or the orientation of the welding surface **88**. In some embodiments, more than three apertures may be used to determine the position and/or the orientation of the welding surface **88**. The first and second apertures **93** and **94** may be positioned at any suitable location on the welding surface **88**, and may be any suitable size. In certain embodiments, the position and/or orientation of the welding surface **88** relative to one or more sensing devices **16** may be calibrated using the first and second apertures **93** and **94**. For example, as described in greater detail below, a calibration device configured to be sensed by one or more sensing devices **16** may be inserted into the first aperture **93**, or touched to the first aperture **93**. While the calibration device is inserted into, or touching, the first aperture **93**, a user input provided to the welding software (or other calibration software) may indicate that the calibration device is inserted into the first aperture **93**. As a result, the welding software may establish a correlation between a first data set (e.g., calibration data) received from one or more sensing devices **16** (e.g., position and/or orientation data) at a first time and the location of first aperture **93**. The calibration device may next be inserted into the second aperture **94**, or touched to the second aperture **94**. While the calibration device is inserted into, or touching, the second aperture **94**, a user input provided to the welding software may indicate that the calibration device is inserted into the second aperture **94**. As a result, the welding software may establish a correlation between a second data set (e.g., calibration data) received from one or more sensing devices **16** at a second time and the location of second aperture **94**. Thus, the welding software may be able to calibrate the position and/or orientation of the welding surface **88** relative to one or more sensing devices **16** using the first data set received at the first time and the second data set received at the second time.

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In certain embodiments, the welding surface **88** also includes a first marker **95** and a second marker **96**. The first and second markers **95** and **96** may be used together to determine a position and/or an orientation of the welding surface **88**. As may be appreciated, in certain embodiments, at least three markers may be used to determine the position and/or the orientation of the welding surface **88**. In some embodiments, more than three markers may be used to determine the position and/or the orientation of the welding surface **88**. The first and second markers **95** and **96** may be formed from any suitable material. Moreover, in certain embodiments, the first and second markers **95** and **96** may be built into the welding surface **88**, while in other embodiments, the first and second markers **95** and **96** may be attached to the welding surface **88**. For example, the first and second markers **95** and **96** may be attached to the welding surface **88** using an adhesive and/or the first and second markers **95** and **96** may be stickers. The first and second markers **95** and **96** may have any suitable shape, size, and/or color. Furthermore, in certain embodiments, the first and second markers **95** and **96** may be a reflector formed from a reflective material. The first and second markers **95** and **96** may be used by the welding system **10** to calibrate the position and/or orientation of the welding surface **88** relative to one or more sensing devices **16** without a separate calibration device. Accordingly, the first and second markers **95** and **96** are configured to be detected by one or more sensing devices **16**. In certain embodiments, the first and second markers **95** and **96** may be positioned at predetermined locations on the welding surface **88**. Furthermore, the welding software may be programmed to use the predetermined locations to determine the position and/or the orientation of the welding surface **88**. In other embodiments, the location of the first and second markers **95** and **96** may be provided to the welding software during calibration. With the first and second markers **95** and **96** on the welding surface **88**, one or more sensing devices **16** may sense the position and/or orientation of the first and second markers **95** and **96** relative to the one or more sensing devices **16**. Using this sensed data in conjunction with the location of the first and second markers **95** and **96** on the welding surface **88**, the welding software may be able to calibrate the position and/or orientation of the welding surface **88** relative to one or more sensing devices **16**. In some embodiments, the welding surface **88** may be removable and/or reversible. In such embodiments, the welding surface **88** may be flipped over, such as if the welding surface **88** become worn.

In the illustrated embodiment, the workpiece **82** includes a first marker **98** and a second marker **99**. The first and second markers **98** and **99** may be used together to determine a position and/or an orientation of the workpiece **82**. As may be appreciated, at least two markers are used to determine the position and/or the orientation of the workpiece **82**. In certain embodiments, more than two markers may be used to determine the position and/or the orientation of the workpiece **82**. The first and second markers **98** and **99** may be formed from any suitable material. Moreover, in certain embodiments, the first and second markers **98** and **99** may be built into the workpiece **82**, while in other embodiments, the first and second markers **98** and **99** may be attached to the workpiece **82**. For example, the first and second markers **98** and **99** may be attached to the workpiece **82** using an adhesive and/or the first and second markers **98** and **99** may be stickers. As a further example, the first and second markers **98** and **99** may be clipped or clamped onto the workpiece **82**. The first and second markers **98** and **99** may have any suitable shape, size, and/or color. Furthermore, in

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certain embodiments, the first and second markers **98** and **99** may be a reflector formed from a reflective material. The first and second markers **98** and **99** may be used by the welding system **10** to calibrate the position and/or orientation of the workpiece **82** relative to one or more sensing devices **16** without a separate calibration device. Accordingly, the first and second markers **98** and **99** are configured to be detected by one or more sensing devices **16**. In certain embodiments, the first and second markers **98** and **99** may be positioned at predetermined locations on the workpiece **82**. Furthermore, the welding software may be programmed to use the predetermined locations to determine the position and/or the orientation of the workpiece **82**. In other embodiments, the location of the first and second markers **98** and **99** may be provided to the welding software during calibration. With the first and second markers **98** and **99** on the workpiece **82**, one or more sensing devices **16** may sense the position and/or orientation of the first and second markers **98** and **99** relative to the one or more sensing devices **16**. Using this sensed data in conjunction with the location of the first and second markers **98** and **99** on the workpiece **82**, the welding software may be able to calibrate the position and/or orientation of the workpiece **82** relative to one or more sensing devices **16**. While the markers **95**, **96**, **98**, and **99** have been described herein as being detected by one or more sensing devices **16**, in certain embodiments, the markers **95**, **96**, **98**, and **99** may indicate locations where a calibration device is to be touched for calibration using the calibration device, as described previously.

In certain embodiments, the welding stand **12** includes a first arm **100** extending vertically from the welding surface **88** and configured to provide support for the one or more sensing devices **16** and the display **32**. A knob **101** is attached to the first arm **100** and may be used to adjust an orientation of the one or more sensing devices **16** relative to the first arm **100**. For example, as the knob **101** is adjusted, mechanical components extending through the first arm **100** may adjust an angle of the one or more sensing devices **16**. In certain embodiments, the display **32** includes a cover **102** to protect the display **32** from welding emissions that may occur during a live welding operation. The cover **102** may be made from any suitable material, such as a transparent material, a polymer, and so forth. By using a transparent material, a welding operator may view the display **32** while the cover **102** is positioned in front of the display **32**, such as before, during, and/or after a welding operation. In certain embodiments, the one or more sensing devices **16** may include a camera **104** coupled to the first arm **100** for recording welding operations. In certain embodiments, the camera **104** may be a high dynamic range (HDR) camera. Furthermore, in certain embodiments, the one or more sensing devices **16** may include one or more emitters **105** coupled to the first arm **100**. The emitters **105** may be used to calibrate the position and/or orientation of the welding surface **88** relative to one or more sensing devices **16**. For example, the one or more emitters **105** may be configured to emit a visible pattern onto the welding surface **88**, the workpiece **82**, the welding tool **14**, or the operator, or any combination thereof. That is, the patterns emitted by the one or more emitters **105** are visible to the camera **104**. The emitter **105** may emit the visible pattern at a desired wavelength, such as a wavelength in the infrared, visible, or ultraviolet spectrum (e.g., approximately 1 mm to 120 nm). The visible patterns may be shown onto the welding surface **88** and/or the workpiece **82**. Furthermore, the visible patterns may be detected by the one or more sensing devices **16** to calibrate the position and/or the orientation of the welding

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surface **88** relative to the one or more sensing devices **16**. For example, based on particular features of the visible pattern alignments and/or orientations may be determined by the one or more sensing devices **16** and/or the welding software. Moreover, the visible patterns emitted by the one or more emitters **105** may be used to facilitate positioning of the workpiece **82** on the welding surface **88**. As discussed in greater detail below, the visible patterns may be detected by the one or more sensing devices **16** (e.g., cameras **104**) to determine a shape (e.g., tube, S-shape, I-shape, U-shape) of the workpiece **82**, the operator, or position of the welding tool **14** prior to welding. In some embodiments, the visible pattern may be detected by the one or more sensing devices **16** during welding to detect workpiece **82**, the operator, the welding tool **14**, or any combination thereof.

In some embodiments, the one or more sensing devices **16** of the welding stand **12** may include a second camera **109** coupled to a third arm **107** for recording welding operations in a similar manner to the camera **104**. Furthermore, a second emitter **113** coupled to the third arm **107** may emit a visible pattern onto the welding surface **88**, the workpiece **82**, the welding tool **14**, or the operator, or any combination thereof. The second emitter **113** may emit the visible pattern at a desired wavelength, such as a wavelength in the infrared, visible, or ultraviolet spectrum. The visible pattern emitted from the second emitter **113** may be approximately the same wavelength or a different wavelength than the visible pattern emitted by the emitter **105**. As may be appreciated, the second camera **109** and the second emitter **113** may be positioned to have a different orientation (e.g., perpendicular) relative to the workpiece **82** than the camera **104** and the emitter **105**, thereby enabling the determination of the shape of the workpiece **82**, the position of the operator, or the position of the welding tool **14** in the event that the sensing device **16** of either arm **100**, **107** is obscured from view of a portion of the welding environment. In some embodiments, the sensing devices **16** may include multiple sets of cameras and emitters arranged at various points about the welding environment on or off the welding stand **12** to facilitate the monitoring of the position and movement of objects in the welding environment if one or more sensing devices **16** are obscured from view of the welding environment. As discussed in greater detail below, the camera **104** and the emitter **105** may be integrated with the welding helmet **41**, thereby enabling the welding system **10** to monitor the position and/or orientation of the welding tool **14** and the workpiece relative to the welding helmet **41**.

In certain embodiments, the welding stand **12** also includes a second arm **106** extending vertically from the welding surface **88** and configured to provide support for a welding plate **108** (e.g., vertical welding plate, horizontal welding plate, overhead welding plate, etc.). The second arm **106** may be adjustable to facilitate overhead welding at different heights. Moreover, the second arm **106** may be manufactured in a number of different ways to facilitate overhead welding at different heights. The welding plate **108** is coupled to the second arm **106** using a mounting assembly **110**. The mounting assembly **110** facilitates rotation of the welding plate **108** as illustrated by arrow **111**. For example, the welding plate **108** may be rotated from extending generally in the horizontal plane (e.g., for overhead welding), as illustrated, to extend generally in the vertical plane (e.g., for vertical welding). The welding plate **108** includes a welding surface **112**. In certain embodiments, the welding surface **112** includes slots **114** that may aid a welding operator in positioning the workpiece **82** on the welding surface **112**, similar to the slots **91** on the welding surface **88**.

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In certain embodiments, the position of the workpiece **82** may be provided to welding software of the welding system **10** to calibrate the welding system **10**. For example, a welding operator may provide an indication to the welding software identifying which slot **114** of the welding surface **112** the workpiece **82** is aligned with. Furthermore, a pre-defined welding assignment may direct the welding operator to align the workpiece **82** with a particular slot **114**. In certain embodiments, the workpiece **82** may include an extension configured to extend into one or more of the slots **114** for alignment of the workpiece **82** with the one or more slots **114**. As may be appreciated, each of the slots **114** may be positioned at a location corresponding to a respective location defined in the welding software.

In certain embodiments, the welding surface **112** also includes a first marker **116** and a second marker **118**. The first and second markers **116** and **118** may be used together to determine a position and/or an orientation of the welding surface **112**. As may be appreciated, at least two markers are used to determine the position and/or the orientation of the welding surface **112**. In certain embodiments, more than two markers may be used to determine the position and/or the orientation of the welding surface **112**. The first and second markers **116** and **118** may be formed from any suitable material. Moreover, in certain embodiments, the first and second markers **116** and **118** may be built into the welding surface **112** (or another part of the welding plate **108**), while in other embodiments, the first and second markers **116** and **118** may be attached to the welding surface **112** (or another part of the welding plate **108**). For example, the first and second markers **116** and **118** may be attached to the welding surface **112** using an adhesive and/or the first and second markers **116** and **118** may be stickers. As a further example, the first and second markers **116** and **118** may be clipped or clamped onto the welding surface **112**. In some embodiments, the first and second markers **116** and **118** may be integrated into a holding clamp that is clamped onto a welding coupon. The first and second markers **116** and **118** may have any suitable shape, size, and/or color. Furthermore, in certain embodiments, the first and second markers **116** and **118** may be a reflector formed from a reflective material.

The first and second markers **116** and **118** may be used by the welding system **10** to calibrate the position and/or orientation of the welding surface **112** relative to the one or more sensing devices **16** without a separate calibration device. Accordingly, the first and second markers **116** and **118** are configured to be detected by the one or more sensing devices **16**. In certain embodiments, the first and second markers **116** and **118** may be positioned at predetermined locations on the welding surface **112**. Furthermore, the welding software may be programmed to use the predetermined locations to determine the position and/or the orientation of the welding surface **112**. In other embodiments, the location of the first and second markers **116** and **118** may be provided to the welding software during calibration. With the first and second markers **116** and **118** on the welding surface **112**, one or more sensing device **16** may sense the position and/or orientation of the first and second markers **116** and **118** relative to the one or more sensing devices **16**. Using this sensed data in conjunction with the location of the first and second markers **116** and **118** on the welding surface **112**, the welding software may be able to calibrate the position and/or orientation of the welding surface **112** relative to one or more sensing devices **16**. Furthermore, the one or more sensing devices **16** may sense and/or track the first and second markers **116** and **118** during a weld to account

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for any movement of the welding plate **108** that may occur during the weld. While the markers **116** and **118** have been described herein as being detected by the one or more sensing devices **16**, in certain embodiments, the markers **116** and **118** may indicate locations where a calibration device is to be touched or inserted for calibration using the calibration device, as described previously.

FIG. **4** is a perspective view of an embodiment of a calibration device **120**. In some embodiments, the calibration device **120** is shaped like a welding tool and may be used for calibrating the position and/or orientation of the welding surfaces **88** and **112** relative to the one or more sensing devices **16**. In other embodiments, the calibration device **120** may be used for calibrating the position and/or orientation of a welding joint. The calibration device **120** includes a handle **122** and a nozzle **124**. The nozzle **124** includes a pointed end **126** that may be used to touch a location for calibration and/or to be inserted into an aperture for calibration. The calibration device **120** also includes a user interface **128** that enables the welding operator to provide input corresponding to a time that the calibration device **120** is touching a location for calibration and/or is being inserted into an aperture for calibration. Moreover, in certain embodiments, the calibration device **120** includes markers **130** configured to be sensed by the one or more sensing devices **16**. As illustrated, the markers **130** extend from the calibration device **120**. However, in other embodiments, the markers **130** may not extend from the calibration device **120**. The markers **130** may be any suitable marker configured to be detected by the one or more sensing devices **16** (e.g., cameras). Moreover, the markers **130** may be any suitable size, shape, and/or color.

During calibration, the sensing devices **16** may sense a position of the calibration device **120** and/or an orientation of the calibration device **120**. The position and/or orientation of the calibration device **120** may be used by the welding software to determine a position and/or orientation of one or more of the welding surfaces **88** and **112** relative to the sensing devices **16**, a position and/or orientation of the workpiece **82** relative to the sensing devices **16**, a position and/or orientation of a fixture relative to the sensing devices **16**, and so forth. Thus, the calibration device **120** may facilitate calibration of the welding system **10**. In some embodiments, a tray may be positioned beneath the welding surface **88** for storing the calibration device **120**. Moreover, in certain embodiments, live welding may be disabled if the calibration device **120** is able to be tracked by the sensing devices **16** (e.g., to block spatter from contacting the calibration device **120**).

FIG. **5** is a perspective view of an embodiment of a fixture assembly **132**. The fixture assembly **132** may be positioned on the welding surface **88** and/or the welding surface **112**, and may secure the workpiece **82** thereon. In certain embodiments, the fixture assembly **132** may be configured to align with one or more of the slots **91** and **114**. In other embodiments, the fixture assembly **132** may be placed at any location on the welding surface **88** and/or the welding surface **112**. The fixture assembly **132** also includes a first marker **134** and a second marker **136**. The first and second markers **134** and **136** may be used together to determine a position and/or an orientation of the fixture assembly **132**. As may be appreciated, at least two markers are used to determine the position and/or the orientation of the fixture assembly **132**. The first and second markers **134** and **136** may be formed from any suitable material. Moreover, in certain embodiments, the first and second markers **134** and **136** may be built into the fixture assembly **132**, while in

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other embodiments, the first and second markers **134** and **136** may be attached to the fixture assembly **132**. For example, the first and second markers **134** and **136** may be attached to the fixture assembly **132** using an adhesive and/or the first and second markers **134** and **136** may be stickers. The first and second markers **134** and **136** may have any suitable shape, size, and/or color. Furthermore, in certain embodiments, the first and second markers **134** and **136** may be a reflector formed from a reflective material. The first and second markers **134** and **136** may be used by the welding system **10** to calibrate the position and/or orientation of the fixture assembly **132** relative to the one or more sensing devices **16** without a separate calibration device. Accordingly, the first and second markers **134** and **136** are configured to be detected by the sensing devices **16**. In certain embodiments, the first and second markers **134** and **136** may be positioned at predetermined locations on the fixture assembly **132**. Furthermore, the welding software may be programmed to use the predetermined locations to determine the position and/or the orientation of the fixture assembly **132**. In other embodiments, the location of the first and second markers **134** and **136** may be provided to the welding software during calibration. With the first and second markers **134** and **136** on the fixture assembly **132**, the one or more sensing devices **16** may sense the position and/or orientation of the first and second markers **134** and **136** relative to the sensing devices **16**. Using this sensed data in conjunction with the location of the first and second markers **134** and **136** on the fixture assembly **132**, the welding software may be able to calibrate the position and/or orientation of the fixture assembly **132** relative to the sensing devices **16**. While the first and second markers **134** and **136** have been described herein as being detected by the sensing devices **16**, in certain embodiments, the first and second markers **134** and **136** may indicate locations where a calibration device is to be touched or inserted for calibration using the calibration device **120**, as described previously.

In the illustrated embodiment, the fixture assembly **132** is configured to secure a lower portion **138** of the workpiece **82** to an upper portion **140** of the workpiece **82** for performing a lap weld. In other embodiments, the fixture assembly **132** may be configured to secure portions of the workpiece **82** for performing a butt weld, a fillet weld, and so forth, to aid a welding operator in performing a weld. The fixture assembly **132** includes vertical arms **142** extending from a base **143**. A cross bar **144** extends between the vertical arms **142**, and is secured to the vertical arms **142**. Adjustment mechanisms **146** (e.g., knobs) may be adjusted to direct locking devices **148** toward the workpiece **82** for securing the workpiece **82** between the locking devices **148** and the base **143** of the fixture assembly **132**. Conversely, the adjustment mechanisms **146** may be adjusted to direct the locking devices **148** away from the workpiece **82** for removing the workpiece **82** from being between the locking devices **148** and the base **143**. Accordingly, the workpiece **82** may be selectively secured to the fixture assembly **132**.

FIG. 6 is a perspective view of an embodiment of a vertical arm assembly **223** of the welding stand **12** of FIG. 3. As illustrated, one or more sensing devices **16** are attached to the first arm **100**. Furthermore, the sensing devices **16** include one or more cameras **224**, and one or more infrared emitters **226**. However, in other embodiments, the sensing device **16** may include any suitable number of cameras, emitters, and/or other sensing devices. A pivot assembly **228** is coupled to the first arm **100** and to the one or more sensing devices **16**, and enables an angle of the one or more sensing

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devices **16** to be adjusted while the one or more sensing devices **16** rotate as illustrated by arrow **229**. As may be appreciated, adjusting the angle of the one or more sensing devices **16** relative to the first arm **100** changes the field of view of the one or more sensing devices **16** (e.g., to change the portion of the welding surface **88** and/or the welding surface **112** sensed by the sensing device **16**). In some embodiments, the one or more sensing devices **16** may be arranged to observe at least a portion (e.g., hands, face) of the operator prior to and/or after completion of a weld process. Observation of the operator by the one or more sensing devices **16**, such as by a camera, may facilitate operator identification and verification that the identified operator performed the observed weld process.

In certain embodiments, cords **230** extend between the knob **101** and the one or more sensing devices **16**. The cord **230** is routed through a pulley **232** to facilitate rotation of the one or more sensing devices **16**. Thus, a welding operator may rotate the knob **101** to manually adjust the angle of the one or more sensing devices **16**. As may be appreciated, the combination of the cord **230** and the pulley **232** is one example of a system for rotating the one or more sensing devices **16**. It should be noted that any suitable system may be used to facilitate rotation of the one or more sensing devices **16**. While one embodiment of a knob **101** is illustrated, it may be appreciated that any suitable knob may be used to adjust the angle of the one or more sensing devices **16**. Furthermore, the angle of the one or more sensing devices **16** may be adjusted using a motor **234** coupled to the cord **230**. Accordingly, a welding operator may operate the motor **234** to adjust the angle of the one or more sensing devices **16**. Moreover, in certain embodiments, control circuitry may be coupled to the motor **234** and may control the angle of the one or more sensing devices **16** based on a desired field of view of the one or more sensing devices **16** and/or based on tracking of an object within the field of view of the one or more sensing devices **16**.

FIG. 7 is a perspective view of an embodiment of an overhead welding arm assembly **235**. The overhead welding arm assembly **235** illustrates one embodiment of a manufacturing design that enables the second arm **106** to have an adjustable height. Accordingly, as may be appreciated, the second arm **106** may be manufactured to have an adjustable height in a number of ways. As illustrated, the overhead welding arm assembly **235** includes handles **236** used to vertically raise and/or lower the second arm **106** as illustrated by arrows **238**. The overhead welding arm assembly **235** includes a locking device **240** to lock the second arm **106** at a desired height. For example, the locking device **240** may include a button that is pressed to disengage a latch configured to extend into openings **242** thus unlocking the second arm **106** from being secured to side rails **243**. With the second arm **106** unlocked from the side rails **243**, the handles **236** may be vertically adjusted to a desired height, thereby adjusting the welding surface **112** to a desired height. As may be appreciated, releasing the button may result in the latch extending into the openings **242** and locking the second arm **106** to the side rails **243**. As may be appreciated, the locking device **240** may operate manually as described and/or the locking device **240** may be controlled by a control system (e.g., automatically controlled). Furthermore, the second arm **106** may be vertically raised and/or lowered using the control system. For example, in certain embodiments, the welding software may control the second arm **106** to move to a desired position automatically. Thus, the welding surface **112** may be adjusted to a desired height for overhead welding.

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FIG. 8 is a block diagram of an embodiment of welding software 244 (e.g., welding training software) of the welding system 10 having multiple modes. As illustrated, the welding software 244 may include one or more of a live-arc mode 246 configured to enable training using a live (e.g., actual) welding arc, a simulation welding mode 248 configured to enable training using a welding simulation, a virtual reality (VR) welding mode 250 configured to enable training using a VR welding simulation, and/or an augmented reality welding mode 252 configured to enable training using augmented reality welding simulation.

The welding software 244 may receive signals from an audio input 254. The audio input 254 may be configured to enable a welding operator to operate the welding software 244 using audible commands (e.g., voice activation). Furthermore, the welding software 244 may be configured to provide an audio output 256 and/or a video output 258. For example, the welding software 244 may provide audible information to a welding operator using the audio output 256. Such audible information may include instructions for configuring (e.g., setting up) the welding system 10, real-time feedback provided to a welding operator during a welding operation, instructions to a welding operator before performing a welding operation, instructions to a welding operator after performing a welding operation, warnings, and so forth.

FIG. 9 is a block diagram of an embodiment of the VR welding mode 250 of the welding software 244. The VR welding mode 250 is configured to provide a welding operator with a VR simulation 260. The VR simulation 260 may be displayed to a welding operator through a VR headset, VR glasses, a VR display, or any suitable VR device. In some embodiments, the display 32 of the helmet 41 of the welding system 10 may facilitate the VR simulation 260. The VR simulation 260 may be configured to include a variety of virtual objects that enable interaction between a welding operator and a selected virtual object of the variety of virtual objects within the VR simulation 260. For example, virtual objects may include a virtual workpiece 262, a virtual welding stand 264, a virtual welding tool 266, virtual wire cutters 268, virtual software configuration 270, virtual training data results 272 and/or a virtual glove 274.

In certain embodiments, the welding operator may interact with the virtual objects without touching a physical object. For example, the one or more sensing devices 16 may detect movement of the welding operator and may result in similar movements occurring in the VR simulation 260 based on the welder operator's movements in the real world. In other embodiments, the welding operator may use a glove or the welding tool 14 to interact with the virtual objects. For example, the glove or the welding tool 14 may be detected by the sensing device 16, and/or the glove or the welding tool 14 may correspond to a virtual object in the VR simulation 260. Furthermore, the welding operator may be able to operate the welding software 244 within the VR simulation 260 using the virtual software configuration 270 and/or the virtual training data results 272. For example, the welding operator may use their hand, the glove, or the welding tool 14 to select items within the welding software 244 that are displayed virtually within the VR simulation 260. Moreover, the welding operator may perform other actions such as picking up wire cutters and cutting virtual welding wire extending from the virtual welding tool 266, all within the VR simulation 260.

FIG. 10 is an embodiment of a method 276 for integrating training results data, non-training results data, simulation results data, and so forth. The method 276 includes the

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welding software 244 of the computer 18 receiving a first set of welding data from a storage device (e.g., storage device 24) (block 278). The first set of welding data may include welding data corresponding to a first welding session (e.g., welding assignment). The method 276 also includes the welding software 244 receiving a second set of welding data from the storage device (block 280). In certain embodiments, the first set and/or second set of welding data may be received from a network storage device. The network storage device may be configured to receive welding data from and/or to provide welding data to the welding system 10 and/or the external welding system 40. The welding software 244 may integrate the first and second sets of welding data into a chart to enable a visual comparison of the first set of welding data with the second set of welding data (block 282). As may be appreciated, the chart may be a bar chart, a pie chart, a line chart, a histogram, and so forth. In certain embodiments, integrating the first set of welding data with the second set of welding data includes filtering the first set of welding data and the second set of welding data to display a subset of the first set of welding data and a subset of the second set of welding data. The welding software 244 may provide the chart to a display device (e.g., the display 32) (block 284). In certain embodiments, providing the chart to the display device includes providing selectable elements on the chart that when selected display data corresponding to a respective selected element of the selectable elements (e.g., selecting wire speed from the chart may change the screen to display the wire speed history for a particular welding session (e.g., welding assignment)).

The first set of welding data and/or the second set of welding data may include a welding tool orientation, a welding tool travel speed, a welding tool position, a contact tip to workpiece distance, an aim of the welding tool, a welding score, a welding grade, and so forth. Moreover, the first set of welding data and the second set of welding data may correspond to training performed by one welding operator and/or by a class of welding operators. Furthermore, the first welding session (e.g., welding assignment) and the second welding session (e.g., welding assignment) may correspond to training performed by one welding operator and/or by a class of welding operators. In certain embodiments, the first welding assignment may correspond to training performed by a first welding operator, and the second welding assignment may correspond to welding performed by a second welding operator. Moreover, the first assignment and the second assignment may correspond to the same welding scenario. Additionally, or in the alternative, the first set of welding data and the second set of welding data may correspond to welding sessions (e.g., welding assignments) performed by one welding operator and/or a class of welding operators outside of a training environment (e.g., production floor).

FIG. 11 is an embodiment of a chart 285 illustrating multiple sets of welding data for a welding operator. The chart 285 may be produced by the welding software 244 and may be provided to the display 32 to be used by a welding instructor to review welding operations performed by a welding student, and/or may be provided to the display 32 to be used by a welding student to review welding operations performed by that welding student. The chart 285 illustrates a bar graph comparison between different sessions (e.g., assignments) of a first set of welding assignments performed by a welding operator. The first set of welding sessions (e.g., welding assignments) includes sessions (e.g., assignments) 286, 288, 290, 292, and 294. The chart 285 also illustrates a bar graph comparison between different assignments of a

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second set of welding sessions (e.g., welding assignments) performed by the welding operator. The second set of welding sessions (e.g., welding assignments) includes sessions (e.g., assignments) 296, 298, 300, 302, and 304. Accordingly, welding sessions (e.g., welding assignments) may be compared to one another for analysis instruction, certification, and/or training purposes. As illustrated, the welding sessions (e.g. welding assignments) may be compared to one another using one of any number of criteria, such as a total score, a work angle, a travel angle, a travel speed, a contact to work distance, an aim, a mode (e.g., live-arc mode, simulation mode, etc.), a completion status (e.g., complete, incomplete, partially complete, etc.), a joint type (e.g., fillet, butt, T, lap, etc.), a welding position (e.g., flat, vertical, overhead, etc.), a type of metal used, a type of

The welding software 244 may associate an operator with welding data (e.g., arc parameters, welding parameters) acquired during a welding session (e.g., live arc welding assignment, simulated welding assignment, and so forth). For example, the welding software 244 may identify the welding operator by an operator name 291, an operator registration number 293, an operator photograph 295, and so forth. For example, the operator identification system 43 discussed above with FIG. 1 may be utilized to determine the operator registration number 293. That is, each operator registration number 293 may correspond to the operator name 291 and a set of identification information (e.g., resettable information 45, biometric information 47, token 49). In some embodiments, the registration number 293 may be reset or reassigned to another operator after a period (e.g., 1, 3, 5, 10, or more years) of inactivity associated with the registration number 293. The registration number 293 may be unique for each operator. In some embodiments, the registration number 293 may be retained by the operator for an extended period of time (e.g., career, life) regardless of activity level associated with the registration number 293. That is, the registration number 293 may be a permanent identifier associated with each operator across one welding system 10 or a network of welding systems 10 coupled via the network 38. Welding data associated with the registration number 293 may be maintained locally or within one or more data storage systems, such as a cloud storage system or database of the network 38 coupled to the welding system 10. The data storage system 318 (e.g., cloud storage system) of the network 38 may be maintained by the manufacturer or another party thereby enabling the welding data associated with a certain registration number 293 to be retained independent of an employment status of the operator with the certain registration number 293. For example, the operator registration number 293 and the data storage system (e.g., cloud storage system) may facilitate the retention of welding data associated with the operator from weld processes performed during training, during a simulation, during a first employment, during a second employment, during personal time, or any combination thereof. In some embodiments, welding data stored within the memory device(s) 22 or the storage device(s) 24 of the computer 18 of the welding system 10 for a particular welding operator (e.g., operator registration number 293) may be selectively or automatically synchronized with the data storage system (e.g., cloud storage system).

Weld history data, such as the data of the chart 285, is associated with each registration number 293. In some embodiments, the weld history data is automatically acquired and stored in the data storage system (e.g., cloud storage system) by the welding software 244 of the welding

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system 10. Additionally, or in the alternative, weld history data may be loaded directly to the data storage system (e.g., cloud storage system) of the network 38 via a remote computer 44. The welding software 244 may facilitate access to the welding history data via a welding history control 297. Additionally, the welding software 244 may enable the operator to associate personal information with the registration number 293 via a personal user control 299. The operator associated with the registration number 293 may input one or more organizations (e.g., training center, school, employer, trade organization) with which the operator is affiliated, experience, certifications for various welding processes and/or welding positions, a résumé, or any combination thereof. Furthermore, the registration number 293 may remain associated with the operator despite changes in affiliated organizations, experience, certifications, or any combination thereof.

FIG. 12 is an embodiment of a chart 305 illustrating welding data for a welder compared to welding data for a class. For example, the chart 305 illustrates a score 306 of a welding operator compared to a score 308 (e.g., average, median, or some other score) of a class for a first assignment. Furthermore, a score 310 of the welding operator is compared to a score 312 (e.g., average, median, or some other score) of the class for a second assignment. Moreover, a score 314 of the welding operator is compared to a score 316 (e.g., average, median, or some other score) of the class for a third assignment. As may be appreciated, scores from one or more welding operators may be compared to scores of the entire class. Such a comparison enables a welding instructor to assess the progress of individual welding students as compared to the class of welding students. Furthermore, scores from one or more welding operators may be compared to scores of one or more other welding operators. In certain embodiments, scores from one class may be compared to scores of another class. Moreover, scores from the first assignment, the second assignment, and/or the third assignment may be selected for comparison.

FIG. 13 is a block diagram of an embodiment of a data storage system 318 (e.g., cloud storage system) for storing welding data 327, such as certification status data 326. The data storage system 318 may include, but is not limited to, the computer 18 of the welding system 10, a remote computer 44 (e.g., server) coupled to the welding system 10 via the internet or a network 38, or any combination thereof. The certification status data may be produced as a welding operator completes various assignments in the welding system 10. For example, a predetermined set of assignments may certify a welding operator for a particular welding device and/or welding process. The data storage system 318 (e.g., cloud storage system) includes control circuitry 320, one or more memory devices 322, and one or more storage devices 324. The control circuitry 320 may include one or more processors, which may be similar to the processor(s) 20. Furthermore, the memory device(s) 322 may be similar to the memory device(s) 22, and the storage device(s) 324 may be similar to the storage device(s) 24. The memory device(s) 322 and/or the storage device(s) 324 may be configured to store certification status data 326 corresponding to a welding certification (e.g., welding training certification) of a welding operator.

The welding data 327 may include any data acquired by the welding system 10 associated with the registration number 293 or the welding operator (e.g., any data that is related to the assignments to certify the welding operator, training welding data, simulated welding data, virtual reality welding data, live welding data), any data related to an

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actual certification (e.g., certified, not certified, qualified, not qualified, etc.), a quantity of one or more welds performed by the welding operator, a timestamp for one or more welds performed by the welding operator, a location and/or facility that the welding operator performs the one or more welds, the components of the welding system utilized by the welding operator for the one or more welds, the organization with which the welding operator is affiliated, the organization for whom the welding operator is performing the one or more welds, welding parameter data for one or more welds performed by the welding operator, a quality ranking of the welding operator, a quality level of the welding operator, a history of welds performed by the welding operator, a history of production welds performed by the welding operator, a first welding process (e.g., a metal inert gas (MIG) welding process, a tungsten inert gas (TIG) welding process, a stick welding process, etc.) certification status (e.g., the welding operator is certified for the first welding process, the welding operator is not certified for the first welding process), a second welding process certification status (e.g., the welding operator is certified for the second welding process, the welding operator is not certified for the second welding process), a first welding device (e.g., a wire feeder, a power supply, a model number, etc.) certification status (e.g., the welding operator is certified for the first welding device, the welding operator is not certified for the first welding device), and/or a second welding device certification status (e.g., the welding operator is certified for the second welding device, the welding operator is not certified for the second welding device).

The control circuitry 320 may be configured to receive a request for the first welding process certification status, the second welding process certification status, the first welding device certification status, and/or the second welding device certification status of the welding operator. Furthermore, the control circuitry 320 may be configured to provide a response to the request. The response to the request may include the first welding process certification status, the second welding process certification status, the first welding device certification status, and/or the second welding device certification status of the welding operator. In certain embodiments, the welding operator may be authorized to use a first welding process, a second welding process, a first welding device, and/or a second welding device based at least partly on the response. Furthermore, in some embodiments, the first welding process, the second welding process, the first welding device, and/or the second welding device or a welding system may be enabled or disabled based at least partly on the response. Moreover, in certain embodiments, the first welding process, the second welding process, the first welding device, and/or the second welding device of a welding system may be enabled or disabled automatically. Thus, a welding operator's certification data may be used to enable and/or disable that welding operator's ability to use a particular welding system, welding device, and/or welding process. For example, a welding operator may have a certification for a first welding process, but not for a second welding process. Accordingly, in certain embodiments, a welding operator may verify their identity at a welding system (e.g., by logging in, by utilizing the operator identification system 43, providing the registration number 293, or some other form of authentication). After the identity of the welding operator is verified, the welding system may check the welding operator's certification status. The welding system may enable the welding operator to perform operations using the first welding process based on the welding operator's certification status, but may block the

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welding operator from performing the second welding process based on the welding operator's certification status.

The storage device 324 of the data storage system 318 (e.g., cloud storage system) may have welding data 327 of multiple operators. The data storage system 318 may be a database that retains welding data 327 associated with registration numbers 293 to enable analysis and tracking of the weld history of the operator over extended durations (e.g., career, lifetime), even across one or more organizations. As may be appreciated, the data storage system 318 (e.g., cloud storage system) may facilitate aggregation of certification status data 326 and/or welding data 327 to identify usage trends, anticipate supply or maintenance issues, and so forth. Moreover, coupling the data storage system 318 to the internet or other network 38 enables instructors or managers to monitor and analyze weld data remote from the operator and the welding system 10.

FIG. 14 is an embodiment of a screen illustrating data corresponding to a weld by an operator identified on the screen by the registration number 293. In some embodiments, each weld session (e.g., weld test, assignment) performed by an operator and monitored by the welding system 10 is assigned a unique serial number 329. The serial number 329 may be associated with the registration number 293 within one or more local and/or remote data storage systems, such as a cloud storage system or database of the network 38 coupled to the welding system 10. The serial number 329 may be used to associate the physical weld sample with the captured weld test results. The format of the serial number 329 may include, but is not limited to a decimal number, a hexadecimal number, or a character string. Moreover, the serial numbers 329 for the same assignment may be different for each operator. In some embodiments, the serial number 329 is affixed to the workpiece 82. For example, the serial number 329 may be attached to, stamped, etched, engraved, embossed, or printed on the workpiece 82. In some embodiments, the serial number 329 is encoded as a barcode affixed to the workpiece 82. Additionally, or in the alternative, the operator may write the serial number 329 on the workpiece 82.

As discussed below, a search feature enables an instructor to enter the serial number 329 to recall the test results for the associated weld session (e.g., weld test, assignment) without the instructor needing to know the user (e.g., registration number 293), the assignment, or any other details about the weld. Accordingly, the instructor may review the data corresponding to each serial number 329, then provide feedback to the respective operator. Furthermore, an inspector or technician may review the serial number 329 of a workpiece 82 to aid in a quality review of the performed weld relative to welding procedure specifications (WPS) and/or to determine a maintenance schedule related to the workpiece 82. That is, the serial number 329 may be utilized to track the workpiece 82, the welding data, the arc data, and the operator (e.g., registration number 293) through a life of the respective workpiece 82. In some embodiments, the serial number 329 may be stored within one or more local and/or remote data storage systems, such as a cloud storage system or database of the network 38 coupled to the welding system 10. The screen may be produced by the welding software 244 and may be displayed on the display 32. The screen illustrates parameters that may be graphically displayed to a welding operator before, during, and/or after performing a welding operation. For example, the parameters may include a work angle 328, a travel angle 330, a contact tip to workpiece distance 332, a welding tool travel speed 334, an aim of the welding tool in relation to the joint of the

workpiece **336**, a welding voltage **337**, a welding current **338**, a welding tool orientation, a welding tool position, and so forth.

As illustrated, graphically illustrated parameters may include an indication **339** of a current value of a parameter (e.g., while performing a welding session). Furthermore, a graph **340** may show a history of the value of the parameter, and a score **341** may show an overall percentage that corresponds to how much time during the welding session that the welding operator was within a range of acceptable values. In certain embodiments, a video replay **342** of a welding session may be provided on the screen. The video replay **342** may show live video of a welding operator performing a real weld, live video of the welding operator performing a simulated weld, live video of the welding operator performing a virtual reality weld, live video of the welding operator performing an augmented reality weld, live video of a welding arc, live video of a weld puddle, and/or simulated video of a welding operation.

In certain embodiments, the welding system may capture video data during a welding session (e.g., welding assignment), and store the video data on the storage device **24** and/or the data storage system **318** (e.g., cloud storage system) via the network **38**. Moreover, the welding software **244** may be configured to retrieve the video data from the storage device **24** or the data storage system **318**, to retrieve welding parameter data from the storage device **24** or the data storage system **318**, to synchronize the video data with the welding parameter data, and to provide the synchronized video and welding parameter data to the display **32**.

In some embodiments, the welding system **10** may receive test data from previously performed welds. Test results **343** based at least in part on the test data may be displayed on the screen. Test data may include properties of the performed welding session (e.g., welding assignment), such as strength, porosity, penetration, hardness, heat affected zone size, appearance, and contamination, or any combination thereof. The test data may be obtained via destructive or non-destructive testing performed after completion of the welding session. For example, strength of a weld may be determined via a destructive test, whereas the porosity and penetration may be obtained via non-destructive testing, such as x-ray or ultrasonic inspection.

In some embodiments, the welding system **10** may determine the test data (e.g., properties of the welding assignment) based at least in part on welding parameter data. Additionally, or in the alternative, the welding system **10** may utilize arc parameter data to determine the test data. The test data (e.g., properties of the welding assignment) may be associated with the welding parameter data and any arc parameter data, such that the test data, welding parameter data, and arc parameter data corresponding to the same welding session (e.g., welding assignment) are stored together. Where the welding session (e.g. welding assignment) is a live welding assignment, the arc parameters (e.g., weld voltage weld current, wire feed speed) may include measured arc parameters and/or set arc parameters. Where the welding session is a simulated, virtual reality, or augmented reality welding assignment, the arc parameters may include simulated arc parameters. In some embodiments, the arc parameters associated with non-live welding sessions (e.g., simulated, virtual reality, augmented reality) may include a null set stored in the data storage.

In some embodiments, the determined properties of the welding session (e.g., welding assignment) are based at least in part on a comparison with welding data (e.g., welding parameters, arc parameters) corresponding to previously

performed welding sessions. The welding data corresponding to previously performed welding sessions may be stored in the data storage system **318**. The welding system **10** may determine (e.g., estimate, extrapolate) properties of a simulated welding assignment, a virtual reality welding assignment, or an augmented reality welding assignment through comparison with welding data (e.g., welding parameters, arc parameters) and associated test data corresponding to previously performed live welding session (e.g., live welding assignments). For example, the welding system **10** may determine the penetration of a virtual reality welding assignment through comparison of the welding parameters (e.g., contact tip to work distance, travel speed) of the virtual reality welding assignment to the welding parameters associated with previously performed live welding assignments. Accordingly, the welding system **10** may facilitate training an operator through providing determined one or more properties of the welding assignment despite the welding assignment (e.g., simulated, virtual reality, augmented reality) being performed without a tangible workpiece produced to test.

The computer **18** of the welding system **10** may determine one or more properties of the welding session (e.g., welding assignment) via executing processor-executable instructions to compare the received welding data with welding data corresponding to previously performed welding sessions. In some embodiments, the one or more properties of the welding session are determined remotely from the welding system **10**, such as on a remote computer **44** or data storage system **318** coupled to the welding system **10** via the network **38**. Additionally, or in the alternative, the one or more determined properties may be transmitted to the data storage system **318**, such as via the network **38**. In some embodiments, the computer **18** may determine properties of the welding session (e.g., welding assignment) while receiving the welding data associated with the welding session. That is, the computer **18** may determine properties (e.g., penetration, porosity, strength, appearance) substantially in real-time while the operator is performing the welding session. The determined properties may be displayed via the display **32** as test results. As may be appreciated, the determined properties may be adjusted upon obtaining results from testing (e.g., destructive testing, non-destructive testing) of the welding session (e.g., welding assignment).

The welding software **244** may analyze welding parameter data to determine a traversed path **344** that may be shown on the display **32**. In some embodiments, a time during a weld may be selected by a welding operator, as shown by an indicator **346**. By adjusting the selected time indicator **346**, the welding operator may view the video replay **342** and/or the traversed path **344** in conjunction with the welding parameters as they were at the selected time in order to establish a correlation between the welding parameters, the video replay **342**, and/or the traversed path **344**. Additionally, or in the alternative, the welding operator may select (e.g., via a cursor on the display **32**) a location of the traversed path **344** displayed to review the welding data **327** corresponding to the one or more times the welding tool **14** traversed the selected location. Moreover, the video replay **342** may show frames of video (e.g., captured images, pictures) corresponding to the selected time **346** and/or selected location. As may be appreciated, a selected location may correspond to multiple frames or captured images when the welding operator utilized a weaving or whipping technique and/or when the welding session includes multiple passes. Accordingly, the display **32** may show the multiple frames (e.g., captured images, pictures), and the welding

operator may select one or more for additional review. In some embodiments, the test results **343** (e.g., one or more determined properties of the welding assignment) displayed may correspond to the selected time shown by the indicator **346** and/or to one or more locations along the traversed path **344**. That is, the test results **343** may display tested characteristics (e.g., porosity, penetration) of the weld corresponding to the selected time indicator **346** and/or the selected location along the traversed path **344**. The welding software **244** may be configured to recreate welding data based at least partly on welding parameter data, to synchronize the video replay **342** with the recreated welding data, and to provide the synchronized video replay **342** and recreated welding data to the display **32**. In certain embodiments, the recreated welding data may be weld puddle data and/or a simulated weld. In some embodiments, the welding software **244** may correlate various aspects (e.g., determined properties, video, non-destructive test results, destructive test results) of the weld data acquired for positions along the traversed path **344** of the weld and/or for selected times during the weld process. The welding software **244** may facilitate correlation of the welding parameters (e.g., work angle **328**, travel angle **330**, CTWD **332**, travel speed **334**, and aim **336** of the welding tool in relation to the joint of the workpiece, a welding tool orientation, a welding tool position) with arc parameters (e.g., voltage **337**, current **338**, wire feed speed), the video replay **342**, and test results **343**, or any combination thereof. The weld data associated with the registration number **293** for an operator may enable the operator, the instructor, or a manager, to review the welding parameters, the arc parameters, the video replay **342**, and the test results **343** (e.g., determined properties) corresponding to the selected time indicator **346** and/or position along the traversed path **344** of the weld process. For example, the operator may review the weld data to identify relationships between changes in the welding parameters (e.g., work angle **328**, CTWD **332**) and changes to the arc parameters (e.g., current, voltage) at the selected time shown by the indicator **346** or a selected position. Moreover, the operator may review the weld data to identify relationships between changes in the welding parameters and changes to the test results **343** of the weld.

In some embodiments, the welding tool **14** (e.g., MIG welding torch, stick welding electrode holder, TIG welding torch) may be utilized as a pointer, where pointing the welding tool **14** at a specific location of the weld displays weld data **327** on the display **32** corresponding to the specific location. In some embodiments, the welding tool **14** may contact the workpiece **82** at the specific location. Moreover, the welding software **244** may determine the specific location from the operator based on the point along the weld that is nearest to where the operator is pointing the welding tool **14** (e.g., electrode). The welding software **244** may produce a location bar **346** (e.g., indicator) to be displayed along the weld data **327** when the welding tool **14** is pointed at locations along the weld upon completion of the session. That is, the location bar may extend across the graphs of the welding parameters (e.g., work angle **328**, travel angle **330**, CTWD **332**, travel speed **334**, and aim **336** of the welding tool in relation to the joint of workpiece) in a similar manner as the selected time line **346** described above. The welding software **244** may be configured to display the video replay **342** (e.g., one or more video frames, captured images) that was captured when the welding tool **14** was at the specific location. For example, the welding software **244** may display between 0 to 30 frames before and/or after when the welding tool **14** was at the specific location. Additionally, or

in the alternative, the welding software **244** may display a cross-sectional view of the weld at the specific location. The cross-sectional view may be based on one or more sets of data including, but not limited to, an x-ray scan, an ultrasonic scan, a generated model based at least in part on the welding data **327**, or any combination thereof. Moreover, the cross-sectional view may enable the welding operator or an instructor to review various quality characteristics of the weld at the specific location, including, but not limited to, porosity, undercut, spatter, underfill, and overfill. While the welding tool **14** may be readily used to point to and select specific locations of the weld before the workpiece **82** is moved upon completion of the session, the welding tool **14** may be used as a pointer for previously completed sessions with moved workpieces **82** upon recalibration of respective workpieces **82**.

In certain embodiments, the storage device **24** may be configured to store a first data set corresponding to multiple welds performed by a welding operator, and to store a second data set corresponding to multiple non-training welds performed by the welding operator. Furthermore, the control circuitry **320** may be configured to retrieve at least part of the first data set from the storage device **24**, to retrieve at least part of the second data set from the storage device **24**, to synchronize the at least part of the first data set with the at least part of the second data set, and to provide the synchronized at least part of the first data set and at least part of the second data set to the display **32**.

FIG. **15** is a block diagram of an embodiment of a welding instructor screen **368** of the welding software **244**. The welding software **244** is configured to provide training simulations for many different welding configurations. For example, the welding configurations may include a MIG welding process **370**, a TIG welding process **372**, a stick welding process **374**, the live-arc welding mode **346**, the simulation welding mode **248**, the virtual reality welding mode **250**, and/or the augmented reality welding mode **252**.

The welding instructor screen **368** may be configured to enable a welding instructor to restrict training of a welding operator **376** (e.g., to one or more selected welding configurations), to restrict training of a class of welding operators **378** (e.g., to one or more selected welding configurations), and/or to restrict training of a portion of a class of welding operators **380** (e.g., to one or more selected welding configurations). Moreover, the welding instructor screen **368** may be configured to enable the welding instructor to assign selected training assignments to the welding operator **382**, to assign selected training assignments to a class of welding operators **384**, and/or to assign selected training assignments to a portion of a class of welding operators **386**. Furthermore, the welding instructor screen **368** may be configured to enable the welding instructor to automatically advance the welding operator (or a class of welding operators) from a first assignment to a second assignment **388**. For example, the welding operator may advance from a first assignment to a second assignment based at least partly on a quality of performing the first assignment. Moreover, the welding instructor screen **368** may be configured to verify the identity of an operator (e.g., to ensure welding data is associated with the proper registration number **293**). In some embodiments, the operator identification system **43** identifies the operator, and the instructor verifies the identity of the operator via the welding instructor screen **368**. For example, the instructor may provide a verification input (e.g., resettable identifier, biometric identifier, physical identifier) to the operator identification system **43** to authorize that the identity of the operator is properly recognized by the opera-

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tor identification system 43. In some embodiments, the instructor (e.g., second operator) provides a second identifier input (e.g., resettable identifier, biometric identifier, token) to the welding system 10, such as via the operator identification system 43, thereby verifying the identity of the operator that provided a first identifier input to the operator identification system 43. The second identifier input may be stored with the welding data (e.g., identity of operator performing the welding session), such as in the memory device 56 of the computer 18 or the data storage system 318). Additionally, or in the alternative, the welding instructor may verify the identity of an operator via a two-step identification process in which the operator identification system 43 separately identifies both the operator and the instructor prior to ensure that welding data is associated with the proper registration number 293.

FIG. 16 is an embodiment of a method 389 for weld training using augmented reality. A welding operator may select a mode of the welding software 244 (block 390). The welding software 244 determines whether the augmented reality mode 252 has been selected (block 392). If the augmented reality mode 252 has been selected, the welding software 244 executes an augmented reality simulation. It should be noted that the welding operator may be wearing a welding helmet and/or some other headgear configured to position a display device in front of the welding operator's view. Furthermore, the display device may generally be transparent to enable the welding operator to view actual objects; however, a virtual welding environment may be portrayed on portions of the display device. As part of this augmented reality simulation, the welding software 244 receives a position and/or an orientation of the welding tool 14, such as from the sensing device 16 (block 394). The welding software 244 integrates the virtual welding environment with the position and/or the orientation of the welding tool 14 (block 396). Moreover, the welding software 244 provides the integrated virtual welding environment to the display device (block 398). For example, the welding software 244 may determine where a weld bead should be positioned within the welding operator's field of view, and the welding software 244 may display the weld bead on the display device such that the weld bead appears to be on a workpiece. After completion of the weld, the augmented reality simulation may enable the welding operator to erase a portion of the virtual welding environment (e.g., the weld bead) (block 400), and the welding software 244 returns to block 390.

If the augmented reality mode 252 has not been selected, the welding software 244 determines whether the live-arc mode 246 has been selected (block 402). If the live-arc mode 246 has been selected, the welding software 244 enters the live-arc mode 246 and the welding operator may perform the live-arc weld (block 404). If the live-arc mode 246 has not been selected and/or after executing block 404, the welding software 244 returns to block 390. Accordingly, the welding software 244 is configured to enable a welding operator to practice a weld in the augmented reality welding mode 252, to erase at least a portion of the virtual welding environment from the practice weld, and to perform a live weld in the live-arc mode 246. In certain embodiments, the welding operator may practice the weld in the augmented reality welding mode 252 consecutively a multiple number of times.

FIG. 17 is an embodiment of another method 406 for weld training using augmented reality. A welding operator may select a mode of the welding software 244 (block 408). The welding software 244 determines whether the augmented

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reality mode 252 has been selected (block 410). If the augmented reality mode 252 has been selected, the welding software 244 executes an augmented reality simulation. It should be noted that the welding operator may be wearing a welding helmet and/or some other headgear configured to position a display device in front of the welding operator's view. Furthermore, the display device may completely block the welding operator's field of vision such that images observed by the welding operator have been captured by a camera and displayed on the display device. As part of this augmented reality simulation, the welding software 244 receives an image of the welding tool 14, such as from the sensing device 16 (block 412). The welding software 244 integrates the virtual welding environment with the image of the welding tool 14 (block 414). Moreover, the welding software 244 provides the integrated virtual welding environment with the image of the welding tool 14 to the display device (block 416). For example, the welding software 244 may determine where a weld bead should be positioned within the welding operator's field of view and the welding software 244 displays the weld bead on the display device with the image of the welding tool 14 and other objects in the welding environment. After completion of the weld, the augmented reality simulation may enable the welding operator to erase a portion of the virtual welding environment (e.g., the weld bead) (block 418), and the welding software 244 returns to block 408.

If the augmented reality mode 252 has not been selected, the welding software 244 determines whether the live-arc mode 246 has been selected (block 420). If the live-arc mode 246 has been selected, the welding software 244 enters the live-arc mode 246 and the welding operator may perform the live-arc weld (block 422). If the live-arc mode 246 has not been selected and/or after executing block 422, the welding software 244 returns to block 408. Accordingly, the welding software 244 is configured to enable a welding operator to practice a weld in the augmented reality welding mode 252, to erase at least a portion of the virtual welding environment from the practice weld, and to perform a live weld in the live-arc mode 246. In certain embodiments, the welding operator may practice the weld in the augmented reality welding mode 252 consecutively a multiple number of times.

FIG. 18 is a block diagram of an embodiment of the welding tool 14. The welding tool 14 includes the control circuitry 52, the user interface 60, and the display 62 described previously. Furthermore, the welding tool 14 includes a variety of sensors and other devices. The welding tool 14 may include a temperature sensor 424 (e.g., thermocouple, thermistor, etc.), a motion sensor 426 (e.g., accelerometer, gyroscope, magnetometer, etc.), a vibration device 428 (e.g., vibration motor), a microphone 429, one or more visual indicators 61 (e.g., LEDs 64), or any combination thereof. In addition, in certain embodiments, the welding tool 14 may include a voltage sensor 425 and/or a current sensor 427 to sense voltage and/or current, respectively, of the arc produced by the welding tool 14. As discussed in detail below, one or more sets of LEDs 64 may be arranged about the welding tool 14 to enable the one or more sensing devices 16 to detect the position and orientation of the welding tool 14 relative to the welding stand 12 and the workpiece 82. For example, sets of LEDs 64 may be arranged on a top side, a left side, and a right side of the welding tool 14 to enable the one or more sensing devices 16 to detect the position and orientation of the welding tool 14 regardless of which side of the welding tool 14 is facing the one or more sensing devices 16. In certain embodiments,

the welding tool **14** may include more than one temperature sensor **424**, motion sensor **426**, vibration device **428**, voltage sensor **425**, current sensor **427**, and/or microphone **429**.

During operation, the welding tool **14** may be configured to use the temperature sensor **424** to detect a temperature associated with the welding tool **14** (e.g., a temperature of electronic components of the welding tool **14**, a temperature of the display **62**, a temperature of a light-emitting device, a temperature of the vibration device, a temperature of a body portion of the welding tool **14**, etc.). The control circuitry **52** (or control circuitry of another device) may use the detected temperature to perform various events. For example, the control circuitry **52** may be configured to disable use of the live-arc mode **246** (e.g., live welding) by the welding tool **14** if the detected temperature reaches and/or surpasses a predetermined threshold (e.g., such as 85° C.). Moreover, the control circuitry **52** may also be configured to disable various heat producing devices of the welding tool **14**, such as the vibration device **428**, light-emitting devices, and so forth. The control circuitry **52** may also be configured to show a message on the display **62**, such as “Waiting for tool to cool down. Sorry for the inconvenience.” In certain embodiments, the control circuitry **52** may be configured to disable certain components or features if the detected temperature reaches a first threshold and to disable additional components or features if the detected temperature reaches a second threshold.

Moreover, during operation, the welding tool **14** may be configured to use the motion sensor **426** to detect a motion (e.g., acceleration, etc.) associated with the welding tool **14**. The control circuitry **52** (or control circuitry of another device) may use the detected acceleration to perform various events. For example, the control circuitry **52** may be configured to activate the display **62** (or another display) after the motion sensor **426** detects that the welding tool **14** has been moved. Accordingly, the control circuitry **52** may direct the display **62** to “wake up,” such as from a sleep mode and/or to exit a screen saver mode to facilitate a welding operator of the welding tool **14** using a graphical user interface (GUI) on the display **62**. Furthermore, the control circuitry **52** may utilize feedback from the one or more motion sensors **426** to determine the position of the welding tool **14** in the welding environment and/or the movement of the welding tool **14** within the welding environment. As discussed in detail below, the sensing devices **16** (e.g., cameras) may utilize markers on the welding tool **14** to determine the position, orientation, and/or movement of the welding tool **14** in the welding environment. In some embodiments, the control circuitry **52** (or control circuitry of another device) may utilize the feedback from the one or more motion sensors **426** to augment the determination with the sensing devices **16** of the position, orientation, and/or movement of the welding tool **14**. That is, the control circuitry **52** may determine the position and orientation of the welding tool **14** based on the feedback from the one or more motion sensors **426** when the workpiece **82** or the operator obscures (e.g., blocks) one or more markers of the welding tool **14** from the view of the sensing device **16**.

In certain embodiments, the control circuitry **52** may be configured to determine that a high impact event (e.g., dropped, used as a hammer, etc.) to the welding tool **14** has occurred based at least partly on the detected motion. Upon determining that a high impact event has occurred, the control circuitry **52** may store (e.g., log) an indication that the welding tool **14** has been impacted. Along with the indication, the control circuitry **52** may store other corresponding data, such as a date, a time of day, an acceleration,

a user name, welding tool identification data, and so forth. The control circuitry **52** may also be configured to show a notice on the display **62** to a welding operator requesting that the operator refrain from impacting the welding tool **14**. In some embodiments, the control circuitry **52** may be configured to use the motion detected by the motion sensor **426** to enable the welding operator to navigate and/or make selections within a software user interface (e.g., welding software, welding training software, etc.). For example, the control circuitry **52** may be configured to receive the acceleration and to make a software selection if the acceleration matches a predetermined pattern (e.g., the acceleration indicates a jerky motion in a certain direction, the acceleration indicates that the welding tool **14** is being shaken, etc.).

The vibration device **428** is configured to provide feedback to a welding operator by directing the welding tool **14** to vibrate and/or shake (e.g., providing vibration or haptic feedback). The vibration device **428** may provide vibration feedback during live welding and/or during simulated welding. As may be appreciated, vibration feedback during live welding may be tuned to a specific frequency to enable a welding operator to differentiate between vibration that occurs due to live welding and the vibration feedback. For example, vibration feedback may be provided at approximately 3.5 H/during live welding. Using such a frequency may enable a welding operator to detect when vibration feedback is occurring at the same time that natural vibration occur due to live welding. Conversely, vibration feedback may be provided at approximately 9 Hz during live welding. However, the 9 Hz frequency may be confused with natural vibration that occurs due to live welding.

The one or more microphones **429** are configured to facilitate determination of the position of the welding tool **14** with a local positioning system. The one or more microphones **429** of the welding tool **14** receive emitted signals (e.g., ultrasonic, RF) from beacons disposed at known locations about the welding environment. As may be appreciated, a local positioning system enables the determination of a location of an object when the object receives the emitted signals (i.e., via unobstructed line of sight) from three or more beacons at known positions. The control circuitry **52** (or control circuitry of another device) may determine the position of the welding tool **14** from the received signals via triangulation, trilateration, or multilateration. In some embodiments, the microphones **429** may facilitate the determination of the position of the welding tool **14** during welding when one or more of the sensing devices **16** (e.g., cameras) are obstructed by the workpiece **82** and/or the operator.

FIG. **19** is an embodiment of a method **430** for providing vibration feedback to a welding operator using the welding tool **14**. The control circuitry **52** (or control circuitry of another device) detects a parameter (e.g., work angle, travel angle, travel speed, tip-to-work distance, aim, etc.) corresponding to a welding operation (block **432**). As may be appreciated, the welding operation may be a live welding operation, a simulated welding operation, a virtual reality welding operation, and/or an augmented reality welding operation. The control circuitry **52** determines whether the parameter is within a first predetermined range (block **434**). As may be appreciated, the first predetermined range may be a range that is just outside of an acceptable range. For example, the parameter may be work angle, the acceptable range may be 45 to 50 degrees, and the first predetermined range may be 50 to 55 degrees. Accordingly, in such an

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example, the control circuitry 52 determine whether the work angle is within the first predetermined range of 50 to 55 degrees.

If the parameter is within the first predetermined range, the control circuitry 52 vibrates the welding tool 14 at a first pattern (block 436). The first pattern may be a first frequency, a first frequency modulation, a first amplitude, and so forth. Moreover, if the parameter is not within the first predetermined range, the control circuitry 52 determines whether the parameter is within a second predetermined range (block 438). The second predetermined range may be a range that is just outside of the first predetermined range. For example, continuing the example discussed above, the second predetermined range may be 55 to 60 degrees. Accordingly, in such an example, the control circuitry 52 determines whether the work angle is within the second predetermined range of 55 to 60 degrees. If the parameter is within the second predetermined range, the control circuitry 52 vibrates the welding tool 14 at a second pattern (block 440). The second pattern may be a second frequency, a second frequency modulation, a second amplitude, and so forth. It should be noted that the second pattern is typically different than the first pattern. In certain 44 embodiments, the first and second patterns may be the same. Furthermore, audible indications may be provided to the welding operator to indicate whether the parameter is within the first predetermined range or within the second predetermined range. In addition, audible indications may be used to indicate a parameter that is not within an acceptable range. In such embodiments, vibration may be used to indicate that a welding operator is doing something wrong, and audible indications may be used to identify what the welding operator is doing wrong and/or how to fix it. The parameter may be any suitable parameter, such as a work angle, a travel angle, a travel speed, a tip-to-work distance, and/or an aim. FIGS. 20 through 22 illustrate embodiments of various patterns.

FIG. 20 is a graph 442 of an embodiment of two patterns each including a different frequency for providing vibration feedback to a welding operator. A first pattern 444 is separated from a second pattern 446 by time 448. In the illustrated embodiment, the first pattern 444 is a first frequency and the second pattern 446 is a second frequency that is different from the first frequency. The first and second frequencies may be any suitable frequency. As may be appreciated, the first and second frequencies may be configured to be different than a natural frequency produced during live welding to facilitate a welding operator differentiating between the natural frequency and the first and second frequencies. Although the illustrated embodiment shows the first frequency being lower than the second frequency, in other embodiments, the second frequency may be lower than the first frequency.

FIG. 21 is a graph 450 of an embodiment of two patterns each including a different modulation for providing vibration feedback to a welding operator. A first pattern 452 is separated from a second pattern 454 by time 456. In the illustrated embodiment, the first pattern 452 is a first modulation and the second pattern 454 is a second modulation that is different from the first modulation. The first and second modulation may be any suitable modulation. For example, the first modulation may include a first number of vibration pulses (e.g., two pulses) and the second modulation may include a second number of vibration pulses (e.g., three pulses). Moreover, the modulation may vary a number of pulses, a time between pulses, etc. In certain embodiments, a number of vibration pulses and/or a time between pulses

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may be configured to gradually increase or decrease as a parameter moves toward or away from acceptable parameter values. Although the illustrated embodiment shows the first modulation as having fewer pulses than the second modulation, in other embodiments, the second modulation may have fewer pulses than the first modulation.

FIG. 22 is a graph 458 of an embodiment of two patterns each including a different amplitude for providing vibration feedback to a welding operator. A first pattern 460 is separated from a second pattern 462 by time 464. In the illustrated embodiment, the first pattern 460 is a first amplitude and the second pattern 462 is a second amplitude that is different from the first amplitude. The first and second amplitudes may be any suitable amplitude. Although the illustrated embodiment shows the first amplitude being lower than the second amplitude, in other embodiments, the second amplitude may be lower than the first amplitude.

The welding tool 14 may provide varied levels of vibration and visual feedback to the operator during simulated welding or live welding. For example, a first feedback mode of the welding tool 14 may provide visual feedback (e.g., via display 62) and vibration feedback to the operator until the operator initiates a simulated or live welding process, and the welding tool 14 may not provide visual or vibration feedback during the simulated or live welding process. A second feedback mode of the welding tool 14 may provide visual and vibration feedback to the operator both prior to and during the simulated or live welding process. A third feedback mode of the welding tool may provide visual and vibration feedback to the operator both prior to and during only simulated welding processes. As may be appreciated, some modes may provide only visual feedback prior to or during a simulated welding process, and other modes may provide only vibration feedback prior to or during a simulated welding process. In some embodiments, an instructor may specify the level of feedback that may be provided to the operator during simulated or live welding sessions to be evaluated. Moreover, the operator may selectively disable vibration and/or visual feedback provided by the welding tool prior to and during simulated or live welding.

FIG. 23 is a perspective view of an embodiment of the welding tool 14 having markers that may be used for tracking the welding tool 14. In some embodiments, the position of the welding tool 14 may be tracked prior to live welding to determine (i.e. calibrate) the shape of the welding joint. For example, the welding tool 14 may be utilized to trace the shape of a workpiece 82 in various positions including, but not limited, to welding positions 1G, 2G, 3G, 4G, 5G, 1F, 2F, 3F, 4F, 5F or 6F. The determined shape of the welding joint may be stored in the data storage system 318 for comparison with a subsequent live welding process along the welding joint. In some embodiments, the position of the welding tool 14 may be tracked during live welding and compared with the shape of the welding joint stored in the data storage system 318. The control circuitry 52 of the welding tool 14 and/or any other component of the welding system 10 may provide approximately real-time feedback to the operator regarding the position (e.g., location) and/or orientation of the welding tool 14 relative to the welding joint. The welding tool 14 includes a housing 466 that encloses the control circuitry 52 of the welding tool 14 and/or any other components of the welding tool 14. The display 62 and user interface 60 are incorporated into a top portion of the housing 466.

As illustrated, a neck 470 extends from the housing 466 of the welding tool 14. Markers for tracking the welding tool 14 may be disposed on the neck 470. Specifically, a mount-

ing bar **472** is used to couple markers **474** to the neck **470**. The markers **474** are spherical markers in the illustrated embodiment; however, in other embodiments, the markers **474** may be any suitable shape (e.g., such as a shape of an LED). The markers **474** are used by the one or more sensing devices **16** for tracking the position and/or the orientation of the welding tool **14**. As may be appreciated, three of the markers **474** are used to define a first plane. Moreover, the markers **474** are arranged such that a fourth marker **474** is in a second plane different than the first plane. Accordingly, the sensing device **16** may be used to track the position and/or the orientation of the welding tool **14** using the four markers **474**. It should be noted that while the illustrated embodiment shows four markers **474**, the mounting bar **472** may have any suitable number of markers **474**.

In certain embodiments, the markers **474** may be reflective markers, while in other embodiments the markers **474** may be light-emitting markers (e.g., light-emitting diodes LEDs). In embodiments in which the markers **474** are light-emitting markers, the markers **474** may be powered by electrical components within the housing **466** of the welding tool **14**. For example, the markers **474** may be powered by a connection **476** between the mounting bar **472** and the housing **466**. Furthermore, the control circuitry **52** (or control circuitry of another device) may be used to control powering on and/or off (e.g., illuminating) the markers **474**. In certain embodiments, the markers **474** may be individually powered on and/or off based on the position and/or the orientation of the welding tool **14**. In other embodiments, the markers **474** may be powered on and/or off in groups based on the position and/or the orientation of the welding tool **14**. It should be noted that in embodiments that do not include the mounting bar **472**, the connection **476** may be replaced with another marker **468** on a separate plane than the illustrated markers **468**. Embodiments of the welding tool **14** are described herein relative to a consistent set of coordinate axes **780**. An X-axis **782** is a horizontal direction along a longitudinal axis of the welding tool **14**, a Y-axis **784** is the vertical direction relative to the longitudinal axis, and a Z-axis **786** is a horizontal direction extending laterally from the welding tool **14**.

FIG. **24** is an embodiment of a neck **800** of the welding tool **14**, taken along line **24-24** of FIG. **23**. Visual markers **802** are arranged at predefined locations on the neck **800** to facilitate detection of the position and orientation of the welding tool **14** by the one or more sensing devices **16**. In some embodiments, the visual markers **802** are LEDs **64**. Additionally, or in the alternative, the visual markers **802** are directional, such that the one or more sensing devices **16** detect visual markers **802** that are oriented toward the one or more sensing devices **16** more readily than visual markers **802** that are less oriented toward the one or more sensing devices **16**. For example, LEDs **64** arranged on a surface may be directed to emit light primarily along an axis substantially perpendicular to the surface. In some embodiments, multiple sets of visual markers **802** are arranged on the neck **800**.

The visual markers **802** of each set may be oriented in substantially the same direction as the other visual markers **802** of the respective set. In some embodiments, a first set **804** of visual markers **802** is directed substantially vertically along the Y-axis **784**, a second set **806** of visual markers **802** is directed in a second direction **808**, and a third set **810** of visual markers **802** is directed in a third direction **812**. That is, the visual markers **802** of each set are oriented to emit light in substantially parallel directions as other visual markers **802** of the respective set. The second direction **808**

is substantially perpendicular to the X-axis **782** along the welding tool **14**, and is offset a second angle **814** from the Y-axis **784**. The third direction **812** is substantially perpendicular to the X-axis **782** along the welding tool **14**, and is offset a third angle **816** from the Y-axis **784**. In some embodiments, the second angle **814** and the third angle **816** have approximately the same magnitude. For example, the second set **806** of visual indicators **802** may be offset from the Y-axis **784** by 45° , and the third set **810** of visual indicators **802** may be offset from the Y-axis **784** by 45° , such that the second angle **814** is substantially perpendicular with the third angle **816**. The second angle **814** and the third angle **816** may each be between approximately 5° to 180° , 15° to 135° , 25° to 90° , or 30° to 75° . As may be appreciated, the neck **800** may have 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more sets of visual markers **802**, with each set oriented in a particular direction to facilitate detection by the one or more sensing devices **16**.

The visual markers **802** of each set may be arranged on the same or substantially parallel planes. For example, the first set **804** of visual markers **802** may be arranged on a first plane **818** or a plane substantially parallel to the first plane **818** that is perpendicular to the Y-axis **784**. The second set **806** of visual markers **802** may be arranged on a second plane **820** or a plane substantially parallel to the second plane **820** that is perpendicular to the second direction **808**. The third set **810** of visual markers **802** may be arranged on a third plane **822** or a plane substantially parallel to the third plane **822** that is perpendicular to the third direction **812**. As used herein, the term “substantially parallel” includes orientations within 10 degrees (or 5 degrees, or 1 degree, or less) of parallel, and the term “substantially perpendicular” includes orientations within 10 degrees (or 5 degrees, or 1 degree, or less) of perpendicular. In addition, as used herein, the term “substantially different”, when referring to orientations, includes angles that differ by more than 10 degrees, more than 15 degrees, more than 20 degrees, more than 30 degrees, more than 45 degrees, or more. The arrangements of the visual markers **802** of each set may facilitate tracking the welding tool **14** during simulated and/or live out of position welding processes including, but not limited to, vertical or overhead welding positions.

Structures **824** of the neck **800** may facilitate the orientation of the sets of the visual markers **802**. For example, a mounting surface of each structure **824** may be substantially parallel to a respective plane for the corresponding set of visual markers **802**. Moreover, the structures **824** may reduce or eliminate the detection of the respective visual marker **802** by the one or more sensing devices **16** when the respective visual marker **802** is oriented relative to the one or more sensing devices **16** at an angle greater than a threshold angle. For example, the second set **806** of visual markers **802** may be configured to be detected by the one or more sensing devices **16** when the operator holds the welding tool **14** with the one or more sensing devices **16** to the left of the operator (i.e., a left-handed operator), and the third set **810** of visual markers **802** may be configured to be detected by the one or more sensing devices **16** when the operator holds the welding tool **14** with the one or more sensing devices **16** to the right of the operator (i.e., a right-handed operator). The neck **800** and/or the structures **824** for the second set **806** of visual markers **802** may reduce or eliminate the detection of the second set **806** of visual markers **802** when a right-handed operator uses the welding tool **14**, and vice versa for the third set **810** of visual markers when a left-handed operator uses the welding tool **14**.

FIG. 25 is a top view of an arrangement of visual markers 802 on the neck 800 of the welding tool 14, similar to the embodiment of the neck 800 illustrated in FIG. 24. The visual markers 802 of the first set 804 (e.g., "A"), the second set 806 (e.g., "B"), and the third set 810 (e.g., "C") are arranged at different predefined positions on the neck 800 that enable the sensing device 16 to determine which side of the welding tool 14 is most directed towards the one or more sensing devices 16 via detecting a distinct pattern or arrangement that corresponds to each side (e.g., top, left 826, right 828, bottom, front) of the welding tool 14. Additionally, or in the alternative, the visual markers 802 (e.g., LEDs 64) of each set may be respectively colored, thereby enabling the one or more sensing devices 16 to determine which side of the welding tool 14 is most directed towards the one or more sensing devices 16 via color detection.

The one or more sensing devices 16 may track the position and orientation of the welding tool 14 relative to the welding stand 12 and the workpiece 82 when the one or more sensing devices 16 detect a threshold quantity of visual markers 802 of a set. The threshold quantity of visual markers 802 of a set may be less than or equal to the quantity of visual markers 802 of the respective set. For example, the one or more sensing devices 16 may detect the right side of the welding tool 14 when detecting the four visual markers 802 of the third set 810, the one or more sensing devices 16 may detect the top side of the welding tool 14 when detecting the five visual markers 802 of the first set 804, and the one or more sensing devices 16 may detect the left side of the welding tool when detecting the four visual markers 802 of the second set. In some embodiments, each set of visual markers 802 may have redundant visual markers, such that the one or more sensing devices 16 may track the position and the orientation of the welding tool 14 when one or more of the redundant visual markers are obscured from view. The one or more sensing devices 16 may track the position and the orientation with substantially the same accuracy, regardless of which set is detected by the one or more sensing devices 16.

The visual markers 802 may be arranged on the neck 800 of the welding tool 14 at positions relative to the X-axis 782 along the welding tool 14, and relative to a baseline 830. For example, the first set 804 may have five visual markers 802: two visual markers 802 along the baseline 830 near a first end 832 of the neck 800 and spaced a first offset 831 from the X-axis 782, a visual marker 802 spaced a first distance 834 from the baseline 830 in a midsection 836 of the neck 800 and spaced a second offset 838 from the X-axis 782 to the left side 826, a visual marker 802 spaced a third distance 840 from the baseline 830 in the midsection 836 and spaced the second offset 838 to the right side 828, and a visual marker 802 near a second end 842 of the neck 800 along the X-axis 782 and spaced a fourth distance 844 from the baseline 830. The second set 806 may have four visual markers 802: a visual marker 802 along the baseline 830 and spaced a third offset 846 from the X-axis 782 on the left side 826, a visual marker 802 spaced a fifth distance 848 from the baseline 830 along the X-axis 782 in the midsection 836, a visual marker 802 spaced a sixth distance 850 from the baseline 830 in the midsection 836 and spaced the second offset 838 from the X-axis 782 on the right side 828, and a visual marker 802 near the second end 842 of the neck 800 spaced the fourth distance 844 from the baseline 830 and spaced the second offset 838 on the left side 826. The third set 810 may have four visual markers 802: a visual marker 802 along the baseline 830 and spaced the third offset 846 from the X-axis 782 on the right side 828, a visual marker

802 spaced a seventh distance 852 from baseline 830 along the X-axis 782 in the midsection 836, a visual marker 802 spaced an eighth distance 854 from the baseline 830 in the midsection 836 and spaced the second offset 838 from the X-axis 782 on the left side 826, and a visual marker 802 near the second end 842 of the neck 800 spaced the fourth distance 844 from the baseline 830 and spaced the second offset 838 on the right side 828.

The arrangements (e.g., distances and offsets relative to the baseline 830 and X-axis 782) of the visual markers 802 for each set 804, 806, 810 may be stored in a memory of the welding system 10. For example, the arrangements may be stored in a memory as calibrations corresponding to a particular welding tool 14 coupled to the welding system 10. As discussed in detail below, the welding system 10 may detect the arrangement of the visual markers 802 directed to the one or more sensing devices 16, and determine the position and orientation of the welding tool 14 relative to the welding stand 12 and the workpiece 82 based at least in part on a comparison of the detected arrangement and the arrangements stored in memory. Each set of visual markers 802 may be calibrated, such as prior to an initial use, after reconnecting the welding tool 14, or at a predetermined maintenance interval. To calibrate a set of visual markers 802, the welding tool 14 may be mounted to the welding stand 12 in a predetermined position and orientation such that the respective set of visual markers 802 is substantially directed toward the one or more sensing devices 16. For example, the first set 804 may be calibrated when the welding tool 14 is mounted such that the Y-axis 784 of the welding tool 14 is generally directed toward the one or more sensing devices 16, the second set 806 may be calibrated when the welding tool 14 is mounted such that the second direction 808 is generally directed toward the one or more sensing devices 16, and the third set 810 may be calibrated when the welding tool 14 is mounted such that the third direction 812 is generally directed toward the one or more sensing devices 16. In some embodiments, the sets of visual markers 802 are calibrated when a calibration tool (e.g., calibration tool 610 discussed below) is coupled to the welding tool 14. The operator may verify the calibrations by moving the welding tool 14 about the welding environment relative to the welding stand 12 and the one or more sensing devices 16.

In certain embodiments, the visual markers 802 described herein, which are detected by the one or more sensing devices 16, may include passive markers (e.g., stickers, reflectors, patterns) and/or active markers (e.g., lights, LEDs). Accordingly, the visual markers 802 may be configured to either emit light that is detected by the one or more sensing devices 16 or reflect light that is detected by the one or more sensing devices 16. Furthermore, as described in greater detail herein, the visual markers 802 may include visible spectrum markers, as well as non-visible spectrum markers such as infrared markers, or some combination thereof, in certain embodiments. In addition, it should be noted that, while the embodiments illustrated in FIGS. 24 and 25 relate to visual markers 802 configured to be detected by the one or more sensing devices 16, in other embodiments, the markers 802 may be other types of markers configured to facilitate tracking of the position, orientation, and/or movement of the welding tool 14. For example, in certain embodiments, the markers 802 may include electromagnetic, acoustic, microelectromechanical (MEMS), or other types of components, that may effectively function as markers on the welding tool 14 to facilitate tracking of

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position, orientation, and/or movement of the welding tool 14 by the one or more sensing devices 16.

FIG. 26 is an embodiment of a method 478 for displaying on a display 62 of a welding tool 14 a welding parameter in relation to a threshold. In the illustrated embodiment, the control circuitry 52 (or control circuitry of another device) receives a selection made by a welding operator of a welding parameter associated with a position, an orientation, and/or a movement of the welding tool 14 (block 480). For example, the welding operator may select a button on the user interface 60 of the welding tool 14 to select a welding parameter. The welding parameter may be any suitable welding parameter, such as a work angle, a travel angle, a travel speed, a tip-to-work distance, an aim, and so forth. As may be appreciated, the welding system 10 may select the welding parameter automatically without input from a welding operator. After the selection is made, the display 62 of the welding tool 14 displays or shows a representation of the welding parameter in relation to a predetermined threshold range and/or target value for the welding parameter (block 482). The displayed welding parameter is configured to change as the position of the welding tool 14 changes, as the orientation of the welding tool 14 changes, and/or as movement of the welding tool 14 changes. Thus, the welding operator may use the welding tool 14 to properly position and/or orient the welding tool 14 while performing (e.g., prior to beginning, starting, stopping, etc.) a welding operation, thereby enabling the welding operator to perform the welding operation with the welding parameter within the predetermined threshold range or at the target value.

For example, the welding operator may desire to begin the welding operation with a proper work angle. Accordingly, the welding operator may select "work angle" on the welding tool 14. After "work angle" is selected, the welding operator may position the welding tool 14 at a desired work angle. As the welding operator moves the welding tool 14, a current work angle is displayed in relation to a desired work angle. Thus, the welding operator may move the welding tool 14 around until the current work angle matches the desired work angle and/or is within a desired range of work angles. As may be appreciated, the display 62 may be turned off and/or darkened so that it is blank during a welding operation. However, a welding operator may select a desired welding parameter prior to performing the welding operation. Even with the display 62 blank, the control circuitry 52 may be configured to monitor the welding parameter and provide feedback to the welding operator during the welding operation (e.g., vibration feedback, audio feedback, etc.).

FIG. 27 is an embodiment of a set of screenshots of the display 62 of the welding tool 14 for showing a welding parameter in relation to a threshold. The set of screenshots illustrate various ways that welding parameters are displayed for a welding operator for performing a welding operation. As may be appreciated, in certain embodiments, the welding parameters may be displayed to the welding operator before, during, and/or after the welding operation. Screen 484 illustrates a work angle that is not within a predetermined threshold range. A parameter portion 486 of the display 62 indicates the selected parameter. Moreover, a range section 488 indicates whether the selected parameter is within the predetermined threshold range. Furthermore, a parameter value section 490 indicates the value of the selected parameter. On the screen 484, the work angle of 38 is out of range as indicated by the arrow extending outward from the central circle. Screen 492 illustrates a work angle

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of 45 that is within the predetermined threshold range as indicated by no arrow extending from the central circle.

As may be appreciated, the one or more sensing devices 16 may be configured to detect whether the travel angle is a drag angle (e.g., the travel angle is ahead of the welding arc) or a push angle (e.g., the travel angle follows behind the welding arc). Accordingly, screen 494 illustrates a drag travel angle of 23 that is outside of a predetermined threshold range as indicated by an arrow extending outward from a central circle. Conversely, screen 496 illustrates a push travel angle of 15 that is within the predetermined threshold range as indicated by no arrow extending from the central circle. Furthermore, screen 498 illustrates a travel speed of 12 that is within of a predetermined threshold range as indicated by a vertical line aligned with the central circle. Conversely, screen 500 illustrates a travel speed of 18 that is outside of (i.e., greater than) the predetermined threshold range as indicated by the vertical line to the right of the central circle. As may be appreciated, a travel speed that is less than a predetermined threshold range may be indicated by a vertical line to the left of the central circle. The travel speed indicator may dynamically move relative to the central circle in real-time during a weld process based at least in part on the determined travel speed, thereby guiding the operator to perform the weld process with a travel speed within the predetermined threshold range.

Screen 502 illustrates a tip-to-work distance of 1.5 that is greater than a predetermined threshold range as indicated by a small circle within an outer band. Moreover, screen 504 illustrates the tip-to-work distance of 0.4 that is less than a predetermined threshold range as indicated by the circle outside of the outer band. Furthermore, screen 506 illustrates the tip-to-work distance of 1.1 that is within the predetermined threshold range as indicated by the circle substantially filling the area within the outer band. Moreover, screen 508 illustrates an aim of 0.02 that is within a predetermined threshold range as indicated by a line 509 aligned with a central circle. Conversely, screen 510 illustrates an aim of 0.08 that is not within the predetermined threshold range as indicated by the line 509 toward the top part of the central circle. In some embodiments, the line 509 of screens 508 and 510 represents the joint relative to the tip of the welding tool 14. For example, screens 508 and 510 illustrate the aim of the welding tool 14 when the welding tool 14 is oriented substantially perpendicular to the joint (as illustrated by the line 509). Screen 511 illustrates the aim of the welding tool 14 when the welding tool 14 is at least partially angled relative to the joint, as indicated by the line 509 and the tilted orientation of the welding tool 14. That is, while the positions of the welding tool 14 relative to the joint (e.g., line 509) corresponding to screens 508 and 511 are substantially the same, the orientation of the line 509 of screen 508 on the display corresponds to a perpendicular orientation of the welding tool 14 relative to the joint and the orientation of the line 509 of screen 511 on the display 62 corresponds to a non-perpendicular orientation of the welding tool 14 relative to the joint. The orientation of the range section 488 (e.g., aim indicator, angle indicator, CTWD indicator) may be rotated on the display by a rotation angle defined as the angle difference between a front edge 513 of the display 62 and the joint. The graphical representations on the display 62 may correspond to the orientation of the welding tool 14 to the joint rather than to the orientation of the display 62 relative to the operator. For example, when the welding tool 14 is positioned near a vertical joint such that the welding tool 14 is substantially parallel with the joint, the line 509 on the display 62 may be oriented vertically. The joint indicator

line 509 may be substantially perpendicular to the travel speed indicator discussed above with screens 498 and 500.

While specific graphical representations have been shown on the display 62 in the illustrated embodiment for showing a welding parameter in relation to a threshold, other embodiments may use any suitable graphical representations for showing a welding parameter in relation to a threshold. Moreover, in certain embodiments individual parameter visual guides may be combined so that multiple parameters are visually displayed together.

Furthermore, in certain embodiments, the welding system may detect if the welding tool 14 is near and/or far from a welding joint. Being near the welding joint is a function of the contact tip-to-work distance (CTWD) and aim parameters. When both the CTWD and aim parameters are within suitable predetermined ranges, the welding system 10 may consider the welding tool 14 near the welding joint. Furthermore, the control circuitry 52 of the welding tool 14 or another device may determine the work angle, the travel angle, and the travel speed based at least in part on the position of the welding tool 14 relative to a known (e.g., calibrated) welding joint of the workpiece 82 when the CTWD and the aim are substantially constant along the welding joint. As may be appreciated, the position and orientation of the welding tool 14 may be determined via the sensing devices 16 and markers on the welding tool 14, the one or more motion sensors 426, and/or the one or more microphones 429 of the welding tool 14. Moreover, when the welding tool 14 is near the welding joint, the visual guides may be displayed on the welding tool 14. When the welding tool 14 is near the welding joint and in the live welding mode, a message (e.g., warning message) may be displayed on a display indicating that proper welding equipment (e.g., welding helmet, etc.) should be in place as a safety precaution for onlookers. However, an external display may continue to display the real-time data at a safe distance from the welding operation. Moreover, in some embodiments, when the welding tool 14 is near the welding joint and in the live welding mode, the display of the welding tool 14 may be changed (e.g., to substantially blank and/or clear, to a non-distracting view, to a predetermined image, etc.) while a welding operator actuates the trigger of the welding tool 14. When the welding tool 14 is far from the welding joint, actuating the trigger of the welding tool 14 will not perform (e.g., begin) a test run. Furthermore, when the welding tool 14 is far from the welding joint, actuating the welding tool 14 will have no effect in a non-live welding mode, and may feed welding wire in the live welding mode without beginning a test run.

FIG. 28 is an embodiment of a method 512 for tracking the welding tool 14 in the welding system 10 using at least four markers. One or more cameras (e.g., such as one or more cameras of the one or more sensing devices 16) are used to detect the markers of the welding tool 14 (block 514). As discussed above, the markers may be reflective markers and/or light-emitting markers. Furthermore, the markers may include four or more markers to facilitate determining an accurate position and/or orientation of the welding tool 14. One or more processors 20 of the computer 18 (or other processors) may be used with the sensing devices 16 to track the position of the welding tool 14 and/or the orientation of the welding tool 14 based on the detected markers (block 516). If the one or more cameras are unable to detect one or more of the markers, the one or more processors 20 (or control circuitry, such as the control circuitry 52) may be configured to block live welding while the one or more cameras are unable to detect the markers

(block 518). However, in some embodiments of the welding system 10, one or more cameras integrated with the helmet 41 may enable detection of four or more markers to facilitate determining an accurate position and/or orientation of the welding tool 14 with respect to the welding helmet 41. Thus, one or more cameras integrated with the helmet 41 may facilitate detection of the position and/or orientation of the welding tool 14 for welding processes that would otherwise obscure the one or more markers from cameras mounted to the welding stand 12. As may be appreciated, the position and/or orientation of the welding helmet 41 in the welding environment may be determined via the one or more sensing devices 16 of the welding system 10 in a similar manner as described above for the welding tool 14 where the markers are observable. In some embodiments, the display 62 of the welding tool 14 may be configured to display a message indicating that the markers are not detected while the one or more cameras are unable to detect the markers of the welding tool 14 (block 520). Accordingly, live welding using the welding tool 14 may be blocked if the welding tool 14 is unable to be tracked by the one or more sensing devices 16.

Some embodiments of the welding system 10 may track the welding tool 14 in the welding environment during periods where one or more of the markers 474 are obscured and not detected. As described above, the welding system 10 may track the position and/or the orientation of the welding tool 14 based at least in part on feedback from one or more motion sensors 426 (e.g., accelerometers, gyroscopes) of the welding tool 14. Moreover, embodiments of the welding system 10 with beacons of a local positioning system and one or more microphones 429 on the welding tool 14 may determine a position of the welding tool 14 within the welding environment when the portions (e.g., markers 474) of the welding tool 14 are obscured from the line of sight of some sensing devices 16 (e.g., cameras). Accordingly, block 518 of method 512 (to block live welding while the markers are not detected) may be optional during intervals when the control circuitry 52 may otherwise determine the position of the welding tool 14 within the welding environment. Additionally, or in the alternative, the welding system 10 may track the welding tool 14 in the welding environment when the welding tool 14 does not have markers 474 as described above. Therefore, in some embodiments, the control circuitry 52 permits live welding while the markers are not detected or not present on the welding tool 14.

FIG. 29 is an embodiment of a method 522 for detecting the ability for the processor 20 (or any other processor) to communicate with the welding tool 14. The welding tool 14 is configured to detect a signal from the processor 20 (block 524). The signal is provided from the processor 20 to the welding tool 14 at a predetermined interval. In certain embodiments, the signal may be a pulsed signal provided from the processor 20 to the welding tool 14 at the predetermined interval. Moreover, the signal is provided to the welding tool 14 so that the welding tool 14 is able to determine that the welding tool 14 is able to communicate with the processor 20. If the welding tool 14 does not receive the signal from the processor 20 within the predetermined interval, control circuitry 52 (or control circuitry of another device) is configured to block live welding using the welding tool 14 while the signal is not detected (block 526). Moreover, the display 62 may be configured to display a message indicating that the signal from the processor 20 is not detected while the live welding is blocked (block 528). Accordingly, the welding tool 14 may detect the ability for the processor 20 to communicate with the welding tool 14.

FIG. 30 is an embodiment of a method 530 for calibrating a curved weld joint that may be used with the welding system 10. One or more cameras (e.g., such as one or more cameras of the one or more sensing devices 16) are used to detect a first position (e.g., first calibration point) of the curved weld joint (block 532). For example, a calibration tool and/or the welding tool 14 may be used to identify the first position of the curved weld joint to the one or more cameras (e.g., such as by touching a tip of the calibration tool and/or the welding tool 14 to the first position). In addition, the one or more cameras may be used to track the calibration tool and/or the welding tool 14 to determine a position and/or an orientation of the calibration tool and/or the welding tool 14 for detecting the first position of the curved weld joint.

Moreover, the one or more cameras are used to detect a second position (e.g., second calibration point) of the curved weld joint (block 534). For example, the calibration tool 120 and/or the welding tool 14 may be used to identify the second position of the curved weld joint to the one or more cameras. In addition, the one or more cameras may be used to track the calibration tool 120 and/or the welding tool 14 to determine a position and/or an orientation of the calibration tool 120 and/or the welding tool 14 for detecting the second position of the curved weld joint. Furthermore, the one or more cameras are used to detect a curved portion of the curved weld joint between the first and second positions of the curved weld joint (block 536). For example, the calibration tool 120 and/or the welding tool 14 may be used to identify the curved weld joint between the first and second positions of the curved weld joint. In addition, the one or more cameras may be used to track the calibration tool 120 and/or the welding tool 14 to determine a position and/or an orientation of the calibration tool 120 and/or the welding tool 14 for detecting the curved portion of the curved weld joint. As may be appreciated, during operation, the first position may be detected, then the curved weld joint may be detected, and then the second position may be detected. However, the detection of the first position, the second position, and the curved weld joint may occur in any suitable order. In certain embodiments, a representation of the curved portion of the curved weld joint may be stored for determining a quality of a welding operation by comparing a position and/or an orientation of the welding tool 14 during the welding operation to the stored representation of the curved portion of the curved weld joint. As may be appreciated, in certain embodiments, the welding operation may be a multi-pass welding operation.

Moreover, calibration for some joints, such as circular weld joints (e.g., Pipe joints) may be performed by touching the calibration tool to three different points around the circumference of the circular weld joint. A path of the circular weld joint may then be determined by calculating a best-fit circle that intersects all three points. The path of the circular weld joint may be stored and used to evaluate welding parameters of training welds. For a more complex geometry, the calibration tool 120 and/or the welding tool 14 might be dragged along the entire joint in order to indicate the joint to the system so that all of the parameters may be calculated.

In some embodiments, the method 530 for calibrating a curved weld joint that may be used with the welding system 10 may not utilize the welding tool 14 or the calibration tool to determine the path of the weld joint. That is, the control circuitry 52 may utilize one or more images captured by cameras (e.g., such as one or more cameras of the one or more sensing devices 16) to detect the first position (block

532), the second position (block 534), and the curved portion (block 536) of the weld joint. Additionally, or in the alternative, the control circuitry 52 may utilize one or more emitters (e.g., emitters 105, 109) to emit a visible pattern (e.g., grid, point field) onto the workpiece 82 and weld joint. Cameras configured to detect the visible pattern may determine the shape of the workpiece 82 and/or the path of the weld joint based on particular features of the shape and orientation of the visible pattern on the workpiece 82 and weld joint. The control circuitry 52 may determine the shape of the weld joint and/or the workpiece 82 utilizing object recognition algorithms (e.g., edge detection) applied to the one or more captured images or visible pattern. The operator may provide input to aid the object recognition, such as selecting a type of joint (e.g., butt, tee, lap, corner, edge) and/or the shape (e.g., planar, tubular, curved) of the workpiece 82.

FIG. 31 is a diagram of an embodiment of a curved weld joint 538. Such a curved weld joint 538 may be calibrated using the method 530 described in FIG. 30. The curved weld joint 538 is on a workpiece 540. Specifically, the curved weld joint 538 includes a first position 542, a second position 544, and a curved portion 546. Using the method 530, a shape of the curved weld joint 538 may be determined and/or stored for evaluating a welding operator performing a welding operation on the curved weld joint 538.

FIG. 32 is a diagram of an embodiment of a complex shape workpiece 539 with a curved weld joint 541. The curved weld joint 541 may be calibrated via markings 543 added to the workpiece 539 near the curved weld joint 541. The markings 543 may include, but are not limited to stickers, reflectors, paints, or pigments applied to the workpiece 539 via a roller tool 545. The operator may roll a marking wheel 547 of the roller tool 545 along the curved weld joint 541, depositing the markings 543 on the workpiece 539. For example, pads 549 on the marking wheel 547 may apply the markings 543 to the workpiece 539 at regular intervals along the curved weld joint 541. Cameras of the one or more sensing devices 16 on the welding stand 12 and/or integrated with the helmet 41 of the welding system 10 may detect the markings 543. Control circuitry of the welding system 10 may determine the shape of the complex shape workpiece 539 and/or the welding system 10 may determine the welding path along the curved weld joint 541 based at least in part on the detected markings 543. The shape of the complex shape workpiece 539 and/or the welding path of the curved weld joint 541 may be stored for evaluating a welding operator performing a welding operation on the curved weld joint 541. While the markings 543 shown in FIG. 39 are discontinuous, some embodiments of the markings 543 may be continuous along the curved weld joint 541.

FIG. 33 is an embodiment of a method 548 for tracking a multi-pass welding operation. One or more cameras (e.g., such as one or more cameras of the one or more sensing devices 16) are used to detect a first pass of the welding tool 14 along a weld joint during the multi-pass welding operation (block 550). Moreover, the one or more cameras are used to detect a second pass of the welding tool 14 along the weld joint during the multi-pass welding operation (block 552). Furthermore, the one or more cameras are used to detect a third pass of the welding tool 14 along the weld joint during the multi-pass welding operation (block 554). The control circuitry 52 (or control circuitry of another device) may be configured to store a representation of the first pass, the second pass, and/or the third pass together as a single welding operation for determining a quality of the multi-

pass welding operation. As may be appreciated, the multi-pass welding operation may be a live welding operation, a training welding operation, a virtual reality welding operation, and/or an augmented reality welding operation.

FIG. 34 is a perspective view of an embodiment of the welding stand 12. The welding stand 12 includes the welding surface 88 supported by the legs 90. Moreover, the welding surface 88 includes one or more slots 91 to facilitate positioning of a workpiece on the welding surface 88. Furthermore, the welding surface 88 includes multiple apertures 556 (e.g., holes or openings) that extend through the welding surface 88. The apertures 556 may be used to enable the one or more sensing devices 16 to determine a position and/or an orientation of the welding surface 88. Specifically, markers may be arranged below the apertures 556, yet within the view of the one or more sensing devices 16 to enable the sensing devices 16 to determine the position and/or the orientation of the welding surface 88. The markers may be arranged below the welding surface 88 to facilitate longer lasting markers and/or to block debris from covering the markers, as explained in greater detail in relation to FIG. 35.

Drawers 558 are attached to the welding stand 12 to enable storage of various components with the welding stand 12. Moreover, wheels 560 are coupled to the welding stand 12 to facilitate easily moving the welding stand 12. Adjacent to the drawers 558, a calibration tool holder 562 and a welding tool holder 564 enable storage of the calibration tool 120 and the welding tool 14. In certain embodiments, the welding system 10 may be configured to detect that the calibration tool 120 is in the calibration tool holder 562 at various times, such as before performing a welding operation. A support structure 566 extending vertically from the welding surface 88 is used to provide structure support to the one or more sensing devices 16 and the display 32. Moreover, a tray 568 is coupled to the support structure 566 to facilitate storage of various components.

The protective cover 102 is positioned over the display 32 to block certain environmental elements from contacting the display 32 (e.g., weld spatter, smoke, sparks, heat, etc.). A handle 570 is coupled to the protective cover 102 to facilitate rotation of the protective cover 102 from a first position (as illustrated) used to block certain environmental elements from contacting the display 32 to a second raised position away from the display 32, as illustrated by arrows 572. The second position is not configured to block the environmental elements from contacting the display 32. In certain embodiments, the protective cover 102 may be held in the first and/or the second position by a latching device, a shock, an actuator, a stop, and so forth.

In certain embodiments, a switch 573 is used to detect whether the protective cover 102 is in the first position or in the second position. Moreover, the switch 573 may be coupled to the control circuitry 52 (or control circuitry of another device) and configured to detect whether the protective cover 102 is in the first or the second position and to block or enable various operations (e.g., live welding, auxiliary power, etc.) while the switch 573 detects that the protective cover 102 is in the first and/or the second position. For example, if the switch 573 detects that the protective cover 102 is in the second position (e.g., not properly covering the display 32), the control circuitry 52 may block live welding and/or simulation welding (with the protective cover 102 in the second position, the one or more sensing devices 16 may be unable to accurately detect markers). As another example, if the switch 573 detects that the protective cover 102 is in the second position, control circuitry of the

welding stand 12 may block the availability of power provided to an outlet 574 of the welding stand 12. In certain embodiments, the display 32 may show an indication that the protective cover 102 is in the first and/or the second position. For example, while the protective cover 102 is in the second position, the display 32 may provide an indication to the welding operator that live welding and/or power at the outlet 574 are unavailable. The welding stand 12 includes speakers 575 to enable audio feedback to be provided to a welding operator using the welding stand 12. Furthermore, in certain embodiments, if the trigger of the welding tool 14 is actuated while the protective cover 102 is in the second position, the welding system 10 may provide visual and/or audio feedback to the operator (e.g., the welding system 10 may provide a visual message and an audible sound effect).

As illustrated, the support structure 566 includes a first arm 576 and a second arm 578. The first and second arms 576 and 578 are rotatable about the support structure 566 to enable the first and second arms 576 and 578 to be positioned at a selected height for vertical and/or overhead welding. In the illustrated embodiment, the first and second arms 576 and 578 are independently (e.g., separately) rotatable relative to one another so that the first arm 576 may be positioned at a first vertical position while the second arm 578 may be positioned at a second vertical position different from the first vertical position. In other embodiments, the first and second arms 576 and 578 are configured to rotate together. Moreover, in certain embodiments, the first and second arms 576 and 578 may be rotated independently and/or together based on a selection by a welding operator. As may be appreciated, in other embodiments, arms may not be coupled to the support structure 566, but instead may be positioned at other locations, such as being positioned to extend vertically above one or more front legs, etc. Furthermore, in some embodiments, a structure may be coupled to the welding stand 12 to facilitate a welding operator leaning and/or resting thereon (e.g., a leaning bar).

Each of the first and second arms 576 and 578 includes a shock 580 (or another supporting device) that facilitates holding the first and second arms 576 and 578 in selected vertical positions. Moreover, each of the first and second arms 576 and 578 includes a braking system 582 configured to lock the first and second arms 576 and 578 individually in selected positions. In certain embodiments, the braking system 582 is unlocked by applying a force to a handle, a switch, a pedal, and/or another device.

As illustrated, the workpiece 82 is coupled to the second arm 578 for overhead and/or vertical welding. Moreover, the first arm 576 includes the welding plate 108 for overhead, horizontal, and/or vertical welding. As may be appreciated, the workpiece 82, the welding plate 108, and/or a clamp used to hold the welding plate 108 may include multiple markers (e.g., reflective and/or light emitting) to facilitate tracking by the one or more sensing devices 16. For example, in certain embodiments, the workpiece 82, the welding plate 108, and/or the clamp may include three markers on one surface (e.g., in one plane), and a fourth marker on another surface (e.g., in a different plane) to facilitate tracking by the one or more sensing devices 16. As illustrated, a brake release 584 is attached to each of the first and second arms 576 and 578 for unlocking each braking system 582. In certain embodiments, a pull chain may extend downward from each brake release 584 to facilitate unlocking and/or lowering the first and second arms 576 and 578, such as while the brake release 584 of the first and second arms 576 and 578 are vertically above the reach of

a welding operator. Thus, the welding operator may pull a handle of the pull chain to unlock the braking system **582** and/or to lower the first and second arms **576** and **578**.

As illustrated, the second arm **578** includes a clamp assembly **588** for coupling the workpiece **82** to the second arm **578**. Moreover, the clamp assembly **588** includes multiple T-handles **590** for adjusting, tightening, securing, and/or loosening clamps and other portions of the clamp assembly **588**. In certain embodiments, the first arm **576** may also include various T-handles **590** for adjusting, tightening, securing, and/or loosening the welding plate **108**. As may be appreciated, the clamp assembly **588** may include multiple markers (e.g., reflective and/or light emitting) to facilitate tracking by the one or more sensing devices **16**. For example, in certain embodiments, the clamp assembly **588** may include three markers on one surface (e.g., in one plane), and a fourth marker on another surface (e.g., in a different plane) to facilitate tracking by the one or more sensing devices **16**. It should be noted that the welding system **10** may include the clamp assembly **588** on one or both of the first and second arms **576** and **578**.

In certain embodiments, the one or more sensing devices **16** may include a removable cover **592** disposed in front of one or more cameras of the sensing device **16** to block environmental elements (e.g., spatter, smoke, heat, etc.) or other objects from contacting the sensing device **16**. The removable cover **592** is disposed in slots **594** configured to hold the removable cover **592** in front of the sensing device **16**. In certain embodiments, the removable cover **592** may be inserted, removed, and/or replaced without the use of tools. As explained in detail below, the removable cover **592** may be disposed in front of the sensing device **116** at an angle to facilitate infrared light passing therethrough.

As illustrated, a linking assembly **596** may be coupled between the first and/or second arms **576** and **578** and the one or more sensing devices **16** to facilitate rotation of the sensing devices **16** as the first and/or second arms **576** and **578** are rotated. Accordingly, as the first and/or second arms **576** and **578** are rotated, the sensing device **16** may also rotate such that one or more cameras of the one or more sensing devices **16** are positioned to track a selected welding surface. For example, if the first and/or second arms **576** and **578** are positioned in a lowered position, the one or more sensing devices **16** may be configured to track welding operations that occur on the welding surface **88**. On the other hand, if the first and/or second arms **576** and **578** are positioned in a raised position, the one or more sensing devices **16** may be configured to track vertical, horizontal, and/or overhead welding operations. In some embodiments, the first and/or second arms **576** and **578** and the one or more sensing devices **16** may not be mechanically linked, yet rotation of the first and/or second arms **576** and **578** may facilitate rotation of the sensing devices **16**. For example, markers on the first and/or second arms **576** and **578** may be detected by the one or more sensing devices **16**, and the sensing devices **16** may move (e.g., using a motor) based on the sensed position of the first and/or second arms **576** and **578**.

In some embodiments, movement of the first and/or second arms **576**, **578** may at least partially invalidate previous calibrations of the one or more sensing devices **16** with components of the welding stand **12**. For example, after the sensing devices **16** are calibrated with the main (e.g., horizontal) welding surface **88** of the welding stand **12**, subsequent movement of the first and second arms **576**, **578** may invalidate the calibration of the main welding surface **88** based at least in part on movement of the sensing devices

16. Accordingly, the one or more sensing devices **16** may be recalibrated with the main welding surface **88** after the operator performs welding sessions that utilize the first and/or second arms **576**, **578**. In some embodiments, the computer **18** notifies the operator via the display **32** and/or audible notifications when the one or more sensing devices **16** are to be recalibrated based on detected movement of the sensing devices **16** relative to the welding surface **88**. Additionally, or in the alternative, the display **62** of the welding tool **14** may notify the operator when the one or more sensing devices **16** are to be recalibrated.

FIG. **35** is a cross-sectional view of an embodiment of the welding surface **88** of the welding stand **12** of FIG. **34**. As illustrated, the welding surface **88** includes multiple apertures **556** extending therethrough between an upper plane **597** of the welding surface **88** and a lower plane **598** of the welding surface **88**. A bracket **599** is positioned beneath each aperture **556**. The brackets **599** may be coupled to the welding surface **88** using any suitable fastener or securing means. In the illustrated embodiment, the brackets **599** are coupled to the welding surface **88** using fasteners **600** (e.g., bolts, screws, etc.). In other embodiments, the brackets **599** may be welded, bonded, or otherwise secured to the welding surface **88**. Moreover, in certain embodiments, the brackets **599** may be mounted to a lateral side of the welding stand **12** rather than the welding surface **88**. Markers **602** are coupled to the brackets **599** and positioned vertically below the apertures **556**, but the markers **602** are horizontally offset from the apertures **556** to block dust and/or spatter from contacting the markers **602** and to enable the one or more sensing devices **16** to sense the markers **602**. In some embodiments, the markers **602** may be positioned within the apertures **556** and/or at any location such that the motion tracking system is positioned on one side of the upper plane **597** and the markers **602** are positioned on the opposite side of the upper plane **597**. As may be appreciated, the markers **602** may be light reflective and/or light-emissive. For example, in certain embodiments, the markers **602** may be formed from a light reflective tape. In some embodiments, the markers **602** may be spherical markers. Accordingly, the one or more sensing devices **16** may detect the markers **602** to determine a position and/or an orientation of the welding surface **88**.

FIG. **36** is a cross-sectional view of an embodiment of a sensing device **16** having the removable cover **592**. As illustrated, the removable cover **592** is disposed in the slots **594**. The sensing device **16** includes a camera **604** (e.g., infrared camera) having a face **605** on a side of the camera **604** having a lens **606**. The removable cover **592** is configured to enable infrared light to pass therethrough and to block environmental elements (e.g., spatter, smoke, heat, etc.) or other objects from contacting the lens **606** of the camera **604**. As may be appreciated, the camera **604** may include one or more infrared emitters **607** configured to emit infrared light. If the removable cover **592** is positioned directly in front of the face **605**, a large amount of the infrared light from the infrared emitters **607** may be reflected by the removable cover **592** toward the lens **606** of the camera **604**. Accordingly, the removable cover **592** is positioned at an angle **608** relative to the face **605** of the camera **604** to direct a substantial portion of the infrared light from being reflected toward the lens **606**. Specifically, in certain embodiments, the removable cover **592** may be positioned with the angle **608** between approximately 10 to 60 degrees relative to the face **605** of the camera **604**. Moreover, in other embodiments, the removable cover **592** may be positioned with the angle **608** between approximately 40 to 50

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degrees (e.g., approximately 45 degrees) relative to the face 605 of the camera 604. The removable cover 592 may be manufactured from any suitable light-transmissive material. For example, in certain embodiments, the removable cover 592 may be manufactured from a polymeric material, or any other suitable material.

FIG. 37 is a perspective view of an embodiment of a calibration tool 610. As may be appreciated, the calibration tool 610 may be used to calibrate a workpiece, a work surface, a weld joint, and so forth, for a welding operation. The calibration tool 610 includes a handle 612 to facilitate gripping the calibration tool 610. Moreover, the calibration tool 610 is configured to be detected by the one or more sensing devices 16 for determining a spatial position that a tip 614 of the calibration tool 610 is contacting. In certain embodiments, the computer 18 coupled to the one or more sensing devices 16 may be configured to determine a calibration point merely by the tip 614 contacting a specific surface. In other embodiments, the computer 18 is configured to determine a calibration point by a welding operator providing input indicating that the tip 614 is contacting a calibration point. Furthermore, in the illustrated embodiment, the computer 18 is configured to detect a calibration point by the tip 614 contacting the calibration point while a downward force is applied to the calibration tool 610 via the handle. The downward force directs a distance between two adjacent markers to decrease below a predetermined threshold thereby indicating a selected calibration point. The one or more sensing devices 16 are configured to detect the change in distance between the two adjacent markers and the computer 18 is configured to use the change in distance to identify the calibration point. The handle 612 is coupled to a light-transmissive cover 616. Moreover, a gasket 618 is coupled to one end of the light-transmissive cover 616, while an end cap 620 is coupled to an opposite end of the light-transmissive cover 616. During operation, as a downward force is applied to the calibration tool 610 using the handle 612, a distance 622 between the tip 613 and the gasket 618 decreases.

FIG. 38 is a perspective view of the calibration tool 610 of FIG. 37 having the outer cover 616 removed. The calibration tool 610 includes a first portion 624 having a first shaft 626. Moreover, the first shaft 626 includes the tip 614 on one end, and a bearing 628 (or mounting structure) on an opposite end. In certain embodiments, the bearing 628 has a cup like structure configured to fit around a contact tip of the welding tool 14. Furthermore, the first shaft 626 includes a first marker 630 and a second marker 632 coupled thereto. The calibration tool 610 also includes a second portion 634 having a second shaft 636 with a third marker 638 coupled thereto. A spring 640 is disposed around the second shaft 636 between the third marker 638 and the bearing 628. As may be appreciated, the spring 640 facilitates the third marker 638 being directed toward the second marker 632. For example, as a downward force is applied to the calibration tool 610 using the handle 612, the spring 640 is compressed to decrease a first distance 642 between the second and third markers 632 and 638. In contrast, as the downward force is removed from the calibration tool 610, the spring 640 is decompressed to increase the first distance 642 between the second and third markers 632 and 638. A second distance 644 between the first and second markers 630 and 632 is fixed, and a third distance 646 between the first marker 630 and the tip 614 is also fixed.

In certain embodiments, the welding system 10 uses the calibration tool 610 to detect calibration points using a predetermined algorithm. For example, the third distance

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646 between the tip 614 and the closest marker to the tip 614 (e.g., the first marker 630) is measured. The third distance 646 is stored in memory. The second distance 644 between two fixed markers (e.g., the first marker 630 and the second marker 632) is measured. The second distance 644 is also stored in memory. Furthermore, a compressed distance between the markers (e.g., the second and third markers 632 and 638) with the spring 640 disposed therebetween is measured. A line is calculated between the two fixed markers using their x, y, z locations. The line is used to project a vector along that line with a length of the third distance 646 starting at the first marker 630 closest to the tip 614. The direction of the vector may be selected to be away from the compressed markers. Accordingly, the three dimensional location of the tip may be calculated using the markers. In some embodiments, only two markers may be used by the calibration tool 610. In such embodiments, an assumption may be made that the marker closest to the tip 614 is the marker closest to the work surface (e.g., table or clamp). Although the calibration tool 610 in the illustrated embodiment uses compression to indicate a calibration point, the calibration tool 610 may indicate a calibration point in any suitable manner, such as by uncovering a marker, covering a marker, turning on an LED (e.g., IR LED), turning off an LED (e.g., IR LED), enabling and/or disabling a wireless transmission to a computer, and so forth.

The first, second, and third markers 630, 632, and 638 are spherical, as illustrated; however, in other embodiments, the first, second, and third markers 630, 632, and 638 may be any suitable shape. Moreover, the first, second, and third markers 630, 632, and 638 have a reflective outer surface and/or include a light-emitting device. Accordingly, the first, second, and third markers 630, 632, and 638 may be detected by the one or more sensing devices 16. Therefore, the one or more sensing devices 16 are configured to detect the first, second, and third distances 642, 644, and 646. As the first distance 642 decreases below a predetermined threshold, the computer 18 is configured to identify a calibration point. As may be appreciated, the first, second, and third distances 642, 644, and 646 are all different to enable the one or more sensing devices 16 and/or the computer 18 to determine a location of the tip 614 using the location of first, second, and third markers 630, 632, and 638.

To calibrate a workpiece 82, the workpiece 82 may first be clamped to the welding surface 88. After the workpiece 82 is clamped to the welding surface 88, a welding operator may provide input to the welding system to signify that the workpiece 82 is ready to be calibrated. In certain embodiments, the clamp used to secure the workpiece 82 to the welding surface 88 may include markers that facilitate the welding system detecting that the workpiece 82 is clamped to the welding surface 88. After the welding system 10 receives an indication that the workpiece 82 is clamped to the welding surface 88, the welding operator uses the calibration tool 610 to identify two calibration points on the workpiece 82. Where the clamp assembly 588 securing the workpiece 82 has markers (e.g., visual markers 802), the measurements of the joint calibration tool 610 may be relative to the markers of the clamp assembly 588. Accordingly, the computer 18 may compensate for movement of the workpiece 82 and/or clamp assembly 588 after the joint has been calibrated based on identification of the clamp markers. Specifically, in the illustrated embodiment, the welding operator touches the tip 614 to a first calibration point and applies downward force using the handle 612 until the welding system 10 detects a sufficient change in distance between adjacent markers, thereby indicating the first cali-

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bration point. Furthermore, the welding operator touches the tip **614** to a second calibration point and applies downward force using the handle **612** until the welding system **10** detects a sufficient change in distance between adjacent markers, thereby indicating the second calibration point. In certain embodiments, the welding system **10** will only detect a calibration point if the calibration tool **610** is pressed and held at the calibration point for a predetermine period of time (e.g., 0.1, 0.3, 0.5, 1.0, 2.0 seconds, and so forth). The welding system **10** may be configured to capture multiple calibration points (e.g., 50, 100, etc.) over the predetermined period of time and average them together. If movement of the multiple calibration points greater than a predetermined threshold is detected, the calibration may be rejected and done over. Furthermore, if a first point is successfully calibrated, a second point may be required to be a minimum distance away from the first point (e.g., 2, 4, 6 inches, etc.). If the second point is not the minimum distance away from the first point, calibration of the second point may be rejected and done over. The welding system **10** uses the two calibration points to calibrate the workpiece **82**.

In certain embodiments, the welding system **10** may determine a virtual line between the first and second calibration points. The virtual line may be infinitely long and extend beyond the first and second calibration points. The virtual line represents a weld joint. Various welding parameters (e.g., work angle, travel angle, contact tip-to-work distance (CTWD), aim, travel speed, etc.) may be in reference to this virtual line. Accordingly, the virtual line may be important for calculating the various welding parameters.

It should be noted that in certain embodiments the first, second, and third markers **630**, **632**, and **638** are all disposed vertically above the handle **612**, while in other embodiments, one or more of the first, second, and third markers **630**, **632**, and **638** are disposed vertically below the handle **612** to enable a greater distance between adjacent markers. In certain embodiments, the first portion **624** may be removed from the calibration tool **610** and coupled to a contact tip of the welding tool **14** for calibrating the welding tool **14**. As may be appreciated, the tip **614** of the calibration tool **610** may be any suitable shape. FIGS. **39** through **41** illustrate a few embodiments of shapes the tip **614** may have.

Specifically, FIG. **39** is a side view of an embodiment of a pointed tip **648** of the calibration tool **610**. Using the pointed tip **648**, the calibration tool **610** may be used for calibrating various joints on the workpiece **82**, such as the illustrated fillet joint, a lap joint, a butt joint with no root opening, and so forth. Moreover, FIG. **40** is a side view of an embodiment of a rounded tip **650** of the calibration tool **610**. Using the rounded tip **650**, the calibration tool **610** may be used for calibrating various joints on the workpiece **82**, such as the illustrated butt joint with no root opening, a filled joint, a lap joint, and so forth. In certain embodiments, the tip of the calibration tool **610** may be removable and/or reversible, such that the tip includes two different types of tips (e.g., one type of tip on each opposing end). Accordingly, a welding operator may select the type of tip used by the calibration tool **610**. In certain embodiments, one or more markers may be coupled to the calibration tool **610** if the calibration tool **610** is reversible. The one or more markers may be used to indicate

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which side of the tip is being used so that the welding system **10** may use a suitable marker-tip distance for calibration calculations.

FIG. **42** is an embodiment of a method **654** for detecting a calibration point. The one or more sensing devices **16** (or another component of the welding system **10**) detect a first marker of the calibration tool **610**, a second marker of the calibration tool **610**, and/or a third marker of the calibration tool **610** (block **656**). Moreover, the welding system **10** determines a first distance between the first marker and the second marker and/or a second distance between the second marker and the third marker (block **658**). Furthermore, the welding system **10** detects whether the first distance or the second distance is within a predetermined distance range (e.g., signifying a compressed distance) (block **660**). The welding system **10** determines a position of a calibration point if the first distance or the second distance is within the predetermined distance range (e.g., signifying a compressed distance) (block **662**). In addition, the welding system **10** determines a location of a calibration tip of the calibration tool **610** relative to at least one of the first, second, and third markers to determine the spatial position of the calibration point (block **664**).

FIG. **43** is an embodiment of a method **666** for determining a welding score based on a welding path. Accordingly, the method **666** may be used for evaluating a welding operation. The one or more sensing devices **16** (or any suitable motion tracking system) detect an initial position of the welding operation (block **668**). Moreover, the one or more sensing devices **16** detect a terminal position of the welding operation (block **670**). In addition, the one or more sensing devices **16** detect a spatial path of the welding operation between the initial position and the terminal position (block **672**). For example, the one or more sensing devices **16** track a position and/or an orientation of the welding operation. The welding system determines a score of the welding operation based at least partly on the spatial path of the welding operation (e.g., whether the welding operation receives a passing score based on the spatial path of the welding operation) (block **674**). For example, in certain embodiments, the spatial path of the welding operation may alone be used to determine whether a welding score fails. In some embodiments, the one or more sensing devices **16** may be used to detect a calibration point that corresponds to the initial position and/or a calibration point that corresponds to the terminal position.

For example, in certain embodiments, the welding system **10** determines whether the welding operation receives a passing score by determining whether: a distance of the path of the welding operation is greater than a predetermined lower threshold, the distance of the path of the welding operation is less than the predetermined lower threshold, the distance of the path of the welding operation is greater than a predetermined upper threshold, the distance of the path of the welding operation is less than the predetermined upper threshold, the path of the welding operation deviates substantially from a predetermined path of the welding operation, the path of the welding operation indicates that multiple welding passes occurred at a single location along a weld joint, a time of welding along the path of the welding operation is greater than a predetermined lower threshold, the time of welding along the path of the welding operation is less than the predetermined lower threshold, the time of welding along the path of the welding operation is greater than a predetermined upper threshold, and/or the time of welding along the path of the welding operation is less than the predetermined upper threshold.

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Moreover, in some embodiments, for the welding system 10 to determine a score, the welding system 10 may disregard a first portion of the path adjacent to the initial position and a second portion of the path adjacent to the terminal position. For example, the first portion of the path and the second portion of the path may include a distance of approximately 0.5 inches. Moreover, in other embodiments, the first portion of the path and the second portion of the path may include portions of the path formed during a time of approximately 0.5 seconds.

FIG. 44 is an embodiment of a method 676 for transitioning between welding modes using a user interface of the welding tool 14. The control circuitry 52 of the welding tool 14 (or control circuitry of another device) detects a signal produced by a user interface of the welding tool 14 indicating a request to change the welding mode (e.g., welding training mode) (block 678). Moreover, the control circuitry 52 determines a length of time that the signal is detected (block 680). The control circuitry 52 is configured to change the welding mode from a simulation mode (e.g., virtual reality mode, augmented reality mode, etc.) to a live welding mode if the length of time that the signal is detected is greater than a predetermined threshold (block 682). Conversely, the control circuitry 52 is configured to change the welding mode from the live welding mode to the simulation mode merely if the signal is detected (block 684) (e.g., there is no length of time that the signal is to be detected before a transition from the live welding mode is made). The control circuitry 52 is configured to direct the welding tool 14 to vibrate after changing to the live welding mode (block 686). For example, the control circuitry 52 may be configured to direct the welding tool 14 to vibrate two or more times (e.g., vibration pulses) to indicate a change to the live welding mode.

Moreover, the control circuitry 52 may be configured to direct the welding tool 14 to vibrate any suitable number of times (e.g., predetermined number of times) to indicate a change to the live welding mode. As may be appreciated, the signal indicating the request to change the welding mode may be produced by pressing a button on the user interface 60 of the welding tool 14. As such, the welding mode may be changed from the live welding mode by pressing and releasing the button (e.g., the button does not have to be held down for a predetermined period of time). In contrast, the welding mode may be changed from the simulation mode to the live welding mode by pressing and holding the button for a predetermined period of time. In certain embodiments, an audible sound may be produced after changing welding modes. Furthermore, in some embodiments an audible sound and a vibration may accompany any change between welding modes. In addition, a display of the welding tool 14 may show the welding mode after changing the welding mode. In some embodiments, the display may flash the welding mode on the display a predetermined number of times.

FIG. 45 is a block diagram of an embodiment of a remote training system, such as a helmet training system 41 (e.g., helmet). In some embodiments, the helmet 41 facilitates acquisition of welding parameters (e.g., a work angle, a travel angle, a contact tip to workpiece distance, a welding tool travel speed, a welding tool orientation, a welding tool position, an aim of the welding tool relative to the joint of the workpiece, and so forth) of a weld process and/or arc parameters (e.g., a welding voltage, a welding current, wire feed speed) without utilizing the welding stand 12 described above. As may be appreciated, operators utilize helmets during welding, and the helmet 41 integrates the one or more

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sensing devices 16 (e.g., emitters, receivers) into the helmet. Various embodiments of the helmet 41 may incorporate the computer 18 (e.g., as a controller), couple to the computer 18 via a wired connection, or couple to the computer via a wireless connection. In some embodiments, the helmet 41 utilizes a lens 700 to shield the operator from the arc during a weld process. In some embodiments, the display 32 is disposed within the helmet 41 such that the operator may view the display 32 and the lens 700 in preparation for or during a weld process. The display 32 may be a heads-up display that is at least partially overlaid with the operator's view through the helmet 41. As may be appreciated, the welding software may utilize the display 32 disposed within the helmet 41 to present information to the operator in a similar manner as described above with the display 32 external to the helmet 41. For example, the display 32 of the helmet 41 may show a visual representation (e.g., number, text, color, arrow, graph) of one or more arc parameters, one or more welding parameters, or any combination thereof. That is, the display 32 of the helmet 41 may display a visual representation of a welding parameter in relation to a predetermined threshold range and/or to a target value for the welding parameter according to a selected welding assignment. In some embodiments, the display 32 may show a graphical representation of a welding parameter or an arc parameter in relation to a threshold similar to the displays 62 of the welding tool 14 described above with FIG. 27. Additionally, the display 32 of the helmet 41 may show one or more parameters (e.g., arc parameters, welding parameters) before, during, or after the operator using the helmet 41 performs a welding session (e.g., welding assignment).

The helmet 41 utilizes one or more integrated sensing devices 16 to determine the welding parameters from observations of the welding tool 14 and the workpiece 82. The one or more sensing devices 16 of the helmet 41 may include one or more receivers 702 including, but not limited to, microphones, cameras, infrared receivers, or any combination thereof. Moreover, in some embodiments, one or more emitters 704 may emit energy signals (e.g., infrared light, visible light, electromagnetic waves, acoustic waves), and reflections of the energy signals may be received by the one or more receivers 702. In some embodiments, fiducial points 706 (e.g., markers) of the welding tool 14 and/or the workpiece 82 are active markers (e.g., LEDs) that emit energy signals, as discussed above with FIGS. 24 and 25. Accordingly, the one or more receivers 702 of the helmet 41 may receive energy signals emitted from active markers. In particular, the receivers 702 may identify fiducial points (e.g., markers) 706 disposed on the workpiece 82, the work environment and/or the welding tool 14, and the receivers 702 may send feedback signals to the computer 18 (e.g., controller) that correspond to the identified fiducial points. As discussed above, arrangements of the identified fiducial points 706 may enable the sensing device 16 to determine the position and orientation of the welding tool 14 in the work environment 708. The computer 18 (e.g., controller) may determine the distances between the fiducial points 706 and may determine the welding parameters based at least in part on the feedback from the receivers 702. Additionally, the computer 18 (e.g., controller) may be coupled to sensors within the welding power supply 28, the wire feeder 30, and/or the welding tool 14 to determine the arc parameters of the welding process.

In some embodiments, the helmet 41 may determine the types of components of the welding system 10 from the identified fiducial points. For example, the fiducial points of a TIG welding tool are different than the fiducial points of

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a MIG welding tool. Moreover, the welding software 244 executed by the computer 18 may control the welding power supply 28 and/or the wire feeder 30 based at least in part on the determined types of components of the welding system 10. For example, the helmet 41 may control the arc parameters (e.g., weld voltage, weld current) based on the type of welding tool 14, the welding position of the workpiece 82, and/or the workpiece material. The helmet 41 may also control the arc parameters based on the experience or certification status of the operator associated with the registration number 293. For example, the helmet 41 may control the welding power supply 28 to reduce the weld current available for selection by an operator with less than a predetermined threshold of experience with weld processes on relatively thin workpieces or in the overhead welding position. In some embodiments, the one or more sensing devices 16 of the helmet 41 include motion sensors 709 (e.g., gyroscopes and accelerometers) that are coupled to the computer 18. The motion sensors 709 may enable the computer 18 to determine the orientation and relative movement of the helmet 41 within the environment.

In some embodiments, the helmet 41 includes the operator identification system 43. The operator identification system 43 may utilize a scanner 710 (e.g., fingerprint scanner, retinal scanner, barcode scanner) or an input/output device 712 (e.g., keyboard, touch screen) to receive the identification information from the operator. As discussed above, the identification information may be associated with the registration number 293 unique to the operator. Welding data received by the computer 18 (e.g., controller) may be stored in the memory device(s) 22 or storage device(s) 24, as discussed above. The computer 18 (e.g., controller) may associate the received and stored welding data with the registration number 293 of the identified operator. The network device 36 couples to the network 38 via a wired or wireless connection to store the welding data 327 from the helmet 41 in the data storage system 318 (e.g., cloud storage system). In some embodiments the helmet 41 may store welding data locally within the storage device(s) 24 of the computer 18 while the helmet 41 is operated remotely (e.g., production floor, worksite). The helmet 41 may be configured to upload stored welding data to the data storage system 318 (e.g., cloud storage system) upon connection with the network 38, such as when the operator stows the helmet 41 at the end of a shift or at the end of a work week. In some embodiments, the network device 36 of the helmet 41 may stream welding data to the data storage system 318 (e.g., cloud storage system) via the network 38 during and/or after the operator performs a welding session.

As may be appreciated, using the systems, devices, and techniques described herein, a welding system 10 may be provided for training welding operators. The welding system 10 may be cost efficient and may enable welding students to receive high quality hands on training. While the welding systems 10 described herein may be utilized for receiving and correlating weld data 327 for training and educational purposes, it may be appreciated that the welding systems 10 described herein may be utilized to monitor operators and obtain weld data 327 from non-training weld processes. That is, weld data obtained from non-training weld processes may be utilized to monitor weld quality and/or weld productivity of previously trained operators. For example, the weld data 327 may be utilized to verify that welding procedures for a particular weld process were executed. As illustrated in FIG. 45, multiple welding systems 10 may be coupled to the data storage system 318 (e.g., cloud storage system) via the network 38. Accordingly, the data storage system 318 may

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receive welding data 327 associated with registration numbers 293 from multiple welding systems 10 (e.g., systems with welding stands 12, helmet training systems 41). Moreover, welding data associated with each registration number 293 may include serial numbers 329 corresponding to other welding sessions performed by the respective operator. Moreover, as utilized herein, the term “assignment” is not to be limited to weld tests performed by the operator for training and educational purposes. That is, assignments may include non-training weld processes, training simulated weld processes, and training live weld processes, among others. Moreover, the term “welding session” may include, but is not limited to, welding assignments, welds performed on a production floor, welds performed at a worksite, or any combination thereof.

The welding data 327 of the data storage system 318 (e.g., cloud storage system) may be monitored and/or managed via a remote computer 44 coupled to the network 38. The stored welding data 327 corresponds to weld processes (e.g., live, simulated, virtual reality) performed by various operators at one or more locations. FIG. 46 illustrates an embodiment of a user viewable dashboard screen 720 that may be utilized by a manager or instructor to monitor and/or analyze the stored welding data 327 in the data storage system 318. The welding data 327 may be organized by characteristics (e.g., filter criteria) of the welding data 327. Characteristics of the welding data 327 that may be utilized for sorting the welding data 327 may include, but are not limited to, one or more organizations 722 (e.g., training center, employer, work site), one or more groups 724 (e.g., shift) within the organization, one or more registration numbers 726 of operators within the selected organizations 722 or groups 724, time (e.g., dates 728, time of day) welding processes were performed, systems 725, and weld identifications 730 (e.g., particular welding assignments, unique identifier associated with a welding session, workpiece part number, or types of welds). For example, welding data 327 associated with one or more registration numbers 293 over a period of time (e.g., dates 728) and across different organizations 722 or different groups 724 may be displayed on the dashboard screen 720. Accordingly, the manager or instructor may track the progress of an operator over time across different organizations via welding data associated with the registration number 293 of the operator. In some embodiments, a welding data type 732 (e.g., live training, live non-training, simulated, virtual reality) may be used to filter the viewed welding data. Moreover, a welding process type 735 (e.g., GMAW, TIG, SMAW) may be used to filter the viewed welding data in some embodiments. As may be appreciated, welding data for each welding session (e.g., welding assignment) may be sorted (e.g., filtered) into various subsets. As illustrated in FIG. 46, live, non-training welds performed by an operator with registration number 58,794 on Jun. 25, 2014 with system I may be displayed on the dashboard screen 720 via selection of one or more of the appropriate fields for registration numbers 726, systems 725, dates 728, and welding data types 732.

Additionally, or in the alternative, the instructor may utilize a search control 733 to search for welding data 327 associated with various parameters (e.g., serial numbers 329, organization 722, group 724, operator name, registration number 726, time, welding data type) corresponding to welding sessions performed by operators. Upon selection of a set of welding data, a section 734 of the dashboard screen 720 may display graphical indicia (e.g., a score) associated with the selected welding data and/or at least a portion of the welding data. Moreover, details of the welding data 327 may

be viewed upon selection of the welding data **327** and a user control **736**. The dashboard screen **720** may enable the manager or instructor to save or edit the arrangement of the welding data on the dashboard screen **720**. Furthermore, the dashboard screen **720** may enable the manager or instructor to export at least a portion of the welding data **327**. For example, the manager may export the welding data **327** corresponding to the sessions performed by a set of operators over the course of a day or a week. The dashboard screen **720** may enable the manager or instructor to export the welding data **327** in various formats, including but not limited to a comma-separated values (CSV) file, a spreadsheet file, and a text file. In some embodiments, the manager or instructor may remove a subset of welding data (e.g., demonstration welding data) from the data storage system (e.g., cloud storage system). Additionally, or in the alternative, the manager or instructor may edit the welding data type **732**, such as to revise training weld data as non-training weld data, revise the operator associated with welding data, revise the time associated with welding data, and so forth.

As may be appreciated, the dashboard screen **720** may enable the manager or instructor to monitor, compare, and analyze the welding data associated with one or more registration numbers **726**. In some embodiments, the performance, experience, and historical data of welding operators may be compared across organizations or groups via the registration numbers **726**. In some embodiments, the dashboard screen **720** may enable the manager or instructor to set goals or provide assignments to desired registration numbers **726**. Furthermore, the manager or instructor may monitor and adjust previously established goals. The dashboard screen **720** may enable notes or comments regarding the welding performance associated with one or more registration numbers to be entered and stored with the welding data.

FIG. **47** illustrates an embodiment of the welding system **10** in the welding environment **11** that may track the position and/or orientation of the welding tool **14** without utilizing markers **474** on the welding tool **14** discussed above with respect to FIGS. **23-25**. The welding system **10** of FIG. **47** may track the position and/or orientation of the welding tool **14** prior to conducting a welding process. In some embodiments, the welding system **10** of FIG. **47** may track the position and/or orientation of the welding tool **14** during the welding process. One or more depth sensors **750** are arranged at various positions in the welding environment **11**, such as a first depth sensor **752** above the workpiece **82**, a second depth sensor **754** integrated with the welding helmet **41** (e.g., helmet training system), or a third depth sensor **756** horizontal with the workpiece **82**, or any combination thereof. Each depth sensor **750** may have an emitter configured to emit a visible pattern at a desired wavelength and a camera configured to monitor the visible pattern in the welding environment **11**. The visible pattern emitted by each depth sensor **750** may be the same or different than the visible pattern emitted by other depth sensors **750**. Moreover, the desired wavelength of the visible pattern for each depth sensor **750** may be the same or different among the depth sensors **750**. FIG. **47** illustrates respective emitted visible patterns from each depth sensor **750** with solid arrows, and illustrates the patterns reflected toward each depth sensor **750** with dashed arrows. The wavelength of the visible patterns may be within the infrared, visible, or ultraviolet spectrum (e.g., approximately 1 mm to 120 nm). The emitter of each depth sensor emits the respective visible pattern into the welding environment **11** onto the welding surface **88**, the workpiece **82**, the welding tool **14**, or the operator, or any combination thereof. By observing the

visible pattern reflected in the welding environment **11**, the computer **18** may track objects (e.g., welding tool **14**, operator) moving within the welding environment. Additionally, the computer **18** may identify the shape of the workpiece **82** or a welding joint path on the workpiece **82** based upon observations of the visible pattern in the welding environment **11**.

As may be appreciated, an arc **758** struck by the welding tool **14** with the workpiece **82** emits electromagnetic radiation. The wavelengths and the intensity of the emissions at each wavelength of the electromagnetic radiation emitted by the arc may be based on a variety of factors including, but not limited to, the workpiece material, the electrode material, the shielding gas composition, the weld voltage, the weld current, the type of welding process (e.g., SMAW, MIG, TIG). In some embodiments, the one or more sensing devices **16** include a light sensor configured to detect the wavelengths electromagnetic radiation of the welding environment **11** prior to and during welding processes. The computer **18** of the welding system **10** may determine the emitted wavelengths and the intensity of the emitted wavelengths from the emitted based on feedback received from the one or more sensing devices **16**. Additionally, or in the alternative, the computer **18** may determine the emitted wavelengths and the intensity of the emitted wavelengths from data stored in memory of the computer **18** or the data storage system **318**, the welding parameters, and the arc parameters. For example, the computer **18** may determine that the arc for steel MIG welding has different predominant wavelengths than the arc for aluminum TIG welding.

In some embodiments, the wavelengths of the one or more visible patterns emitted by the depth sensors **750** may be selected to reduce noise from the arc **758** during welding processes. Furthermore, in some embodiments, the depth sensors **750** can vary the wavelength of the emitted visible pattern. Accordingly, the computer **18** may adaptively control the wavelengths of the emitted visible patterns to improve the accuracy of the position and orientation determinations from the depth sensor feedback. That is, the computer **18** may control the depth sensors **750** to emit the visible pattern in a first range for steel MIG welding, and to emit the visible pattern in a different second range for aluminum TIG welding. Additionally, or in the alternative, the computer **18** may filter the signals received by the depth sensors **750** to reduce or eliminate the effects of the emissions by the arc **758**.

Furthermore, the arc **758** may not be continuous during the weld formation for some welding processes (e.g., short circuit MIG). The emitted electromagnetic radiation when the arc **758** is out (e.g., during a short circuit phase of the welding process) may be substantially less than the emitted electromagnetic radiation when the arc **758** is live. The computer **18** may control the depth sensors **750** to emit the respective visible patterns when the arc **758** rather than when the arc **758** is live, thereby enabling the depth sensors **750** to track the position and/or orientation of the welding tool **14** during the weld process. That is, the computer **18** may synchronize the emitted visible patterns to substantially coincide with the short circuit phases of the welding process. The short circuit frequency may be greater than 30 Hz, thereby enabling the computer **18** to determine the position and/or the orientation of the welding tool **14** in the welding environment **11** at approximately 30 Hz or more.

Additionally, or in the alternative to the depth sensors **750**, the welding system **10** may utilize a local positioning system **762** to determine the position of the welding tool **14** within the welding environment **11**. Beacons **764** of the local

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positioning system 762 are arranged at known locations about the welding environment and emit signals 766 (e.g., ultrasonic, RF) received via one or more microphones 429 on the welding tool. The computer 18 coupled to the one or more microphones 429 may determine the location of the welding tool 14 within the welding environment 11 based at least in part on received signals from three or more beacons 764. The computer may determine the position of the welding tool 14 via triangulation, trilateration, or multilateration. More than three beacons 764 of the local positioning system 762 distributed about the welding environment 11 increase the robustness of the local positioning system 762 and increase the likelihood that the welding tool 14 is within a line of sight of at least three beacons 764 at any point along a workpiece 82 having a complex shape (e.g., pipe). In some embodiments, beacons 764 may be positioned with depth sensors 750 or components of the welding system 10, such as the welding power supply 28.

Returning to FIGS. 24 and 25, embodiments of the welding tool 14 may have multiple sets of visual markers 802 to facilitate detection of the position and the orientation of the welding tool 14 relative to the welding stand 12 and to the workpiece 82. In some embodiments, the visual markers 802 are LEDs 64 that may be independently controlled. For example, each set (e.g., first set 804, second set 806, third set 810) of LEDs 64 may be separately controlled so that only one set is turned on and emits light at a time. Reducing the quantity of visual markers 802 detectable by the one or more sensing devices 16 may reduce the complexity of the determination of the position and the orientation of the welding tool 14. That is, the one or more sensing devices 116 may readily determine which side (e.g., top, left, right) of the welding tool 14 is facing the one or more sensing devices 16 based on the arrangement of the detected LEDs 64 when only one set of LEDs 64 is turned on at a time. The control circuitry 52 of the welding tool 14 may control the LEDs 64 so that at least one set of the LEDs 64 is detectable by the one or more sensing devices 16 during a simulated or live welding session (e.g., live welding assignment).

The processor 20 coupled to the one or more sensing devices 16 and/or the control circuitry 52 may determine which set of LEDs 64 to turn on to track the movement and position of the welding tool 14 utilizing a method 860 illustrated in FIG. 48. As may be appreciated, the method 860 may be performed by a controller, which includes, but is not limited to the processor 20, the control circuitry 52, or a combination thereof. Generally, the controller may turn on each set of LEDs 64 sequentially for a detection interval, then compare the response detected by the one or more sensing devices 16 from each set to determine which set of LEDs 64 enables better tracking data. For example, the controller may turn on (block 862) the left set (e.g., second set 806) of LEDs 64. The controller determines (node 864) whether the left set of LEDs 64 is detected within the detection interval (e.g., approximately 50 to 500 ms). If the left set of LEDs 64 is not detected at node 864, the controller may turn on (block 866) the top set (e.g., first set 802) of LEDs 64. The controller then determines (node 868) whether the top set of LEDs 64 is detected. If the top set of LEDs 64 is not detected at node 868, the controller may turn on (block 870) the right set (e.g., third set 810) of LEDs 64. The controller then determines (node 872) whether the right set of LEDs 64 is detected. If the right set of LEDs 64 is not detected at node 872, then the controller may return to the start of the method 860, and turn on (block 862) the left set of LEDs 64. In some embodiments, the controller may

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repeat method 860 to turn on each set of LEDs 64 in sequence until at least one set of LEDs 64 is detected during the detection interval.

As discussed herein, when the controller determines whether a set of LEDs 64 is detected (e.g., nodes 864, 868, 872), the controller may determine whether the threshold quantity of LEDs 64 for the respective set is detected. As discussed above, the threshold quantity may be less than or equal to the total quantity of visual markers (e.g., LEDs 64) of a respective set. In some embodiments, the controller is configured to determine a rigid body (RB) model of the welding tool 14 upon detection of the threshold quantity of LEDs 64. The controller determines (nodes 874) which rigid body model corresponding to tracked sets of LEDs 64 is the closest to an ideal model. As may be appreciated, the ideal model may correspond to when a set of LEDs 64 is directed directly towards the one or more sensing devices 16 within a predetermined range of angles (e.g., approximately 20, 30, 45, or 60 degrees). Furthermore, each set of LEDs 64 side may have its own predetermined range of angles, such as approximately 45 degrees for the top set of LEDs 64 and approximately 30 degrees for the left and right sets of LEDs 64. In some embodiments, the first set 802 of LEDs 64 may approximate the ideal model when the Y-axis 784 relative to the welding tool 14 is directed to the one or more sensing devices 16. If the determined rigid body model of the welding tool 14 corresponding to one set or LEDs 64 (e.g., second set 806) does not approximate the ideal model, the controller may turn off the one set and turn on the next set (e.g., first set 802) of LEDs 64 to determine if an approximately ideal rigid body model may be detected with the next set. Additionally, or in the alternative, the controller may utilize the detected non-ideal angle of one set (e.g., first set 804) of LEDs 64 and the predetermined relative angles of the other sets (e.g., second set 806, third set 810) of LEDs 64 to determine which set (e.g., third set 810) of LEDs 64 corresponds closest to the ideal model, thereby enabling the controller to turn on that set (e.g., third set 810) of LEDs 64 directly without turning on other sets (e.g., second set 806). The controller may be configured to latch to a set of turned on LEDs 64 when the determined rigid body model approximates the ideal model.

In some embodiments, a set of LEDs 64 may approximate the ideal model when LEDs 64 are oriented within approximately 20 to 60 degrees or approximately 30 to 50 degrees of the one or more sensing devices 16. Accordingly, based on the orientation of the sets of LEDs 64, some embodiments of the controller may be able to determine a rigid body model corresponding to more than one set of LEDs 64 at a time. Where multiple rigid body models may be determined, the controller may determine which set of LEDs 64 is most oriented toward the one or more sensing devices 16. Moreover, the controller may utilize a hysteresis control when the welding tool orientation fluctuates near an angle threshold where multiple rigid body models may be determined respective sets of LEDs 64. As discussed above, the first set 802 of LEDs 64 may be oriented approximately along the Y-axis 784, and the second set 806 of LEDs 64 may be oriented so that the second direction 808 is offset approximately 45 degrees from the Y-axis 784. In some embodiments, rigid body models may be determined for each respective set of LEDs 64 oriented within approximately 30° of the one or more sensing devices 16, such that rigid body models for each respective set may be determined for an overlapping range of approximately 15°. Utilizing the hysteresis control, the controller may remain latched to the first set 802 of LEDs 64 when the first set 802 is oriented within

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approximately 25° offset from the Y-axis **784** and within approximately 20° offset from the second direction **808**. That is, the hysteresis control may reduce the turning off and on sets of LEDs **64** when multiple sets of LEDs **64** may be detectable by the one or more sensing devices **16** and prevents rapid oscillation between sets of LEDs **64** when the welding tool **14** is oriented near the threshold between sets of LEDs **64**.

Upon latching to a set of LEDs **64** that approximate the ideal model, the controller (blocks **876**) may update the items displayed on the display **32** of the welding system **10**, the display **32** of the helmet **41**, and/or the display **62** of the welding tool **14** based at least in part on the position and orientation determined from the tracked set of LEDs **64**. The controller may maintain the status (e.g., on, off) of each set of LEDs **64** while the determined rigid body model approximates the ideal model. In some embodiments, the controller may repeat method **860** at intervals during operation, thereby turning on each set of LEDs **64** sequentially to verify that the determined rigid body model of the latched set of LEDs **64** most approximates the ideal model. For example, the controller may repeat method **860** every 1, 5, or 15 minutes. Additionally, or in the alternative, the controller may repeat method **860** upon receipt of an assignment, selection of an assignment, upon lifting the welding tool **14** from the welding stand **12**, or any combination thereof.

As discussed above, various elements of the welding system **10** may have markers that for utilization to track movement of the respective element within the welding environment in real-time and/or to calibrate the position and orientation of the element relative to the welding stand **12** or to the workpiece **82**. For example, the welding stand **12** of FIG. 3 may have the first and second markers **95**, **96**, the welding surface **112** may have the markers **116**, **118**, the calibration tool **120** of FIG. 4 may have the markers **130**, the fixture assembly **132** of FIG. 5 may have the first and second markers **134**, **136**, and the welding tool **14** of FIG. 23 may have the visual markers **802**. FIG. 49 illustrates a cross-sectional view of a base component **880** that may be provided with visual markers **882**. The base component **880** may include, but is not limited to, the welding stand **12**, the workpiece **82**, the welding surface **112**, the calibration tool **120**, the fixture assembly **132**, the welding tool **14**, the clamp assembly **588**, or any combination thereof.

The base component **880** may be coated with a thermally insulating layer **884** (e.g., plastic, fabric, ceramic, resin, glass). The thermally insulating layer **884** may be wrapped about, molded to, mechanically fastened to, or bonded to the base component **880**. As may be appreciated, the base component **880** may receive or conduct thermal heat from the welding process. The visual markers **882** may be positioned at distinct locations on the insulating layer **884** of the base component **880**. The visual markers **882** may be readily detectable by the one or more sensing devices **16**. For example, the visual markers **882** may be reflective to one or more electromagnetic waves. For example, the visual markers **882** may reflect visible and/or infrared (IR) light. The position of the each visual marker **882** may be configured to enable the one or more sensing devices **16** to determine the position and the orientation of the base component **880** within the welding environment. The visual markers **882** may be positioned on one or more faces of the base component **880**. Different quantities and/or arrangements of the visual markers **882** on each side of the base component **880** may facilitate identification of the respective sides based on detection of the arrangement of the visual markers **882**.

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A cover layer **886** (e.g., cover plate) is coupled to the insulating layer **884** and to the visual markers **882**. The cover layer **886** may cover the visual markers **882**, thereby shielding the visual markers **882** from some environmental factors, such as spatter, dust, unintentional removal, and so forth. In some embodiments, the cover layer **886** does not cover or only partially covers the visual markers **882**. In some embodiments, the cover layer **886** is a plastic, such as polycarbonate. The cover layer **886** may be a material that is not substantially reflective of one or more electromagnetic waves that are reflected by the markers **882**. Additionally, or in the alternative, the cover layer **886** may be conditioned to reduce or eliminate reflections of electromagnetic waves. For example, the cover layer **886** may be painted, coated, or roughened (e.g., sandblasted), or any combination thereof. In some embodiments, the cover layer **886** is substantially non-reflective except in an area immediately covering the visual markers **882**.

FIG. 50 is a perspective view of an embodiment of the welding stand **12**, the arms **576**, **578**, and the clamp assembly **588**. As discussed above, the first and second arms **576**, **578** are rotatable about the support structure **566** to enable the first and second arms **576**, **578** to be positioned at a selected height for vertical and/or overhead welding. As illustrated, the second arm **578** includes a clamp assembly **588** for coupling the workpiece **82** to the second arm **578**. The second arm **578** and the clamp assembly **588** may be positioned at various heights relative to the welding stand **12**. Additionally, or in the alternative, the clamp assembly **588** may be coupled to each arm **576**, **578**, and the clamp assembly **588** may be oriented in various directions relative to the sensing device **16**. As may be appreciated, the clamp assembly **588** may include multiple visual markers **802** markers (e.g., reflective and/or light emitting) to facilitate tracking by the sensing device **16**. For example, in certain embodiments, the clamp assembly **588** may include three markers on one surface (e.g., in one plane) of a clamp body **889**, and a fourth marker on another surface (e.g., in a different plane) to facilitate tracking by the one or more sensing devices **16**. A clamp face **890** of the clamp body **889** may be substantially parallel to a sensing device **16**, or oriented at an offset angle from a sensing device **16**. A mount **892** couples the clamp assembly **588** to the second arm **578**.

FIG. 51 is a top view of an embodiment of the mount **892** of the clamp assembly **588** of FIG. 50, taken along line 51-51. A clamp axle **900** couples the mount **892** to the clamp body **889**. In some embodiments, a retaining feature **902** of the clamp axle **900** may limit the movement of the clamp axle **900** along a clamp axis **904** in at least one direction. Furthermore, a clamp fastener **906** may interface with the retaining feature **902** and the mount **892** to retain the clamp axle **900** in a desired position along the clamp axis **904**. The mount **892** may rotate about an axis **908**, thereby adjusting the orientation of the clamp body **889** and the clamp face **890** relative to a sensing device **16**. In some embodiments, a fastener **910** (e.g., pin) may couple the mount **892** to the second arm **578** at a desired orientation. The fastener **910** may be fixedly coupled to the mount **892**, thereby preventing removal of the fastener **910** from the welding system **10**. In some embodiments, the retaining feature **902** and/or the fastener **910** may be biased (e.g., spring loaded) with respect to the clamp assembly **588**, thereby enabling automatic engagement with the clamp assembly **588** in one or more predetermined positions. For example, inserting the fastener **910** into a first recess **912** orients the clamp face **890** in a first direction **914** substantially parallel to a sensing device **16**, inserting the fastener **910** into a second recess **916** orients

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the clamp face **890** in a second direction **918**, and inserting the fastener **910** into a third recess **920** orients the clamp face **890** in a third direction **922**. The second and third directions **918** and **922** may be oriented within approximately 10, 20, 30, 40, or 50 degrees of direction **914** (e.g., towards a sensing device **16**). The second and third directions **918** and **922** of FIG. **51** are approximately 30° offset from the first direction **914**. When the clamp assembly **588** is mounted on the second arm **578** and the clamp face is oriented in the second direction **918**, the clamp assembly **588** may be configured for welding in positions in which a portion of the workpiece **82** may obscure part of the joint from view of the one or more sensing devices **16**. For example, welds performed in the 3F position (e.g., vertical fillet welds of T and lap joints) may be readily observed by the one or more sensing devices **16** when the workpiece **82** is coupled to the clamp assembly **588** on the second arm **578** such that the clamp face **890** is oriented in the second direction **918**.

The position and the orientation of the arms and respective clamp assemblies are calibrated to enable the one or more sensing devices **16** to track the movement of the welding tool **14** relative to a joint of the workpiece **82** coupled to the clamp assembly **588**. As illustrated in FIG. **52**, a calibration block **930** may be coupled to the clamp assembly **588** to facilitate the calibration of the clamp assembly **588**. In some embodiments, the calibration tool **610** of FIGS. **37** and **38** is coupled to the calibration block **930** such that the calibration tool **610** extends from the calibration block **930** at a predefined angle (e.g., perpendicular). The calibration block **930** and the calibration tool **610** may enable the one or more sensing devices **16** to calibrate the normal vector of the clamp assembly **588**, to calibrate the normal vector of workpieces **82** secured to the clamp assembly **588**, and/or to calibrate the true vertical (i.e., zenith) vector relative to the floor. The one or more sensing devices **16**, via the computer **18**, may determine a rigid body model and/or a centroid of clamp markers for the clamp assembly **588** when mounted to each arm **576**, **578**, during which different sides of the clamp assembly **588** are in view of the one or more sensing devices **16** where each side of the clamp assembly **588** has a unique configuration of markers. The one or more sensing devices **16** may be coupled to the arms **576**, **578** so that as each arm is raised and lowered, a y-value or a centroid of the clamp markers or the respective side changes. As discussed above, movement of each arm **576**, **578** may adjust the orientation of the one or more sensing devices **16**. Accordingly the one or more sensing devices **16** may determine the y-value of the centroid of clamp markers for the clamp assembly **588** at multiple heights of the respective arms **576**, **578**. The computer **18** may determine the zenith vector for each of the centroids at the respective heights, thereby enabling the computer **18** to determine (e.g., interpolate) the zenith vector for any height using the y-value of the centroid of clamp markers when the clamp assembly **588** is coupled to each arm **576**, **578**. A level may be utilized with the clamp calibration block **930** during calibration at each height to ensure the orientation of calibration tool **610** accurately represents the zenith vector. The y-value of the centroid of clamp markers can also be used to determine the height of the clamp and to provide the operator with feedback on correct height positioning for welding session. The height of the clamp assembly **588** during a welding session may be stored with the welding data **327** for each welding session. In some embodiments, the welding system **10** may determine the orientation of the clamp assembly **588** relative to the sensing device **16**, thereby enabling the welding system

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10 to notify the operator if the workpiece **82** is in an improper orientation for the welding session. For example, the welding system **10** may notify the operator when the clamp assembly **588** and workpiece **82** are oriented such that the visual markers **802** of the welding tool **14** would be at least partially obscured from view of the one or more sensing devices **16** during the welding session, thereby enabling the operator to adjust the clamp assembly **588** so that all of the visual markers **802** may be observed.

FIG. **53** is a flowchart **940** that illustrates the set up and execution of assignment welding session utilizing one of the arms for a vertical or overhead (e.g., out of position) session. The operator selects (block **942**) an out of position session (e.g., 2G, 3G, 3F, 4G, 4F) and tacks (block **944**) the workpiece together. The operator then sets up (block **946**) the desired arm to the height corresponding to the session and adjusts the clamp assembly for calibration with the sensing device. Upon setup of the arm and clamp assembly, the operator couples (block **948**) the workpiece to the clamp assembly. Then the operator may adjust (block **950**) the clamp orientation, such as if the workpiece at least partially obscures the joint from the sensing device, if markers of the workpiece or clamp assembly are obscured from the sensing device, or if the clamp assembly is not substantially perpendicular to the ground, or any combination thereof. After adjusting the clamp orientation, the operator, an instructor, or an administrator may calibrate (block **952**) the clamp assembly. In some embodiments, the calibration may be performed once for each occasion that the arm is moved or for each occasion that the clamp assembly is attached to the arm, such that the clamp assembly may not be calibrated prior to each session. The calibration of the clamp assembly may validate that the clamp assembly is detected in the configuration and/or orientation specified for the session. The operator calibrates (block **954**) the joint ends, thereby establishing the 2 points in a line representing the joint. In some embodiments, such as for welding sessions in the 3F position, the operator calibrates (block **954**) the joint ends utilizing the calibration tool **610** described above with FIGS. **37** and **38**, where an axis of the calibration tool is held within approximately 5° of parallel to the sensing device. As may be appreciated, welding sessions in other positions may be calibrated with the calibration tool having other orientations relative to the sensing device. Additionally, or in the alternative, the computer may compensate for orientations of the calibration tool during calibrations where the markers of the calibration tool are observed at a skewed angle. For example, the computer may determine the angle of the calibration tool relative to the clamp assembly, then utilize the determined angle to adjust calibration values of the joint ends. After the calibration of the joint ends, then the operator performs (block **956**) the welding session and reviews (block **958**) the results. In some embodiments, the display of the welding stand **12** and/or the display of the welding tool **14** may provide instructions to the operator to guide the setup for the welding session.

The one or more sensing devices **16** may track the position and orientation of the clamp assembly **588**, the workpiece **82**, and the welding tool **14** prior to performing assignment welding session, during the welding session, and after performing the welding session. As discussed above, the one or more sensing devices **16** may include a camera that detects visual markers **802**, such as visual markers of the clamp assembly **588**, the workpiece **82**, and the welding tool **14**. In some embodiments, the computer **18** may utilize data corresponding to the visual markers **802** of fixed surfaces (e.g., the clamp assembly **588**, the workpiece **82**) for refer-

ence with respect to other tracked objects in the welding environment whenever the visual markers **802** of the fixed surfaces are detectable. That is, the visual markers **802** of the fixed surfaces facilitate real-time tracking of other objects (e.g., welding tool **14**, calibration tool **610**) within the welding environment. The visual markers **802** detected by the camera of the sensing device **16** may include passive markers (e.g., stickers, reflectors, patterns) and/or active markers (e.g., lights, LEDs). The passive markers may be best observed with a first exposure setting of the cameras of the one or more sensing devices **16**, and the active markers may be best observed with a second exposure setting of the camera, which may be different than the first exposure setting. In some embodiments, the visual markers **802** of the clamp assembly **588** and the workpiece **82** may be passive markers, and the visual markers **802** of the welding tool **14** may be active markers (e.g., LEDs **64**). Moreover, the passive markers may be illuminated by lights (e.g., LEDs **64**) of the sensing device **16**, where light (e.g., infrared light) from the lights reflects off the passive markers and is observed by cameras of the one or more sensing devices **16**. Accordingly, the exposure setting of the camera may be adjusted based at least in part on the type of visual marker to be observed. As may be appreciated, the second exposure setting for sampling the active markers that emit light may be less than the first exposure setting for sampling the passive markers that reflect light.

The computer **18** may alternately track the visual markers **802** of the welding tool **14** and the fixed surfaces of the welding environment prior to performing and during performance of a welding session (e.g., simulated welding assignment, live welding assignment). Accordingly, the computer **18** may track in real-time the position and the orientation of the welding tool **14**, the clamp assembly **588**, and the workpiece **82** relative to each other and to the welding stand **12**. Prior to live welding, the computer **18** may primarily track the visual markers **802** of welding tool **14** when detecting the position and orientation of objects in the welding environment about the welding stand **12**, and the computer **18** may secondarily track the visual markers **802** of the fixed surfaces (e.g., main welding surface **88**, clamp assembly **588**, clamped workpiece **82**). The active markers of the welding tool **14** may be turned on substantially continuously before, during, and after a simulated or live welding session (e.g., welding assignment). The computer **18** may control the exposure setting of the cameras of the one or more sensing devices **16** to control the respective sampling rates of the fixed surfaces and the welding tool **14**. For example, the visual markers **802** of the welding tool **14** may be sampled 1.5, 2, 3, 4, 5, or more times than the visual markers **802** of the fixed surfaces are sampled. That is, the computer **18** cycles the exposure setting of the camera between the second exposure setting (e.g., low exposure value to track the active markers of the welding tool **14**) and the first exposure setting (e.g., high exposure value to track the passive markers of the fixed surfaces).

Prior to initiating a simulated welding session (e.g., welding assignment), the computer **18** may control the lights of the one or more sensing devices **16** (e.g., LEDs **64**) to be turned on, thereby enabling the computer **18** to track the passive markers of the fixed surface and the active markers of the welding tool **14** prior to initiating the simulated welding session, during the simulated welding session, and after the simulated welding session. As described above, the computer **18** may cycle the exposure setting of the camera to sample the passive markers with the first exposure setting and to sample the active markers with the second exposure

setting. During live welding (e.g., while the trigger of the welding tool **14** is actuated), the computer **18** may control the lights of the one or more sensing devices **16** to pulse at an increased brightness level, thereby cyclically increasing the reflected light from the passive markers. Pulsing the lights may enable the cameras of the one or more sensing device **16** to readily track the passive markers with a reduced exposure setting during live welding with the bright arc and spatter. The computer **18** may control the exposure setting of the camera to be synchronized with the pulsing of the lights of the sensing device **16**, such that the lights pulse more brightly when the exposure setting is at the first (e.g., high) exposure setting, and the lights dim when the exposure setting is at the second (e.g., low) exposure setting. Additionally, or in the alternative, the computer **18** may control the lights of the one or more sensing devices **16** to turn off during calibration of the clamp assembly **588**, thereby distinguishing the active markers of the welding tool **14** from the passive markers of the clamp assembly **588**. In some embodiments, a pulsed brightness level of the lights of the one or more sensing devices **16** may be greater than when the lights turned on substantially continuously. The one or more sensing devices **16** may more readily detect the passive markers at the greater brightness level of the lights than at the lower brightness level. However, pulsing the lights of the one or more sensing devices **16** during a simulated weld may unintentionally activate an auto-darkening circuit of a welding helmet. Accordingly, the lights of the one or more sensing devices **16** may be pulsed during live welding when the welding helmet is darkened due to the arc, yet the lights of the one or more sensing devices **16** are turned continuously on during simulated welding when the welding helmet is not darkened.

In some embodiments, the welding system **10** may track a multi-pass (e.g., multirun) session, thereby recording welding data **327** for each pass (e.g., run) of the multi-pass session. As discussed above, the control circuitry **52** of the welding system **10** may record the welding data **327** for each run of the multi-run session as a single welding operation for determining a quality of the multi-run session or for otherwise reviewing the multi-run session. In some embodiments, the control circuitry **52** of the welding system **10** may record welding data **327** for a multi-run session as a group of runs that correspond to a serial number or other identifier for the multi-run session. That is, the welding data **327** for a multi-run session may be reviewed and evaluated as a group, or each run of the multi-run session may be reviewed and evaluated separately. Multi-run sessions may include, but are not limited to a live process, a simulated process, a virtual reality process, or any combination thereof.

FIG. **54** is a flowchart **970** that illustrates the selection and execution of a multipass (e.g., multi-run) welding session (e.g., welding assignment). The operator selects (block **972**) a multi-run session and sets up (block **974**) the workpiece **82** together on the welding stand **12**. Set up of the workpiece **82** may include clamping the workpiece **82** to the welding stand **12**. The operator calibrates (block **976**) the joint, such as by utilizing the joint calibration tool **610** to calibrate the position of a first end of the joint and the second end of the joint. As may be appreciated, the joint calibration tool **610** may directly interface with the workpiece **82** for the calibration (block **976**) prior to the first run of the multi-run session. The operator selects (node **978**) whether to perform the next (i.e., first) run of the multi-run session in a simulated welding mode or a live welding mode. In some embodiments, the selected welding session (e.g., welding assignment) may prohibit or limit the quantity of simulated welds that may be

performed prior to live welds. In some embodiments, the selected session may prohibit the live welding mode until completion (e.g., satisfactory completion) of a simulated weld. When the simulated weld mode is selected, the operator performs (block 980) the simulated run. The control circuitry 52 may display (block 982) the results of the simulated run via the display 32 of the welding stand 12 and/or the display 62 of the welding tool 14. For example, the control circuitry 52 may display the weld data 327 from the simulated run and the target specifications for the simulated run. Additionally, or in the alternative, the control circuitry may display the weld score for the simulated run. After completing the simulated run, the operator again selects (nodes 978) whether to perform the next run in the simulated welding mode or in the live welding mode.

When the live welding mode is selected, the operator performs (block 984) the live weld run on the calibrated joint. The control circuitry 52 may display (block 986) the results of the live run via the display 32 of the welding stand 12 and/or the display 62 of the welding tool 14. For example, the control circuitry 52 may display the weld data 327 from the live run and the target specifications for the live run. Additionally, or in the alternative, the control circuitry 52 may display the weld score for the live run. The displayed results for the live run may be displayed with results of any previous simulated runs for the same joint.

Each run (e.g., simulated or live) of the multi-run welding session (e.g., welding assignment) may be evaluated separately based at least in part on target specifications (e.g., minimum, goal, maximum) for tool position parameters (e.g., work angle, travel angle, CTWD, travel speed, aim) and/or electrical parameters (e.g., weld voltage, weld current, wire feed speed). For example, a rootpass run may have different specification parameters than subsequent runs. After a run of the multi-run session is completed, the control circuitry 52 may determine whether the completed run of the session satisfies the target parameter values for the respective run. For example, the welding data 327 for a run of the multi-run session may be compared with the target parameter values to generate a score for each parameter and/or a total score for the respective run. The control circuitry 52 may determine whether the run passes the target specifications for the respective run.

The control circuitry 52 determines (node 988) whether all of the runs of the selected welding session (e.g., welding assignment) have been completed. If all of the runs of the selected multi-run session have not been completed, then the operator selects (block 990) the next run. In some embodiments, the operator may proceed to the next run of the multi-run session regardless of whether the previous run passes the target specifications. Additionally, or in the alternative, the operator may proceed to the next run of the multi-run session regardless of whether the weld data 327 for the previous run is complete. For example, if the one or more sensing devices 16 cannot track the position and the orientation of the welding tool 14 for at least a portion of a run of the multi-run session, the operator may continue performing each run of the multi-run session. The operator calibrates (block 976) the joint for each run of a multi-run session, such as by utilizing the joint calibration tool 610 to calibrate the position of a first end of the joint and the second end of the joint. As may be appreciated, joint calibration tool 610 may have directly interfaced with the workpiece 82 for the initial calibration of the joint prior to the first run. Subsequent calibrations may directly interface the joint calibration tool 610 with the previously formed weld bead of one or more previous runs. Accordingly, the calibrated ends

of the joint for each run may have a different position relative to the one or more sensing devices 16 of the welding system 10. When the subsequent calibration for the next run is completed, the operator again selects (nodes 978) whether to perform the next run in the simulated welding mode or in the live welding mode.

If all of the runs of the selected multi-run session have been completed, then the control circuitry 52 may display (block 992) the results of each of the live runs via the display 32 of the welding stand 12 and/or the display of the welding tool 14. For example, the control circuitry 52 may display the weld data 327 from each of the live runs and the target specifications for each of the live runs. Additionally, or in the alternative, the control circuitry 52 may determine whether the group of runs passes the target specifications for the multi-run session based on one or more evaluations of the runs. For example, the control circuitry 52 may evaluate the group of runs based on a geometric mean of the scores for each run, an arithmetic mean of the scores for each run, whether each run was completed with a passing score, or any combination thereof. In some embodiments, a threshold quantity (e.g., 1, 2, or 3) of runs with untracked welding tool position and orientation may not affect the evaluation of the multi-run session. That is, the one or more runs with untracked welding tool position and orientation may not be counted in the geometric and/or arithmetic mean. Upon display of the session results (block 992), the operator may select (block 994) to retest with selected session. The operator removes the previously tested joint, and sets up (block 974) a new joint for the retest. The control circuitry 52 may assign a different serial number to the new joint for the retest than the serial number of the previously tested joint, thereby enabling the operator and an instructor to review and evaluate the weld data 327 from each joint.

As described herein, various parameters may be tracked (e.g., detected, displayed, and stored) during operation of the welding system (e.g., in real-time while the welding system 10 is being used) including, but not limited to, tool position parameters (e.g., work angle, travel angle, CTWD, travel speed, aim) and arc parameters (e.g., weld voltage, weld current, wire feed speed). The arc parameters, for example, may be detected in the welding tool 14 (e.g., using the voltage sensor 425, the current sensor 427, or other sensors, as illustrated in FIG. 18), converted using analog-to-digital conversion (ADC) circuitry, and communicated to the computer 18 via a communication interface 68 (e.g., RS-232 communication channel), as discussed herein with respect to FIG. 1. Alternatively to, or in addition to, being detected in the welding tool 14 (e.g., in the handle of the welding tool 14), the arc parameters may be detected in the weld cable 80, the welding power supply 28, the wire feeder 30, or some combination thereof, each of which are illustrated in FIG. 2.

The welding system 10 may detect and display (e.g., numerically, graphically, and so forth) the arc parameters via a screen viewable on the display 32 of the welding system 10. An exemplary screen 996 having a weld mode indicator 998 that indicates that the welding system 10 is in a live-arc weld mode may be displayed on the display 32 is illustrated in FIG. 55. As illustrated in FIG. 55, the arc parameters may be displayed on the screen 996. For example, in the illustrated screen 996, a voltage graph 340 may display a time series of voltage 337 of the arc produced by the welding tool 14, and an amperage graph 340 may display a time series of the current 338 produced by the welding tool 14. In certain embodiments, filters may be applied to at least some of the

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arc parameters and the tool position parameters to smooth out noise in the time series graphs **340** of the values detected by the welding tool **14**.

It will be appreciated that the arc parameters may be time synchronized by the welding software **244** in real-time with the tool position parameters that is captured through the motion tracking system (e.g., the one or more sensing devices **16**). In other words, the arc parameters and the tool position parameters may all be graphed on their respective graphs **340** such that data points for each of the time series are vertically aligned with data points from each of the other time series that are captured at approximately the same time (e.g., within 100 milliseconds, within 10 milliseconds, or even closer in time, in certain embodiments). This enables the user to correlate the arc parameters with the tool position parameters. Although not illustrated in FIG. **55**, in certain embodiments, wire feed speed may also be detected in real-time in the same manner as voltage and current.

As illustrated in FIG. **55**, in certain embodiments, each arc parameter (as well as each tool position parameter) may be individually scored in relation to a pre-defined upper limit, lower limit, and/or target value, and the scores **341** may be depicted on the screen **996**. In addition, in certain embodiments, a total score may be determined by the welding software **244** and depicted on the screen **996**. In addition, in certain embodiments, the total score **1000**, indications of target total scores **1002** and high total scores **1004** (for example, of an entire class) may be determined by the welding software **244** and depicted on the screen **996**. In addition, in certain embodiments, an indication **1006** of whether the test was successful or not successful may also be determined by the welding software **244** and depicted on the screen **996**. In certain embodiments, the total score **1000** may be based on the individual scores **341** for the tool position parameters, but not based on the individual scores **341** for the arc parameters.

In addition, as illustrated in FIG. **55**, in certain embodiments, an overall status bar **1008** may be depicted on the screen **996**. The overall status bar **1008** may include indications of whether all of the tool position parameters are within their respective upper and lower limits or not. For example, if one of the tool position parameters are not within their respective upper and lower limits, the overall status bar **1008** may indicate, at the same vertical position on the screen **996** as the corresponding tool position parameter values, a red status. Conversely, if all of the tool position parameters are within their respective upper and lower limits, the overall status bar **1008** may indicate, at the same vertical position on the screen **996** as the corresponding tool position parameter values, a green status. It will be appreciated that other status colors may be used in other embodiments.

As illustrated, in certain embodiments, the value **339** for each of the parameters (e.g., the tool position parameters and the arc parameters) may be displayed as an average value over the course of a test period. For example, as illustrated in FIG. **55**, the average voltage and amperage over the test period depicted are 18.7 volts and 146 amps, respectively. FIG. **56** is another illustration of the screen **996** depicted in FIG. **55**. In this instance, the average voltage and amperage is depicted as being 0.1 volts and 2 amps, respectively, which are on the order of noise, indicating that an actual welding arc is not being detected. In such a situation, the amperage and voltage can be used by the welding software **244** to determine whether or not welding took place during a given "weld mode" test period. If the value of either voltage or amperage is below a certain predetermined

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threshold (e.g., the average voltage is less than 10 volts) or between a certain predetermined minimum and maximum threshold (e.g., the average voltage is between -8 volts and +10 volts), the welding software **244** may determine that a weld actually did not take place during the time period. In such a scenario, the welding software **244** may automatically mark a test as failed (or "unsuccessful") and/or the test may be flagged by the welding software **244** as having no welding detected. For example, as illustrated, in certain embodiments, if the average voltage and/or the average amperage for a given test period do not meet certain predetermined threshold(s) or fall within certain predetermined range(s), the indication **1006** of whether the test was successful or not successful may depict that the test was "Unsuccessful" (which may also be displayed for other reasons, such as the total score does not meet a specific requirement, for example). In addition, as also illustrated, in certain embodiments, when the average voltage and/or the average amperage for a given test period do not meet certain predetermined threshold(s) or fall within certain predetermined range(s), instead of depicting the total score **1000** on the screen **996**, an "Arc Not Detected" message **1010** may be depicted instead.

FIG. **57** illustrates an exemplary screen **1012** that may be displayed as part of the assignment development routines of the welding software **244**. In particular, FIG. **57** illustrates a screen **1012** that enables input of completion criteria for a series of weld tests and length requirements associated with the testing. As illustrated, the screen **1012** is displayed when the Completion Criteria/Length Requirements tab **1014** of the assignment development routines is selected (and, therefore, highlighted on screen **1012**). As illustrated, other tabs associated with configuration settings of the assignment development routines of the welding software **244** may include, but are not limited to, an Assignment Name tab **1016** that causes a screen to be displayed where the assignment name and other general information relating to the assignment may be entered; a Joint Design tab **1018** that causes a screen to be displayed where properties of the joint to be welded upon (e.g., type of joint, length, etc.) may be entered; a Base Metals tab **1020** that causes a screen to be displayed where properties related to the base metals to be welded upon may be entered; a Filler Metals/Shielding tab **1022** that causes a screen to be displayed where properties relating to the filler metals (e.g., of the welding electrode) and shielding gas(es) may be entered; a Position/Electrical Char. tab **1024** that causes a screen to be displayed where properties (e.g., upper limits, lower limits, target values, etc.) of the tool position parameters and the arc parameters, respectively, may be entered; a Preheat/Postweld Heat Tr. tab **1026** that causes a screen to be displayed where properties relating to preheating and postweld heating, respectively, may be entered; a Welding Procedure/1 Pass tab **1028** that causes a screen to be displayed where properties relating to the welding procedure (e.g., process type, etc.) and the number of passes in the test (e.g., one pass or more than one pass); and a Real-Time Feedback tab **1030** that causes a screen to be displayed where properties relating to real-time feedback may be entered. It will be appreciated that, in certain embodiments, all of the properties relating to an assignment may be entered on the described screens, may be automatically detected by the welding software **244** (e.g., based on specific equipment of the welding system **10**, based on other properties that are set, and so forth), or some combination thereof.

As illustrated in FIG. **57**, the screen **1012** relating to the Completion Criteria/Length Requirements tab **1014** includes

a first section **1032** specifically dedicated to the completion criteria properties and a second section **1034** specifically dedicated to length requirements associated with the testing. In certain embodiments, in the completion criteria section **1032** of the screen **1012**, a series of inputs **1036** enables a target score (e.g., 90 as illustrated), a number of weld tasks in a set of weld tasks (e.g., 5 as illustrated), a number of successful weld test required per weld set (e.g., 3 as illustrated), and whether a weld test will be failed if an arc is not detected (e.g., as shown in FIG. **56**) to be entered. In addition, as illustrated, in certain embodiments, a depiction **1038** of what these selections of completion criteria will look like to the user (e.g., as illustrated in FIG. **55** in the Actions section **1040** of the screen **996**). In addition, in certain embodiments, in the length requirements section **1034** of the screen **1012**, a series of inputs **1042** enables a length of a start section (A) of a weld that will be ignored in the score compilations, an end section (B) of a weld that will be ignored in the score compilations, and a maximum length (C) of the test, which may be less than the coupon length (which may, for example, be entered via the screen relating to the Joint Design tab **1018**) to be entered. In addition, in certain embodiments, respective illustrations **1044** of relative dimensions of the entered properties relating to the length requirements may also be depicted to aid the user in setting the length requirements.

FIG. **58** illustrates an exemplary screen **1046** that may be displayed when the Welding Procedure/1 Pass tab **1028** is selected. As described above, this screen **1046** enables properties relating to the welding procedure and the number of passes in the test (e.g., one pass or more than one pass) to be entered. As illustrated, in certain embodiments, a first series of inputs **1048** enables a process type (e.g., FCAW-G as illustrated), a class and diameter of the filler metals (e.g., the welding electrode) (e.g., E71T-8JD H8 and 0.072 inches, respectively, as illustrated), a weld pattern (e.g., stringer vs. weave; stringer as illustrated), a vertical progression (e.g., up vs. down; up as illustrated), and any comments related to the welding procedure to be entered. In addition, as illustrated, in certain embodiments, a second series of inputs **1050** enables minimum, target, and maximum values for the arc parameters (e.g., volts, wirefeed speed, and amps), labeled as Welding Power Source Settings, and the tool position parameters (e.g., work angle, travel angle, CTWD, travel speed, and aim), labeled as Tool Technique Parameters, to be entered. Also as illustrated, in certain embodiments, a third series of inputs enable more detailed input relating to minimum, target, and maximum values (e.g., relating to how much deviation from target values are allowed for the upper and lower limits, and so forth) for a highlighted arc parameter or tool position parameter (e.g., volts as illustrated). In certain embodiments, when more than one pass is selected for a given assignment, the minimum, target, and maximum values for the arc parameters and/or the tool position parameters may be individually set for each pass within the assignment. In certain embodiments, entry of properties for multiple passes for a given assignment may be enabled via an Add Pass button **1054**, as illustrated.

As discussed above with respect to FIGS. **55** and **56**, the arc parameters may be displayed when the welding software **244** is in a live-arc weld mode. Conversely, FIG. **59** illustrates an exemplary screen **1056** that depicts the welding software **244** when in a simulated weld mode, as indicated by the weld mode indicator **998**. As illustrated, when the welding software **244** is in a simulated weld mode, the arc

parameters are not displayed since actual welding is disabled in this mode, and a message indicating as much may be displayed instead.

In certain embodiments, the arc parameters are not displayed by default below the tool position parameters, such as illustrated in FIGS. **55** and **56**. Rather, FIG. **60** illustrates an exemplary screen **1058** that is depicted by default (i.e., before a weld test has been initiated). As illustrated, instead of the arc parameters, a welding procedure summary pane **1060** is illustrated to summarize for the user what the overall properties (e.g., target properties) for a given test weld are. In certain embodiments, from the welding procedure summary pane **1060**, a user may select a View WPS button **1062**, which will cause the screen **1064** illustrated in FIG. **61** to be displayed. As illustrated, FIG. **61** is a summary of all of the information relating to all of the parameters of a weld test session or a weld test assignment (e.g., which may be entered via selection of the various assignment development tabs **1014-1030** illustrated in FIGS. **57** and **58**).

Returning now to FIG. **60**, once the user has completed pre-test procedures and is prepared to begin a weld test, upon activation of the trigger **70** of the welding tool **14** to start a weld test, the welding procedure summary pane **1060** is replaced by the information relating to the arc parameters to display the real-time graphing of the arc parameters during performance of the weld test (see, e.g., FIG. **62**), allowing the user to view all graphs relating to the tool position parameters and the arc parameters in real-time during the weld test. Indeed, in certain embodiments, upon activation of the trigger **70** of the welding tool **14** to start a weld test, whatever screen is currently being displayed may be replaced with, for example, the screen **996** illustrated in FIG. **62** such that all of the tool position parameters and arc parameters may be graphically displayed in real-time.

FIG. **63** illustrates an alternative screen **1066** that may be displayed following the performance of a test weld. As illustrated, in certain embodiments, in addition to the arc parameters (e.g., voltage, amperage, wire feed speed), heat input **1068** may be displayed and, as with all of the other tool position parameters and the arc parameters, is time synchronized along their respective time series. In general, the detected voltage and amperage data and the detected travel speed data may be used to compute the heat input in real-time for each point in time along the time series (e.g., time-based) or at each location along the weld joint (e.g., distance-based). In particular, in certain embodiments, the heat input (in kilojoules) may be calculated as a function of the voltage, the amperage, and the travel speed (in inches per minute) as:

$$\text{HeatInput} = \frac{\text{Amps} \times \text{Volts} \times 60}{1000 \times \text{TravelSpeed}}$$

In addition, although not illustrated in FIG. **63**, in certain embodiments, the weld size (fillet size; in millimeters) can be computed in real-time using the wire feed speed (WFS; in inches per minute), which may either be detected or specified by a user, travel speed (in meters per minute), and a predetermined value for efficiency (0%), and wire diameter (in millimeters) as:

$$\text{FilletSize} = \sqrt{\frac{\left(\frac{\pi}{4} \times \text{WireDiameter}^2\right) \times (25.4 \times \text{WFS}) \times \text{Efficiency}}{\left(\frac{1000 \times \text{TravelSpeed}}{2}\right)}}$$

In certain embodiments, the predetermined value for efficiency may take into account any detected spatter, which may be determined using the techniques disclosed in “Devices and Methods for Analyzing Spatter Generating Events”, U.S. Patent Application No. 2013/0262000, filed on Mar. 30, 2012 in the name of Richard Martin Hutchison et al. which is hereby incorporated into its entirety. For example, the predetermined value of efficiency may be adjusted to, for example, lower the predetermined value of efficiency when more spatter generating events are determined to occur, increase the predetermined value of efficiency when fewer spatter generating events are determined to occur, and so forth.

As discussed above, the welding tool **14** may either be an actual welding tool configured to facilitate the creation of an actual welding arc between the actual welding tool and an actual workpiece **82** or a simulated welding tool configured to simulate the creation of a simulated welding arc between the simulated welding tool and a simulated workpiece **82** (e.g., a workpiece **82** on which an actual weld is not created, but which serves as a guide for the user during a simulated welding process). For example, in certain embodiments, the welding tool **14** may be an actual MIG welding torch configured to facilitate the creation of an actual welding arc between actual welding wire delivered by the actual MIG welding torch and an actual workpiece **82**, or the welding tool **14** may be a simulated MIG welding torch configured to simulate the creation of a simulated welding arc between simulated welding wire delivered by the simulated MIG welding torch and a simulated workpiece **82**. Furthermore, in certain embodiments, the welding tool **14** may be an actual stick welding electrode holder configured to facilitate the creation of an actual welding arc between an actual stick welding electrode held by the actual stick welding electrode holder and an actual workpiece **82**, or the welding tool **14** may be a simulated stick welding electrode holder configured to simulate the creation of a simulated welding arc between a simulation stick welding electrode held by the simulation stick welding electrode holder and a simulated workpiece **82**, or the welding tool **14** may enable both actual and simulated stick welding processes, as described in greater detail herein. In addition, in certain embodiments, the welding tool **14** may be an actual TIG welding torch configured to facilitate the creation of an actual welding arc between an actual tungsten electrode held by the actual TIG welding torch and an actual workpiece **82**, or the welding tool **14** may be a simulated TIG welding torch configured to simulate the creation of a simulated welding arc between a simulated tungsten electrode held by the simulated TIG welding torch and a simulated workpiece **82**.

In certain embodiments, the welding tool **14** may be configured both to facilitate the creation of an actual welding arc between an actual stick welding electrode and an actual workpiece **82**, and to simulate the creation of a simulated welding arc between a simulation stick welding electrode and a simulated workpiece **82**. For example, in certain embodiments, the simulation stick welding electrode holder **1070**, in addition to being configured to mechanically retract the simulation stick welding electrode **1072** toward the stick electrode holding assembly **1078** to simulate consumption of the simulation stick welding electrode **1072** during a simulated stick welding process, the stick electrode holding assembly **1078** of the simulation stick welding electrode holder **1070** may be configured to deliver an electrical current through an actual stick welding electrode **1076** held by the stick electrode holding assembly **1078** (e.g., via conductive properties of the stick electrode holding assem-

bly **1078**), wherein the electrical current is sufficient to generate a welding arc to a workpiece **82** through a tip of the actual stick welding electrode **1076** during an actual stick welding process. Furthermore, it will be appreciated that the welding software **244** may be configured to function interoperably with all of the different types of welding tools **14** described herein.

FIGS. **(64A, 64B, 65A, and 65B)** illustrate embodiments of the welding tools **14** that are stick welding electrode holders. More specifically, FIGS. **64A and 64B** illustrate an embodiment of a simulation stick welding electrode holder **1070** configured to simulate the creation of a simulated welding arc between a simulation stick welding electrode **1072** held by the simulation stick welding electrode holder and a simulated workpiece **82** (see FIG. **66A**), and FIGS. **65A and 65B** illustrate an embodiment of an actual stick welding electrode holder **1074** configured to facilitate the creation of an actual welding arc **1075** between an actual stick welding electrode **1076** held by the actual stick welding electrode holder **1074** and an actual workpiece **82** (see FIG. **66B**). In certain embodiments, both stick welding electrode holders **1070, 1074** are coupled to communication lines. In certain embodiments, the communication lines may be disposed within the weld cable **80** through which power may be provided to the stick welding electrode holders **1070, 1074**. In other embodiments, the communication lines may be directly coupled to the weld cable **80** (e.g., disposed in a sleeve that may be tied to the weld cable **80**). It will be appreciated that the embodiments of the stick welding electrode holders **1070, 1074** may include any and all of the relevant components of the welding tool **14** described herein, for example, as illustrated in FIG. **2**.

In certain embodiments, both stick welding electrode holders **1070, 1074** include visual markers **802** disposed on an outer structure **1077** that is configured to at least partially surround respective stick electrode holding assemblies **1078, 1080** of the stick welding electrode holders **1070, 1074**. In certain embodiments, the visual markers **802** disposed on the outer structure **1077** are substantially similar to (and substantially similarly disposed on outer surfaces of the stick welding electrode holders **1070, 1074**) as the visual markers **802** disposed on the neck **800** of the welding tool **14** described with respect to FIGS. **24 and 25**, which facilitate detection by the one or more sensing devices **16**. For example, as discussed above with respect to FIG. **48**, in certain embodiments, the visual markers **802** may be LEDs **64** configured to emit light that is detected by the one or more sensing devices **16** to determine position, orientation, and/or movement of the stick welding electrode holders **1070, 1074**. For example, similar to the embodiments illustrated in FIGS. **24 and 25**, in certain embodiments, the stick welding electrode holders **1070, 1074** may include multiple sets **804, 806, 810** of visual markers **802**, each set **804, 806, 810** of visual markers **802** including multiple LEDs **64** that emit light in different directions from the stick welding electrode holders **1070, 1074** which may be detected by the one or more sensing devices **16** to determine position, orientation, and/or movement of the stick welding electrode holders **1074**. In particular, in certain embodiments, four or more sets **804, 806, 810** of visual markers **802**, each set including multiple LEDs **64**, may be used.

In other embodiments, the multiple sets **804, 806, 810** of visual markers **802** may be disposed on the stick electrode holding assembly **1078** of the simulation stick welding electrode holder **1070**, and may be disposed on the stick electrode holding assembly **1078** such that the visual markers **802** may be detected by the one or more sensing devices

16. As described in greater detail herein, the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070 may be used to retract a simulation stick welding electrode 1072 (or an actual stick welding electrode 1076, in certain situations where an actual stick welding electrode 1076 is used instead of a simulation stick welding electrode 1072, but during a simulated stick welding process where an actual welding arc is not generated via the actual stick welding electrode 1076) to simulate consumption of the stick welding electrode 1072, 1076 and, as such, may function as a stick welding electrode retraction assembly having visual markers 802 disposed thereon, the position, orientation, and/or movement of which may be tracked by the one or more sensing devices 16, thereby enabling the position, orientation, and/or movement of a stick welding electrode 1072, 1076 being held by the stick electrode holding assembly 1078, 1080 to be inferred by the welding software 244.

As such, the visual markers 802 disposed on the outer structure 1077 of the stick welding electrode holders 1070, 1074 enable the welding software 244 to determine the position, orientation, and/or movement of the stick welding electrode holders 1070, 1074 using the one or more sensing devices 16. Knowing the position, orientation, and/or movement of the stick welding electrode holders 1070, 1074 enables the welding software 244 to infer the orientation of a tip of the respective stick welding electrodes 1072, 1076. The position and/or movement of the tip of the stick welding electrode 1072, 1076 can be inferred by, for example, estimating the actual consumption of the actual stick welding electrode 1076 during an actual stick welding process performed by the actual stick welding electrode holder 1074 (or, in the case of the simulation stick welding electrode 1072, directly tracking the retraction of the simulation stick welding electrode 1072, as described in greater detail herein). Alternatively, or in addition to, as illustrated in FIG. 66A, in the case of the simulation stick welding electrode holder 1070, in certain embodiments, one or more visual markers 802 may be disposed directly on the simulation stick welding electrode 1072 along an axis of the simulation stick welding electrode 1072 such that the position, orientation, and/or movement of the simulation stick welding electrode 1072 may be directly detected using the one or more sensing devices 16. In addition to, or in place of, having the visual markers 802 directly on the simulation stick welding electrode 1072, in certain embodiments, the visual markers 802 may be connected (e.g., indirectly) to the simulation stick welding electrode by, for example, being attached to some rigid body extending from the simulation stick welding electrode 1072. As will be appreciated, embodiments described herein as having the visual markers 802 disposed on other parts of the stick welding electrode holders 1070, 1074 (e.g., on the outer structure 1077 and/or the stick electrode holding assemblies 1078, 1080) may instead also similarly have the visual markers 802 connected (e.g., indirectly) to these other parts of the stick welding electrode holders 1070, 1074.

As illustrated in FIGS. 64A, 64B, 65A, and 65B, in certain embodiments, the outer structure 1077 for the stick welding electrode holders 1070, 1074 may be substantially similar to each other. Indeed, in certain embodiments, many of the components of the stick welding electrode holders 1070, 1074 will be substantially similar to each other. For example, in certain embodiments, both stick welding electrode holders 1070, 1074 include substantially similar handles 1082 through which electrical current may be delivered to the respective stick electrode holding assemblies

1078, 1080 (and, ultimately, to the respective stick welding electrodes 1072, 1076), and each of these handles 1082 may be coupled to the respective stick electrode holding assemblies 1078, 1080 and the outer structure 1077 at a distal end 1084 of the handle 1082.

In certain embodiments, both stick welding electrode holders 1070, 1074 include a trigger 1086 that may be pressed by a user to bring the trigger 1086 closer to the handle 1082. For each of the stick welding electrode holders 1070, 1074, the effect of pressing the trigger 1086 may be different. For example, for the simulation stick welding electrode holder 1070, pressing the trigger 1086 may cause the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070 to cause the simulation stick welding electrode 1072 to be extended from the stick electrode holding assembly 1078 to a simulated unconsumed position, thereby enabling a new simulated weld procedure to be performed by the simulation stick welding electrode holder 1070. In certain embodiments, a release mechanism may be used to reset the simulation stick welding electrode 1072 to its unconsumed position. In contrast, for the actual stick welding electrode holder 1074 pressing the trigger 1086 may cause a more conventional effect for the actual stick welding electrode holder 1074, namely, that opposing jaws 1088 of the stick electrode holding assembly 1080 of the actual stick welding electrode holder 1074 are opened, such that an actual stick welding electrode 1076 may be inserted between the opposing jaws thereby enabling a new actual weld procedure to be performed by the actual stick welding electrode holder 1074.

As depicted in FIG. 66B, the actual stick welding electrode holder 1074 illustrated in FIGS. 65A and 65B is configured to facilitate the delivery of actual welding power through an actual stick welding electrode 1076 held by the actual stick welding electrode holder 1074 such that an actual welding arc 1075 at a distal tip of the actual stick welding electrode 1076 is created when the tip of the actual stick welding electrode is brought into proximity with an actual workpiece 82, thereby completing an electrical circuit between the welding power supply 28, the actual stick welding electrode holder 1074 (as the welding tool 14), the actual stick welding electrode 1076, and the actual workpiece 82 (e.g., facilitated by the weld cable 80 and the work cable 84). As will be appreciated, as the actual welding is performed using the actual stick welding electrode holder 1074, the actual stick welding electrode 1076 is consumed such that the tip of the actual stick welding electrode 1076 gradually moves toward the jaws 1088 of the actual stick welding electrode holder 1074 that holds the actual stick welding electrode 1076 in place with respect to the actual stick welding electrode holder 1074.

In contrast, as depicted in FIG. 66A, the simulation stick welding electrode holder 1070 illustrated in FIGS. 64A and 64B is not configured to facilitate delivery of actual welding power through the simulation stick welding electrode 1072 to a distal tip 1090 of the simulation stick welding electrode 1072. Rather, in certain embodiments, the simulation stick welding electrode holder 1070 is configured to simulate the consumption of the simulated stick welding electrode 1072 by retracting the simulation stick welding electrode 1072 back toward the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070 as the simulated welding process is performed by the simulation stick welding electrode holder 1070. For example, in certain embodiments, the simulation stick welding electrode holder 1070 includes a motor that facilitates retraction of the simulation stick welding electrode 1072 toward the stick

electrode holding assembly 1078 of the simulation stick welding electrode holder 1070.

As discussed above, in certain embodiments, the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070 is configured to retract the simulation stick welding electrode 1072 toward the stick electrode holding assembly 1078 to simulate consumption of the simulation stick welding electrode 1072 during a simulated stick welding process performed by the simulation stick welding electrode holder 1070. With reference to FIGS. 64A and 64B, assuming that the tip 1090A is the tip 1090 being used as the simulated tip of the simulation stick welding electrode 1072 at which the simulated welding arc is to be created when the tip 1090A is brought into proximity with a simulated workpiece 82, the stick electrode holding assembly 1078 is configured to retract the simulation stick welding electrode 1072 in a direction toward the stick electrode holding assembly 1078 during the simulated stick welding process, as illustrated by arrow 1092.

FIG. 67 illustrates an embodiment of the stick electrode holding assembly 1078 of the simulation stick welding electrode 1072. In the illustrated embodiment, the stick electrode holding assembly 1078 includes three distinct tracks 1094 within which the simulation stick welding electrode 1072 may be held. In certain embodiments, each track 1094 comprises a respective drive wheel 1096, which may be configured to rotate, thereby driving translation of the simulation stick welding electrode 1072 when the simulation stick welding electrode 1072 is being held within the respective track 1094. In alternative embodiments, instead of the drive wheels 1096, each track 1094 may include a screw drive configured to mate with threading on the simulation stick welding electrode 1072 (e.g., as shown on the simulation stick welding electrode 1072 illustrated in FIGS. 64A and 64B) such that the screw drive drives the translation of the simulation stick welding electrode 1072.

In certain embodiments, each track 1094 may include at least one guide 1098 that guides the simulation stick welding electrode 1072 through the respective track 1094. It will be appreciated that each discrete track 1094 is defined by the respective guiding features (e.g., drive wheels 1096 and guides 1098) that define a respective axis of the track 1094 through which the simulation stick welding electrode 1072 may move. In the illustrated embodiment, a motor assembly 1100 causes rotation of a central shaft 1102, as illustrated by arrow 1104. Rotation of the central shaft 1102 directly causes rotation of at least one of the drive wheels 1096 (i.e., drive wheel 1096A). In addition, in certain embodiments, each track 1094 may be associated with a respective gear 1106 of a gear assembly 1108. More specifically, as illustrated, in certain embodiments, the central shaft 1102 may be coupled to a first gear 1106A that is associated with the first drive wheel 1096A. The first gear 1106A may be directly coupled to the second and third gears 1106B and 1106C, which may in turn be directly coupled to the second and third drive wheels 1096B and 1096C. In the illustrated embodiment, the second and third gears 1106B and 1106C are generally oriented at approximately 45° angles with respect to the first gear 1106A, however, other orientations of the gears 1106 are contemplated. Moreover, although the illustrated embodiment depicts the first track 1094A being generally oriented transverse to a central axis 1110 of the simulation stick welding electrode holder 1070, and the second and third tracks 1094 being generally offset from the central axis 1110 by approximately 45° angles, other orientations of the tracks 1094 (as well as the associated drive wheels 1096 and guides 1098) are contemplated.

While FIG. 67 illustrates one embodiment where a plurality of discrete tracks 1094 may be used that remain in a relatively fixed position with respect to the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070, in other embodiments, a single track 1094 may be used, and the stick electrode holding assembly 1078 may be rotatable with respect to the simulation stick welding electrode 1072, thereby enabling any continuous angular orientation of the simulation stick welding electrode 1072 with respect to the simulation stick welding electrode holder 1070. For example, in certain embodiments, only the first track 1094A, the first drive wheel(s) 1096A, the first set of guides 1098A, etc., may exist in the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070, and the entire stick electrode holding assembly 1078 may be rotatable (e.g., pivotable) with respect to the simulation stick welding electrode holder 1070, as illustrated by arrow 1112, thereby enabling the various angular orientations (e.g., any continuous angular orientation, as opposed to the discrete angular orientations enabled by the fixed tracks 1094 illustrated in FIG. 67) of the simulation stick welding electrode 1072 with respect to the simulation stick welding electrode holder 1070 (e.g., with respect to the handle 1082 of the simulation stick welding electrode holder 1070). In other embodiments, a combination of both the plurality of tracks 1094, as illustrated in FIG. 67 and a rotatable (e.g., pivotable) stick electrode holding assembly 1078 may be utilized, enabling even greater control and customization of the angular orientation of the simulation stick welding electrode holder 1070, enabling either a relatively small number of discrete angular orientations or enabling an infinite number of continuous angular orientations.

As opposed to enabling continuous angular orientations of the simulation stick welding electrode holder 1070 via the rotatable (e.g., pivotable) stick electrode holding assembly 1078, in certain embodiments, the stick electrode holding assembly 1078 may be configured to be fixed into a discrete number of angular orientations when rotated with respect to the simulation stick welding electrode holder 1070 (e.g., with respect to the handle 1082 of the simulation stick welding electrode holder 1070) by, for example, utilizing grooves and mating indentations on surfaces of the stick electrode holding assembly 1078 and the outer structure 1077. In certain embodiments, a spring-loaded pin may hold the stick electrode holding assembly 1078 in place with respect to the outer structure 1077 and, when the pin is removed, the stick electrode holding assembly 1078 may be moveable with respect to the outer structure 1077. In addition, in certain embodiments, the stick electrode holding assembly 1078 may be rotated such that the simulation stick welding electrode 1072 aligns generally parallel to the central axis 1110 of the simulation stick welding electrode holder 1070. In such an alignment, the motor assembly 1 may be configured to retract the simulation stick welding electrode 1072 into an inner volume of the handle 1082 of the simulation stick welding electrode holder 1070 (for storage purposes, for example).

In certain embodiments, once the user selects one of the discrete angular orientations of the simulation stick welding electrode 1072 with respect to the simulation stick welding electrode holder 1070 (e.g., by selecting a discrete track 1094 of the stick electrode holding assembly 1078), the user may enter which discrete angular orientation was selected. In other embodiments, the selected angular orientation of the simulation stick welding electrode with respect to the simulation stick welding electrode holder 1070 may be detected

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by a sensor assembly 1113 disposed in the simulation stick welding electrode holder 1070. For example, the sensor assembly 1113 may include a sensor (e.g., an optical sensor) configured to detect when the simulation stick welding electrode 1072 is inserted into a particular discrete track 1094 of the stick electrode holding assembly 1078 and/or a sensor (e.g., position sensor) configured to detect an angular orientation (e.g., discrete or continuous angular orientation) of the stick electrode holding assembly 1078 in embodiments that include a rotatable (e.g., pivotable) stick electrode holding assembly 1078.

In certain embodiments where only one track 1094 is utilized in the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070, the simulation stick welding electrode 1072 may be permanently captured in the track 1094 and, therefore, not removable from the stick electrode holding assembly 1078. In other embodiments, the simulation stick welding electrode 1072 may be removable from the track 1094 (for storage purposes, for example). In embodiments where multiple tracks 1094 are utilized in the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070, the simulation stick welding electrode 1072 will also be removable, for example, to change the track 1094 within which the simulation stick welding electrode 1072 is held. In addition, again, even in embodiments having only one track 1094, the stick electrode holding assembly 1078 may still be configured to be rotated into various angular orientations with respect to the simulation stick welding electrode holder 1070 (e.g., with respect to the handle 1082 of the simulation stick welding electrode holder 1070).

FIGS. 64A and 64B illustrate another embodiment of the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070. In the illustrated embodiment, the simulation stick welding electrode 1072 includes threading that are manipulated by gears of a gear assembly 1108 to cause the simulation stick welding electrode 1072 to retract back toward the stick electrode holding assembly 1078 in the direction of arrow 1092 (as well as to extend back away from the stick electrode holding assembly 1078, e.g., opposite to the direction of arrow 1092). The gear assembly 1108 may be driven by a motor assembly 1100 substantially similar to the motor assembly 1100 described with respect to FIG. 67. In certain embodiments, a coupling mechanism, such as a pulley or gearing, may couple rotational shafts associated with the motor and gear assemblies 1100, 1108.

As described above, the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070 is configured to retract the simulation stick welding electrode 1072 toward the stick electrode holding assembly 1078 to simulate consumption of the simulation stick welding electrode 1072 during a simulated welding process performed using the simulation stick welding electrode holder 1070. The welding software 244 determines the rate at which the simulation stick welding electrode 1072 should be retracted, and sends control signals to the stick electrode holding assembly 1078 to adjust the rate of retraction accordingly. For example, in certain embodiments, the welding software 244 determines the rate of retraction of the simulation stick welding electrode 1072, and sends control signals to the motor assembly 1100 of the stick electrode holding assembly 1078 to effectuate the rate of retraction of the simulation stick welding electrode 1072.

The welding software 244 may determine the rate of retraction of the simulation stick welding electrode 1072 in several different ways, each selectable via the screens dis-

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played on the displays described herein. For example, in certain embodiments, the rate of retraction may be set to a constant retraction rate by an instructor. The constant retraction rate may be directly entered by the instructor, or may be indirectly set based on assignment parameters set by the instructor (or, in certain embodiments, set by the user himself), such as a type (e.g., E7018, E6010, etc.) of the simulation stick welding electrode 1072, a diameter (e.g., $\frac{3}{32}$ ", $\frac{1}{8}$ ", $\frac{5}{32}$ ", etc.) of the simulation stick welding electrode 1072, a length (e.g., up to 14", in certain embodiments) of the simulation stick welding electrode 1072, a simulated welding current, a desired simulated arc length, etc. Furthermore, in certain embodiments, instead of causing a constant rate of retraction of the simulation stick welding electrode 1072, the welding software 244 may instead dynamically change the rate of the retraction of the simulation stick welding electrode 1072 based on welding parameters (e.g., work angle, travel angle, arc length, travel speed, aim, arc length, and so forth) detected in real-time during performance of the simulated welding process. In other words, the welding software 244 may continuously adjust (e.g., control in real-time) the rate of retraction (e.g. update and implement a new rate of retraction every 1 second, every 0.1 second, every 0.01 second, or even more frequently, in certain embodiments) during performance of the simulated welding process. In certain embodiments, the continuous adjustment of the rate of retraction may be based at least in part on the welding parameters (e.g., work angle 328, travel angle 330, travel speed 334, and aim 336) relating to the simulation stick welding electrode 1072, which may be determined based at least in part on tracking of position, orientation, and/or movement data relating to the visual markers 802 disposed on (or otherwise fixedly connected to) the simulation stick welding electrode holder 1070. As used herein, the term "fixedly connected" is intended to mean connected to in a fixed manner, for example, not movable with respect to. In other embodiments, the welding software 244 may look up the rate of retraction, for example, in a lookup table stored in the memory device(s) 22.

In certain embodiments, the welding software 244 may continuously adjust (e.g., control in real-time) the rate of retraction of the simulation stick welding electrode 1072 based at least in part on a simulated arc length, which may be determined by the welding software 244 based at least in part on tracking of position, orientation, and/or movement data relating to the visual markers 802 disposed on (or otherwise fixedly connected to) the simulation stick welding electrode holder 1070. The simulated arc length determined by the welding software 244 represents the distance of the tip 1090 of the simulation stick welding electrode 1072 from the simulated workpiece 82. As such, in addition to being based on the position, orientation, and/or movement data relating to the simulation stick welding electrode holder 1070, the simulated arc length determined by the welding software 244 may also be based at least in part on the continuously adjusted rate of retraction of the simulation stick welding electrode 1072 (e.g., such that the welding software 244 tracks position, orientation, and/or movement of the simulation stick welding electrode holder 1070, as well as position, orientation, and/or movement of the simulation stick welding electrode 1072 with respect to the simulation stick welding electrode holder 1070), as well as the other parameters mentioned previously.

In certain embodiments, the simulation stick welding electrode holder 1070 may be configured to simulate an arc start for the simulated welding process performed by the simulation stick welding electrode holder 1070. In particu-

lar, in certain embodiments, the simulated welding process may be started when a tip 1090 of the simulation stick welding electrode 1072 comes into contact (e.g., either electrically or physically) with the simulated workpiece 82. The determination that the tip 1090 of the simulation stick welding electrode 1072 has contacted the simulated workpiece 82 may be accomplished in various ways. For example, in certain embodiments, the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070 may be configured to deliver a low-level current, which is not for the purpose of establishing a welding arc, through the tip 1090 of the simulation stick welding electrode 1072 when the tip 1090 is brought into contact with the simulated workpiece 82 (by closing an electrical circuit similar to conventional welding processes). Detection of the low-level current (e.g., via current sensing circuitry 1242 disposed in a connection box 1194, as illustrated in FIG. 84, in certain embodiments) initiates the start of a simulation test, during which welding parameters are captured and the simulation stick welding electrode is retracted. In certain embodiments, once a test has begun, subsequent detection of the low-level current causes the test to end and the retraction of the simulation stick welding electrode to cease. In certain embodiments, the welding software 244 may determine that the tip 1090 of the simulation stick welding electrode 1072 has contacted (or is, at least, in close proximity to) the simulated workpiece 82 based at least in part on a difference in voltages of the simulated workpiece 82 and of the simulation stick welding electrode 1072 (e.g., at least in part via voltage sensing circuitry 1242 disposed in the connection box 1194, as illustrated in FIG. 84, in certain embodiments).

Alternatively, or in addition to, in certain embodiments, the welding software 244 may infer the location of the tip 1090 of the simulation stick welding electrode 1072 using the known length of the simulation stick welding electrode 1072 and the known position, orientation, and/or movement of the simulation stick welding electrode holder 1070 detected by the one or more sensing devices 16 (e.g., by tracking the visual markers 802 on the simulation stick welding electrode holder 1070, for example). The welding software 244 may then compare the location of the tip 1090 of the simulation stick welding electrode 1072 to the location of the simulated workpiece 82, which may either be known by the welding software 244 or detected using the one or more sensing devices 16. In other embodiments, the welding software 244 may determine that the tip 1090 of the simulation stick welding electrode 1072 has contacted the simulated workpiece 82 based on actuation of a mechanical feature of the simulation stick welding electrode 1072 (e.g., a push button disposed in the tip 1090 of the simulation stick welding electrode 1072, in certain embodiments) when the mechanical feature of the simulation stick welding electrode 1072 contacts the simulated workpiece 82. In other embodiments, the welding software 244 may determine that the tip 1090 of the simulation stick welding electrode 1072 has contacted the simulated workpiece 82 based on feedback from other types of sensors (e.g., force sensors, accelerometers, etc.) disposed in the simulation stick welding electrode 1072 that are configured to detect certain position, orientation, and/or movement of the simulation stick welding electrode 1072 with respect to the simulated workpiece 82.

Regardless, once the welding software 244 determines that the tip 1090 of the simulation stick welding electrode 1072 has contacted the simulated workpiece 82, the welding software 244 sends appropriate control signals to the stick electrode holding assembly 1078 to start retraction of the

simulation stick welding electrode 1072. In certain embodiments, the simulation stick welding electrode holder 1070 may be configured to generate vibrations (e.g., via haptic feedback mechanisms) and/or audible feedback (e.g., via speakers) to simulate the arc starting when retraction starts. It should be noted that, in certain embodiments, an additional function that the mechanisms of the stick welding electrode holders 1070, 1074 for generating the audible feedback may serve is to generate a sound effect when the stick welding electrode holder 1070, 1074 is not tracked by the one or more sensing devices 16 for a given threshold time period (e.g., 1 second, in certain embodiments), in other words, when the stick welding electrode holder 1070, 1074 has been moved from the detection area of the one or more sensing devices 16 or when one or more visual markers 802 on the stick welding electrode holder 1070, 1074 are obstructed from view of the one or more sensing devices 16. In addition, in certain embodiments, if the actual stick welding electrode holder 1074 goes untracked for the given threshold time period, weld power to the actual stick welding electrode holder 1074 may be disabled by a contactor 1212 (e.g., contactor 1212 illustrated in FIG. 84), but a warning screen may remain on-screen for a given period of time (e.g., 5 seconds, in certain embodiments, otherwise known as a "5 second latch") before disappearing.

Once retraction of the simulation stick welding electrode 1072 has begun, the retraction may continue until either the simulation stick welding electrode 1072 has been completely retracted (i.e., cannot be retracted any further) or when the tip 1090 of the simulation stick welding electrode 1072 exceeds a threshold distance (e.g., 1" in certain embodiments) from the simulated workpiece 82. Detection of the distance of the tip 1090 of the simulation stick welding electrode 1072 from the simulated workpiece 82 may be performed using any of the position detection techniques described herein with respect to establishment of the simulated welding arc. In addition, in certain embodiments, other types of sensors (e.g., accelerometers, in certain embodiments) disposed in the simulation stick welding electrode 1072 (or elsewhere in the simulation stick welding electrode holder 1070) may be used to determine when the tip 1090 of the simulation stick welding electrode 1072 has moved away from the simulated workpiece 82 by the threshold distance, such that retraction of the simulation stick welding electrode 1072 may be stopped. In addition, in certain embodiments, in addition to being based on the distance of the tip 1090 of the simulation stick welding electrode 1072 from the simulated workpiece 82, retraction of the simulation stick welding electrode 1072 may be stopped based on a detected angle of the simulation stick welding electrode 1072, which may be detected by an angle sensor disposed in the simulation stick welding electrode 1072 (or elsewhere in the simulation stick welding electrode holder 1070 for example).

FIG. 68A illustrates an embodiment of the actual stick welding electrode holder 1074 with the outer structure 1077 removed for illustration purposes. As illustrated, in certain embodiments, the actual stick welding electrode holder 1074 includes a discrete number of distinct slots 1114 formed on an inner surfaces of at least one of the jaws 1088 of the actual stick welding electrode holder 1074 that are configured to grip the actual stick welding electrode 1076. FIG. 68B illustrates an exemplary inner surface 1116 of a jaw 1088 of the actual stick welding electrode holder 1074 having two transverse slots 1114A and 1114B and two

diagonal slots 1114C and 1114D, however, other numbers, sizes, and configurations of discrete slots 1 114 are also contemplated.

One benefit of using the discrete number of slots 1114 in the stick electrode holding assembly 1080 of the actual stick welding electrode holder 1074 is that the welding software 244 can more readily know exactly where the actual stick welding electrode 1076 is with respect to the actual stick welding electrode holder 1074. For example, since the length of a particular actual stick welding electrode 1076 is known by the welding software 244, when the welding software 244 also knows the slot 1114 of the stick electrode holding assembly 1080 that is holding the actual stick welding electrode 1076 in place, the welding software 244 may calculate the location of the distal tip of the actual stick welding electrode 1076 prior to consumption of the actual stick welding electrode 1076. To that end, the user may be guided via an on-screen prompt 1117, as illustrated in FIG. 68C, to insert the actual stick welding electrode 1076 into one of the discrete slots 1114 (labeled slots 1, 2, 3, and 4 in the illustrated embodiment). In certain embodiments, corresponding labels (e.g., 1, 2, 3, and 4) may be on an outer surface of one or both of the jaws 1088 of the actual stick welding electrode holder 1074 to help guide the user. Once the user inserts the actual stick welding electrode 1076 into one of the slots 1114, the user may select which slot 1114 into which the actual stick welding electrode 1076 was inserted. The welding software 244 uses this information to accurately locate the distal tip of the actual stick welding electrode 1076, and to aid in tracking the positioning of the actual stick welding electrode 1076 as it is consumed during an actual stick welding process. For example, in certain embodiments, the welding software 244 may use the information relating to the selected slot 1114 into which the actual stick welding electrode 1076 is inserted to determine from which side of the actual stick electrode holder 1074 that actual stick welding electrode 1076 is extending. For example, for any given slot 1114, knowing the length of the actual stick welding electrode 1076 (before consumption of the actual stick welding electrode 1076), there are only two possible locations of the tip of the actual stick welding electrode 1076 (which can both be determined by the welding software 244 at any given time). The welding software 244 may use other information relating to position, orientation, and/or movement of the actual stick electrode holder 1074 (e.g., by tracking the visual markers 802) to determine which of those two determined points makes sense (e.g., by estimating a relative location from an actual workpiece 82 for both two determined points, and determining which one is closer). To make the system more accurate, in certain embodiments, the user may be prompted to not bend the actual stick welding electrode. In addition, in certain embodiments, if the user selects a slot 1114 that is not appropriate for a particular type of weld (e.g., certain types of slot positions for an overhead weld), the user may be notified via an on-screen prompt. Alternatively, or in addition to, in certain embodiments, certain slot selection options are not provided to the user via the on-screen prompt for particular types of welds. In addition, in certain embodiments, the welding software 244 stores a user-selected slot option for a particular type of weld in memory device(s) (e.g., the memory device(s) 22 or storage device(s) 24 of the computer 18), and automatically defaults to this user-selected slot option during subsequent tests on the same type of welds.

In another embodiment, the stick electrode holding assembly 1080 of the actual stick welding electrode holder

1074 may only have one slot 1114 in which the actual stick welding electrode 1076 may be inserted. In this case, the stick electrode holding assembly 1080 may be rotatable with respect to the handle 1082 into a plurality of angular orientations (substantially similar to the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070). In certain embodiments, visual markers 802 may also be attached to this rotating stick electrode assembly to facilitate tracking of the actual stick welding electrode 1076 (as similarly discussed herein with respect to the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070).

It will be appreciated that, in embodiments of the stick electrode holding assembly 1078 of the simulation stick welding electrode holder 1070 having more than one track 1094 into which the simulation stick welding electrode 1072 may be inserted, similar on-screen prompts may be used to guide the user to insert the simulation stick welding electrode 1072 into an appropriate track 1094, and enter which track 1094 into which the simulation stick welding electrode 1072 was inserted. It should be noted that the user may freely switch between using the simulated stick welding electrode holder 1070 and the actual stick welding electrode holder 1074. As such, certain features of the stick welding electrode holders 1070, 1074 may help prevent the user from inserting inappropriate stick welding electrodes 1072, 1076 into either the tracks 1094 of the simulation stick welding electrode holder 1070 or the slots 1114 of the actual stick welding electrode holder 1074. For example, in certain embodiments, the simulation stick welding electrodes 1072 may be larger than the actual stick welding electrodes 1076 (e.g., have larger diameters) such that the simulation stick welding electrodes 1072 do not fit into the slots 1114 of the actual stick welding electrode holder 1074 (i.e., because they are too large), and the actual stick welding electrodes 1076 cannot be firmly held within the tracks 1094 of the simulated stick welding electrode holder 1070 (i.e., because they are too small). It should be noted that, in other embodiments, an actual stick welding electrode 1076 may be used with the simulation stick welding electrode holder 1070. For example, in such embodiments, the tracks 1094 of the simulated stick welding electrode holder 1070 may be appropriately sized such that actual stick welding electrodes 1076 may be inserted into them.

FIGS. 69A and 69B illustrate embodiments of button panels 1118, 1120 that may be disposed on the simulation stick welding electrode holder 1070 and the actual stick welding electrode holder 1074, respectively. As primarily described herein as including buttons, in other embodiments, the buttons of the stick welding electrode holder 1070, 1074 described herein may include other various input devices or input elements including, but not limited to, touch screens, sliders, scroll wheels, switches, knobs, liquid crystal displays, or any other suitable input devices or input elements configured to enable operator interaction with the welding software 244 via the stick welding electrode holder 1070, 1074. In certain embodiments, the button panels 1118, 1120 may include translucent membranes with LEDs underneath to glow certain icons as illustrated in FIGS. 69A and 69B. Locating the button panels 1118, 1120 on the stick welding electrode holders 1070, 1074 facilitates easier selection of certain options by the user. Both button panels 1118, 1120 include two navigation buttons 1122, 1124 (although, in certain embodiments, the button panels 1118, 1120 may include only one navigation button) that may be manipulated by the user to help navigate the screens, menus, etc. that are displayed on the display devices described herein. In addi-

tion, both button panels **1118**, **1120** include a power button **1126**, however, manipulation of the power button **1126** causes slightly different functionality with respect to the stick welding electrode holders **1070**, **1074**. For example, in certain embodiments, when the user presses the power button **1126** of the simulation stick welding electrode holder **1070**, the simulation stick welding electrode holder **1070** is activated by the welding software **244**, and a simulation stick welding process indicator **1128** (e.g., a blue LED in certain embodiments) is turned on. It will be appreciated that, in certain embodiments, the buttons described herein enable context-sensitive navigation through the screens, menus, etc. that are displayed on the display devices described herein. In other words, the navigational functions of the buttons described herein may change based on the context of the information currently being displayed on the display devices. In such situations, context-sensitive prompts may be displayed on the display devices to aid the user in navigating the screens, menus, etc.

In contrast, in certain embodiments, when the user first presses the power button **1126** of the actual stick welding electrode holder **1074**, the actual stick welding electrode holder **1074** is activated by the welding software **244**, and an actual stick welding process indicator **1130** (e.g., an orange LED in certain embodiments) is turned on, but weld power is not yet provided to the actual stick welding electrode holder **1074**. After the actual stick welding electrode holder is activated, the power button **1126** may subsequently be pressed for a certain amount of time (e.g., two second in certain embodiments) to enable weld power through the actual stick welding electrode holder **1074**, at which time an actual stick welding power indicator **1132** (e.g., an orange LED in certain embodiments) begins blinking (to indicate that weld power is enabled through the actual stick welding electrode holder **1074**). In certain embodiments, subsequent pressing of the power button **1126** causes the weld power to be disabled. In certain embodiments, the actual weld power is only enabled through the actual stick welding electrode holder **1074** when the welding software **244** determines that the actual stick welding electrode **1076** is proximate (e.g., within an inch, for example) the actual workpiece **82**, using the position detection techniques described herein. In certain embodiments, both of the stick welding electrode holders **1070**, **1074** may only be activated when appropriate screens are being displayed to the user. In addition, in certain embodiments, the stick welding electrode holders **1070**, **1074** may be configured to generate vibrations (e.g., via haptic feedback mechanisms) to confirm to the user that a button has been pressed.

The button panels **1118**, **1120** illustrated in FIGS. **69A** and **69B** may take various forms and may be disposed at various locations of the stick welding electrode holders **1070**, **1074**. For example, as illustrated in FIG. **70**, the button panel **1120** of the actual stick welding electrode holder **1074** may be located near a proximal end **1134** of the handle **1082** of the actual stick welding electrode holder **1074**. In addition, as illustrated in FIG. **71**, the button panel **1120** of the actual stick welding electrode holder **1074** may be located on an outer surface of the outer structure **1077** of the actual stick welding electrode holder **1074**. It will be appreciated that, in certain embodiments, the button panel **1118** may be similarly located on the simulation stick welding electrode holder **1070**.

In certain embodiments, the stick welding electrode holders **1070**, **1074** may include certain features to facilitate real-time feedback to the user during operation of the stick welding electrode holders **1070**, **1074**. For example, in

certain embodiments, the stick welding electrode holders **1070**, **1074** may include a status indicator **1136** that may indicate to a user whether a specific parameter of interest (e.g., welding parameters such as work angle, travel angle, aim, and so forth, arc parameters such as voltage and current), or combination of parameters, is acceptable or not (e.g., within an acceptable range of values, for example, predetermined upper and lower limits). Other parameters of interest that may be indicated by the status indicator **1136** may include whether the visual markers **802** of the stick welding electrode holder **1070**, **1074** are blocked from being detected by the one or more sensing devices **16**, whether the stick welding electrode holder **1070**, **1074** is in a particular mode of operation (e.g., a welding mode versus a setup mode, for example), and so forth. It will be appreciated that the specific parameter of interest may be selected by the user via the various screens presented to the user as described herein. As illustrated in FIGS. **72A** and **72B**, in certain embodiments, the status indicator **1136** may be disposed proximate a distal end of the stick welding electrode holder **1070**, **1074**.

In certain embodiments, the status indicator **1136** may be an LED capable of illuminating either green or red and, if the specific parameter of interest is acceptable (e.g., whether the specific parameter is within a predetermined upper and lower limit), the status indicator **1136** may be illuminated green, if the specific parameter of interest is not acceptable, the status indicator **1136** may not be illuminated at all, and when the visual markers **802** are not being tracked, the status indicator may be illuminated red. In addition, in certain embodiments, the color, intensity, and/or patterns of illumination of the status indicator **1136** may be varied based on a relation of the parameter of interest to the limits. In certain embodiments, so as to minimize the potential for distracting the user, the status indicator **1136** may not be illuminated at all unless the stick welding electrode holder **1070**, **1074** is proximate the workpiece **82**, using the position detection techniques described herein.

In addition, in certain embodiments, the status indicator **1136** may also be capable of generating vibrations (e.g., via haptic feedback mechanisms) and/or audible feedback (e.g., via speakers) to indicate to the user whether the specific parameter of interest is acceptable or not. With respect to the actual stick welding electrode holder **1074** (and sometimes with the simulation stick welding electrode holder **1070**), in certain embodiments, so as to minimize the potential to distract the user, these real-time feedback features may be disabled during the actual stick welding process performed by the actual stick welding electrode holder **1074** whereas in most embodiments, these real-time feedback features may be left on during the simulated stick welding process performed by the simulation stick welding electrode holder **1070**.

In addition, in certain embodiments, the stick welding electrode holders **1070**, **1074** may include a plurality of status indicators **1136** that may indicate a plurality of statuses. In addition, each of the plurality of status indicators **1136** may indicate multiple different statuses (e.g., depending on color, intensity, and/or patterns of illumination of the status indicators **1136** (in the case of LEDs), depending on intensity and/or patterns of vibrations (in the case of haptic feedback mechanisms), depending on volume, tone, and/or patterns of audible feedback (in the case of speakers), and so forth). In certain embodiments, any combination of the plurality of status indicators **1136** may be enabled or disabled at any given time.

In certain embodiments, the real-time feedback features described with respect to FIGS. 72A and 72B may be particularly beneficial when the user is starting the particular stick welding process (e.g., bringing the stick welding electrode holder 1070, 1074 into position to perform the particular stick welding process). FIG. 73 illustrates an exemplary screen 1138 that may be displayed when the actual stick welding electrode holder 1074 is activated, before an actual welding process has been initiated, and while the actual stick welding electrode holder 1074 is not in position (e.g., the welding software 244 has determined that the actual stick welding electrode 1076 is not proximate the actual workpiece 82). When this is the case, the welding procedure summary pane 1060 is illustrated to summarize for the user what the overall properties (e.g., target properties) for a given test weld are. It will be appreciated that the screen 1138 illustrated in FIG. 73 will be substantially similar when the simulation stick welding electrode holder 1070 is activated, before the simulated welding process has been initiated, and while the simulation stick welding electrode holder 1070 is not in position (e.g., the welding software 244 has determined that the simulation stick welding electrode 1072 is not proximate (e.g., within an inch, for example) the simulated workpiece 82).

It should be noted that for the stick welding processes performed by the stick welding electrode holders 1070, 1074, contact tip to workpiece distance 332 may be replaced by arc length index 1140 in the case of actual stick welding processes performed by the actual stick welding electrode holder 1074 (and may be arc length in the case of simulated stick welding processes performed by the simulation stick welding electrode holder 1070). In the case of actual stick welding processes performed by the actual stick welding electrode holder 1074, arc length index is an index that attempts to approximate the arc length (i.e., the length of the arc from the tip of the actual stick welding electrode 1076 to the workpiece 82 along the axis of the actual stick welding electrode 1076). The necessity for the approximation is that the actual arc length between the tip of the actual stick welding electrode 1076 and the workpiece 82 cannot be directly determined as accurately as with simulated stick welding processes performed by the simulation stick welding electrode holder 1070 (e.g., by tracking visual markers 802 on the simulation stick welding electrode 1072 in certain embodiments). This is primarily attributable to the fact that the position of the tip 1090 of the simulation stick welding electrode 1072 with respect to the simulated workpiece 82 is more readily determined than the position of the tip of the actual stick welding electrode with respect to an actual workpiece 82 (due at least in part to the actual consumption of the actual stick welding electrode during the actual stick welding process). In certain embodiments, the arc length index may be calculated as a function of the voltage through the actual stick welding electrode 1076 during the actual stick welding process. More specifically, in certain embodiments, the arc length index may be calculated as 2.5 times the voltage through the actual stick welding electrode 1076 during the actual stick welding process, and the arc length index may be depicted on a scale of 0 to 100 in certain embodiments.

In the case of simulated arc welding, instead of arc length index, an arc length may be calculated. For example, in certain embodiments, visual markers 802 disposed on the simulation stick welding electrode 1072 may be tracked, and the arc length may be calculated based on a position of the tip 1090 of the simulation stick welding electrode 1072 relative to a position of the simulation workpiece 82 (which

may be similarly tracked, as described herein). In this scenario, the arc length is the difference in distance between the tip 1090 of the simulation stick welding electrode 1072 and the simulation workpiece 82 as measured along the axis of the simulation stick welding electrode 1072. In certain embodiments where the visual markers 802 are instead disposed on the outer structure 1077 of the simulation stick welding electrode holder 1070, the relative position of the tip 1090 of the simulation stick welding electrode 1072 may be determined by detecting when current flows from the tip 1090 of the simulation stick welding electrode 1072 to the simulation workpiece 82. At that time, the position of the tip 1090 of the simulation stick welding electrode 1072 is known via the known position of the simulation workpiece 82. Using this known location, as well as the known rate of retraction of the simulation stick welding electrode 1072, the continually updated location of the tip 1090 of the simulation stick welding electrode 1072 may be determined, and the arc length may be calculated based on this continually updated location. In turn, the calculated arc length may be used by the welding software 244 to dynamically update the rate of retraction of the simulation stick welding electrode 1072.

In certain embodiments, during a weld, if a remaining length of the actual stick welding electrode 1076 is determined to be shorter than a threshold electrode length (e.g., 3 inches, in certain embodiments), weld power to the actual stick welding electrode holder 11074 may be disabled, thereby protecting the actual stick welding electrode holder 11074 from heat/spatter, and prolonging the life of the actual stick welding electrode holder 11074. In addition, in certain embodiments, the actual stick welding electrode holder 1074 may include one or more sensors configured to detect temperatures of the actual stick welding electrode holder 1074, and the user may be notified (and weld power disabled) when the detected temperatures exceed certain threshold temperatures.

FIG. 74 illustrates an exemplary screen 1142 that may be displayed when the actual stick welding electrode holder 1074 is activated, before the actual welding process has been initiated, while the actual stick welding electrode holder 1074 is in position (e.g., the welding software 244 has determined that the actual stick welding electrode 1076 is proximate (e.g., within an inch, for example) the actual workpiece 82), but before a contactor (e.g., contactor 1212 illustrated in FIG. 84) has been enabled to provide actual welding power through the actual stick welding electrode holder 1074 (and, thus, the actual stick welding electrode 1076). As illustrated, a targeting graphic 1144 may be displayed (that functions as visual guides) to provide an indication to the user how the positioning of the actual stick welding electrode holder 1074 should be adjusted. More specifically, the depicted solid round circle is intended to represent the current targeting 1146 of the actual stick welding electrode 1076, the larger open circle is intended to represent the desired work and travel angle targets 1148, and the horizontal line is intended to represent the desired aim 1150. The purpose of the targeting graphic 1144 is to help the user correct the positioning of the actual stick welding electrode holder 1074 such that the current targeting 1146 is moved inside the desired work and travel angle targets 1148 and level with the desired aim 1150. When this is done, the positioning of the actual stick welding electrode 1076 is considered to be acceptable. In certain embodiments, the targeting graphic 1144 may include a vertical line that is intended to represent a desired travel speed.

FIG. 75 illustrates an exemplary screen 1152 that may be displayed when the actual stick welding electrode holder 1074 is activated, before the actual welding process has been initiated, while the actual stick welding electrode holder 1074 is in position (e.g., the welding software 244 has determined that the actual stick welding electrode 1076 is proximate the actual workpiece 82), and after a contactor (e.g., contactor 1212 illustrated in FIG. 84) has been enabled to provide actual welding power through the actual stick welding electrode holder 1074 (and, thus, the actual stick welding electrode 1076). As illustrated, a message may be displayed to bystanders that the actual welding process is about to begin, and that the bystanders should not be looking at the screen 1152, but rather should be paying attention to the impending actual welding process. FIG. 76 illustrates an exemplary screen 1154 that may be displayed when the actual stick welding electrode holder 1074 is activated, just after the actual welding process has been initiated. As will be appreciated, this is the next step in the process after screen 1152 of FIG. 75 and that, once the actual welding process has been initiated, the targeting graphic 1144 is removed to further reduce the possibility of distracting the user.

To further aid in drawing the user's attention away from the stick welding electrode holders 1070, 1074, in certain embodiments, the targeting graphic 1144 illustrated in FIGS. 74 and 75 may be positioned more conveniently for the user. More specifically, FIGS. 77 A through 77C depict various embodiments of display devices whereby a display of the targeting graphic 1144 may be produced at various locations more proximate the welding process being performed by the stick welding electrode holder 1070, 1074. For example, FIG. 77 A illustrates an embodiment where a display 1156 is integrated into the stick welding electrode holder 1070, 1074. In the illustrated embodiment, the display 1156 is integrated into the outer structure 1077 of the stick welding electrode holder 1070, 1074. More specifically, the illustrated embodiment includes the display 1156 disposed on an extension of the outer structure 1077 of the stick welding electrode holder 1070, 1074. However, in other embodiments, the display 1156 may be integrated into the handle 1082 or some other component of the stick welding electrode holder 1070, 1074. As illustrated in FIG. 77 A, the display 1156 may be configured to display the targeting graphic 1144 such that the user may receive real-time feedback relating to the positioning of the stick welding electrode holder 1070, 1074 without having to direct the user's attention away from the stick welding electrode holder 1070, 1074. It will be appreciated that the welding software 244 may be configured to send control signals through the stick welding electrode holder 1070, 1074 to the display 1156 in substantially real-time to adjust the display of the targeting graphic 1144.

FIG. 77B illustrates another embodiment where a handheld device 1158 having its own display 1160 is configured to display the targeting graphic 1144. It will be appreciated that the illustrated handheld device 1158 may be located at various locations near the workpiece 82 and/or the stick welding electrode holder 1070, 1074 such that the user may receive real-time feedback relating to the positioning of the stick welding electrode holder 1070, 1074 with respect to the workpiece 82 without having to direct the user's attention away from the workpiece 82 and/or the stick welding electrode holder 1070, 1074. It will be appreciated that, in certain embodiments, the welding software 244 may be configured to send control signals to the display 1160 wirelessly (e.g., via the network device 36 illustrated in FIG.

1) in substantially real-time to adjust the display of the targeting graphic 1144. In certain embodiments, the handheld device 1158 may be a device specifically dedicated to displaying the targeting graphic 1144 and/or other graphical representations relating to the welding system 10. However, in the embodiment illustrated in FIG. 77B, the handheld device 1158 may instead be a device not specifically dedicated to displaying the targeting graphic 1144 and/or other graphical representations relating to the welding system 10, but rather a multi-purpose device (such as a smart phone) having a software application installed thereon that is configured to display the targeting graphic 1144 and/or other graphical representations relating to the welding system 10. As illustrated in FIG. 77B, in certain embodiments, a stand 1162 may be used to help orient the handheld device 1158 at a convenient orientation for the user to view the display 1160 of the handheld device 1158. In certain embodiments, the stand 1162 may be integrated into the workpiece 82 or any other component of the welding system 10. In certain embodiments, the display 1156 of the stick welding electrode holder 1070, 1074 of FIG. 77A and/or the display 1160 of the handheld device 1158 of FIG. 77B may be configured to display the screenshots illustrated in FIG. 27.

FIG. 77C illustrates yet another embodiment where the stick welding electrode holder 1070, 1074 includes a projection system 1164 configured to project the targeting graphic 1144 directly onto the workpiece 82. In the illustrated embodiment, the projection system 1164 is integrated into the outer structure 1077 of the stick welding electrode holder 1070, 1074. However, in other embodiments, the projection system 1164 may be integrated into the handle 1082 or some other component of the stick welding electrode holder 1070, 1074. It may be appreciated that, in such an embodiment, the projection system 1164 may need to be configured to direct the projected images around the stick welding electrode 1072, 1076. For example, the projection system 1164 may require at least two projection subsystems disposed on opposite sides of the stick welding electrode 1072, 1076. Due at least in part to the fact that, in certain embodiments, the stick welding electrode 1072, 1076 may be located in various locations and orientations with respect to the stick welding electrode holder 1070, 1074, numerous projection sub-systems of the projection system 1164 may need to be located about the stick welding electrode holder 1070, 1074, and the welding software 244 may be configured to control which of the various projection sub-systems of the projection system 1164 are used to project the targeting graphic 1144 onto the workpiece 82 by, for example, selectively activating and controlling the projection sub-systems of the projection system 1164. It will also be appreciated that, in certain embodiments, the targeting graphic 1144 may be displayed on an internal display 32 of the welding helmet 41 described herein.

In addition, in certain embodiments, the stick welding electrode holder 1070, 1074 may include other graphical indicators (e.g., visual guides) for providing real-time feedback to the user. For example, as illustrated in FIG. 78, in certain embodiments, the stick welding electrode holder 1070, 1074 may include one or more graphical range indicators 1166 (e.g. disposed on an extension of the outer structure 1077) for depicting where a specific parameter of interest currently is with respect to an acceptable range (e.g., between an acceptable lower and upper limit). Although illustrated in FIG. 78 as relating to travel angle and work angle, the one or more graphical range indicators 1166 may be related to any of the parameters described herein. In the illustrated embodiment, the one or more graphical range

indicators **1166** are depicted as being aligned substantially parallel. However, in other embodiments, the one or more graphical range indicators **1166** may be aligned generally crosswise with each other, thereby representing a cross-like representation. In the illustrated embodiment, the one or more graphical range indicators **1166** are integrated into the outer structure **1077** of the stick welding electrode holder **1070**, **1074**. However, in other embodiments, the one or more graphical range indicators **1166** may be integrated into the handle **1082** or some other component of the stick welding electrode holder **1070**, **1074**. In certain embodiments, the graphical range indicators **1166** may be used in conjunction with the status indicators **1136** discussed with respect to FIGS. **72A** and **72B**. For example, if the value for a parameter of interest tracked by one of the graphical range indicators **1166** begins to fall outside of the depicted range, corresponding status indicators **1136** may be appropriately activated.

As described with respect to FIGS. **68A** through **68C**, in certain embodiments, the actual stick welding electrode holder **1074** may include multiple discrete slots **1114** into which an actual stick welding electrode **1076** may be inserted and held. As described herein, the user may be prompted that the actual stick welding electrode **1076** should not be bent (so as to improve the accuracy of the tracking of the actual stick welding electrode **1076** during the actual stick welding process. However, in certain circumstances, the user may have reasons for bending the actual stick welding electrode **1076**. In such situations, as illustrated in FIG. **79**, a calibration device **1168** may be slid over the tip of the actual stick welding electrode **1076** after the actual stick welding electrode **1076** has been bent. In general, in certain embodiments, the calibration device **1168** may be, for example, a sleeve that includes a generally cylindrical body having an inner bore that is specifically sized to receive (e.g., fit circumferentially around) tips (e.g., distal ends) of actual stick welding electrodes **1076**. In certain embodiments, the calibration device **1168** may include a detection mechanism (e.g., a force sensor, in certain embodiments) disposed in the calibration device **1168** and configured to detect when the calibration device **1168** has been secured onto the actual stick welding electrode **1076**.

In certain embodiments, the calibration device **1168** may include two or more visual markers **802**, which may be either active or passive markers, the position of which is capable of being detected by the one or more sensing devices **16** to calibrate where, exactly the tip of the actual stick welding electrode **11076** is (e.g., relative to the actual stick welding electrode holder **1074**) and to determine an axis of the actual stick welding electrode **1076** for use by the welding software **244**. The visual markers **802** on the calibration device **1168** function in substantially the same manner as the visual markers **802** on the simulation stick welding electrode **1072**, in certain embodiments, to locate the position of the tip **1090** of the simulation stick welding electrode **1072**. It will be appreciated that, in certain embodiments, the calibration device **1168** may be used instead of having the user select which discrete slot **1114** of the actual stick welding electrode holder **1074** that the actual stick welding electrode **1076** has been inserted into. In certain embodiments, the calibration of the calibration device **1168** may be initiated via a user input, or when circuitry internal to the calibration device **1168** detects that the calibration device **1168** has been fully positioned over the tip of the actual stick welding electrode **1076**. In other embodiments, the calibration of the calibration device **1168**

may be initiated based on compression or expansion of a distance between visual markers **802** of the calibration device **1168**.

As described herein, many various types of simulation welding tools and realworld welding tools may be used with the welding system **10** described herein. Accordingly, many various screens may be presented to the user when using the welding system **10** described herein. FIGS. **80A** and **80B** illustrate exemplary screens **1170**, **1172** relating to assignment lists for MIG welding torches ("SmartGuns") and stick welding electrode holders ("SmartStingers"), respectively. Both screens **1170**, **1172** show assignment lists **1174** for the respective type of welding tool **14**. In addition, both screens show filters for joint type **1176** (e.g., butt joint, lap joint, T-joint, etc.) and position **1178** (e.g., horizontal, vertical, etc.). The main difference between the data shown in the assignment lists **1174** and the filters on the screens **1170**, **1172** is that, for MIG welding torches, the assignment lists **1174** and the filters include data relating to process type **1180** (e.g., GMAW, GMAW-S, FCAW-G, etc.) and, for the stick welding electrode holders, the assignment lists **1174** and the filters include data relating to electrode class **1182** (e.g., E6010, E6013, E7018, etc.). In certain embodiments, for any given user, up to six electrode class **1182** filter options may be displayed,

In certain embodiments, when a user views his assignments, the default screen **1170**, **1172** will be that of the last welding tool **14** the user had selected. For example, if the last welding tool **14** selected by the user was a stick welding electrode holder **1070**, **1074** screen **1172** will be the default assignment screen for the user. In certain embodiments, if a user selects an assignment for a welding tool **14** that is not connected to the welding system **10**, a message may be displayed on-screen informing the user to connect that type of welding tool **14** before proceeding. In certain embodiments, the selected welding tool type **1184** may be displayed in the top left corner of the assignment selection screens **1170**, **1172**, and all subsequent setup and test screens. The selected welding tool type **1184** may also be displayed on history screens corresponding to the particular type of welding tool **14**, as well as Assignment Management screens.

In addition to the assignment selection screens **1170**, **1172** illustrated in FIGS. **80A** and **81B**, welding tool calibration screens **1186** may be displayed for each type of welding tool **14**. FIG. **81** illustrates an exemplary calibration screen **1186** for a MIG welding gun ("SmartGun"). As illustrated, the calibration screens **1186** may provide step-by-step procedures for calibration the selected type of welding tool **14**. As illustrated in FIG. **81**, the first step for calibrating a MIG welding gun is selecting whether the gun is new to the welding system **10**. The second step for calibrating the MIG welding gun is confirming and updating dimensions of a gun axis calibration tool (e.g., calibration tool **610** illustrated in FIGS. **37** and **38**), if necessary. The third step for calibrating the MIG welding gun is attaching the gun axis calibration tip (e.g., tip **614** of the calibration tool **610** illustrated in FIGS. **37** and **38**) as shown. The fourth step for calibrating the MIG welding gun is mounting the gun and selecting Calibrate for multiple positions. As illustrated, there are also additional options for advanced users, which can be selected. As illustrated in FIG. **82**, additional help screens **1188** may be shown to help users know how to use, among other things, the targeting graphics **1144** (i.e., visual guides).

As discussed herein with respect to FIG. **73**, in certain embodiments, arc length or arc length index may be determined by the welding software **244** and displayed on-screen

as a tracked parameter. However, in other embodiments, as illustrated in FIG. 83, instead of arc length or arc length index, feed rate 1192 may be determined and displayed on-screen as a tracked parameter. In general, feed rate 1192 (e.g., an approximation of the rate of consumption of the stick welding electrode 1072, 1076) is the rate of change of distance between the stick welding electrode holder 1070, 1074 and the workpiece 82 along an axis of the stick welding electrode 1072, 1076. As such, it will be appreciated that since the position, orientation, and/or movement of the stick welding electrode holder 1070, 1074 is tracked by the one or more sensing devices 16, and the position of the workpiece 82 is either tracked by the one or more sensing devices 16 or known, then the rate of change of the distance between the stick welding electrode holder 1070, 1074 and the workpiece 82 along an axis of the stick welding electrode 1072, 1076 is a fairly straightforward calculation.

As described herein, various different types of actual and simulation welding tools 14 may be used with the welding system 10 described herein. Accordingly, it will be appreciated that, in certain embodiments, power sources configured to provide power to these various actual and simulation welding tools 14 may also be used in conjunction with the welding system described herein. As will also be appreciated, management of the power and data between these various power sources, actual and simulation welding tools 14, and the welding system 10 can become somewhat cumbersome, due at least to the sheer number of possible connections, cables, and so forth. Accordingly, in certain embodiments, as illustrated in FIG. 84, a dedicated connection (router) box 1194 may be used to connect the various power sources, actual and simulation welding tools 14, the welding system 10, and other related components and devices.

As illustrated in FIG. 84, in certain embodiments, the connection box 1194 may include a connector 1196 configured to connect to a data cable 1198 configured to connect to the welding system 10 such that data (from the welding tools 14 and other components and devices connected to the connection box 1194) can be communicated from the connection box 1194 to the welding system 10 for processing. It will be appreciated that, in certain embodiments, control signals may also be communicated from the welding system 10 to the welding tools 14 and other components and devices connected to the connection box 1194 via the data cable 1198. As such, the connection box 1194 is used to route data between the welding system 10 and the welding tools 14.

In certain embodiments, the various power sources that provide power to the welding tools 14 may interact with the welding tools 14 and the connection box 1194 in various ways, depending on the particular needs of the welding tools 14. For example, as illustrated in FIG. 84, in certain embodiments, a MIG power source 1200 may provide welding power directly to a MIG welding tool 14, and the MIG welding tool 14 may communicate directly with the MIG power source 1200, for example, to provide trigger control signals, and so forth. As such, in certain embodiments, the connection box 1194 may include only a single connector 1202 configured to connect to a data cable 1204 configured to connect to the MIG welding tool 14 such that data can be communicated from the MIG welding tool 14 through the connection box 1194 to the welding system 10 for processing. It will be appreciated that, in certain embodiments, control signals may also be communicated from the welding system 10 (through the connection box 1194) to the MIG welding tool 14 via the data cable 1204.

In contrast, in certain embodiments, the connection box 1194 may include a connector 1206 configured to connect to a weld power cable 1208 configured to connect to a stick welding power source 1210 configured to provide power suitable for a stick welding process performed by an actual stick welding electrode holder 1074. In certain embodiments, the connection box 1194 may include a contactor 1212 configured to be enabled (e.g., closed) or disabled (e.g., opened) to provide welding power to an actual stick welding electrode holder 1074 that is connected to the connection box 1194 or prevent the welding power from being provided to the actual stick welding electrode holder 1074. To that end, in certain embodiments, the connection box 1194 may also include a connector 1214 configured to connect a weld power cable 1216 configured to connect to an actual stick welding electrode holder 1074 such that the weld power provided by the stick welding power source 1210 may be provided to the actual stick welding electrode holder 1074 through the connection box 1194 when the contactor 1212 of the connection box 1194 is enabled. In certain embodiments, after the contactor 1212 has been turned on, the amount of power used to drive the contactor 1212 may vary based on the state of the contactor 1212. For example, significantly more power is used to turn the contactor 1212 on than to maintain the contactor 1212 in the on state.

In certain embodiments, when it is determined that the length of the actual stick welding electrode 1076 is under a certain length (e.g., under 3 inches, in certain embodiments), the contactor 1212 may be opened to stop welding. In addition, in certain embodiments, the contactor 1212 will be configured to be normally open (e.g., in a safe mode). In certain embodiments, the welding system 10 and/or control circuitry 1234 of the connection box 1194 may track the number of times the contactor 1212 opens mid-weld and, when the life limit (e.g., 100,000 cycles in certain embodiments) of the contactor 1212 approaches, the user (or, perhaps more commonly, the instructor) may be alerted (e.g., via a display associated with the welding system 10 and/or the connection box 1194).

In addition, in certain embodiments, the connection box 1194 may include a connector 1218 configured to connect to a data cable 1220 configured to connect to the actual stick welding electrode holder 1074 such that data from the actual stick welding electrode holder 1074 may be communicated back through the connection box 1194 to the welding system 10. It will be appreciated that, in certain embodiments, control signals may also be communicated from the welding system 10 to the actual stick welding electrode holder 1074 via the data cable 1220. Similarly, in certain embodiments, the connection box 1194 may include a connector 1222 configured to connect to a data cable 1224 configured to connect to a simulation welding electrode holder 1070 such that data from the simulation stick welding electrode holder 1070 may be communicated back through the connection box 1194 to the welding system 10. It will be appreciated that, in certain embodiments, control signals may also be communicated from the welding system 10 to the simulation stick welding electrode holder 1070 via the data cable 1224.

It should be noted that, in certain embodiments, an actual or simulation TIG welding torch may be connected to the welding system via a separate TIG connection box 1226 that connects to the connection box 1194 and has specific circuitry configured to control the flow of weld power and data to and from the actual or simulation TIG welding torch. To

that end, in such embodiments, the connection box **1194** may include a connector **1228** for connecting to the TIG connection box **1226**.

In addition, in certain embodiments, the connection box **1194** may include a connector **1230** configured to connect to a DC power source **1232** such that DC power may be provided to the connection box **1194** to provide power for control circuitry **1234** (as well as other circuitry) of the connection box **1194**. It will be appreciated that, in certain embodiments, the power source **1232** may instead be an AC power source, and the connection box **1194** may include AC-to-DC conversion circuitry configured to convert the AC power to DC power for the control circuitry **1234** (and other circuitry) of the connection box **1194**. In certain embodiments, the control circuitry **1234** of the connection box **1194** may include, among other things, one or more processors **1236**, one or more memory devices **1238**, and one or more storage devices **1240**. In other embodiments, the control circuitry **1234** may not include the processor(s) **1236**, the memory device(s) **1238**, and/or the storage device(s) **1240**. The processor(s) **1236** may be used to execute software algorithms as described herein. Moreover, the processor(s) **1236** may be similar to the processor(s) **20** described previously. Furthermore, the memory device(s) **1238** may be similar to the memory device(s) **22**, and the storage device(s) **1240** may be similar to the storage device(s) **24**. It will be appreciated that, in certain embodiments, the control circuitry **1234** of the connection box **1194** may function in cooperation with the welding software **244** of the welding system **10**, for example, sharing certain processing of information.

It will be appreciated that the control circuitry **1234** of the connection box **1194** may obviate the need for having processing circuitry disposed in some of the actual and simulation welding tools **14** (e.g., the stick welding electrode holders **1070**, **1074** illustrated in FIG. **84**). For example, in certain embodiments, the control circuitry **1234** may be configured to control all, or at least most, of the local operational features of the stick welding electrode holders **1070**, **1074** described herein without the need of processing circuitry disposed in the stick welding electrode holders **1070**, **1074**. For example, the control functions for controlling the stick electrode holding assemblies **1078**, **1080** (see, e.g., FIGS. **64A**, **64B**, **65A**, **65B**, and **67**), the control functions for controlling the simulated arc starting feature, the control functions for interacting with the button panels **1118**, **1120** (see, e.g., FIGS. **69A** and **69B**), the control functions for controlling the status indicators **1136** (see, e.g., FIGS. **72A** and **72B**), the targeting graphics **1144** (see, e.g., FIGS. **77A**, **77B**, and **77C**), and graphical range indicators **1166** (see, e.g., FIG. **78**), and so forth, may be performed by the control circuitry **1234** of the connection box **1194**, thereby obviating the need to have processing circuitry in the stick welding electrode holders **1070**, **1074** for performing these control functions. In addition, in certain embodiments, the connection box **1194** may include one or more status indicators **1241** (e.g., light emitting diodes, in certain embodiments) disposed on an outer housing **1243** of the connection box **1194**. The status indicators **1241** may indicate one or more statuses relating to operations of the connection box **1194** including, but not limited to, a state of the contactor **1212** (e.g., closed or open, enabling or disabling welding power, and so forth), whether communications are established with the welding system **10**, whether the connection box **1194** is powered on or off, and so forth. It should be noted that all other components of the connection box **1194** illustrated in FIG. **84** as being disposed within,

or disposed on, the housing **1243** of the connection box **1194** are, indeed, disposed within, or disposed on, the common housing **1243** of the connection box **1194**.

Furthermore, having a dedicated connection box **1194** for connecting the various power sources, actual and simulation welding tools **14**, the welding system **10**, and other related components and devices, also facilitates certain functionality relating to power and data management that might otherwise be difficult, if not impossible, without the connection box **1194** described herein. For example, in certain embodiments, the connection box **1194** may include current sensing or voltage sensing circuitry (C/V sensing circuitry) **1242**, which may sense the current or voltage being delivered to the actual stick welding electrode holder **1074** via the weld power cable **1216**. An additional benefit of having the C/V sensing circuitry **1242** in the connection box **1194** is that the connection box **1194** can determine when there is no more current flowing to the actual stick welding electrode holder **1074**, and may notify the welding software **244** that the test has begun or has ended based on this determination (e.g., a sensed current exceeding, or not exceeding, a threshold). In certain embodiments, the C/V sensing circuitry **1242** may be configured to sense the voltage, and the sensed voltage may be used to detect the polarity of the welding power (e.g., alternating current (AC), direct current electrode negative (DCEN), direct current electrode positive (DCEP), and so forth), and the user may be notified whether the detected polarity is correct (i.e., that the actual stick welding electrode holder **1074** is connected properly). In other words, the control circuitry **1234** (and/or the welding software **244**) may generate a prompt via the status indicator(s) **1241** of the connection box **1194**, via the status indicator(s) **1136** of the welding tools **14**, and/or via any other status indicator (or other output device) described herein, when the detected polarity is not the same as a polarity setting for the welding system **10** (i.e., a desired polarity set by a user via the welding system **10** or otherwise programmed into the welding system **10**).

In addition, in certain embodiments, the control circuitry **1234** of the connection box **1194** (in conjunction with the C/V sensing circuitry **1242**) may execute a stick-stuck algorithm (e.g., stored in the memory device(s) **1238** and/or the storage device(s) **1240** of the control circuitry **1234** and executable by the processor(s) **1236** of the control circuitry **1234**) that is configured to determine when an actual stick welding electrode **1076** is stuck to the workpiece **82**. Exemplary logic for such a stick-stuck algorithm is presented in U.S. Pat. No. 6,750,427, which is incorporated herein by reference in its entirety. In addition, in certain embodiments, the control circuitry **1234** of the connection box **1194** may include a stick-stuck algorithm (e.g., stored in the memory device(s) **1238** and/or the storage device(s) **1240** of the control circuitry **1234** and executable by the processor(s) **1236** of the control circuitry **1234**) that is configured to estimate when a simulated stick welding electrode **1072** would have become stuck to the simulated workpiece **82** (e.g., if it were an actual stick welding electrode **1076** and the workpiece **82** were a real workpiece). Upon stick-stuck detection, various responses may be generated including, but not limited to, providing messages on-screen, recording the occurrence of the stick-stuck event (e.g., in the memory device(s) **1238** and/or the storage device(s) **1240** of the control circuitry **1234**), automatically failing the test, determining a state of the contactor **1212** (e.g., closed or open, enabling or disabling welding power, and so forth), and so forth. In addition, in certain embodiments, upon detection of a stick-stuck event, the control circuitry **1234** may cause the

contactor **1212** to open to disable welding. In certain embodiments, once unstuck, the user may push a button (or otherwise activate another input element) to reset the stick-stuck algorithm. One possible way of determining whether the stick-stuck event is still occurring may be to send a low voltage signal through the actual stick welding electrode **1076**, and if the signal returns, it may be presumed that the actual stick welding electrode **1076** is still stuck to the workpiece **82**. In certain embodiments, once a state of the contactor **1212** (e.g., closed or open, enabling or disabling welding power, and so forth) has been determined by the control circuitry **1234** (and/or the welding software **244**), the control circuitry **1234** (and/or the welding software **244**) may determine a condition of the contactor **1212** (e.g., whether the contactor **1212** is functioning properly, whether the contactor **1212** has failed, and so forth), for example, by comparing a driven state of the contactor **1212** (e.g., as set by the welding system **10**) versus the actual state of the contactor **1212**.

The contactor **1212** and the C/V sensing circuitry **1242** of the connection box **1194** enable various additional functionality. For example, in certain embodiments, the control circuitry **1234** (and/or the welding software **244**) may calibrate the current and/or voltage of the welding power provided to a connected welding tool **14** based at least in part on the current and/or voltage sensed by the C/V sensing circuitry **1242**. For example, in certain embodiments, the control circuitry **1234** (and/or the welding software **244**) may command the current (or voltage) of the welding power provided to the connected welding tool **14** to be lowered (raised) if the current (or voltage) sensed by the C/V sensing circuitry **1242** is greater (less) than a desired current (or voltage) (e.g., a current (or voltage) set via the welding system **10**). In addition, in certain embodiments, the C/V sensing circuitry **1242** may detect an open circuit voltage (OCV), and the control circuitry **1234** (and/or the welding software **244**) may generate a prompt via the status indicator (s) **1241** of the connection box **1194**, via the status indicator (s) **1136** of the welding tools **14**, and/or via any other status indicator (or other output device) described herein, upon detection of the OCV. For example, in certain 24 embodiments, a user may be prompted to properly connect a welding tool **14**, and so forth. In addition, in certain embodiments, if a particular type of welding tool **14** (e.g., stick, MIG, TIG, and so forth) is used that is not compatible with a selected type of welding process (e.g., stick, MIG, TIG, and so forth) for the welding system **10** (i.e., a desired type of welding process set by a user via the welding system **10** or otherwise programmed into the welding system **10**), the control circuitry **1234** (and/or the welding software **244**) may generate a prompt via the status indicator(s) **1241** of the connection box **1194**, via the status indicator(s) **1136** of the welding tools **14**, and/or via any other status indicator (or other output device) described herein.

In certain embodiments, a state machine may be implemented (e.g., in the welding software **244**) to help control operation of the actual stick welding electrode holder **1074**. FIG. **85** depicts a summary of an exemplary state machine. In general, the items to the left of the black bar represent current operating conditions, and the items to the right of the black bar represent responses to the particular combinations of operating conditions. For example, the operating conditions that are taken into account are whether a screen cover is opened or closed (e.g., **1244**), whether the test is in a pre-test mode (e.g., if the welding process is not yet being performed) or a mid-test mode (e.g., if the welding process is currently being performed) (e.g., **1246**), whether the

position, orientation, and/or movement of the actual stick welding electrode holder **1074** is currently being tracked (e.g., **1248**), whether the actual stick welding electrode holder **1074** is proximate (near/far) the workpiece **82** (e.g., **1250**), and whether an arc has been detected (e.g., **1252**). Depending on these operating conditions, the contactor **1212** of the connection box **1194** may be opened or closed (e.g., **1254**), certain graphics may be displayed to the user (e.g., **1256**), the targeting graphics **1144** may be displayed and/or the status indicators **1136** may be activated, as described herein (e.g., **1258**), certain sounds effects may be generated (e.g., **1260**), a warning may be displayed via the primary display (e.g., **1262**), and the test may either be started (e.g., **1264**) or ended (e.g., **1266**).

It should be noted that while described herein as a connection box **1194** that includes a variety of components enclosed within a common housing **1243**, and that the connection box **1194** is separate from the welding system **10**, in certain embodiments, the components illustrated in FIG. **84** as being part of the connection box **1194** may be integrated into the welding system **10** illustrated in FIG. **1** and may be configured to communicate with the control circuitry (e.g., the welding software **244**, among other control circuitry) of the welding system **10** and, as such, these components may collectively form a welding training system interface for interfacing with the welding system **10**.

As used herein, the term "predetermined range" may mean any of the following: a group of numbers bounded by a predetermined upper limit and a predetermined lower limit, a group of number greater than a predetermined limit, and a group of numbers less than a predetermined limit. Moreover, the range may include numbers equal to the one or more predetermined limits.

While only certain features of the present disclosure have been illustrated and described herein, many modifications and changes will occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the disclosure.

The invention claimed is:

1. A method for providing a welding score, comprising: determining an initial position of a welding operation; determining a terminal position of the welding operation; determining a welding path between the initial position and the terminal position; and determining the welding score based on the welding path, wherein the welding score takes into account a time used for welding, and wherein the welding score is based on comparing a time period for welding the welding path to a welding time upper threshold or a welding time lower threshold.
2. The method of claim 1, wherein the initial position or the terminal position is sensed by one or more sensing devices configured to detect a position or an orientation of: a welding tool, a welding stand, or a welding surface.
3. The method of claim 2, wherein at least one of the sensing devices is configured to detect one or more markers.
4. The method of claim 2, wherein at least one of the sensing devices is a camera.
5. The method of claim 2, wherein at least one of the sensing devices is configured to receive electromagnetic or sound energy.
6. The method of claim 1, wherein the initial position or the terminal position is a calibration point.
7. The method of claim 1, wherein the welding score is further based on: comparing a distance of the welding path to an upper threshold or a lower threshold.

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8. A welding system, comprising:
 a welding torch; and
 a computer configured to:
 determine an initial position of a welding operation;
 determine a terminal position of the welding operation;
 determine a welding path between the initial position
 and the terminal position; and
 determine a welding score based on the welding path,
 wherein the welding score takes into account a time
 used for welding or a length of the welding path, and
 determining the welding score comprises disregarding
 a first portion of the welding path starting at the
 initial position or a second portion of the welding
 path starting at the terminal position.
9. The welding system of claim 8, further comprising a
 sensing device configured to detect a position or an orientation
 of a welding tool, a welding stand, or a welding
 surface.
10. The welding system of claim 9, wherein the sensing
 device is configured to detect one or more markers.
11. The welding system of claim 9, wherein the sensing
 device is a camera.
12. The welding system of claim 9, wherein the sensing
 device is configured to receive electromagnetic or sound
 energy.
13. The welding system of claim 8, wherein the computer
 is further configured to determine whether the welding score
 is a passing welding score by comparing the welding score
 to a score threshold.
14. The welding system of claim 8, wherein the computer
 is configured to determine the welding score based on:
 comparing a distance of the welding path to a welding
 path upper threshold or a welding path lower threshold;
 or

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- comparing a time period for welding the welding path to
 one or both of a welding time upper threshold or a
 welding time lower threshold.
15. A method for providing a welding score, comprising:
 determining an initial position of a welding operation;
 determining a terminal position of the welding operation;
 determining a welding path between the initial position
 and the terminal position; and
 determining the welding score based on the welding path,
 wherein the welding score takes into account a time used
 for welding or a length of the welding path, and
 determining the welding score comprises disregarding
 a first portion of the welding path welded during a first
 time period and a second portion of the welding path
 welded during a second time period.
16. The method of claim 15, wherein the initial position
 or the terminal position is sensed by one or more sensing
 devices configured to detect a position or an orientation of
 a welding tool, a welding stand, or a welding surface.
17. The method of claim 16, wherein at least one of the
 sensing devices is configured to detect one or more markers.
18. The method of claim 16, wherein at least one of the
 sensing devices is a camera.
19. The method of claim 16, wherein at least one of the
 sensing devices is configured to receive electromagnetic or
 sound energy.
20. The method of claim 18, wherein the welding score is
 determined by:
 comparing a distance of the welding path to a welding
 path upper threshold or a welding path lower threshold,
 or
 comparing a time period for welding the welding path to
 a welding time upper threshold or a welding time lower
 threshold.

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